



WSQ COURSE SLIDES

Security Operations for Autonomous AI Agents

WSQ Course Code: TGS-2024042604

4 days · 32 hours · Intermediate

Version v1.0 · 12 September 2026

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W



COURSE ADMINISTRATION

Welcome and course orientation

Attendance · outcomes · assessment · learner systems



Digital attendance (mandatory)

1

AM

Scan the trainer-displayed SSG attendance QR code.

2

PM

Complete the second attendance event after lunch.

3

Assessment

Record attendance for the assessment session.

4

Minimum

At least 75% attendance is required.

5

Evidence

Confirm the successful attendance screen.

6

Escalation

Tell the trainer immediately if the scan fails.

About the trainer — template

1 Name

2 Designation

3 Qualifications

4 Industry experience

5 Security expertise

6 Contact

About the trainer — Dr Alfred Ang

1 Role

Principal Trainer and courseware developer at Tertiary Infotech Academy.

2 Focus

Applied AI, cybersecurity, software engineering and hands-on technical learning.

3 Teaching approach

Mechanism-led explanation, reproducible activities and evidence-based verification.

4 Learner support

Clarification during class and post-course support through the academy.

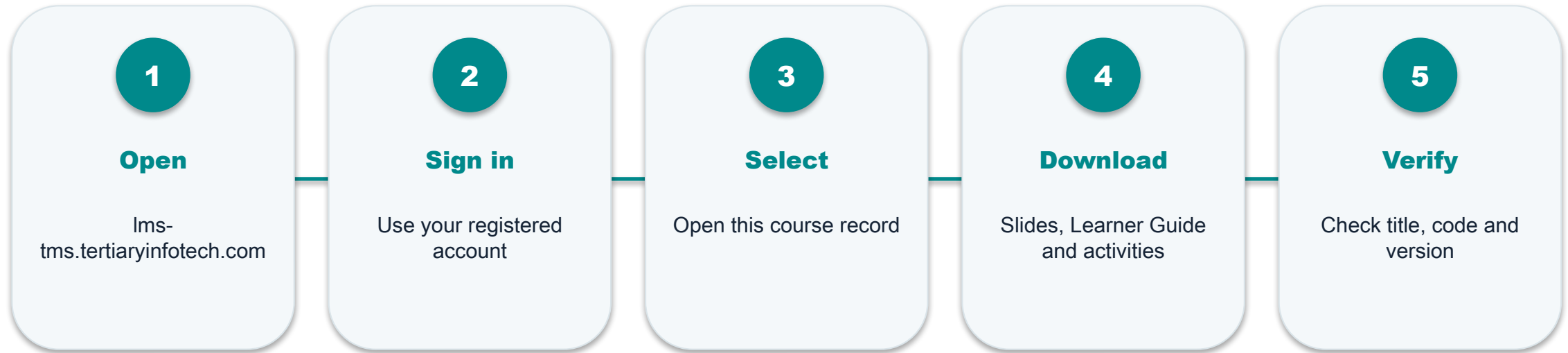
5 Course role

Connect security strategy and governance to practical autonomous-agent operations.

6 Contact

enquiry@tertiaryinfotech.com

Download your course material



Use only files labelled Security Operations for Autonomous AI Agents · TGS-2024042604.

Skills Framework alignment

1

K1

Goal setting and objectives of organisation security

2

K2

Information security and assurance strategy

3

K3

Best practices in information security policies

4

K4

Best practices and emerging technologies in security control

5

K5

Knowledge of security management benchmarks

6

K6

Gap analysis in organisation security

Skills Framework alignment — implementation

1

K7

Implications and impact of security gaps

2

A1

Formulate security goals and objectives from business priorities, security vision, strategy directions and management benchmarks

3

A2

Establish standards and practices that protect information integrity, authenticity and confidentiality

4

A3

Manage compliance with information-security guidelines and classification or permission rules

5

A4

Lead communication of security goals and objectives to the organisation

6

A5

Review existing security controls against current and future business costs and risks

Skills Framework alignment — strategy and operations

1

A6

Develop strategies and plans to resolve security gaps

2

A7

Drive organisation-wide security initiatives in line with internal and external standards

3

Assessment

WA: K1–K7 · PP: A1–A7

4

Course evidence

Ten activities produce reproducible control artifacts.

5

Alignment gate

Nothing assessed is absent from the slides and activities.

6

Terminology

Every hands-on lab is presented as an Activity.

Learning outcomes

1 LO1

Formulate comprehensive security goals and establish business standards with an overarching security vision for autonomous AI agents.

2 LO2

Communicate effective agent-security policies and practices, and manage compliance with best practices and technological advancements.

3 LO3

Evaluate existing agent-security controls against business risks and costs, and develop strategies to resolve identified security gaps.

4 LO4

Implement organisation-wide agent-security initiatives, assessing and addressing the impact of security gaps.

Course outline

1

Topic 01

Security Strategy and Standards for Autonomous AI Agents
— Agent inventory · security objectives · identity · tools · data · resilience

2

Topic 02

Agent Security Policies, Governance and Compliance —
Policy-as-code · evidence · permissions · assurance · communication

3

Topic 03

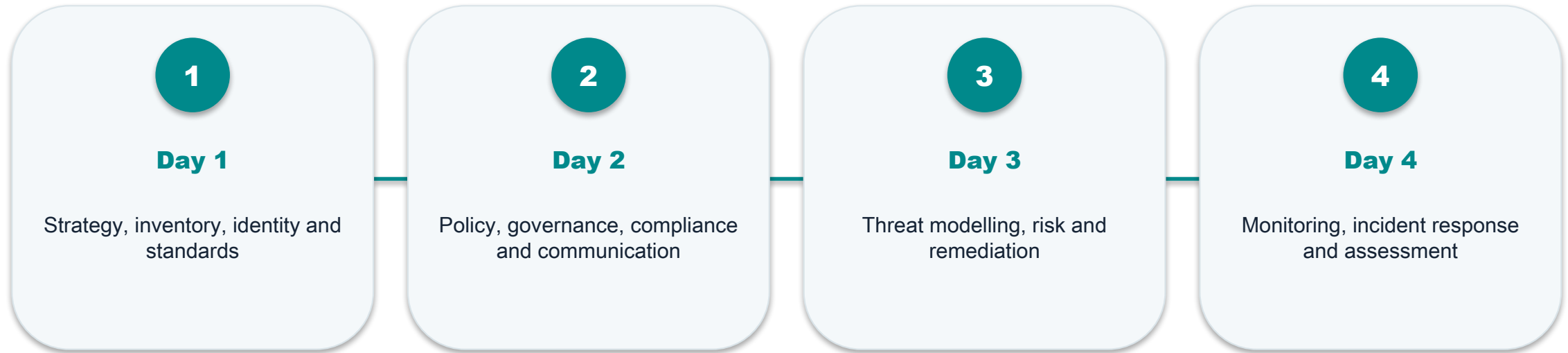
AI Agent Risk Assessment and Security Control Evaluation
— Threat paths · risk quantification · control effectiveness · remediation

4

Topic 04

Implementing and Monitoring Organisation-Wide Agent Security Operations — Telemetry · detection · incident response · containment · recovery

Four-day learning journey



Each day contains 8 scheduled hours excluding lunch; detailed timing is in the Lesson Plan.

Briefing for assessment

1

Individual

Complete each instrument independently.

2

Open book

Use the course slides, Learner Guide and approved materials.

3

No recording

Do not photograph or copy the assessment scripts.

4

Submission

Upload the completed documents to the LMS/TMS.

5

Decision

Results are Competent or Not Yet Competent.

6

Clarification

Oral clarification may close minor gaps where authorised.

Assessment instruments

1

WA (SAQ)

7 open-ended questions · K1–K7 · 1 hour.

2

PP

6 practical tasks · A1–A7 · 3 hours as stated by the verified legacy paper.

3

Question papers

Learner-facing candidate documents only.

4

Answer keys

Trainer-only; never attached to GitHub or learner-facing LMS fields.

5

Alignment

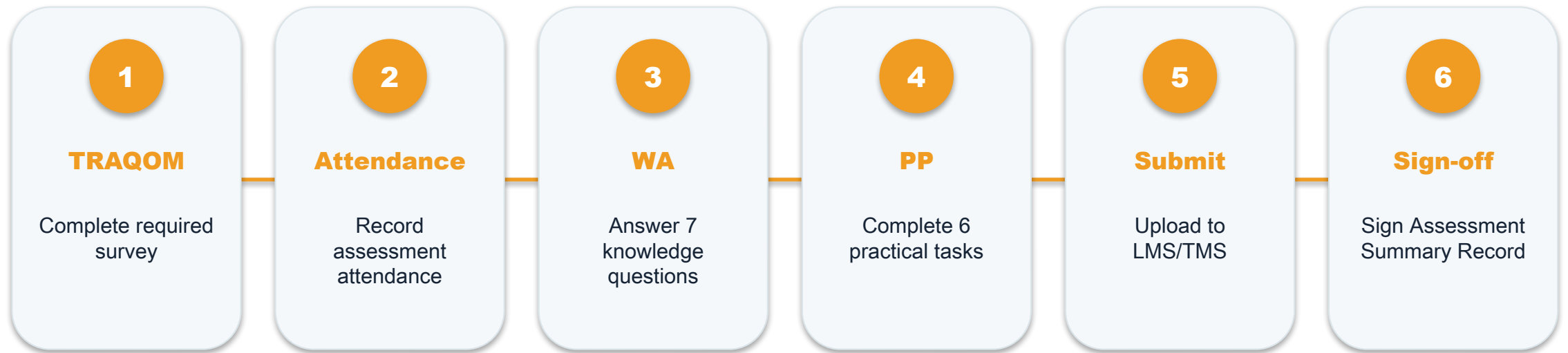
Every WA answer traces to slides; every PP task traces to an activity.

6

Evidence

Scripts, mock data and generated artifacts support the PP tasks.

Assessment flow



Question papers only are learner-facing; answer keys remain trainer-only.

Assessment and funding criteria

1 Attendance

At least 75% recorded attendance.

2 Assessment

Complete both WA and PP.

3 Competence

Meet every required assessment criterion.

4 Identity

Use the registered learner identity.

5 Evidence

Retain successful digital-attendance confirmations.

6 Appeal

Follow the documented appeal process if needed.

Activity environment

1 Python

Python 3.10 or later; standard library only.

2 Editor

VS Code or any text editor.

3 Data

Synthetic JSON, JSONL and CSV files.

4 Execution

Run activity.py inside each activity folder.

5 Evidence

Generated Markdown, CSV, JSON or HTML outputs.

6 Safety

No live credentials, agents, production systems or external APIs.

Practice exam portal

1 Portal

<https://exams.tertiaryinfotech.com>

2 Practice mode

Use untimed practice to identify gaps.

3 Exam mode

Use timed mode after completing the activities.

4 Review

Return to the exact slide/activity mapped to each missed criterion.

5 Scope

K1–K7 knowledge and A1–A7 practical evidence.

6 Reminder

The portal supports preparation; the official assessment remains WA + PP.

TOPIC 01

Security Strategy and Standards for Autonomous AI Agents

Agent inventory · security objectives · identity · tools · data · resilience

LO1 · K1, K2, A1, A2



L01 and assessment criteria

1

K1

Goal setting and objectives of organisation security

2

K2

Information security and assurance strategy

3

A1

Formulate security goals and objectives from business priorities, security vision, strategy directions and management benchmarks

4

A2

Establish standards and practices that protect information integrity, authenticity and confidentiality

5

L01

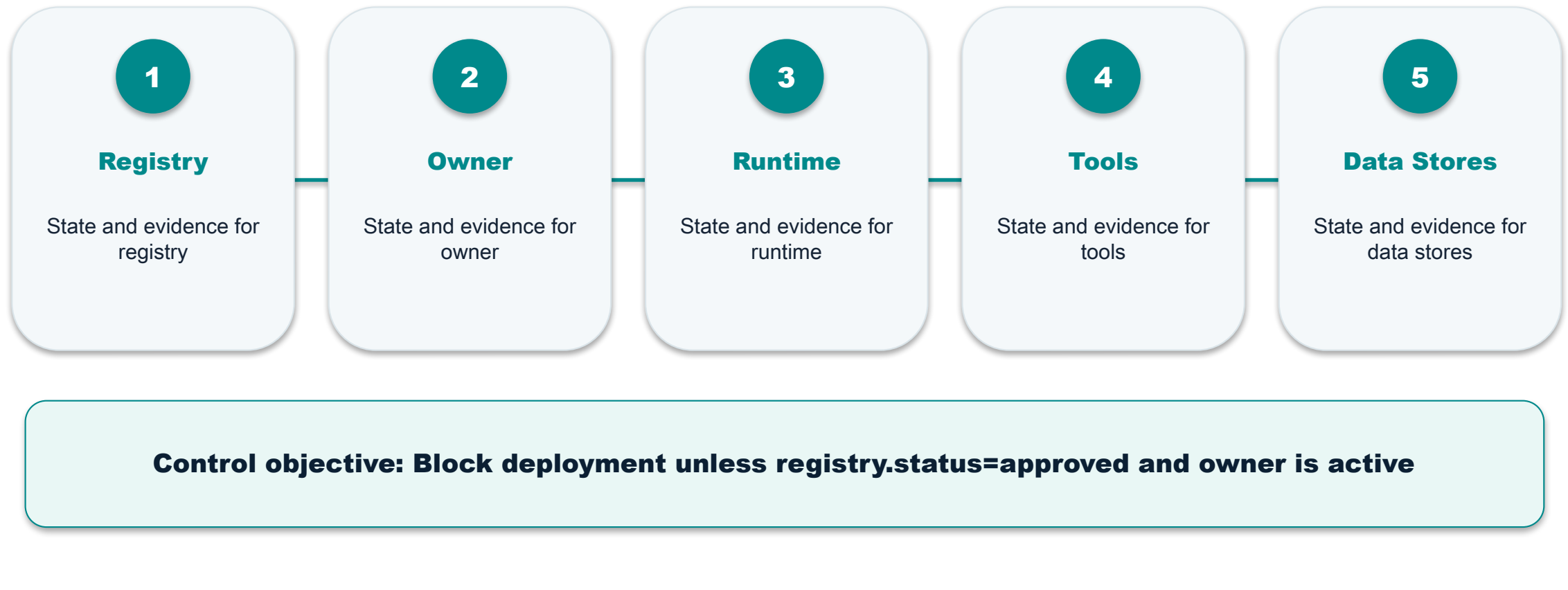
L01: Formulate comprehensive security goals and establish business standards with an overarching security vision for autonomous AI agents.

Security Strategy and Standards for Autonomous AI Agents: operating model

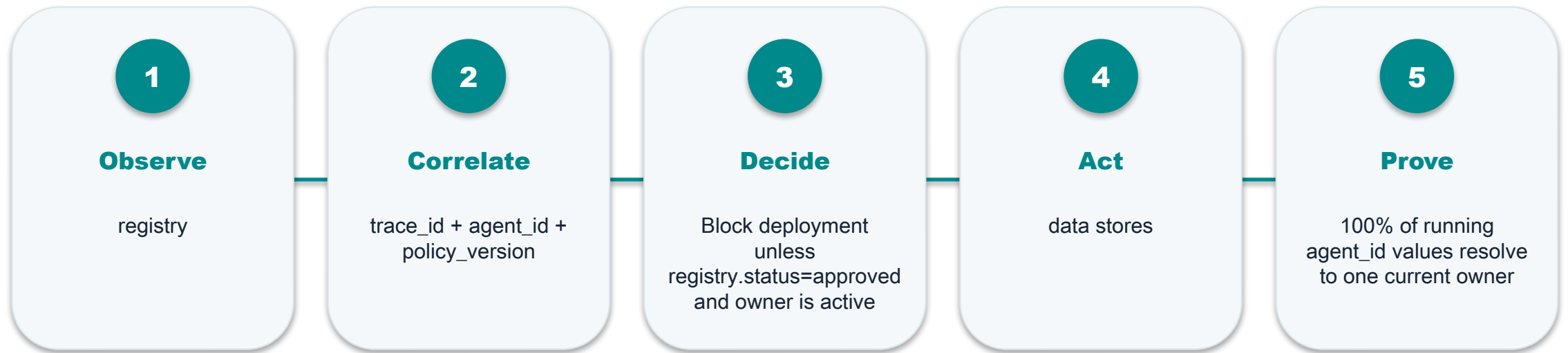


Every mechanism in this topic is taught through all five layers.

Agent inventory: architecture and control boundary



Agent inventory: runtime decision trace



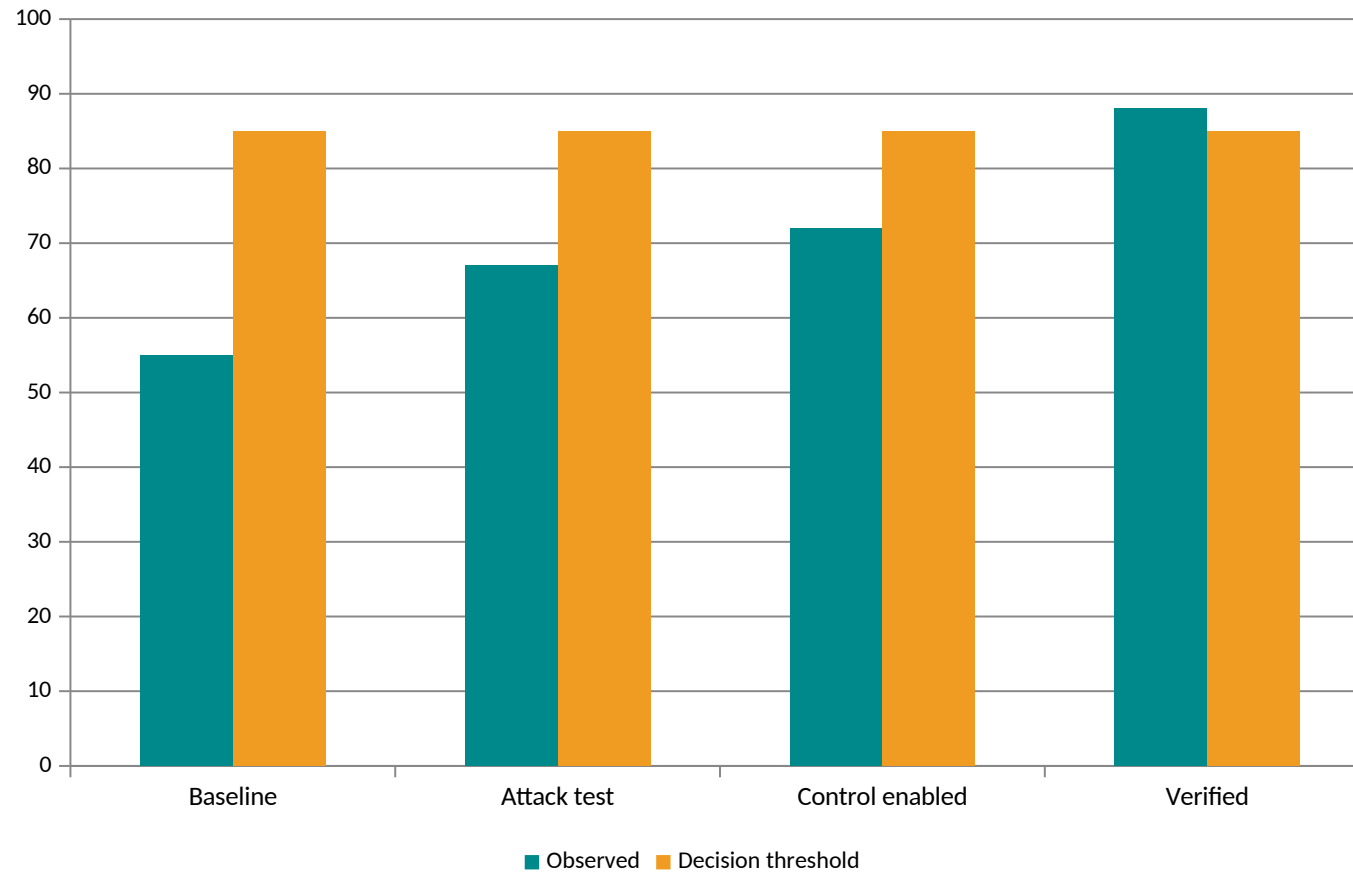
Failure to detect: Unregistered shadow agent reaches a production connector

Agent inventory: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agent_id	Required in decision and audit evidence
field_2	owner	Required in decision and audit evidence
field_3	purpose	Required in decision and audit evidence
field_4	model	Required in decision and audit evidence
field_5	tools[]	Required in decision and audit evidence
field_6	data_classes[]	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Agent inventory: evidence and acceptance decision



DECISION RULE

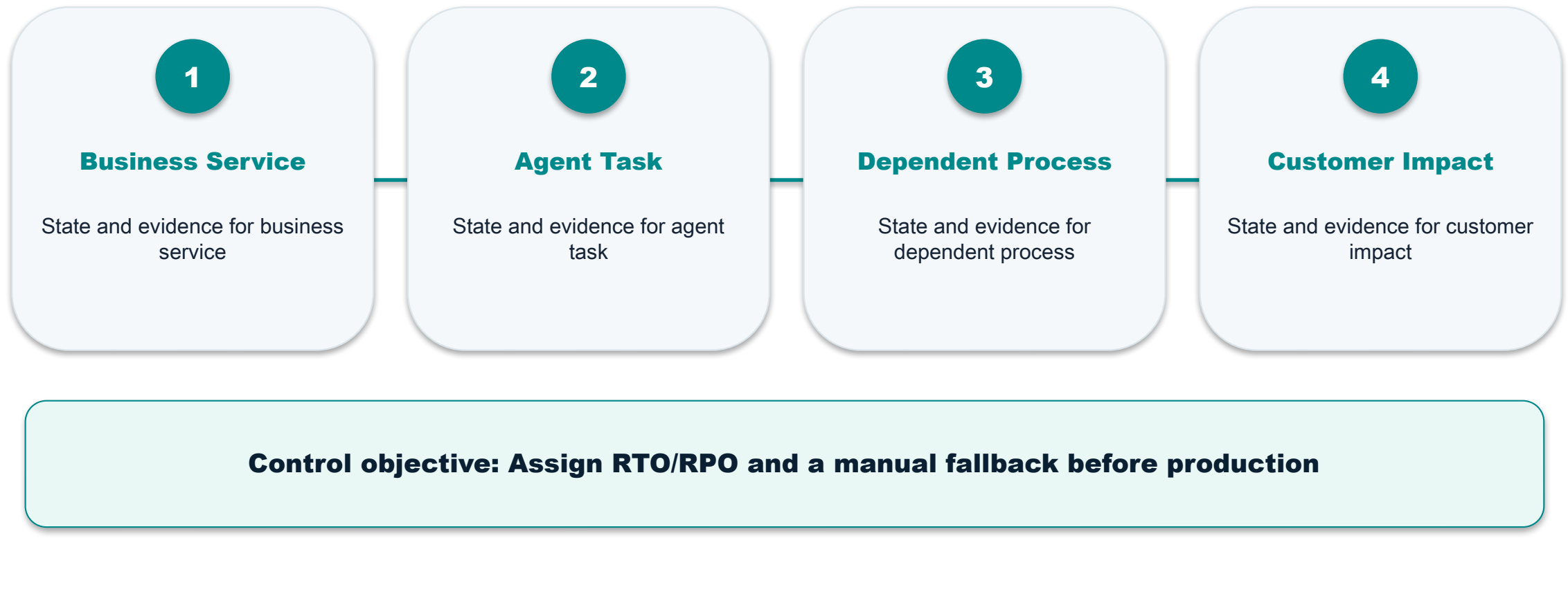
Accept only when 100% of running agent_id values resolve to one current owner. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

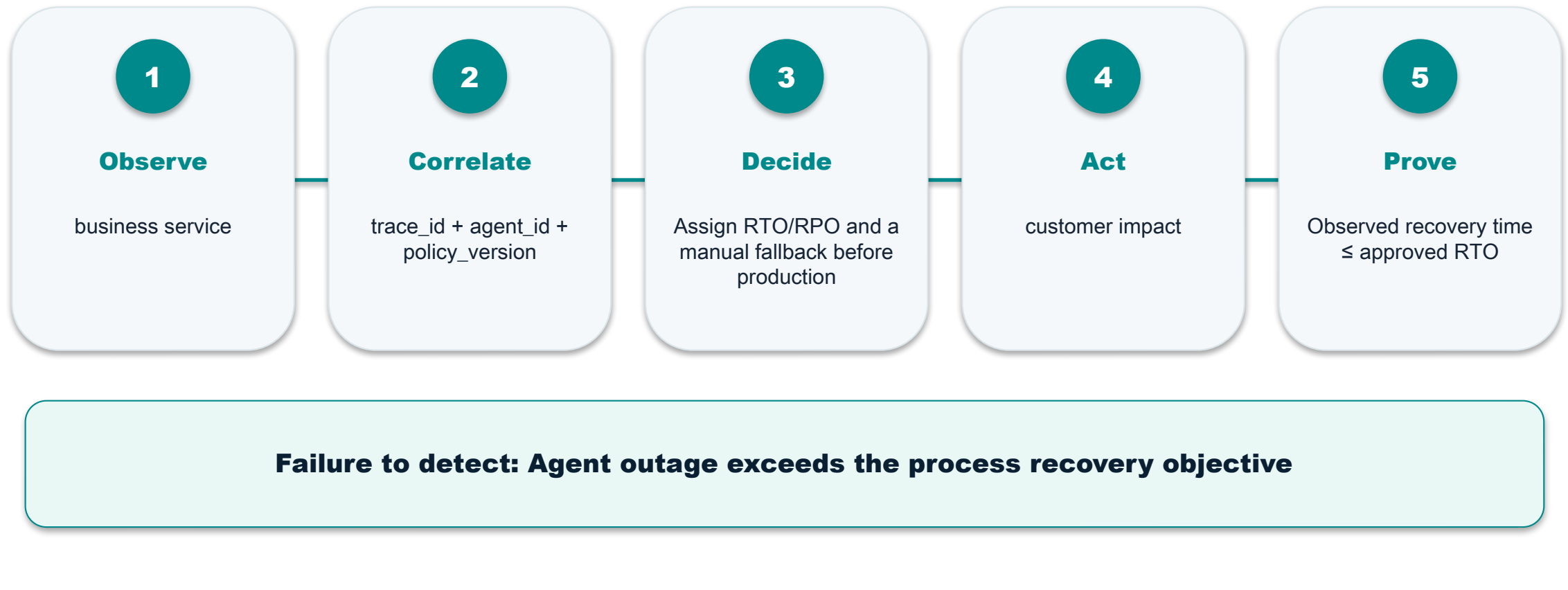
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Business impact mapping: architecture and control boundary



Business impact mapping: runtime decision trace

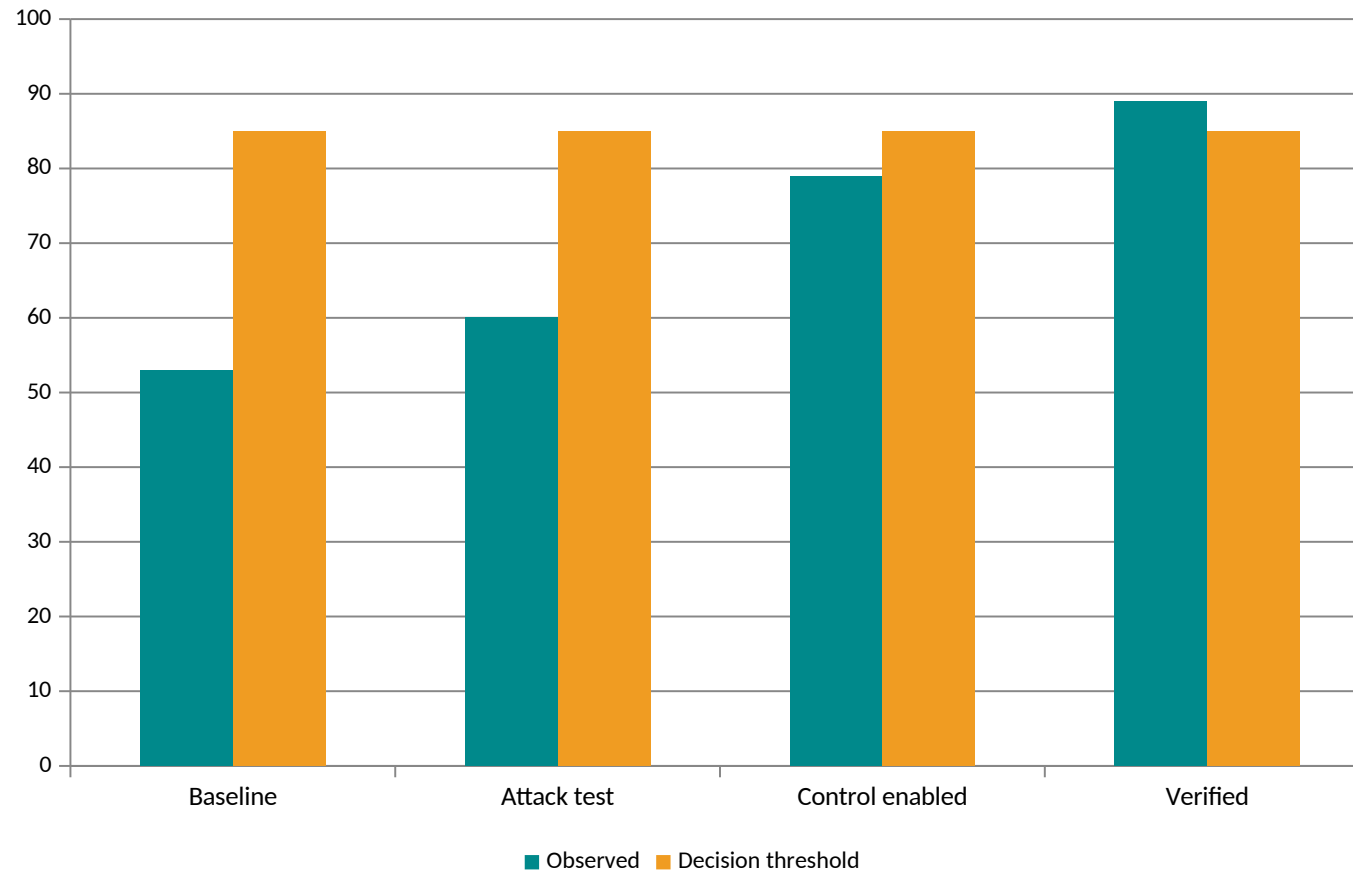


Business impact mapping: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	service_id	Required in decision and audit evidence
field_2	process_rto	Required in decision and audit evidence
field_3	data_rpo	Required in decision and audit evidence
field_4	maximum_loss_sgd	Required in decision and audit evidence
field_5	criticality	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Business impact mapping: evidence and acceptance decision



DECISION RULE

**Accept only when
Observed recovery time
 $\leq$ approved RTO.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

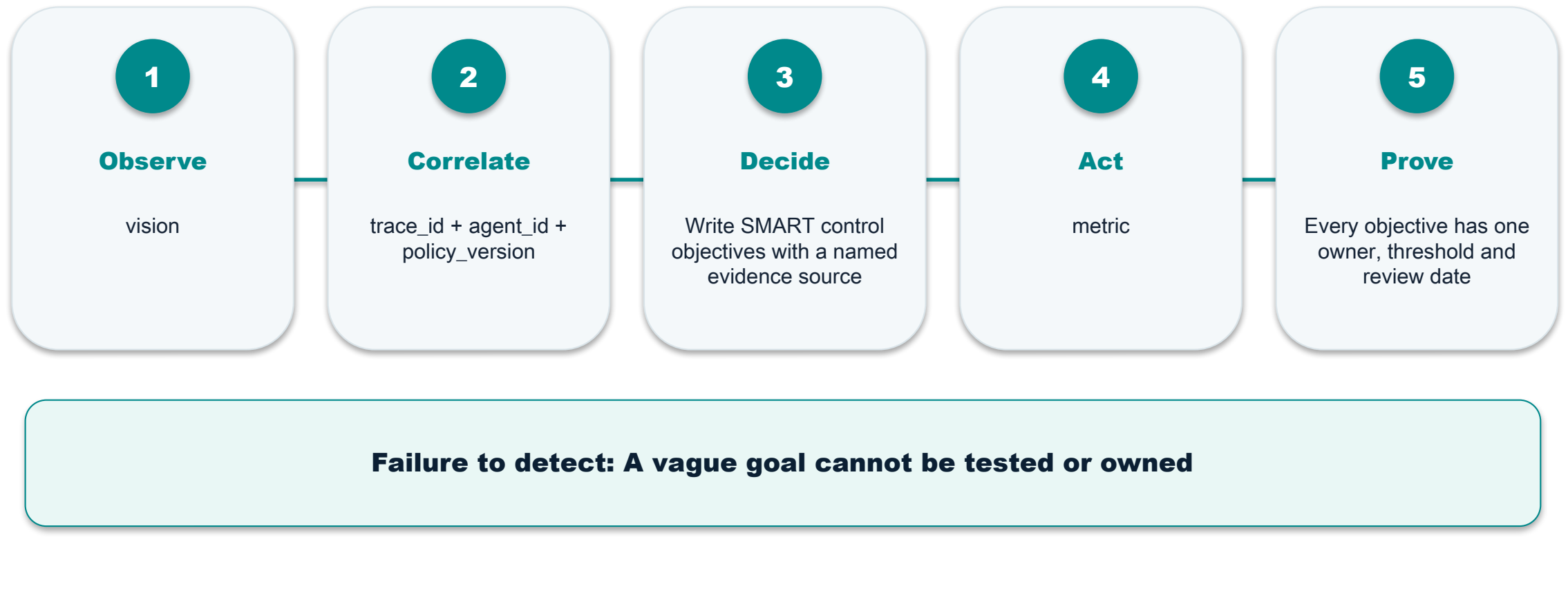
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Security objective hierarchy: architecture and control boundary



Security objective hierarchy: runtime decision trace

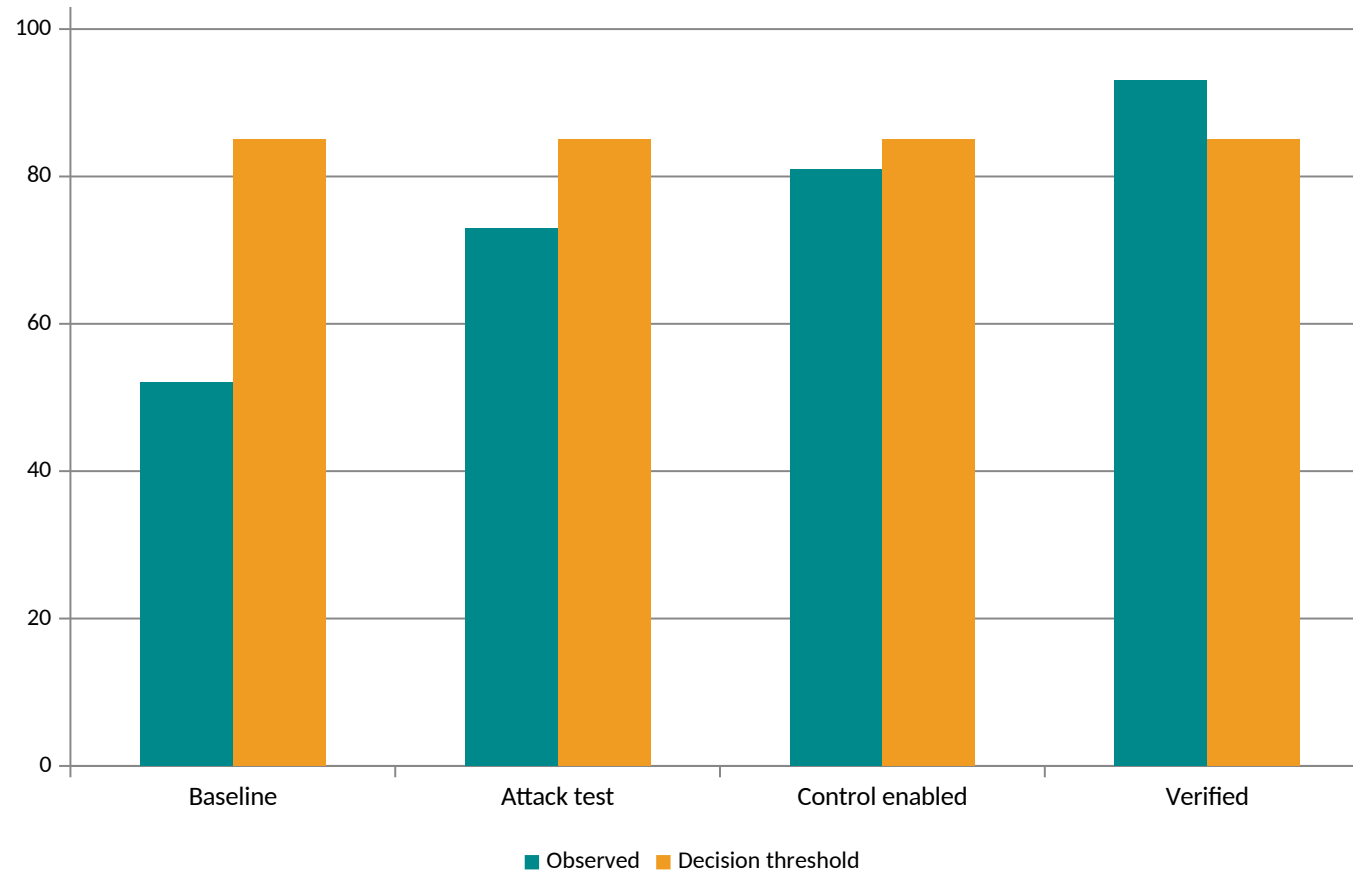


Security objective hierarchy: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	objective_id	Required in decision and audit evidence
field_2	sponsor	Required in decision and audit evidence
field_3	target	Required in decision and audit evidence
field_4	threshold	Required in decision and audit evidence
field_5	due_date	Required in decision and audit evidence
field_6	evidence_source	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Security objective hierarchy: evidence and acceptance decision



DECISION RULE

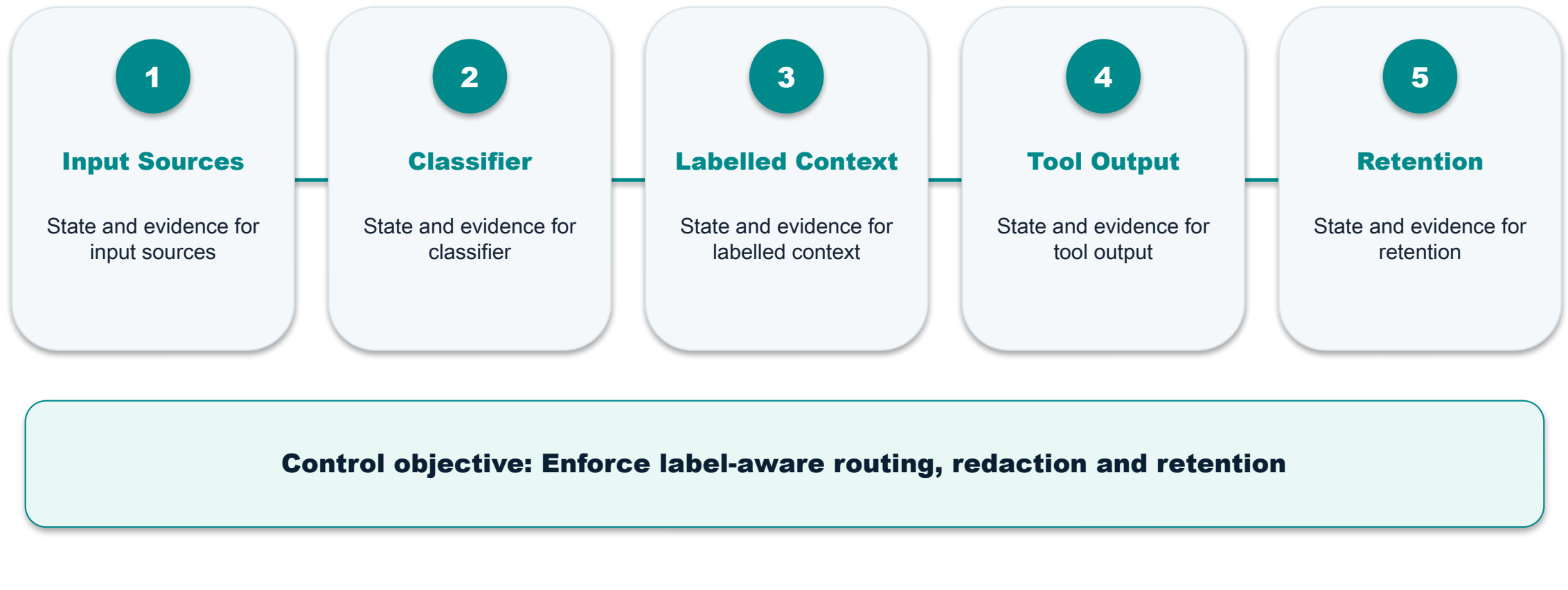
Accept only when Every objective has one owner, threshold and review date. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

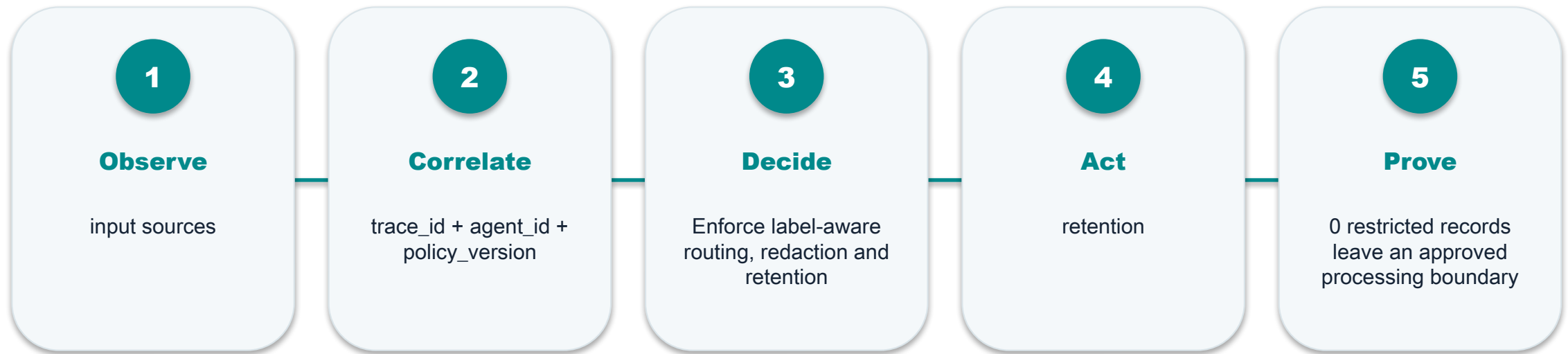
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Data classification: architecture and control boundary



Data classification: runtime decision trace



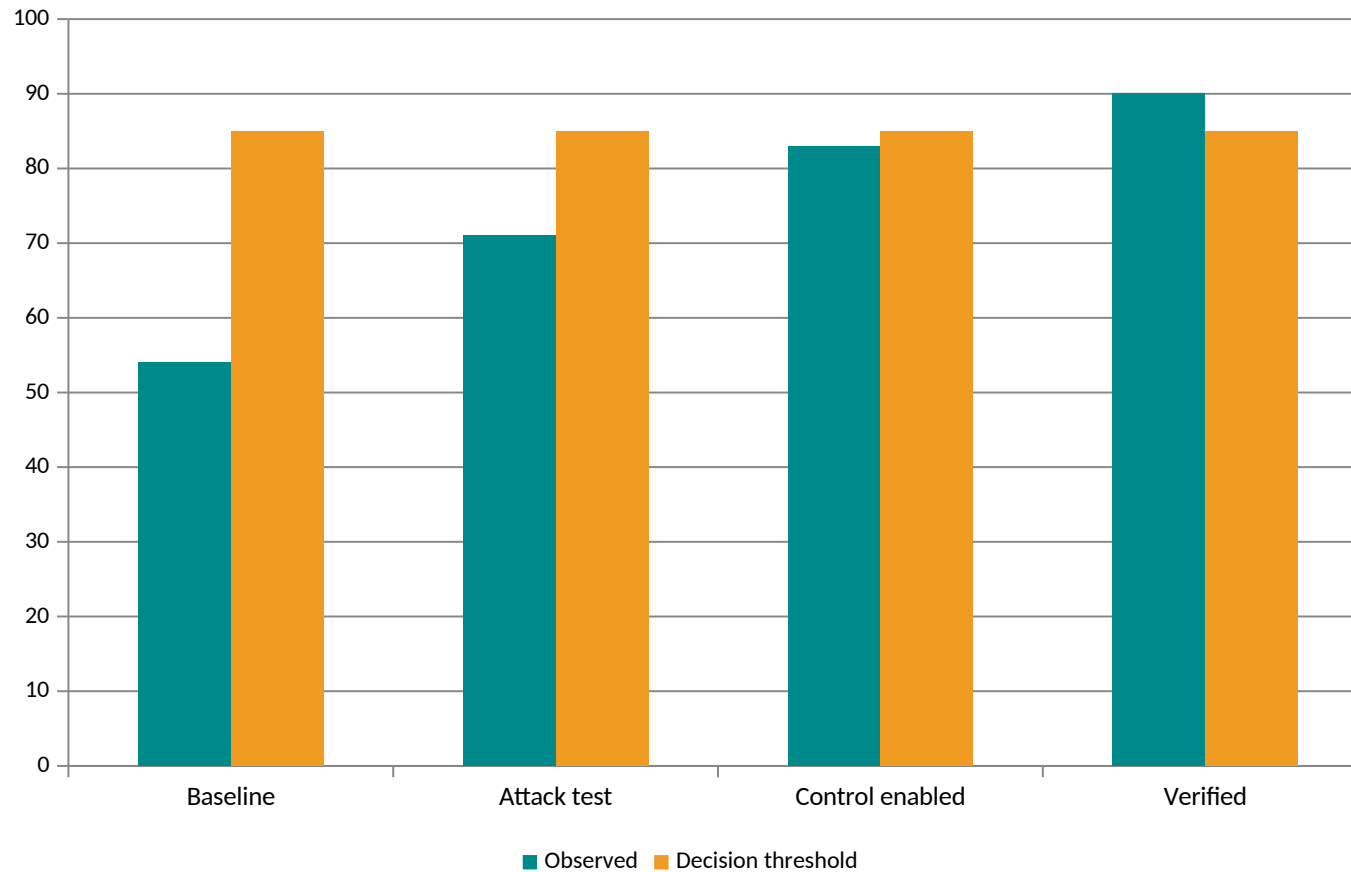
Failure to detect: Restricted data is inserted into an unapproved prompt

Data classification: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	data_class ∈ {public	Required in decision and audit evidence
field_2	internal	Required in decision and audit evidence
field_3	confidential	Required in decision and audit evidence
field_4	restricted}	Required in decision and audit evidence
field_5	purpose	Required in decision and audit evidence
field_6	retention_days	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Data classification: evidence and acceptance decision



DECISION RULE

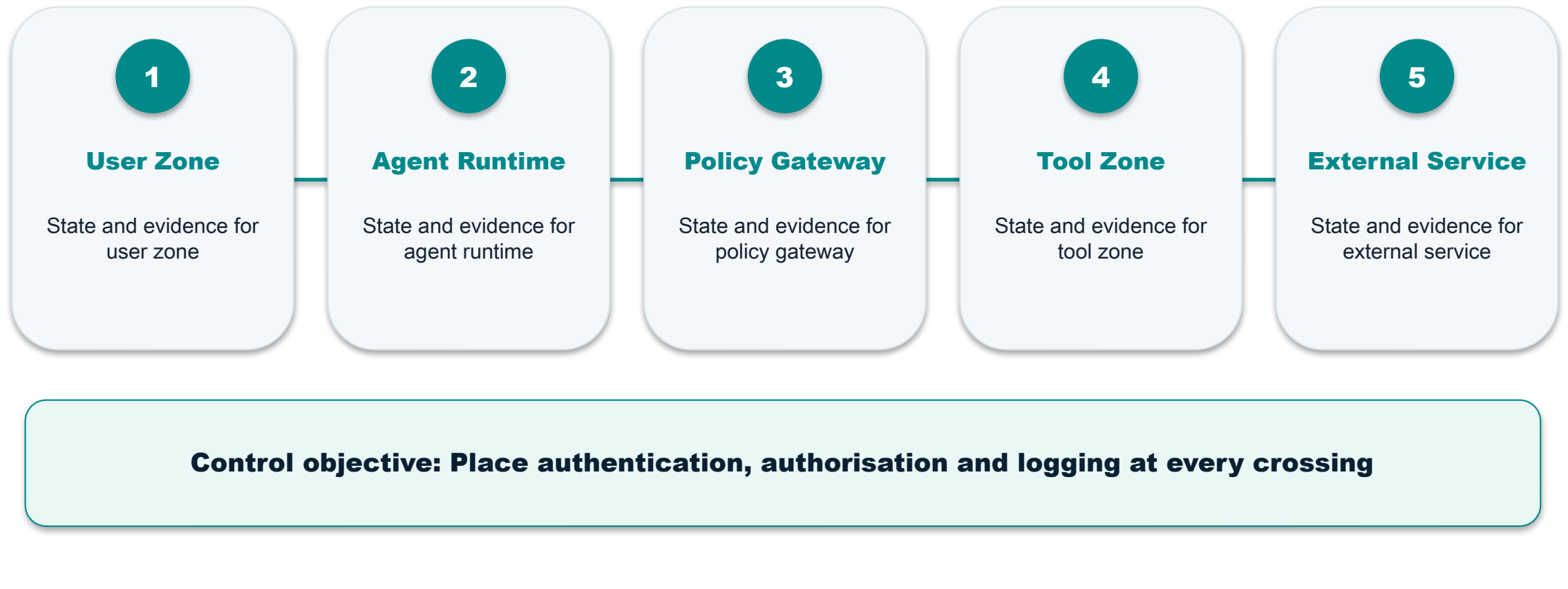
Accept only when 0 restricted records leave an approved processing boundary. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

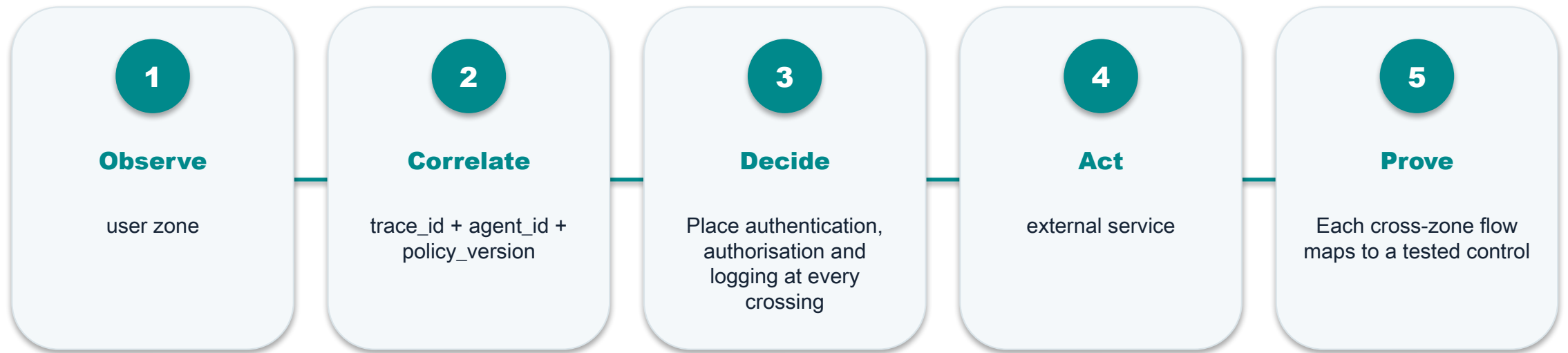
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Trust-boundary map: architecture and control boundary



Trust-boundary map: runtime decision trace



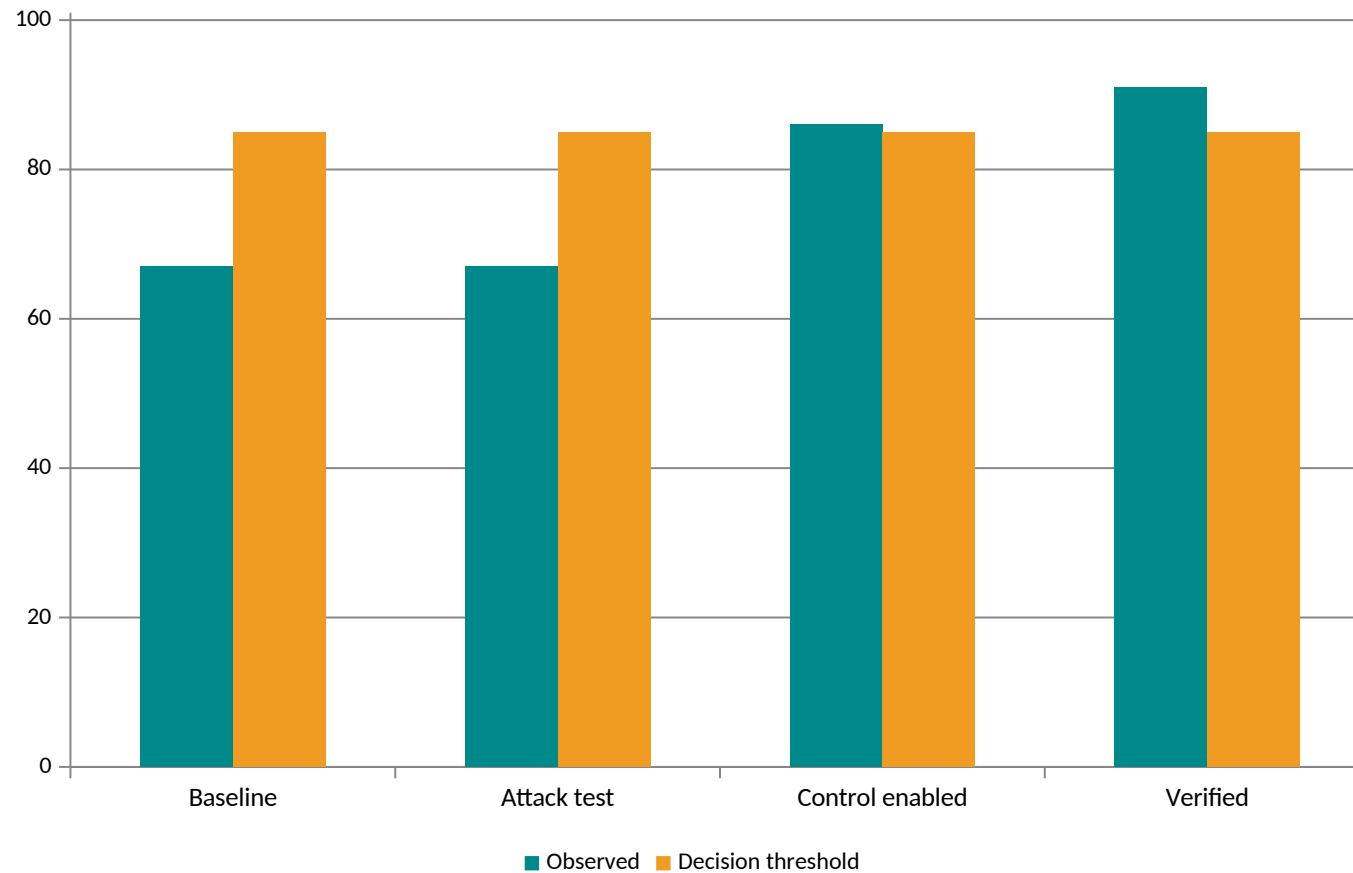
Failure to detect: A tool call crosses a boundary without an enforcement point

Trust-boundary map: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	boundary_id	Required in decision and audit evidence
field_2	source_zone	Required in decision and audit evidence
field_3	target_zone	Required in decision and audit evidence
field_4	protocol	Required in decision and audit evidence
field_5	authn	Required in decision and audit evidence
field_6	authz	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Trust-boundary map: evidence and acceptance decision



DECISION RULE

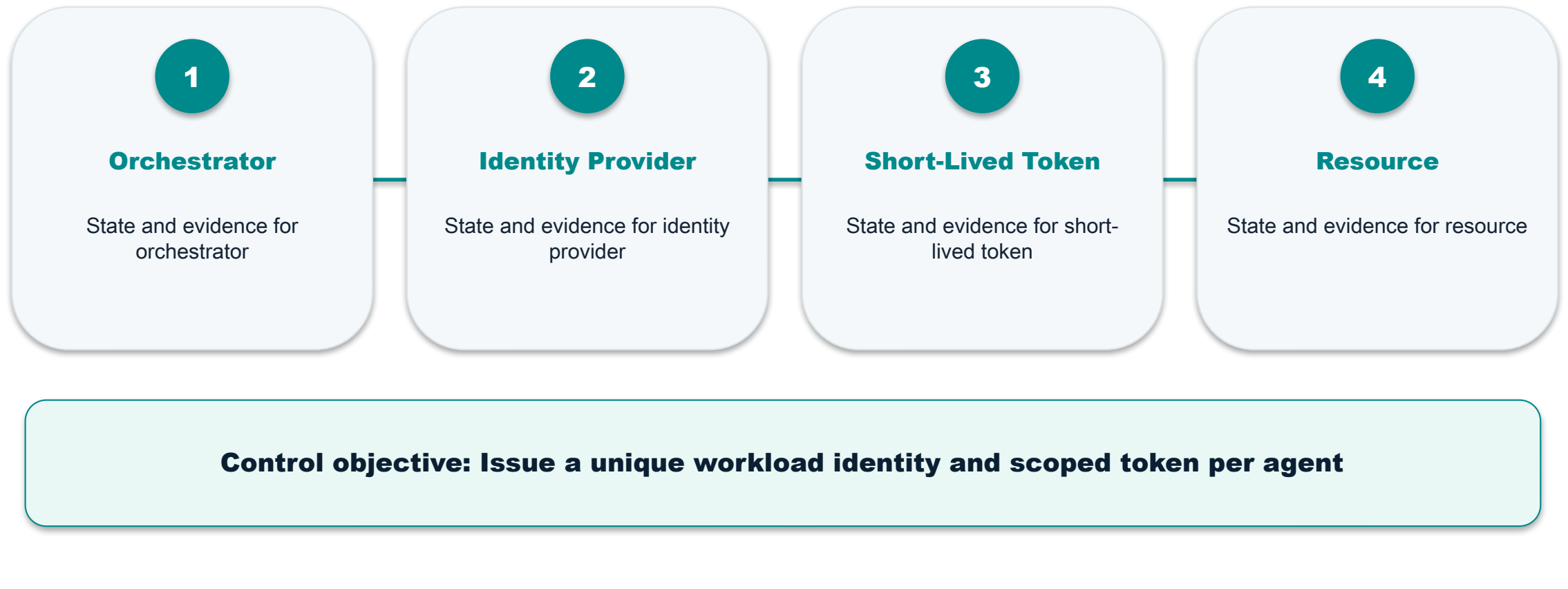
Accept only when Each cross-zone flow maps to a tested control. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

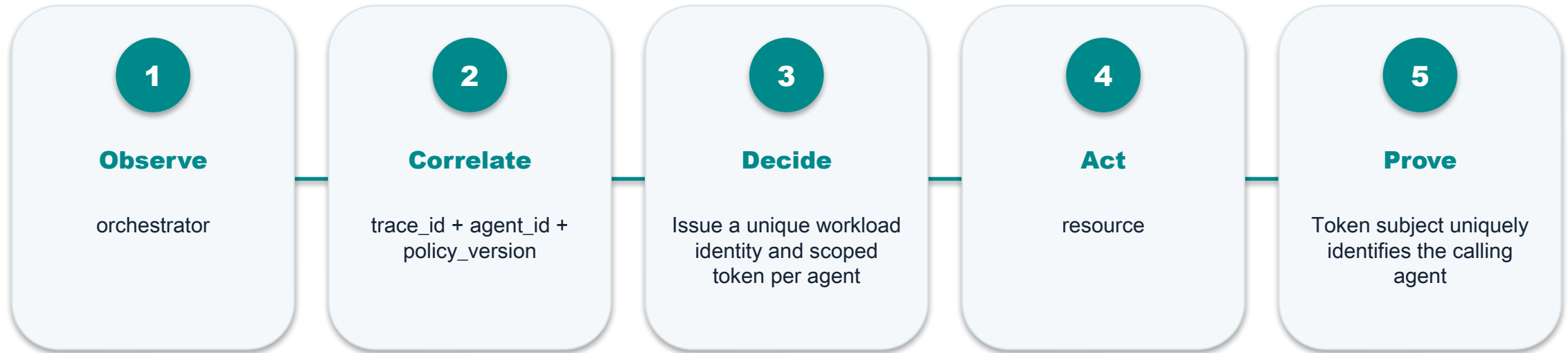
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Workload identity: architecture and control boundary



Workload identity: runtime decision trace



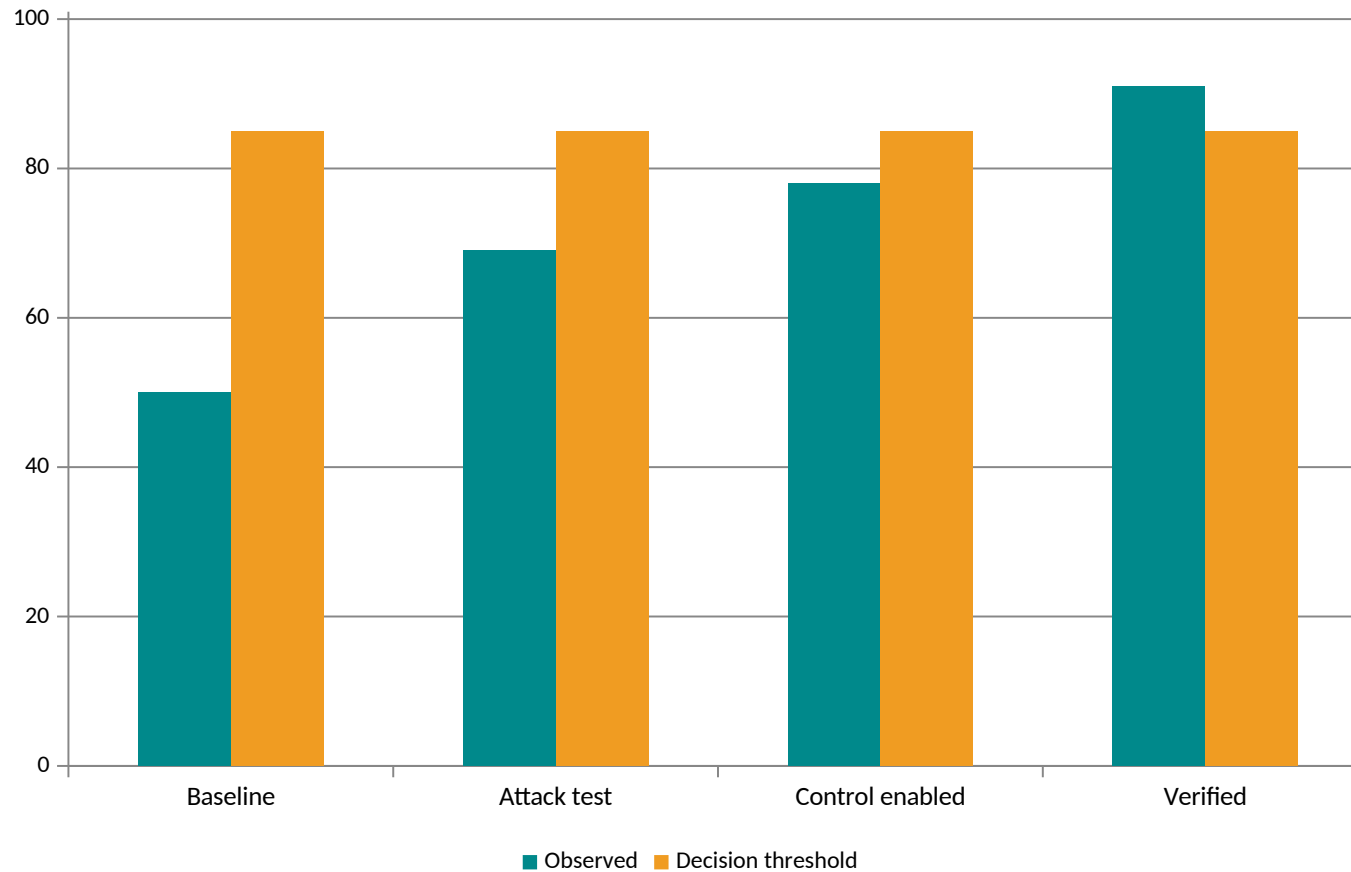
Failure to detect: Several agents share one service-account credential

Workload identity: concrete control contract

Field / key	Expected value or type	Enforcement
subject	agent_id	Required in decision and audit evidence
audience	tool_id	Required in decision and audit evidence
field_3	ttl_seconds≤900	Required in decision and audit evidence
field_4	scopes[]	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Workload identity: evidence and acceptance decision



DECISION RULE

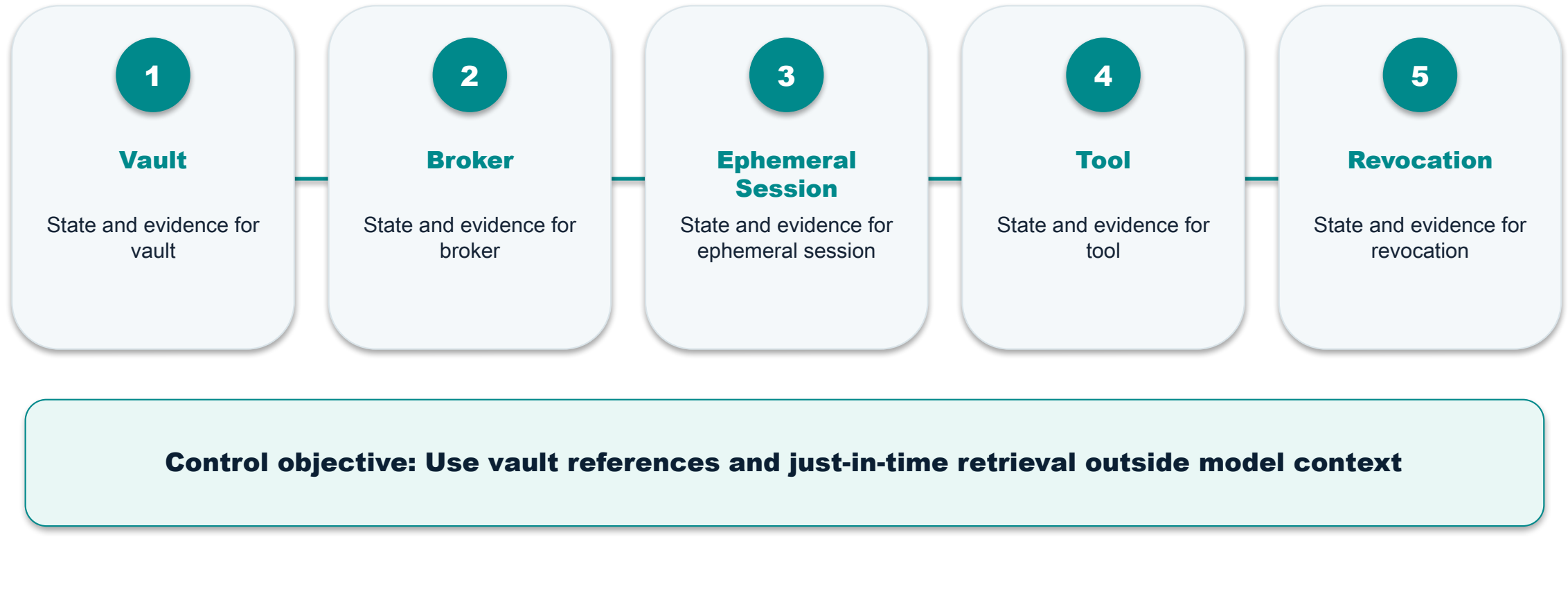
Accept only when Token subject uniquely identifies the calling agent. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

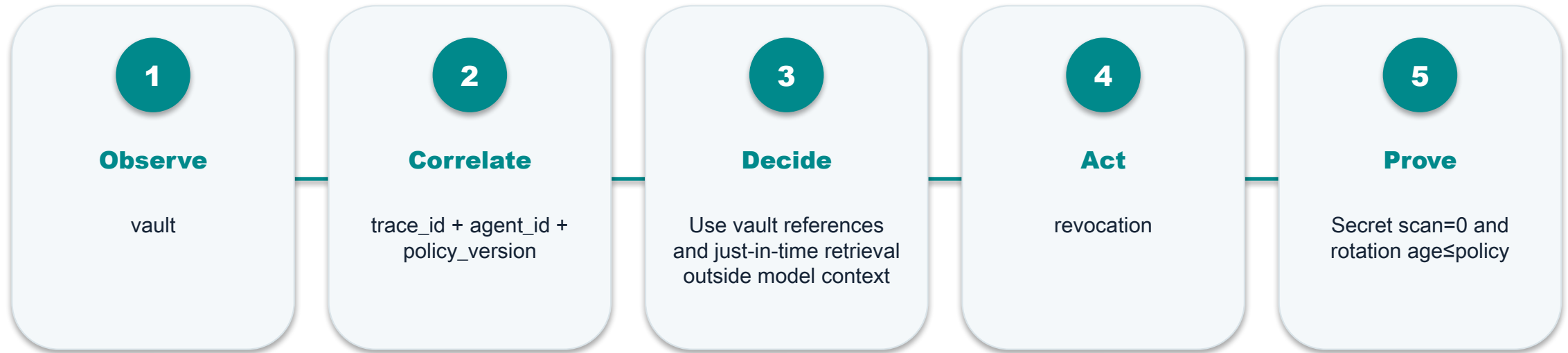
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Credential lifecycle: architecture and control boundary



Credential lifecycle: runtime decision trace



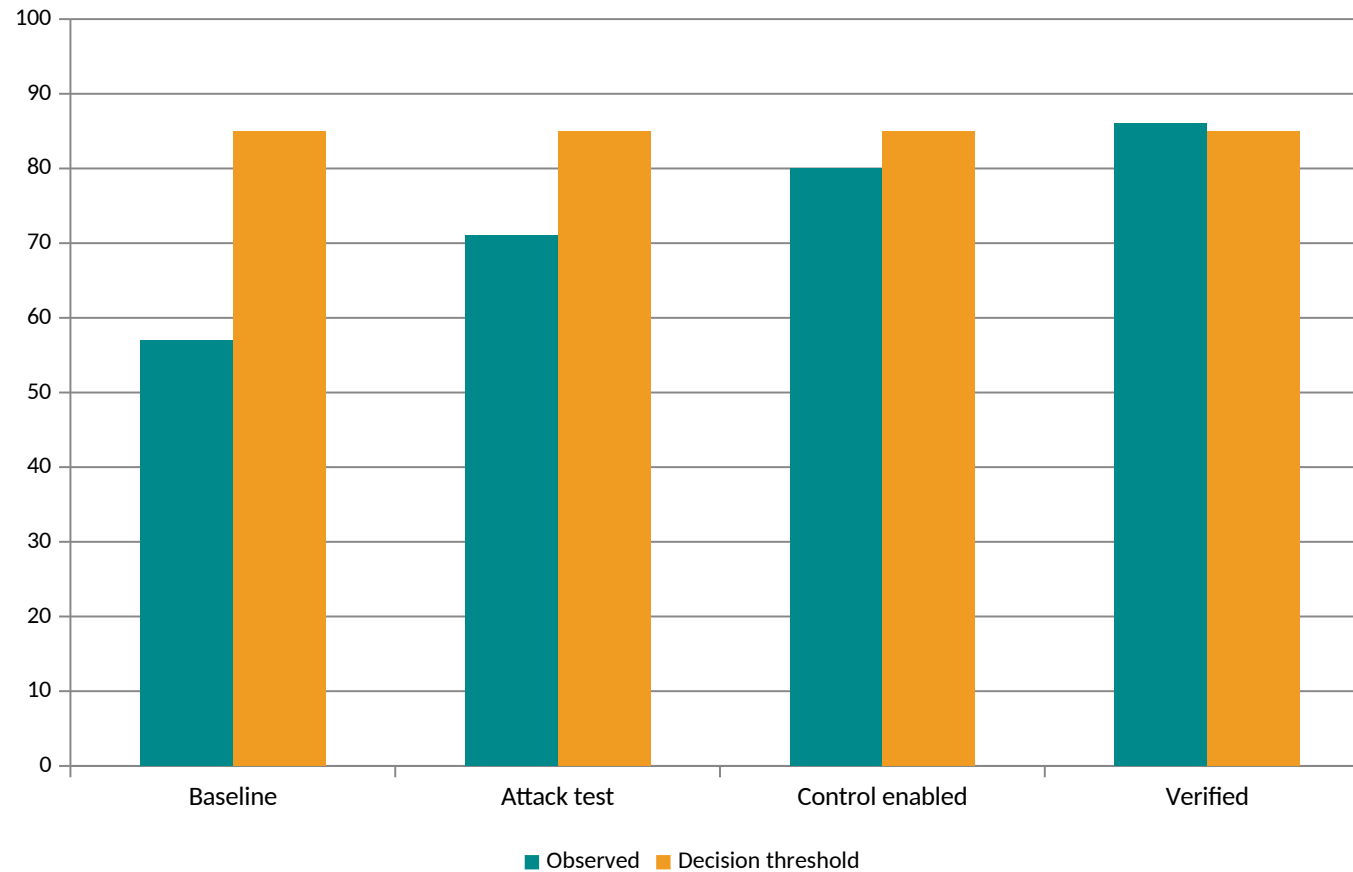
Failure to detect: A credential appears in prompt context or source code

Credential lifecycle: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	secret_ref	Required in decision and audit evidence
field_2	version	Required in decision and audit evidence
field_3	issued_at	Required in decision and audit evidence
field_4	expires_at	Required in decision and audit evidence
field_5	rotation_days	Required in decision and audit evidence
field_6	last_access	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Credential lifecycle: evidence and acceptance decision



DECISION RULE

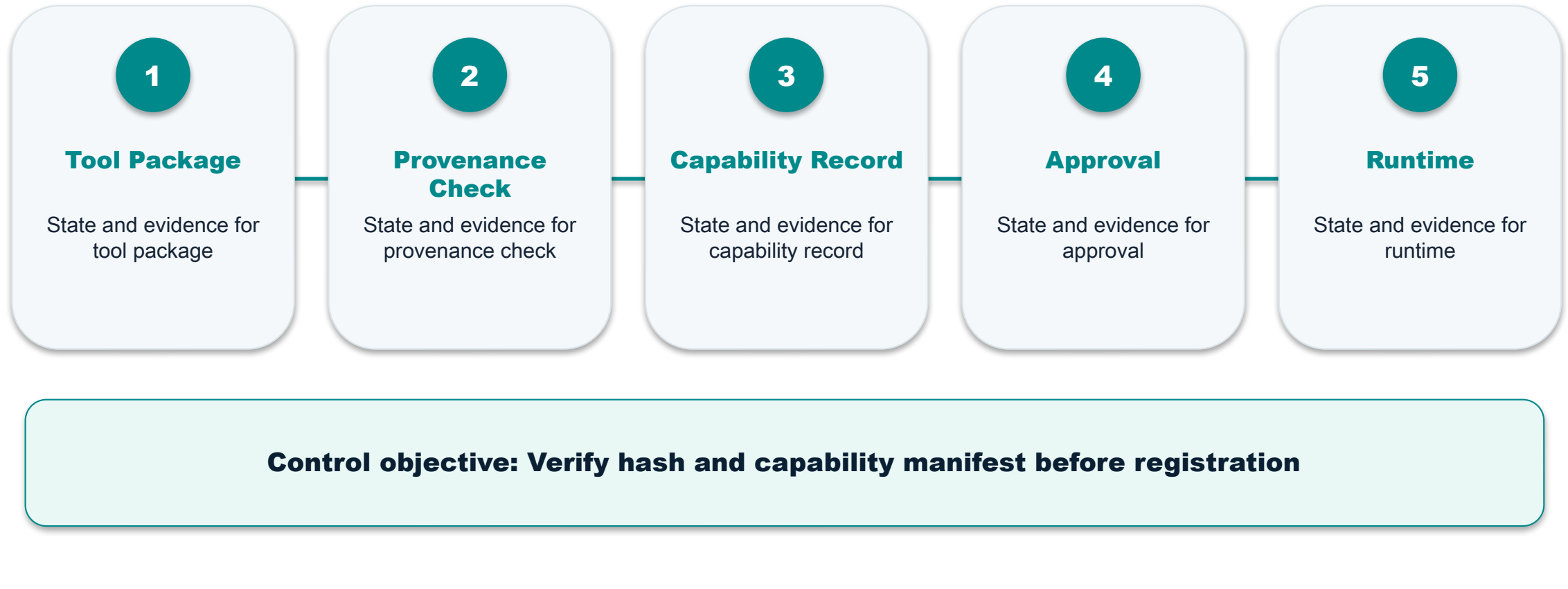
**Accept only when
Secret scan=0 and
rotation age ≤ policy.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

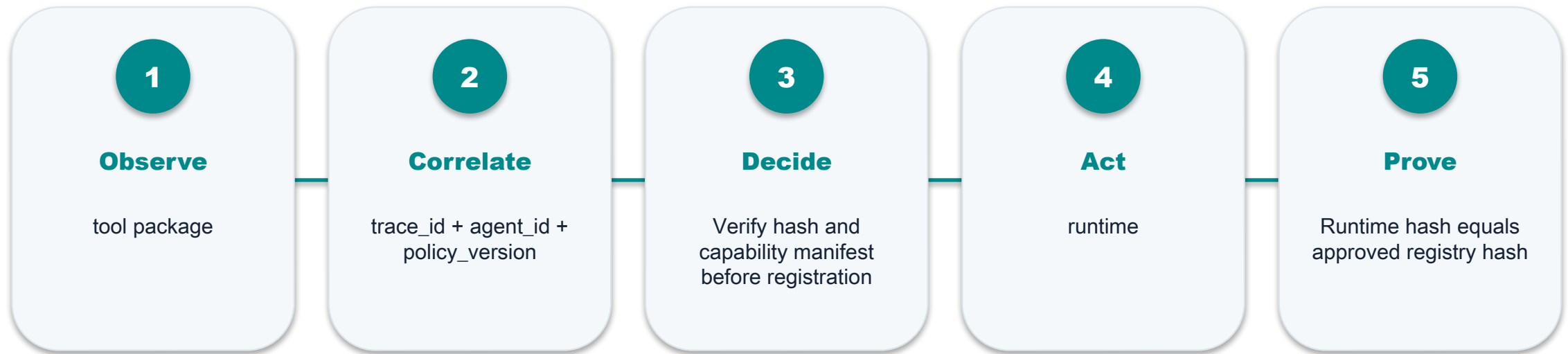
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Tool registry: architecture and control boundary



Tool registry: runtime decision trace



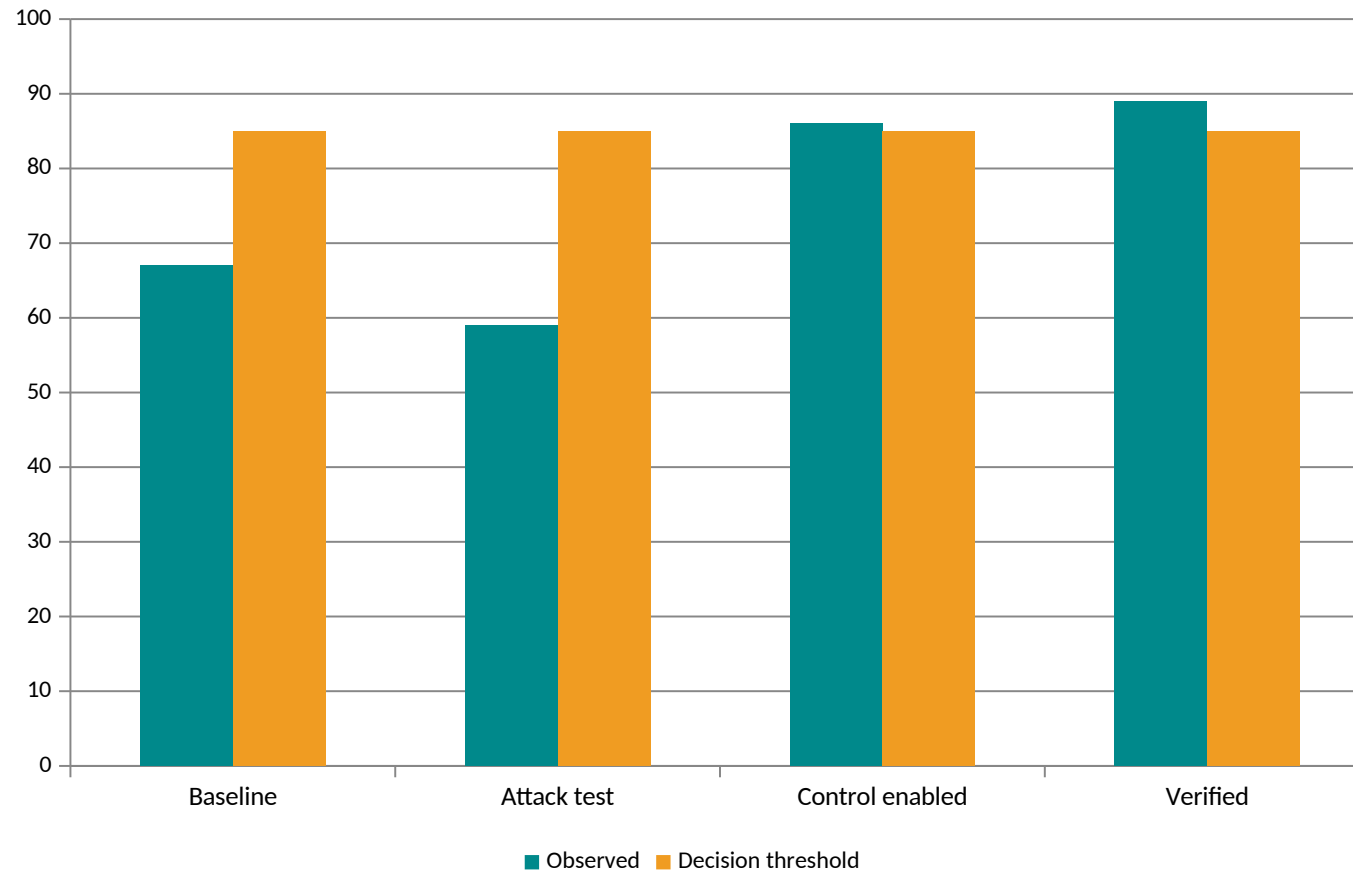
Failure to detect: A renamed tool adds an undeclared destructive capability

Tool registry: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	tool_id	Required in decision and audit evidence
field_2	version	Required in decision and audit evidence
field_3	publisher	Required in decision and audit evidence
field_4	hash	Required in decision and audit evidence
field_5	capabilities[]	Required in decision and audit evidence
field_6	risk_tier	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Tool registry: evidence and acceptance decision



DECISION RULE

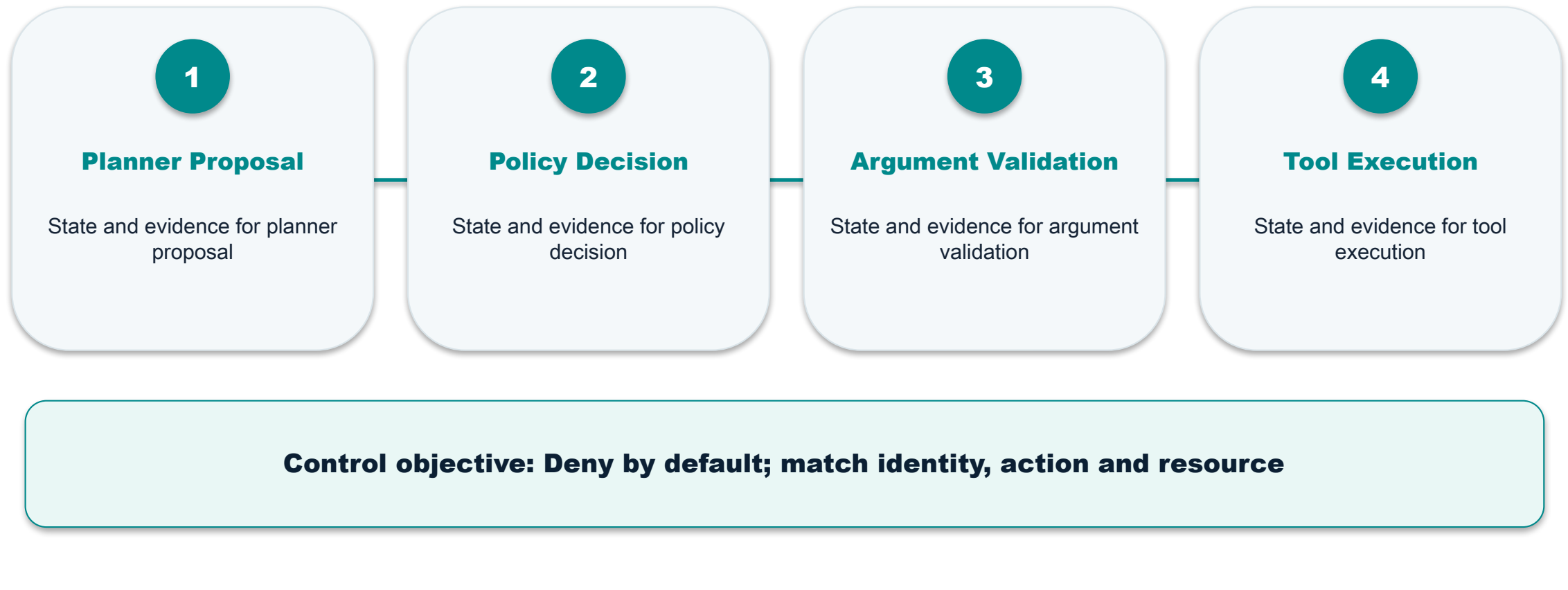
Accept only when Runtime hash equals approved registry hash. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

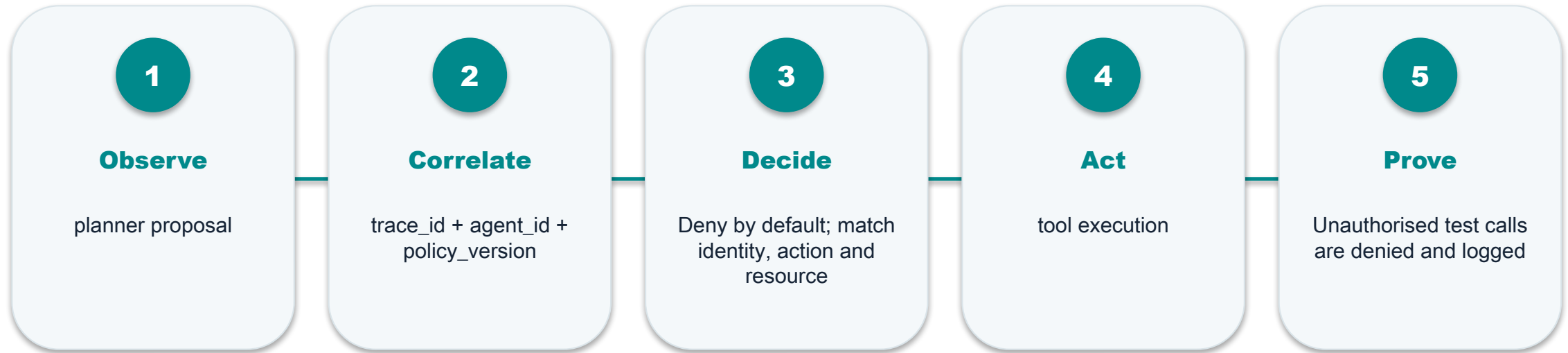
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Tool allowlist: architecture and control boundary



Tool allowlist: runtime decision trace



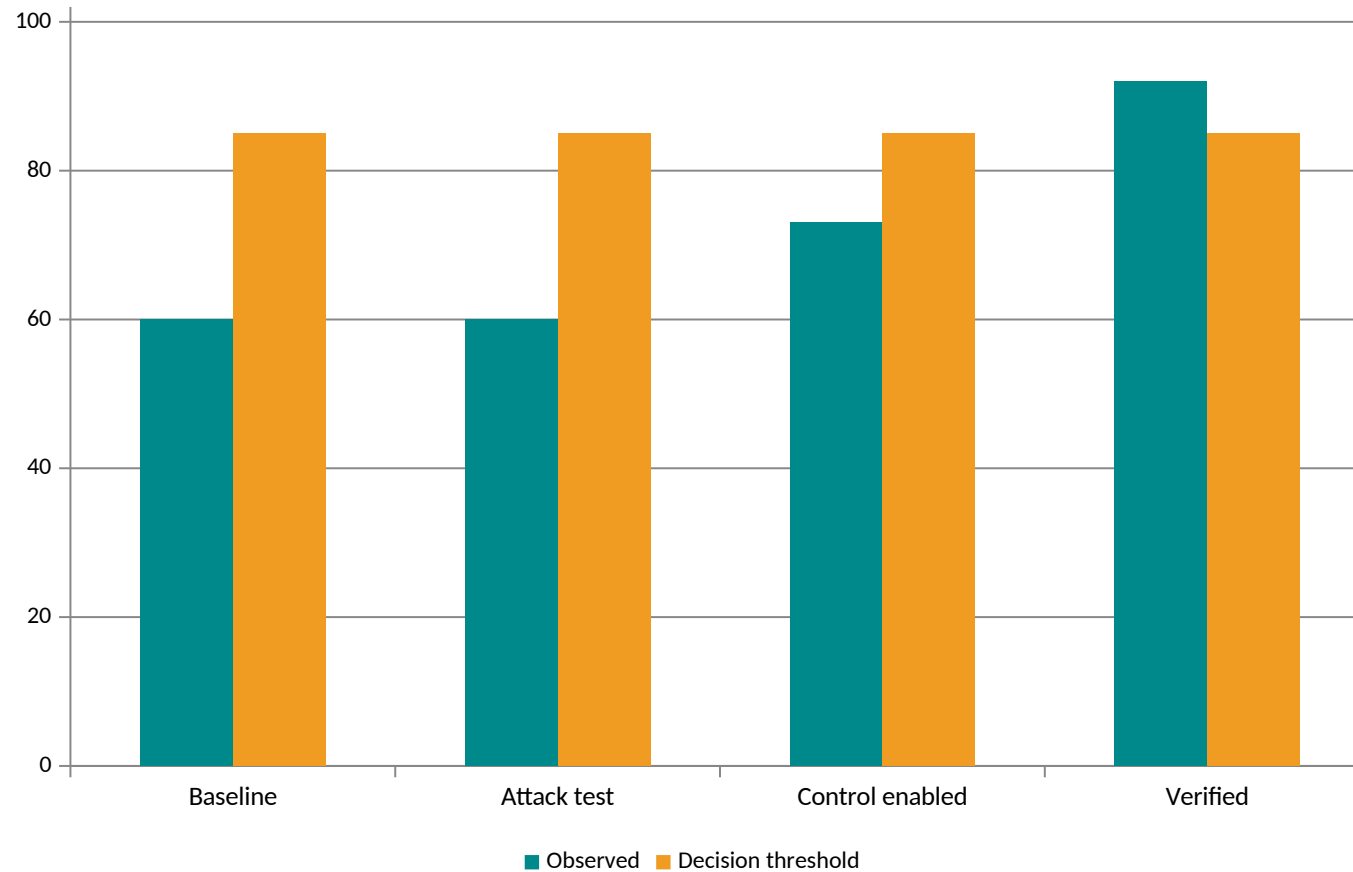
Failure to detect: Model selects a valid but unauthorised administrative tool

Tool allowlist: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agent_id	Required in decision and audit evidence
field_2	tool_id	Required in decision and audit evidence
field_3	action	Required in decision and audit evidence
field_4	resource	Required in decision and audit evidence
field_5	decision	Required in decision and audit evidence
field_6	policy_version	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Tool allowlist: evidence and acceptance decision



DECISION RULE

**Accept only when
Unauthorised test calls
are denied and logged.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Approval taxonomy: architecture and control boundary



Approval taxonomy: runtime decision trace



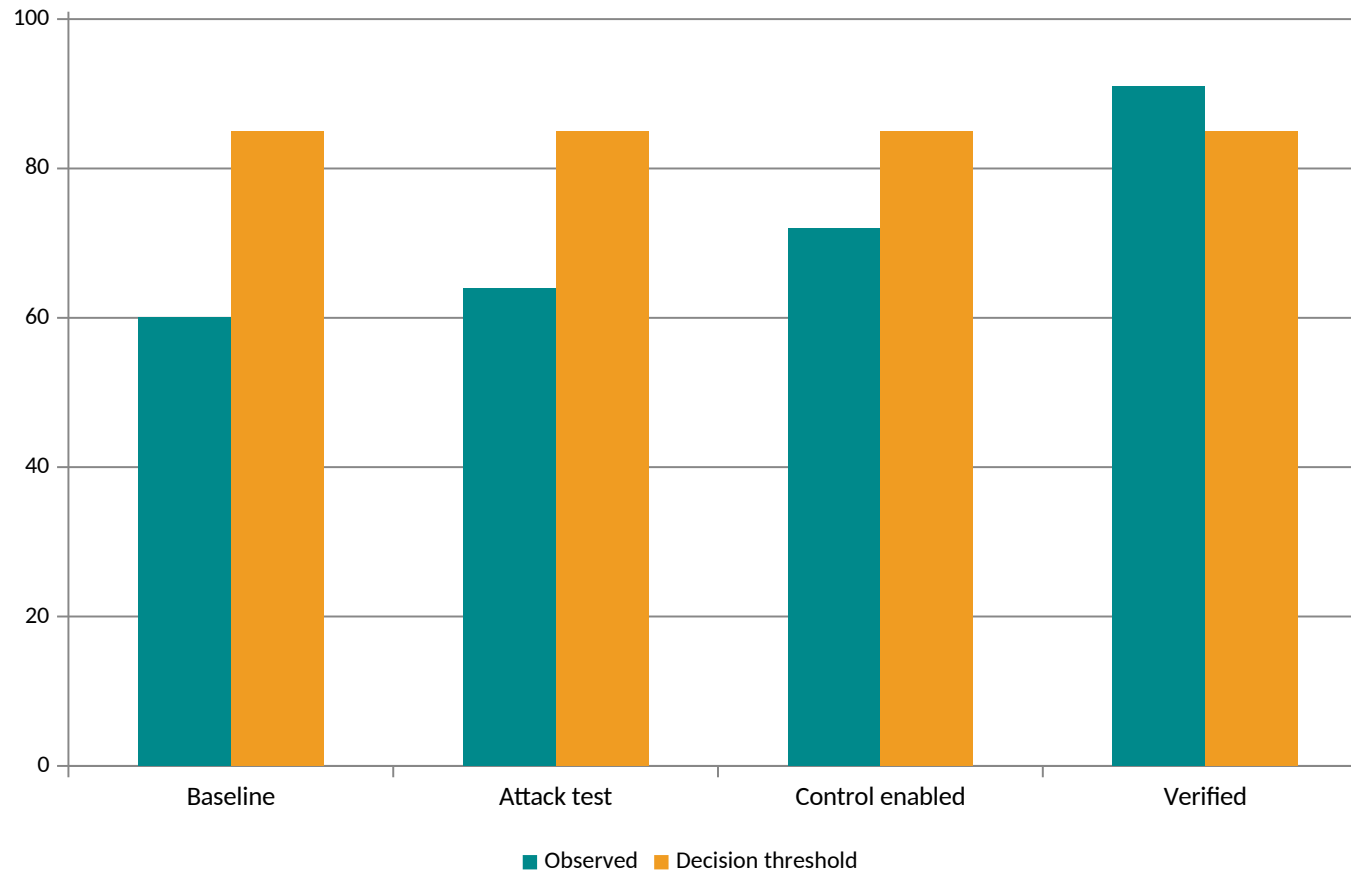
Failure to detect: High-impact action runs without human confirmation

Approval taxonomy: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	risk_tier	Required in decision and audit evidence
field_2	approval_required	Required in decision and audit evidence
field_3	approver_role	Required in decision and audit evidence
field_4	timeout_seconds	Required in decision and audit evidence
field_5	reason	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Approval taxonomy: evidence and acceptance decision



DECISION RULE

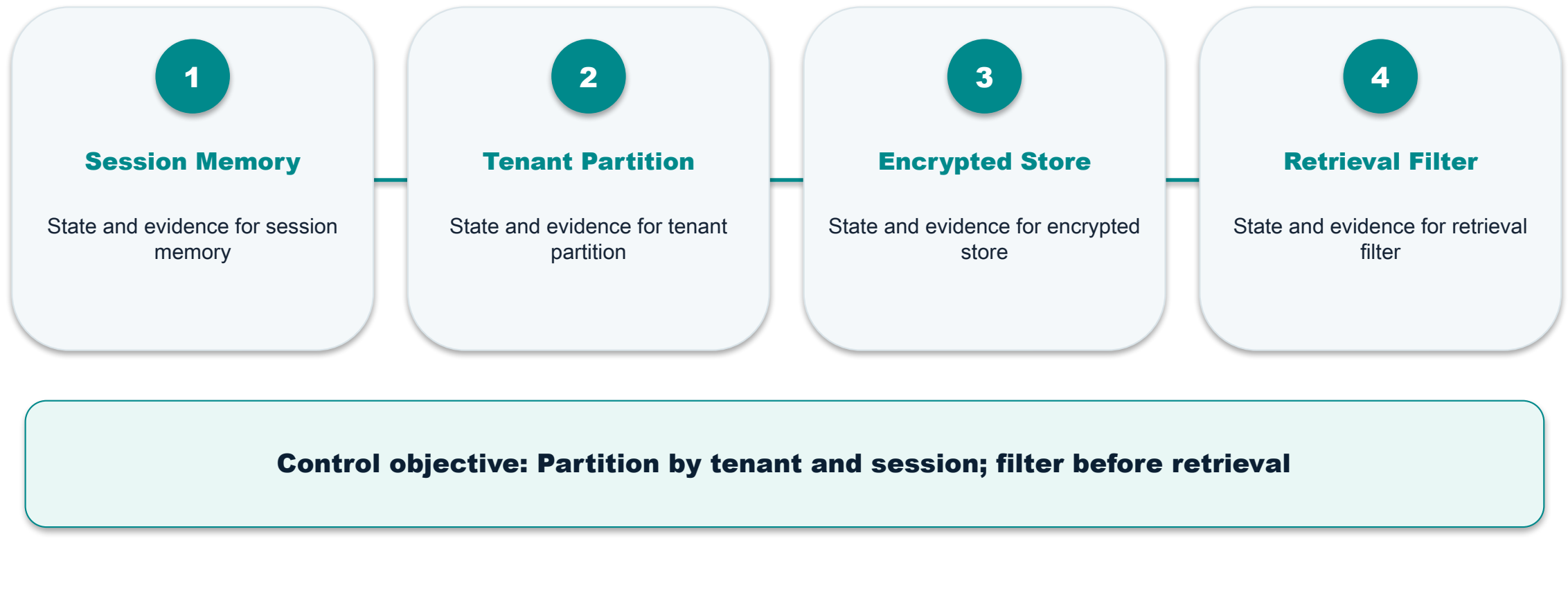
Accept only when 100% tier-3 executions carry a valid approval_id. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

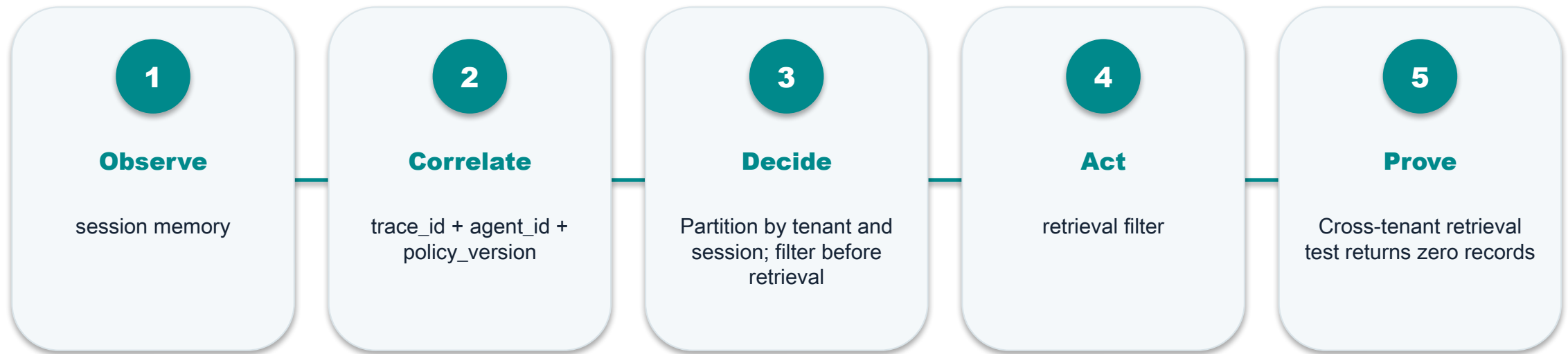
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Memory isolation: architecture and control boundary



Memory isolation: runtime decision trace



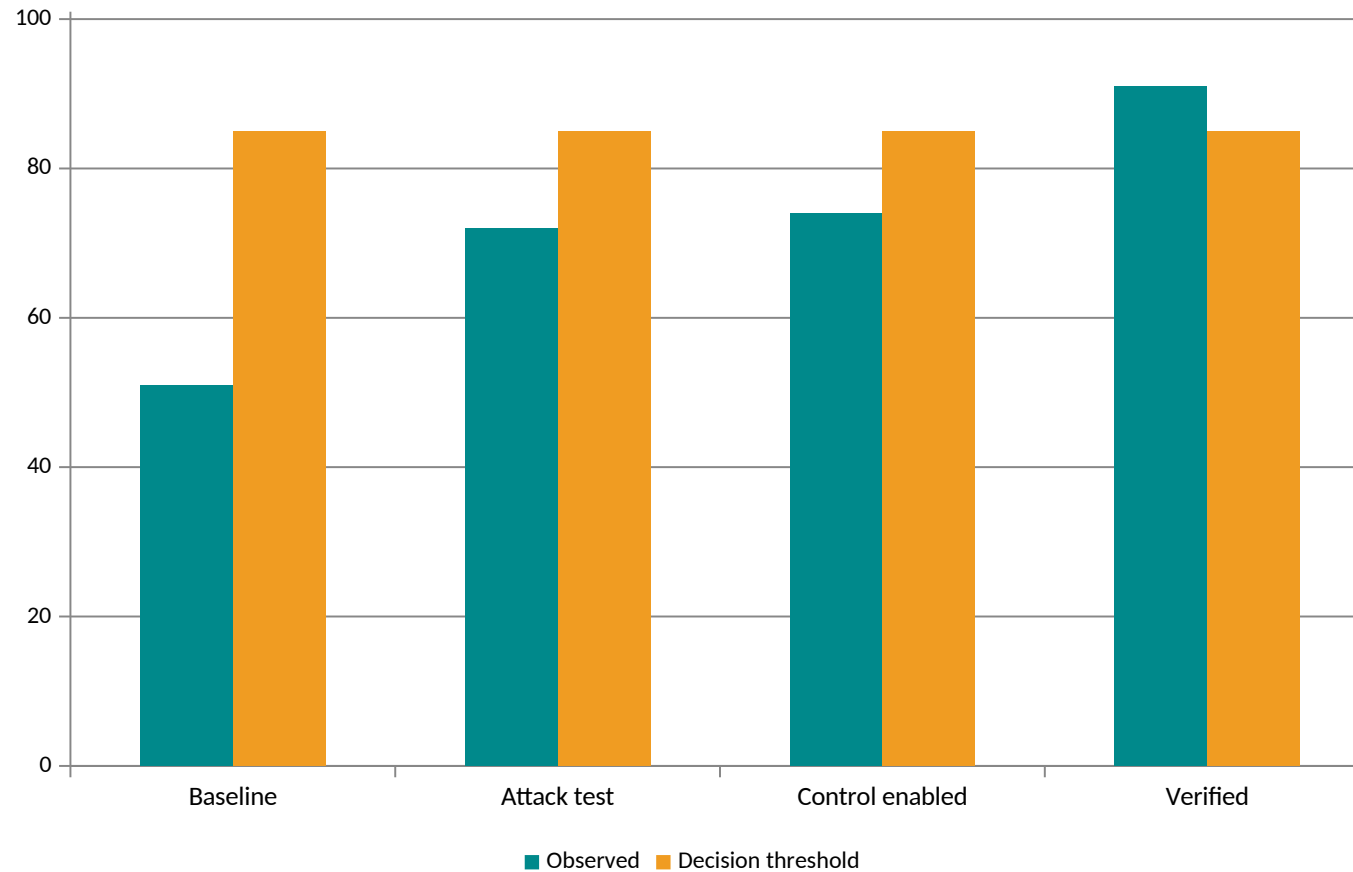
Failure to detect: One tenant retrieves another tenant's memory

Memory isolation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	tenant_id	Required in decision and audit evidence
field_2	session_id	Required in decision and audit evidence
field_3	namespace	Required in decision and audit evidence
field_4	classification	Required in decision and audit evidence
field_5	expires_at	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Memory isolation: evidence and acceptance decision



DECISION RULE

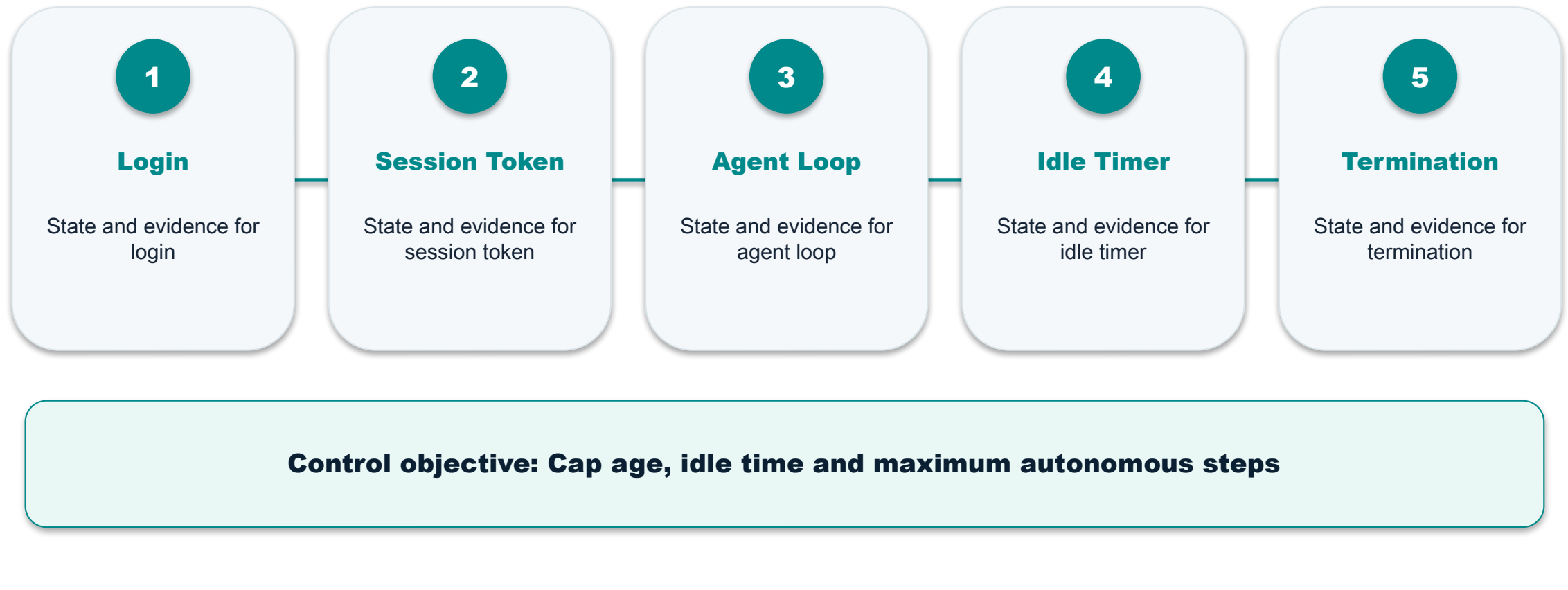
Accept only when Cross-tenant retrieval test returns zero records. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Session lifetime: architecture and control boundary



Session lifetime: runtime decision trace



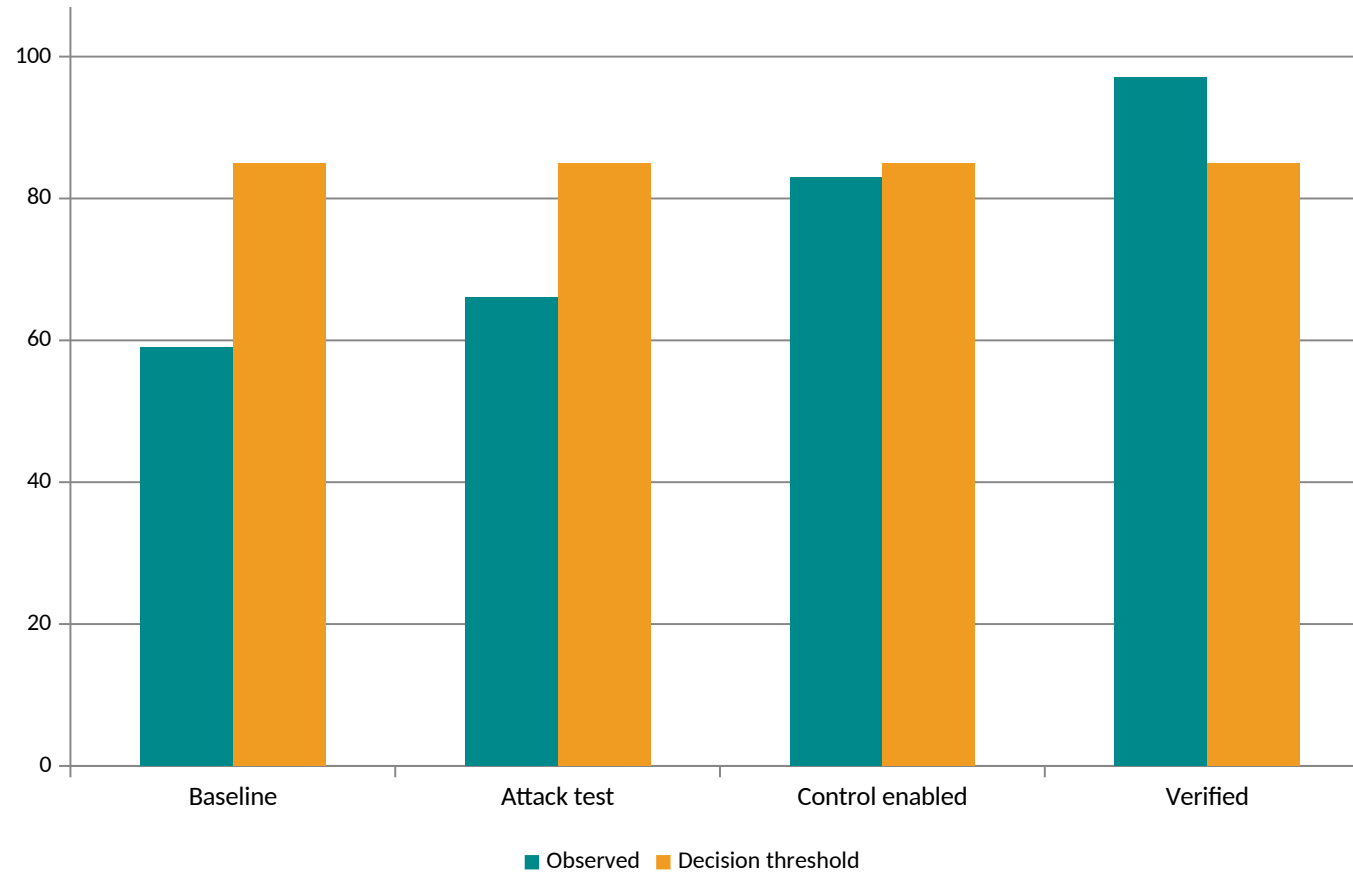
Failure to detect: A forgotten session remains capable overnight

Session lifetime: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	session_id	Required in decision and audit evidence
field_2	created_at	Required in decision and audit evidence
field_3	last_active	Required in decision and audit evidence
field_4	max_age	Required in decision and audit evidence
field_5	idle_timeout	Required in decision and audit evidence
field_6	step_count	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Session lifetime: evidence and acceptance decision



DECISION RULE

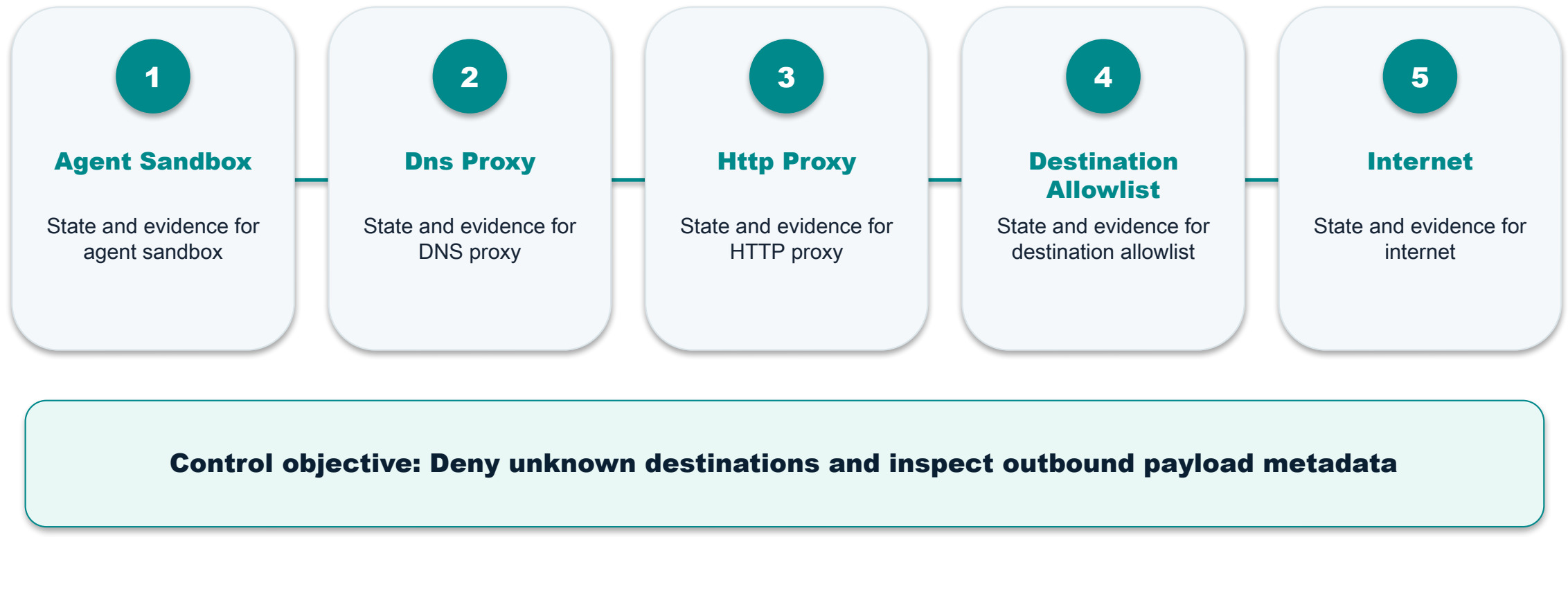
**Accept only when
Expired sessions reject
the next tool call.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

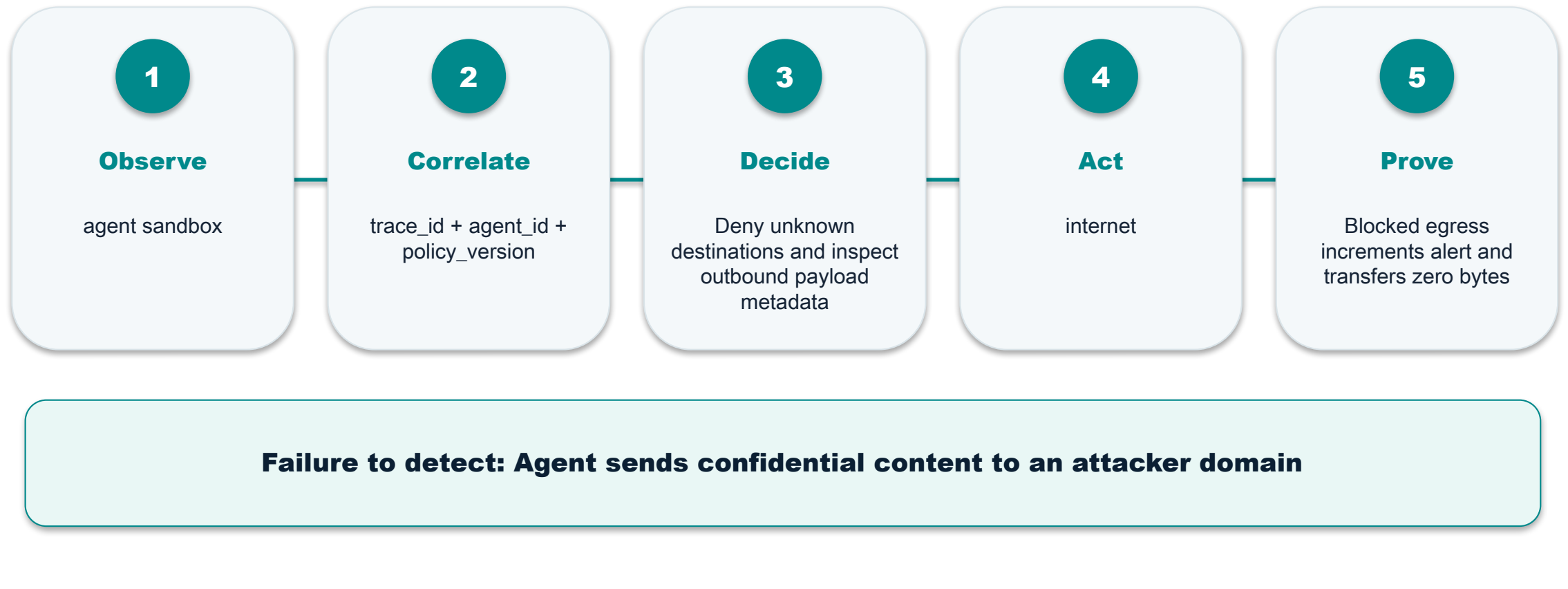
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Egress control: architecture and control boundary



Egress control: runtime decision trace

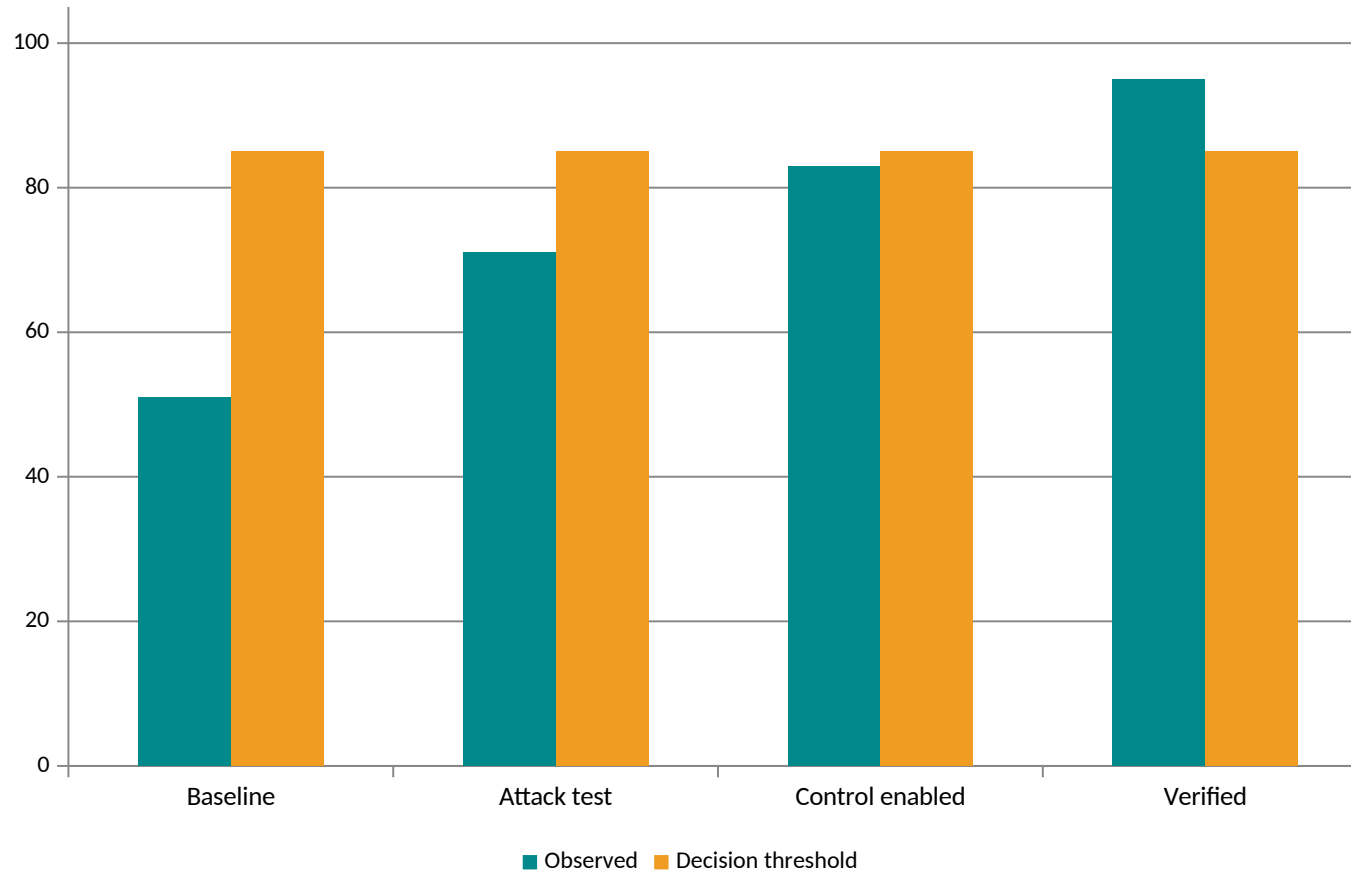


Egress control: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	destination	Required in decision and audit evidence
field_2	port	Required in decision and audit evidence
field_3	method	Required in decision and audit evidence
field_4	bytes_out	Required in decision and audit evidence
field_5	data_class	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Egress control: evidence and acceptance decision



DECISION RULE

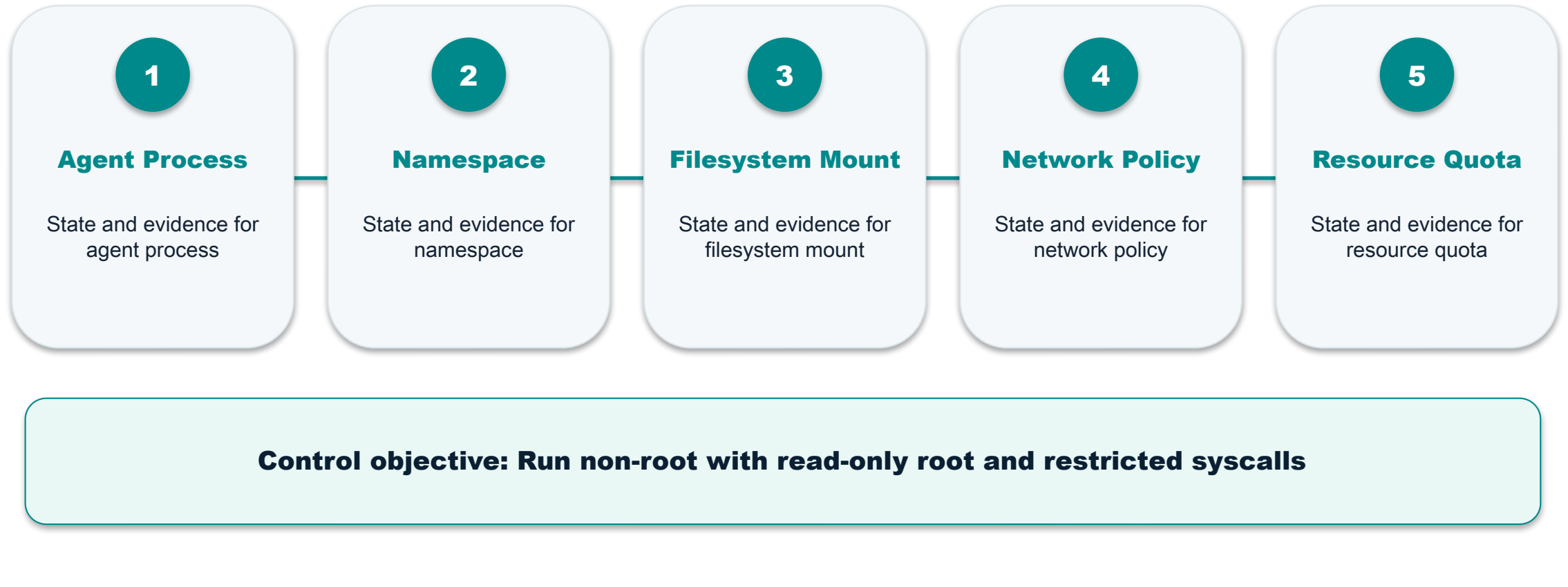
**Accept only when
Blocked egress
increments alert and
transfers zero bytes.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

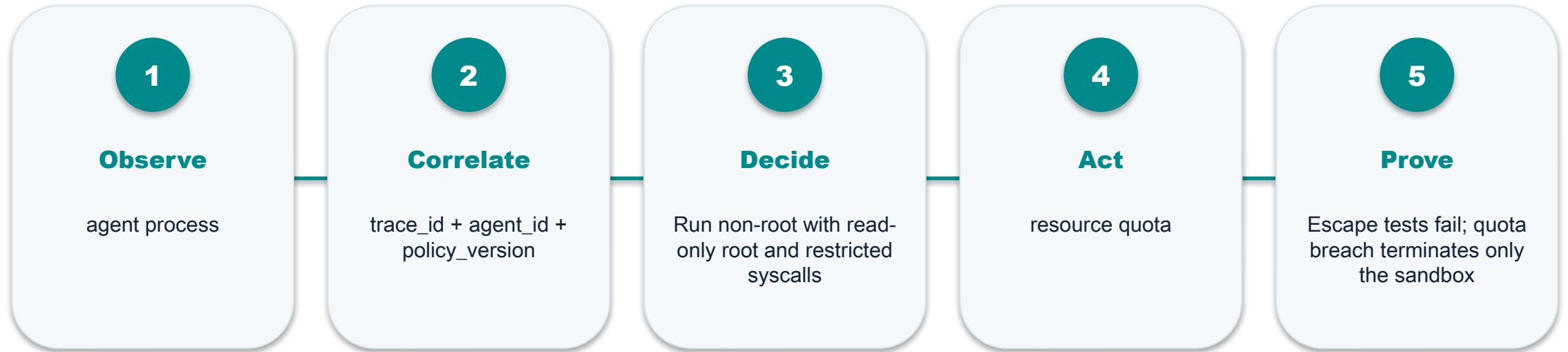
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Sandbox profile: architecture and control boundary



Sandbox profile: runtime decision trace



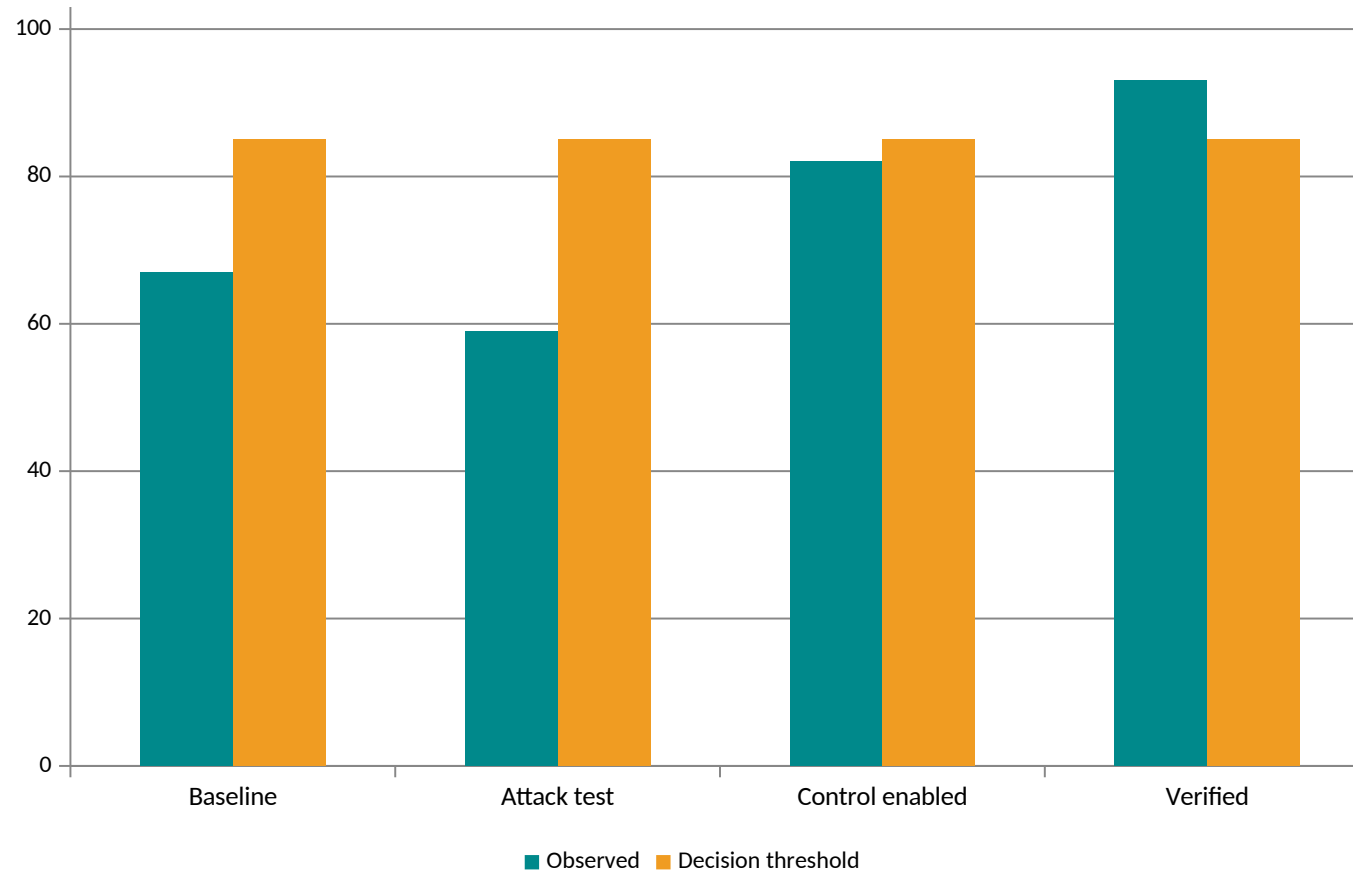
Failure to detect: Generated code writes outside its workspace or escapes to host

Sandbox profile: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	image_digest	Required in decision and audit evidence
field_2	uid	Required in decision and audit evidence
field_3	readonly_root	Required in decision and audit evidence
field_4	cpu_limit	Required in decision and audit evidence
field_5	memory_mb	Required in decision and audit evidence
field_6	network_profile	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Sandbox profile: evidence and acceptance decision



DECISION RULE

Accept only when Escape tests fail; quota breach terminates only the sandbox. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Artifact provenance: architecture and control boundary



Control objective: Require signed provenance tied to source and builder

Artifact provenance: runtime decision trace



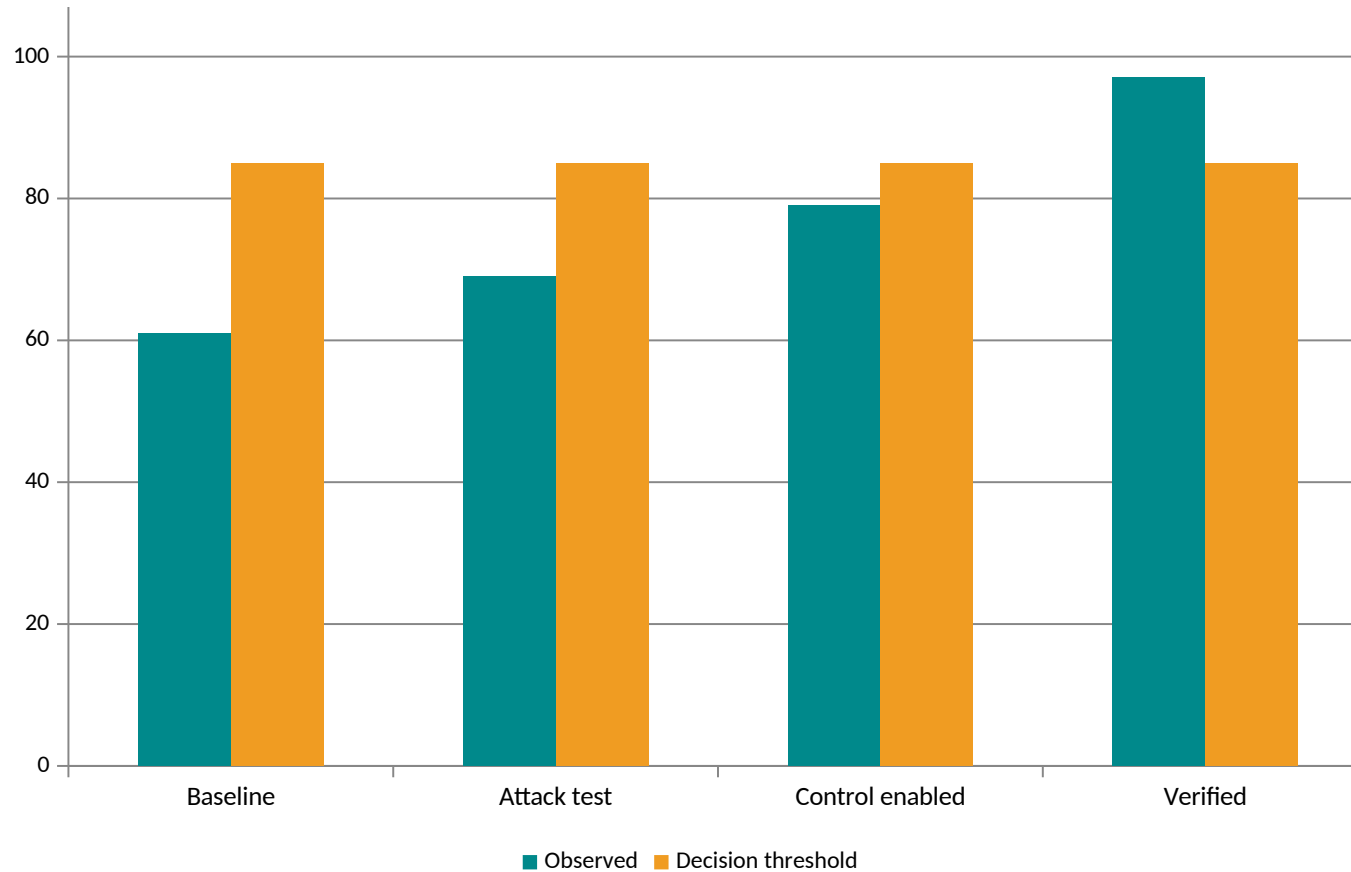
Failure to detect: An unsigned prompt/tool bundle reaches production

Artifact provenance: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	artifact_digest	Required in decision and audit evidence
field_2	source_commit	Required in decision and audit evidence
field_3	builder_id	Required in decision and audit evidence
field_4	signature	Required in decision and audit evidence
field_5	predicate_type	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Artifact provenance: evidence and acceptance decision



DECISION RULE

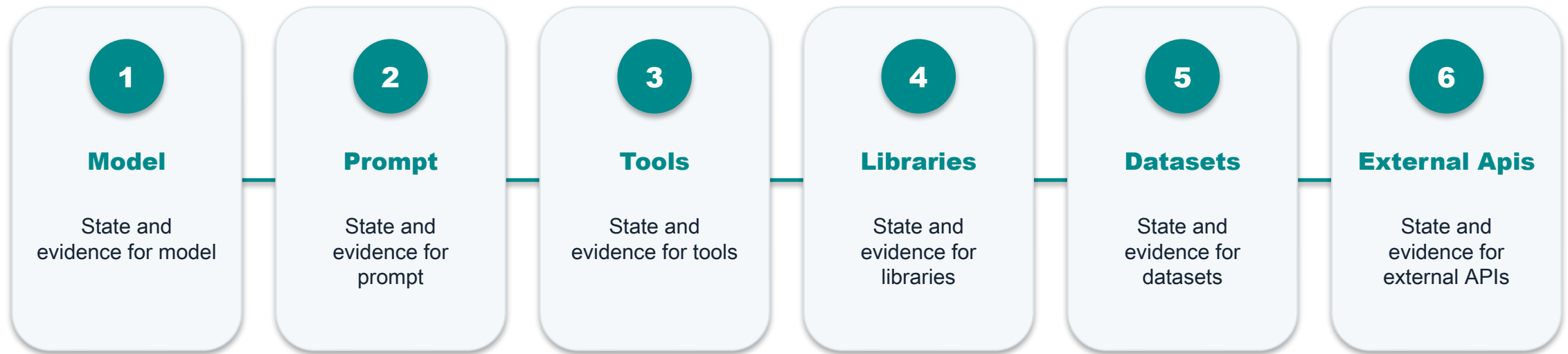
Accept only when Deploy verifier rejects missing or mismatched attestations. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

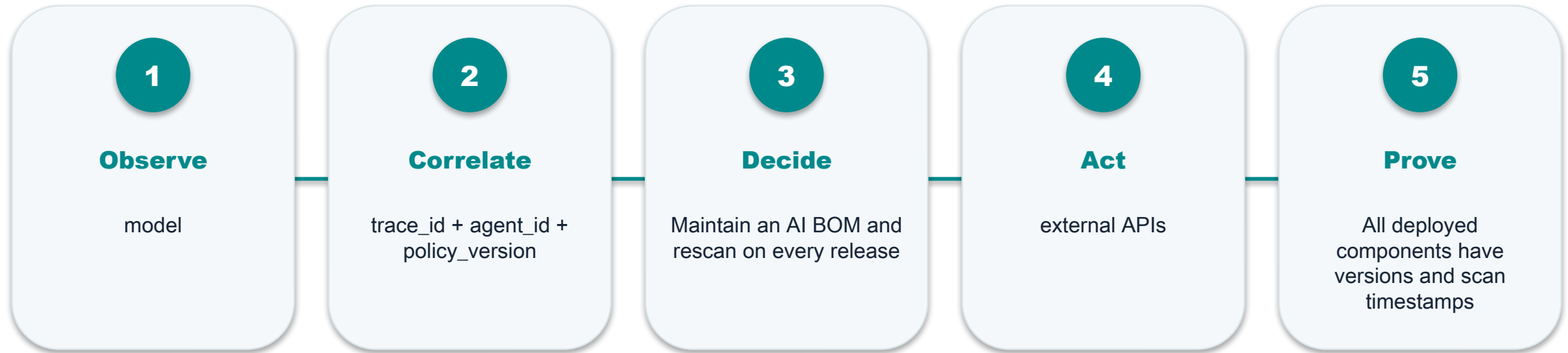
Synthetic training evidence; values are illustrative, not external performance claims

AI bill of materials: architecture and control boundary



Control objective: Maintain an AI BOM and rescan on every release

AI bill of materials: runtime decision trace



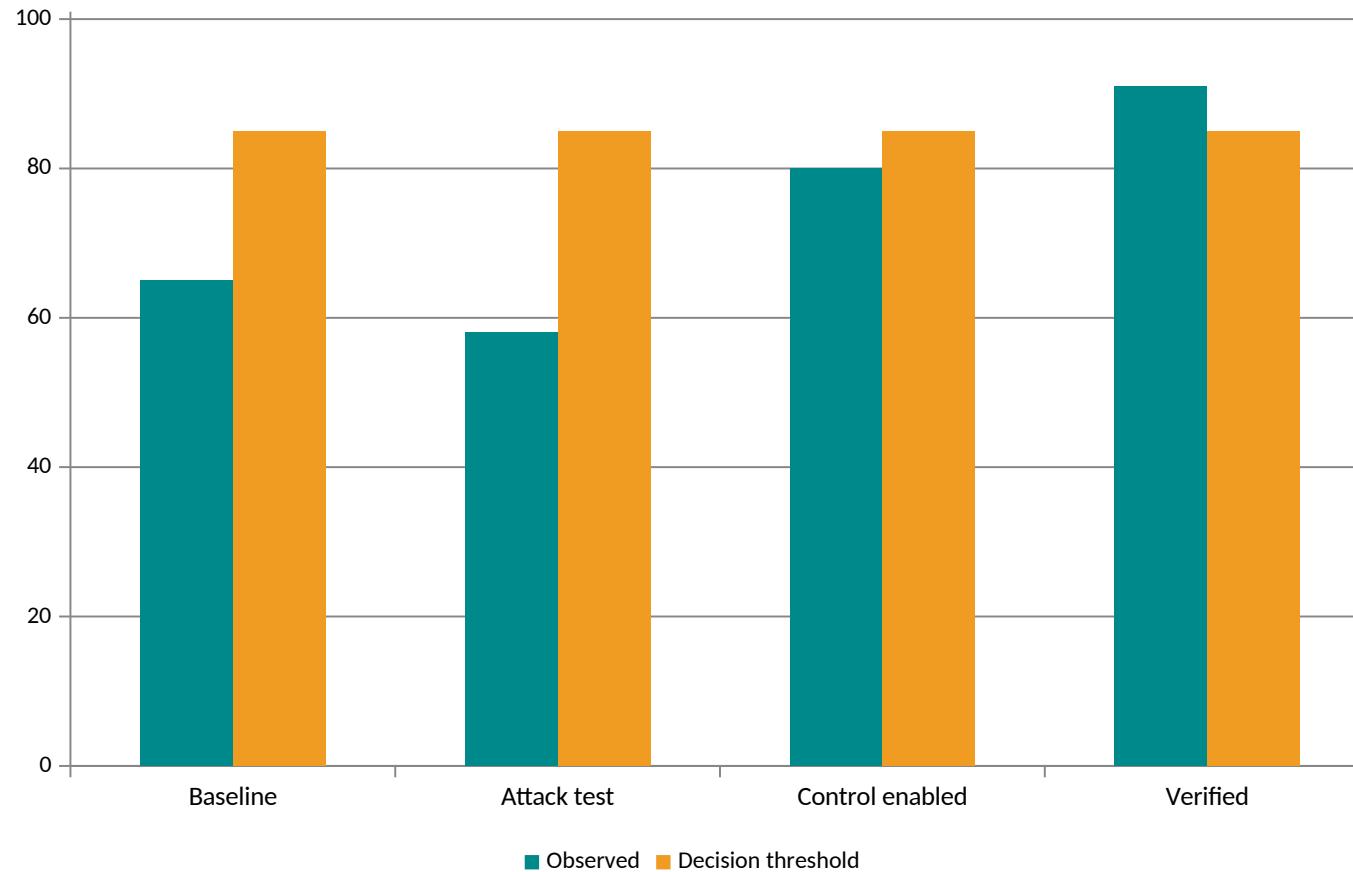
Failure to detect: Vulnerable dependency or changed model is invisible

AI bill of materials: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	component	Required in decision and audit evidence
field_2	supplier	Required in decision and audit evidence
field_3	version	Required in decision and audit evidence
field_4	licence	Required in decision and audit evidence
field_5	hash	Required in decision and audit evidence
field_6	vulnerability_status	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

AI bill of materials: evidence and acceptance decision



DECISION RULE

Accept only when All deployed components have versions and scan timestamps. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

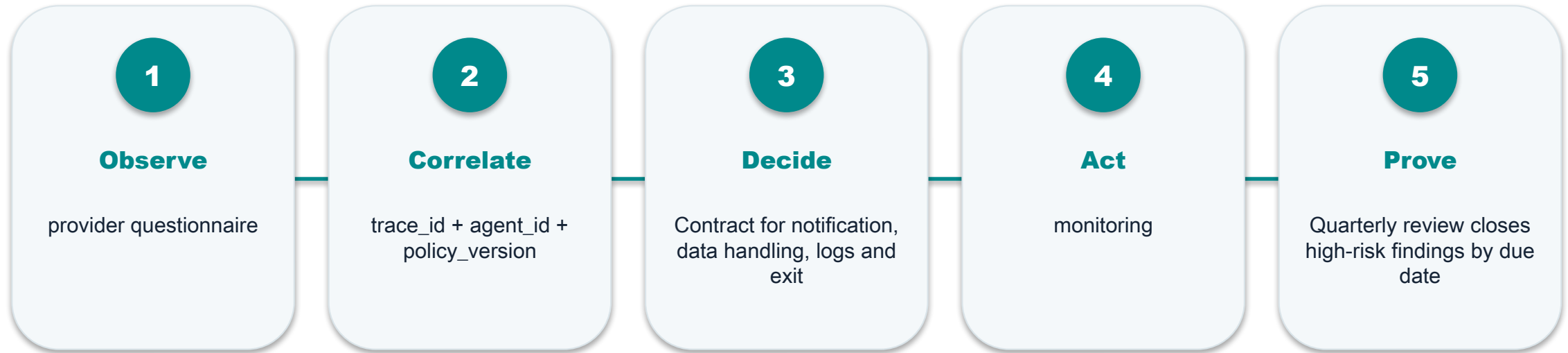
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Supplier assurance: architecture and control boundary



Supplier assurance: runtime decision trace



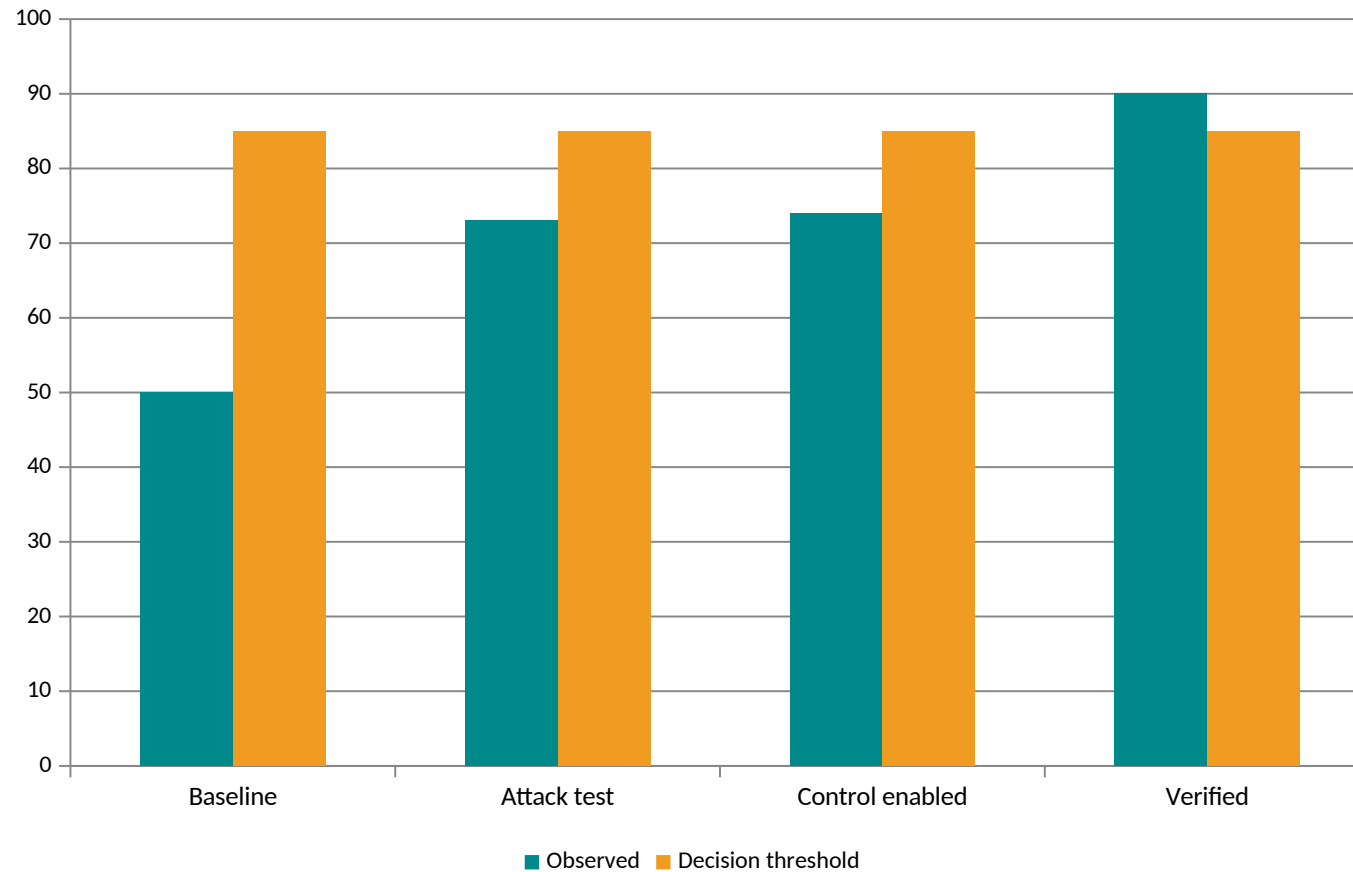
Failure to detect: External model provider changes retention or subprocessor terms

Supplier assurance: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	supplier_id	Required in decision and audit evidence
field_2	service	Required in decision and audit evidence
field_3	data_region	Required in decision and audit evidence
field_4	subprocessors[]	Required in decision and audit evidence
field_5	SLA	Required in decision and audit evidence
field_6	exit_plan	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Supplier assurance: evidence and acceptance decision



DECISION RULE

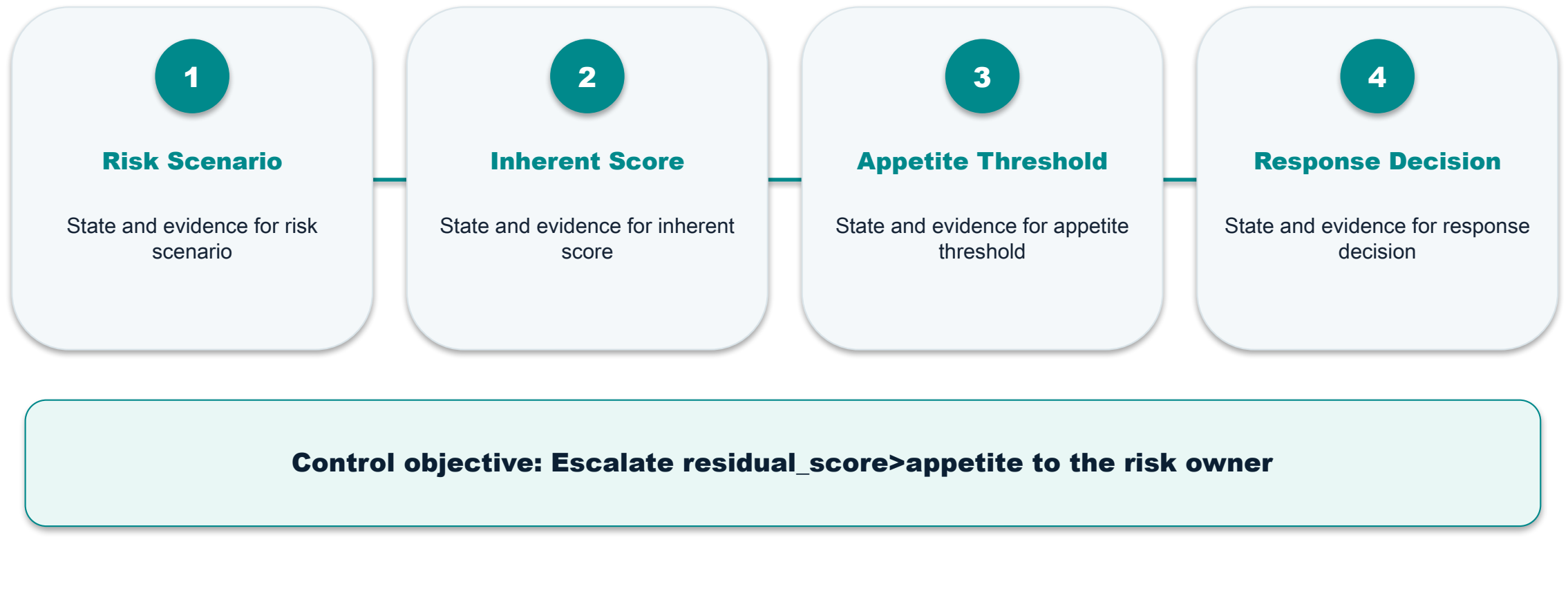
Accept only when Quarterly review closes high-risk findings by due date. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

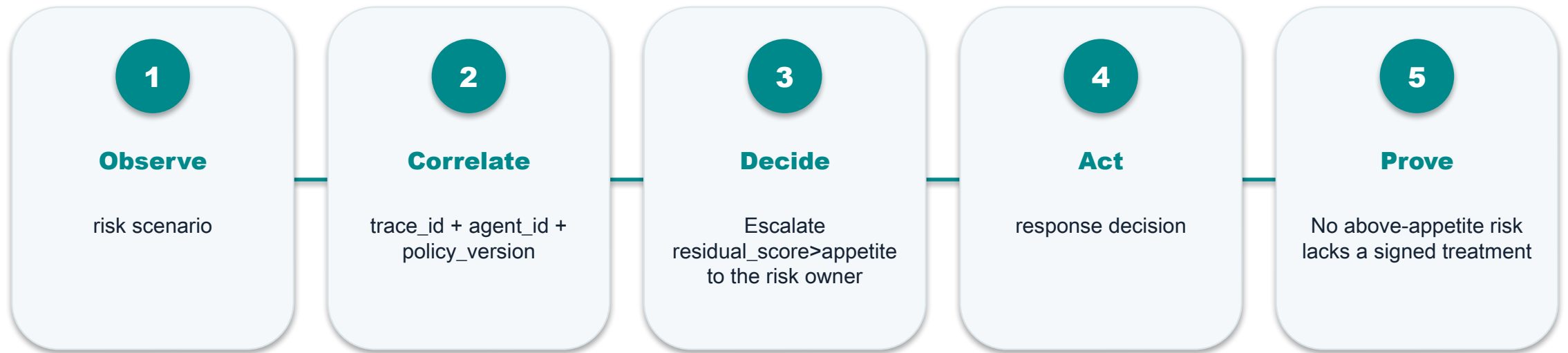
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Risk appetite: architecture and control boundary



Risk appetite: runtime decision trace



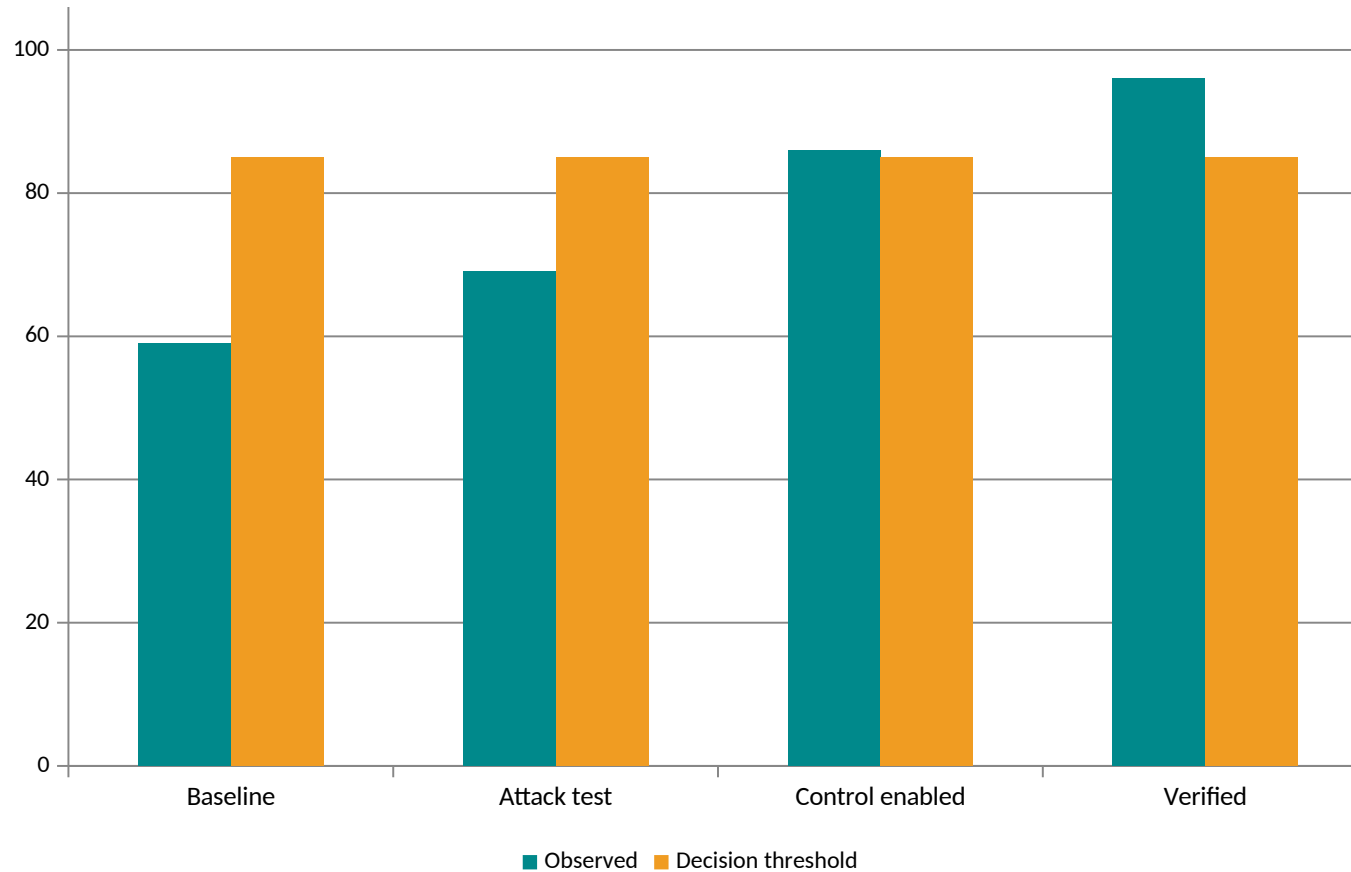
Failure to detect: Teams accept high residual risk without authority

Risk appetite: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	likelihood_1_5	Required in decision and audit evidence
field_2	impact_1_5	Required in decision and audit evidence
field_3	inherent_score	Required in decision and audit evidence
field_4	appetite	Required in decision and audit evidence
field_5	treatment	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Risk appetite: evidence and acceptance decision



DECISION RULE

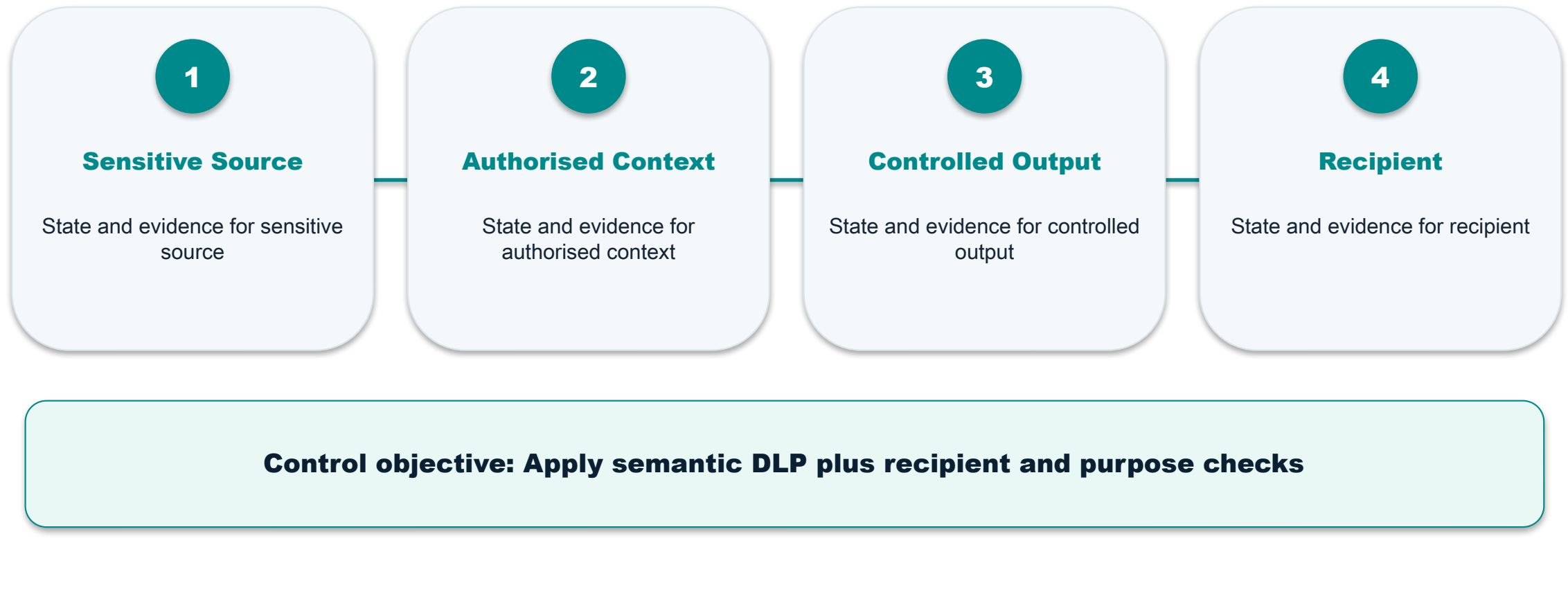
Accept only when No above-appetite risk lacks a signed treatment. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

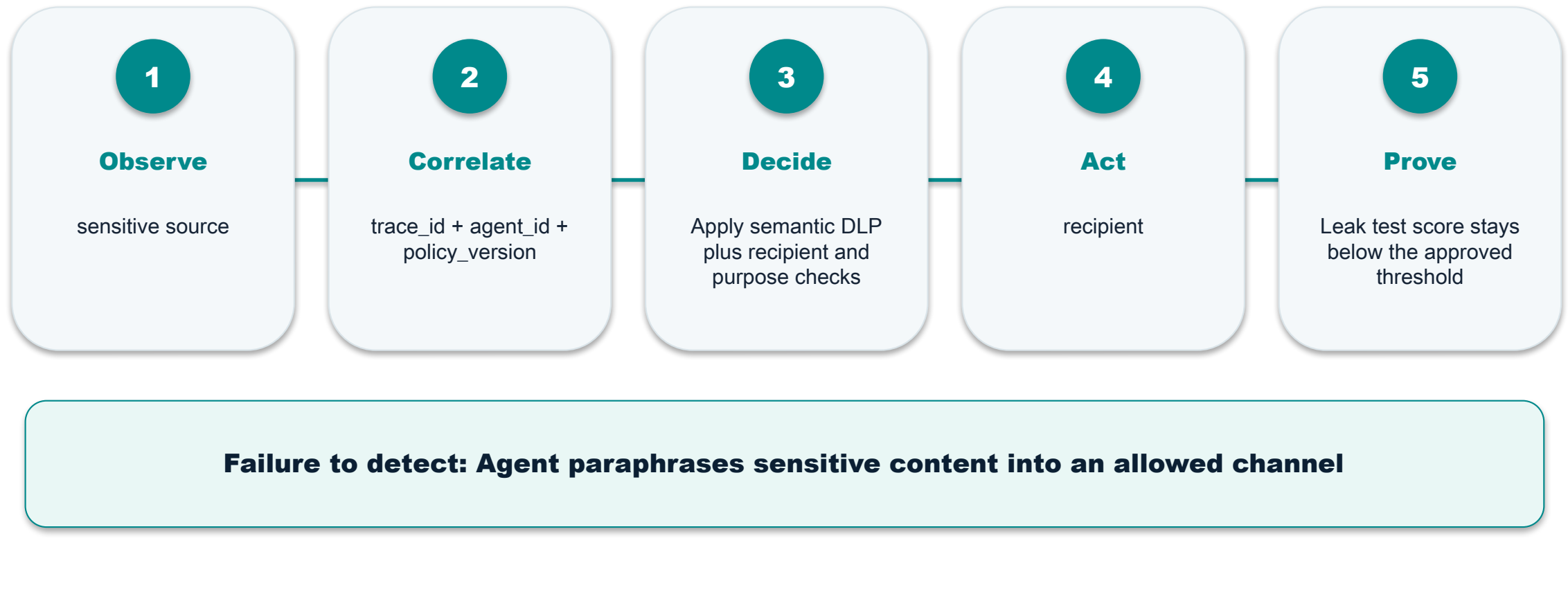
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Confidentiality objective: architecture and control boundary



Confidentiality objective: runtime decision trace

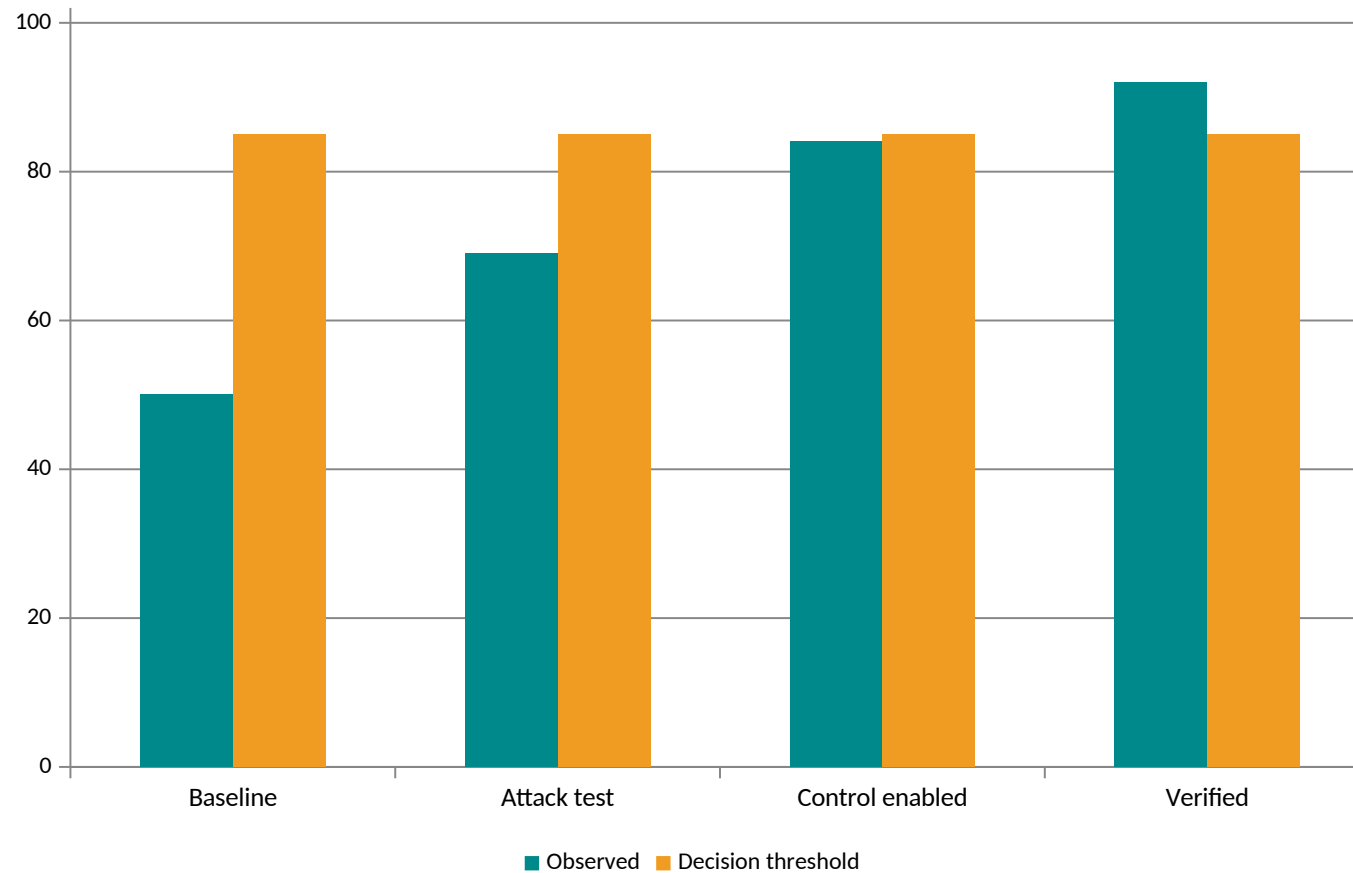


Confidentiality objective: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	classification	Required in decision and audit evidence
field_2	authorised_purposes[]	Required in decision and audit evidence
field_3	recipient	Required in decision and audit evidence
field_4	redactions[]	Required in decision and audit evidence
field_5	leak_score	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Confidentiality objective: evidence and acceptance decision



DECISION RULE

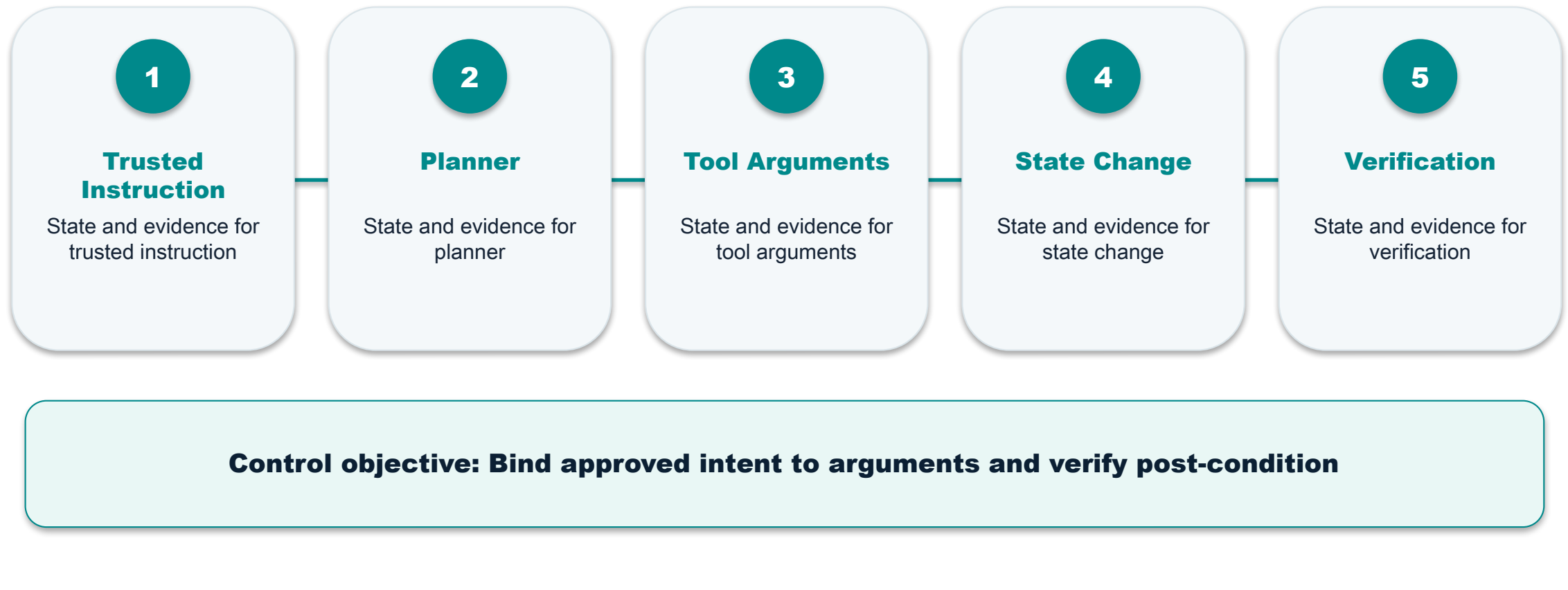
Accept only when Leak test score stays below the approved threshold. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

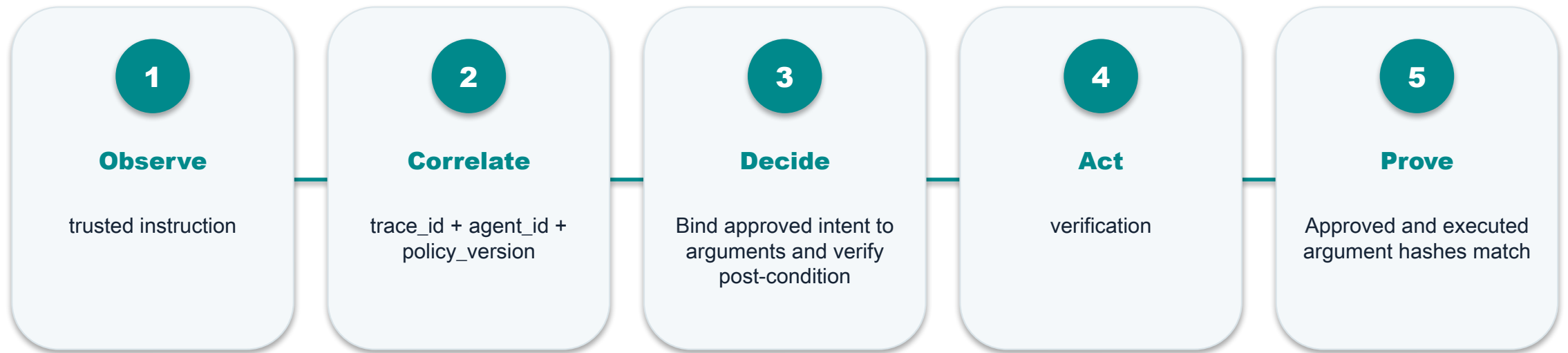
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Integrity objective: architecture and control boundary



Integrity objective: runtime decision trace



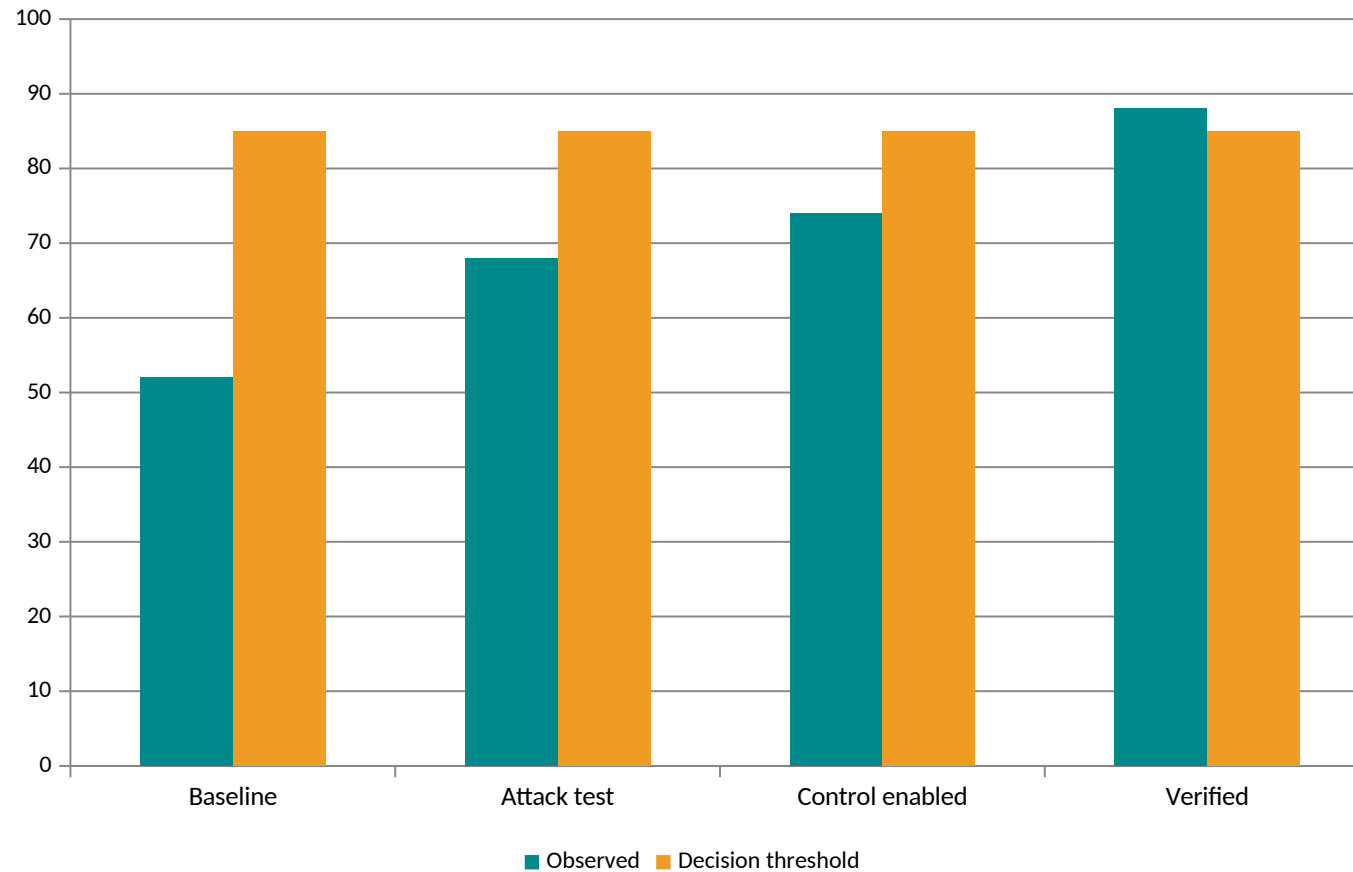
Failure to detect: Prompt injection changes the intended transaction parameters

Integrity objective: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	request_hash	Required in decision and audit evidence
field_2	policy_hash	Required in decision and audit evidence
field_3	argument_hash	Required in decision and audit evidence
field_4	result_hash	Required in decision and audit evidence
field_5	actor	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Integrity objective: evidence and acceptance decision



DECISION RULE

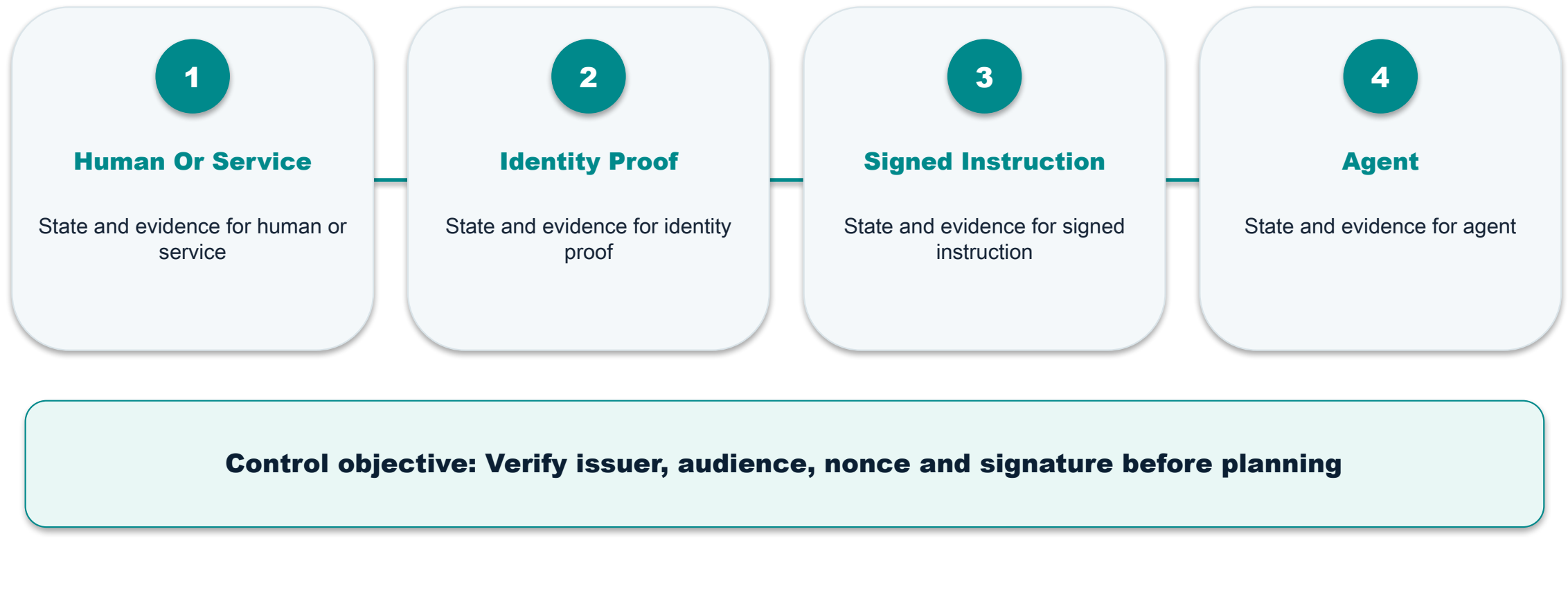
Accept only when Approved and executed argument hashes match. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Authenticity objective: architecture and control boundary



Authenticity objective: runtime decision trace



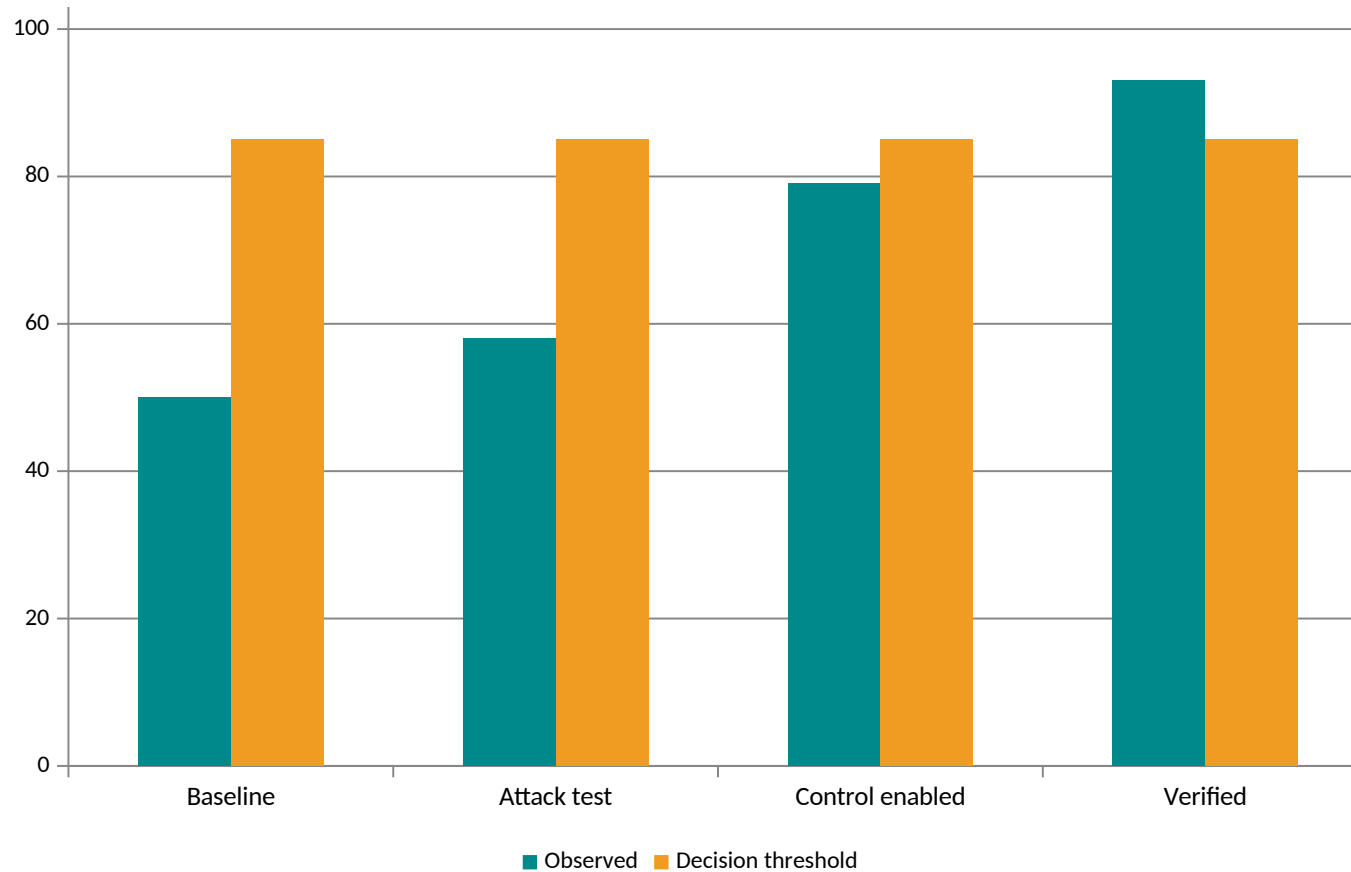
Failure to detect: Spoofed upstream service injects a privileged request

Authenticity objective: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	issuer	Required in decision and audit evidence
field_2	subject	Required in decision and audit evidence
field_3	audience	Required in decision and audit evidence
field_4	issued_at	Required in decision and audit evidence
field_5	nonce	Required in decision and audit evidence
field_6	signature	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Authenticity objective: evidence and acceptance decision



DECISION RULE

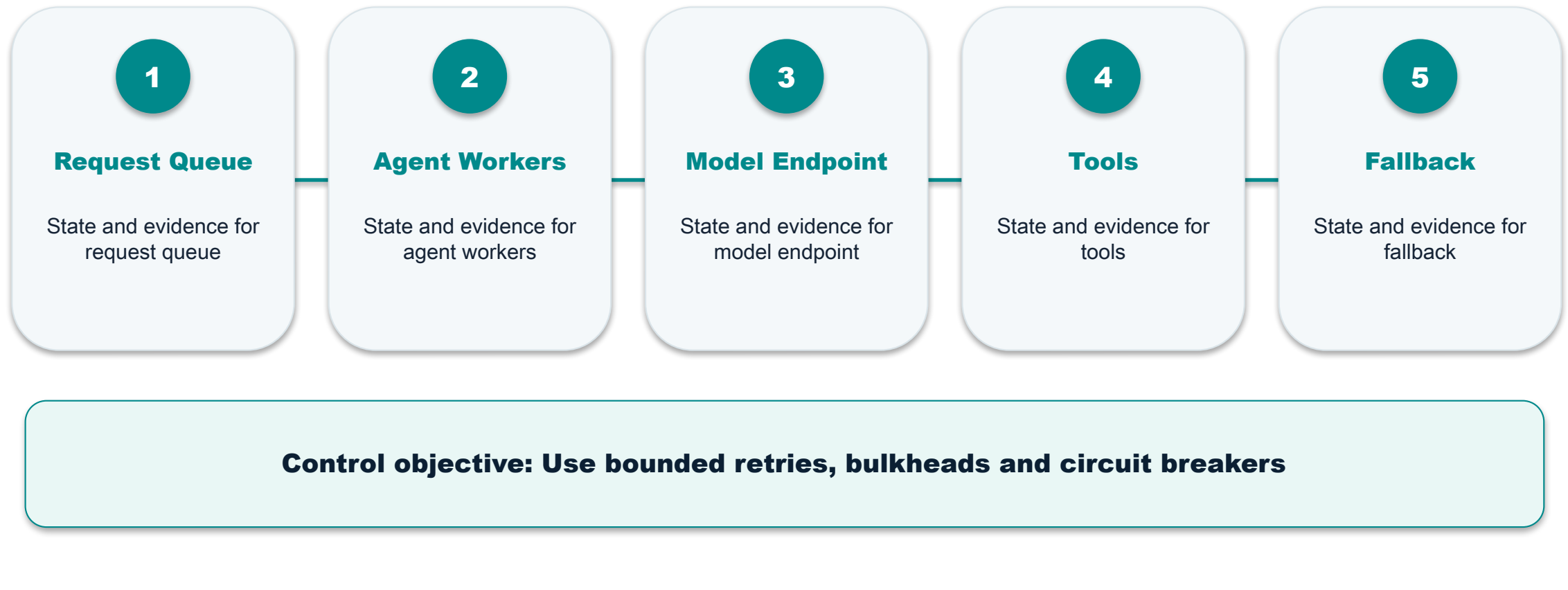
Accept only when Replay and wrong-audience tests are rejected. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

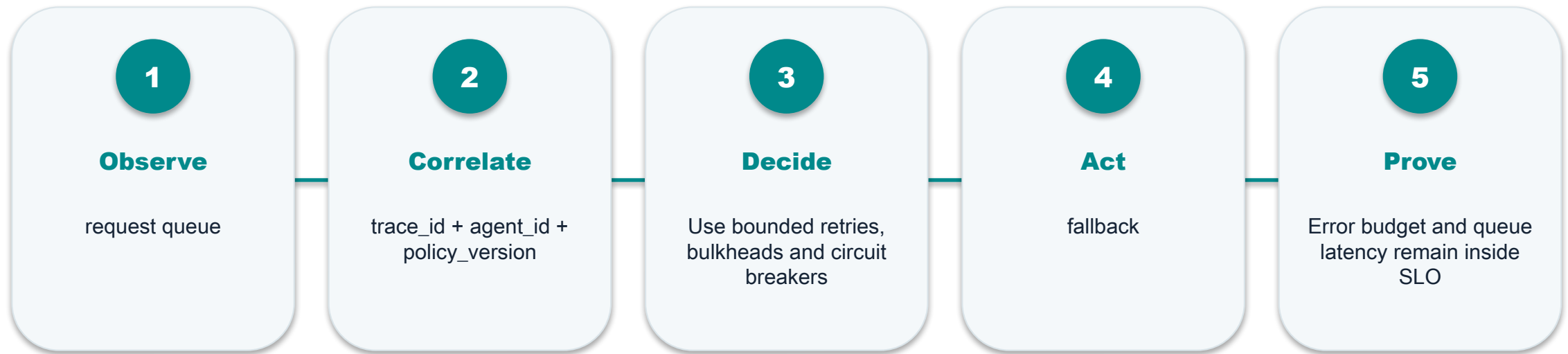
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Availability objective: architecture and control boundary



Availability objective: runtime decision trace



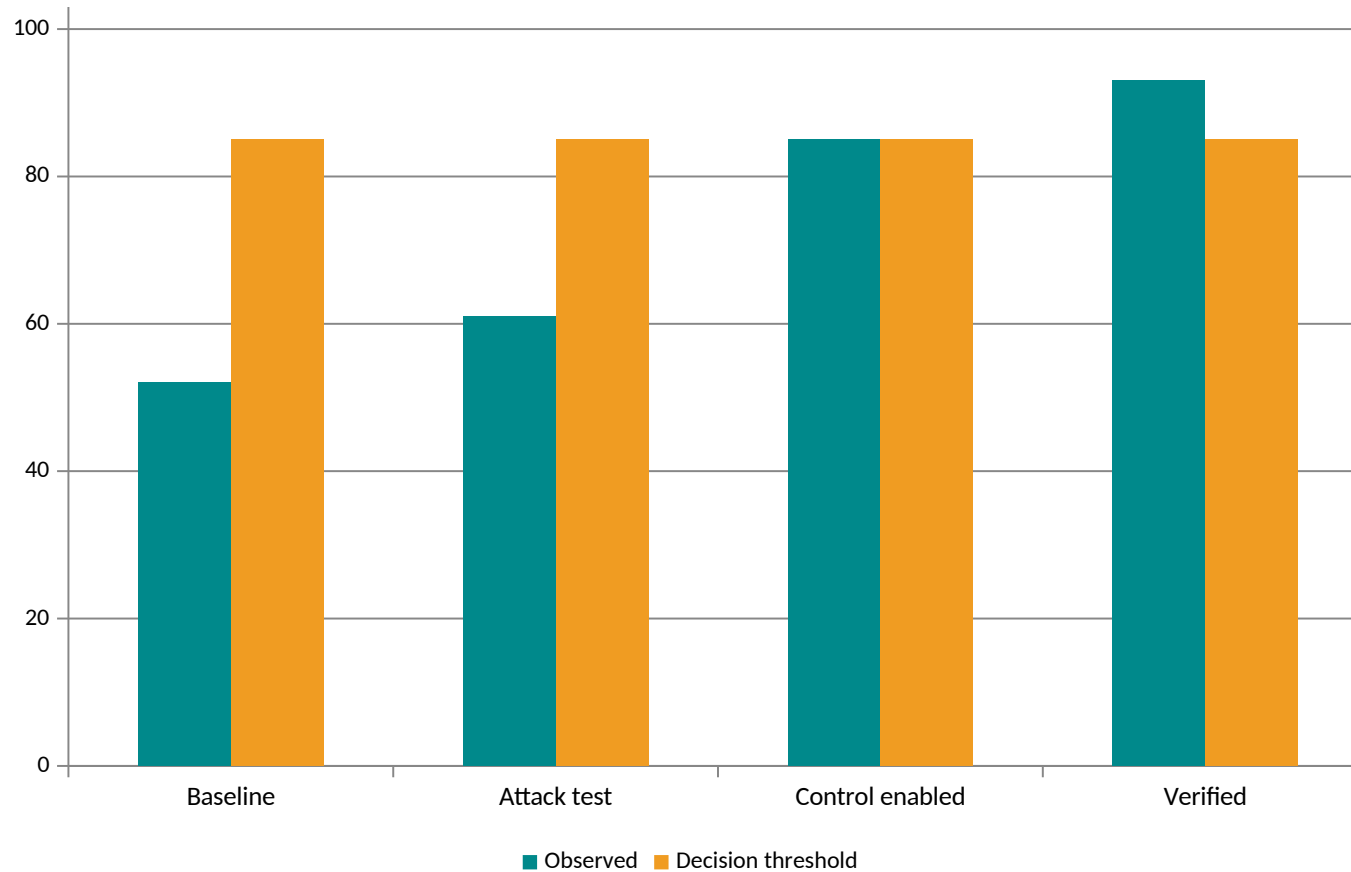
Failure to detect: Retry storm exhausts model and tool quotas

Availability objective: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	queue_depth	Required in decision and audit evidence
field_2	concurrency	Required in decision and audit evidence
field_3	timeout_ms	Required in decision and audit evidence
field_4	retry_budget	Required in decision and audit evidence
field_5	circuit_state	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Availability objective: evidence and acceptance decision



DECISION RULE

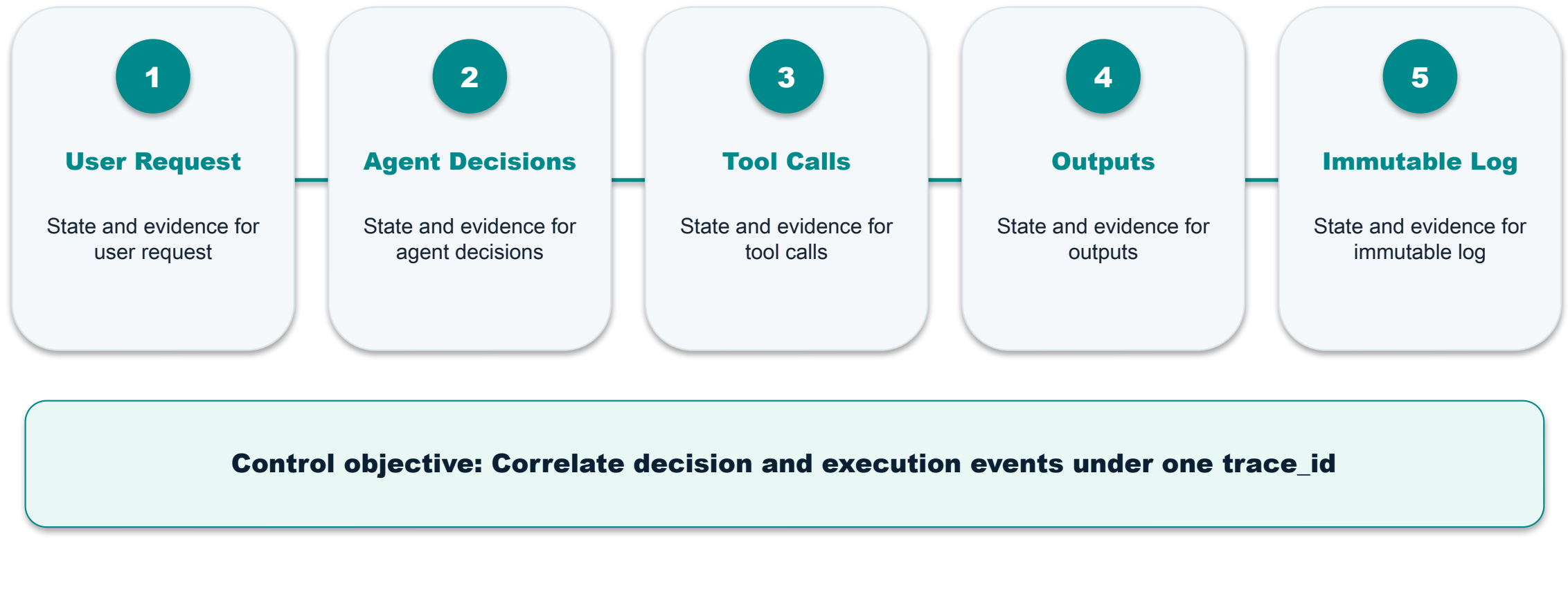
Accept only when Error budget and queue latency remain inside SLO. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

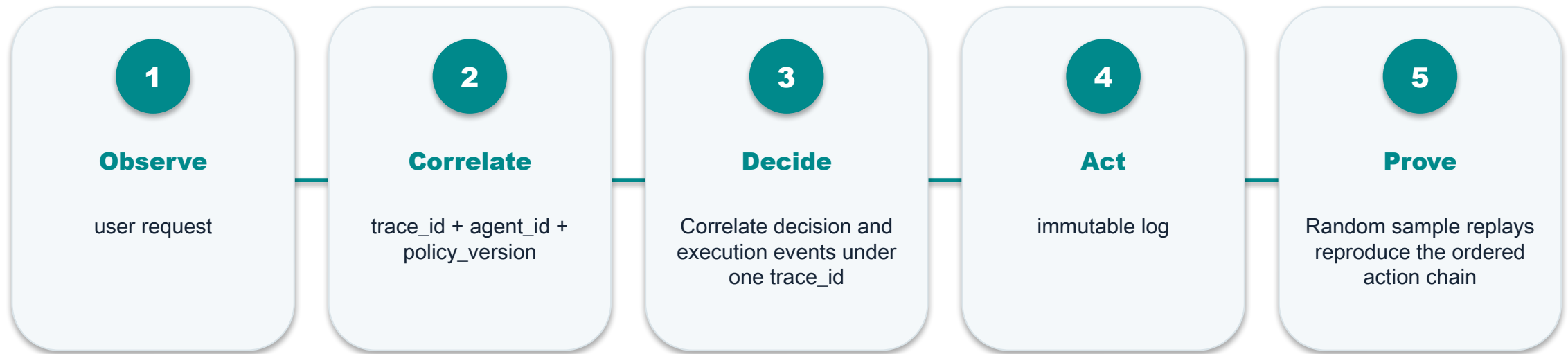
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Auditability objective: architecture and control boundary



Auditability objective: runtime decision trace



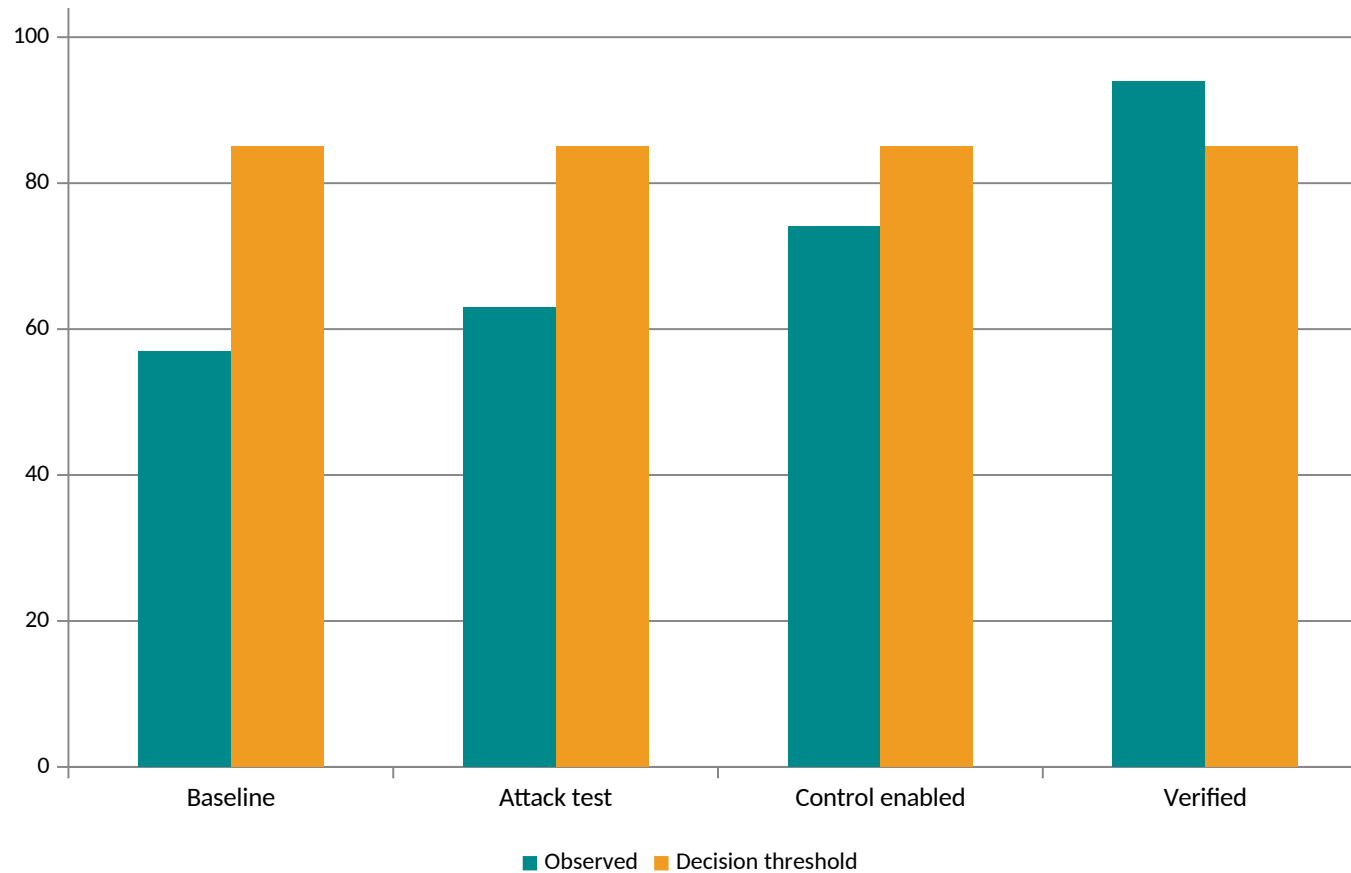
Failure to detect: Incident team cannot reconstruct why a tool ran

Auditability objective: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	trace_id	Required in decision and audit evidence
field_2	parent_span	Required in decision and audit evidence
field_3	actor	Required in decision and audit evidence
field_4	action	Required in decision and audit evidence
field_5	resource	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Auditability objective: evidence and acceptance decision



DECISION RULE

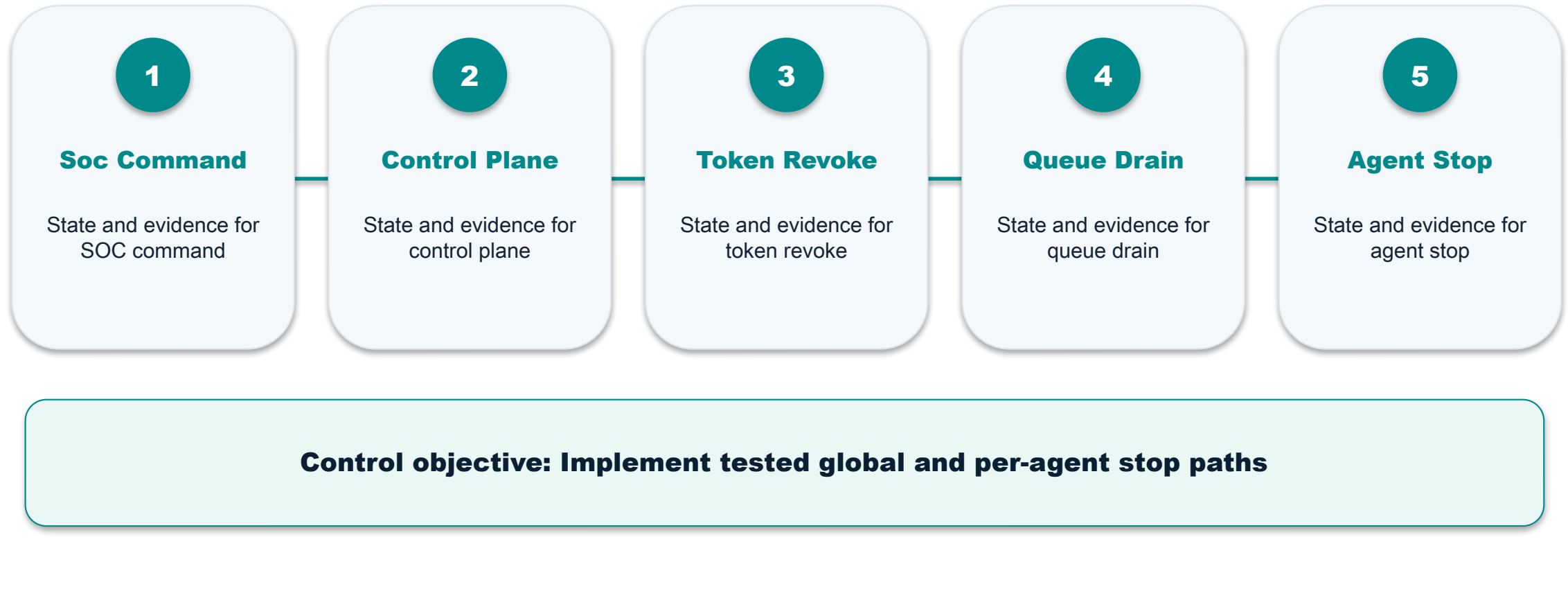
Accept only when Random sample replays reproduce the ordered action chain. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Kill-switch baseline: architecture and control boundary



Kill-switch baseline: runtime decision trace



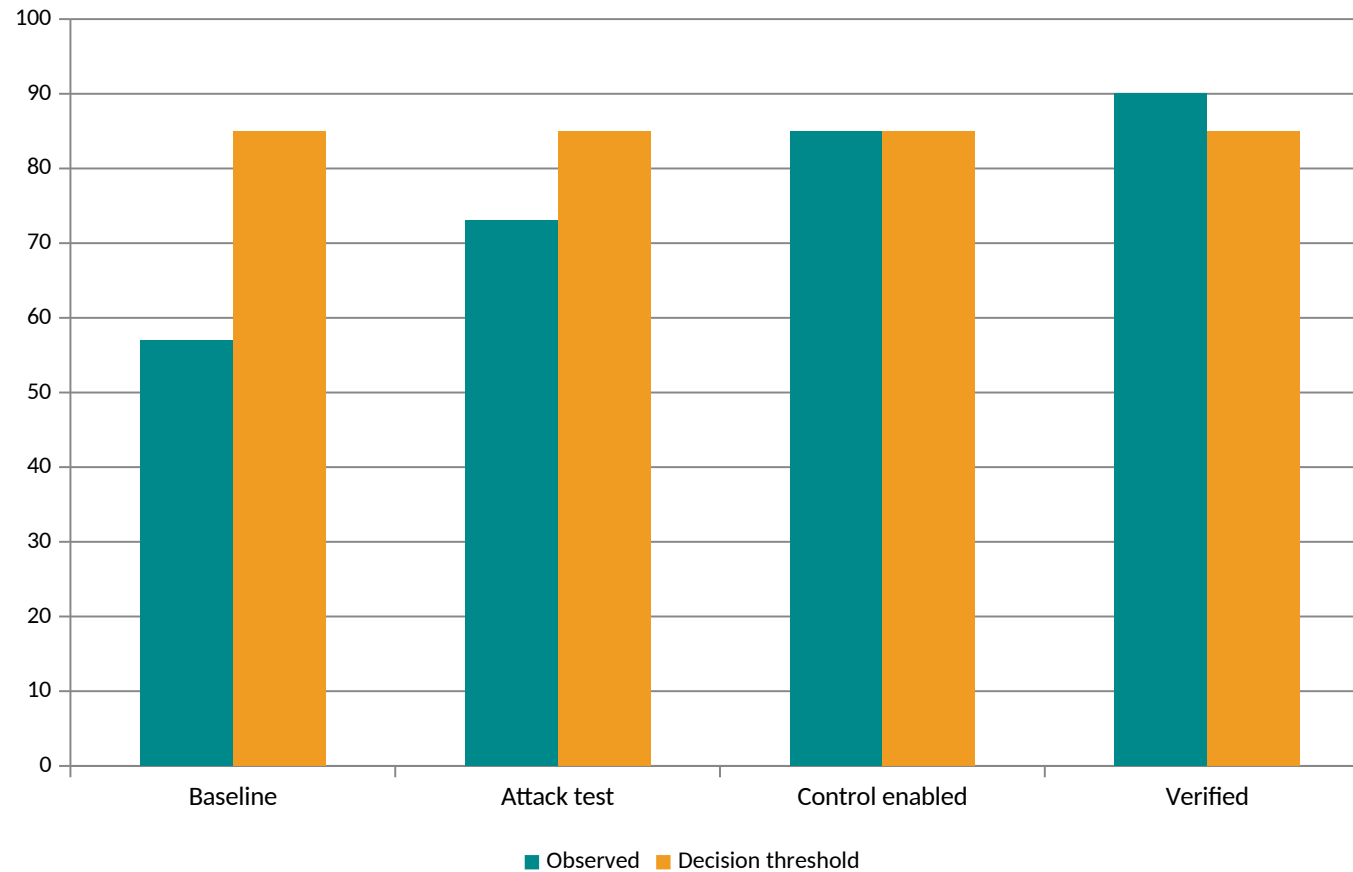
Failure to detect: Agent continues acting after a critical alert

Kill-switch baseline: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agent_id	Required in decision and audit evidence
field_2	scope	Required in decision and audit evidence
field_3	initiated_by	Required in decision and audit evidence
field_4	reason	Required in decision and audit evidence
field_5	effective_at	Required in decision and audit evidence
field_6	confirmation	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Kill-switch baseline: evidence and acceptance decision



DECISION RULE

Accept only when P95 containment time meets the defined target. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Activity 01 — Build an Agent Inventory and Security Objectives

1 Scenario

Use synthetic data in activity-01-agent-inventory to make a security-operations decision.

2 Input contract

mock_agents.json

3 Runnable evidence

activity.py

4 Output artifact

agent_security_baseline.md

5 Acceptance

All active agents have an owner, purpose, data class, risk tier and measurable security objective.

6 Safety boundary

Offline simulation only; no live credentials, systems or external services.

Activity 01: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Activity 01: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_agents.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	agent_security_baseline.md	Generated locally and reproducible
Criterion	A1	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete agent_security_baseline.md only	Retain script and mock data

Aligned activity files in labs/

Activity 02 — Threat-Model Agent Trust Boundaries

1 Scenario

Use synthetic data in activity-02-trust-boundary-threat-model to make a security-operations decision.

2 Input contract

mock_dataflows.json

3 Runnable evidence

activity.py

4 Output artifact

threat_model.json

5 Acceptance

Every cross-zone flow has authentication, authorisation, encryption and logging controls.

6 Safety boundary

Offline simulation only; no live credentials, systems or external services.

Activity 02: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Activity 02: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_dataflows.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	threat_model.json	Generated locally and reproducible
Criterion	A1	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete threat_model.json only	Retain script and mock data

Aligned activity files in labs/

Activity 03 — Enforce Tool and Data Guardrails

1 Scenario

Use synthetic data in activity-03-policy-as-code to make a security-operations decision.

2 Input contract

mock_requests.json

3 Runnable evidence

activity.py

4 Output artifact

policy_decisions.csv

5 Acceptance

Allowed requests execute; restricted data, unlisted tools and missing approvals are denied with reasons.

6 Safety boundary

Offline simulation only; no live credentials, systems or external services.

Activity 03: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Activity 03: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_requests.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	policy_decisions.csv	Generated locally and reproducible
Criterion	A2, A3	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete policy_decisions.csv only	Retain script and mock data

Aligned activity files in labs/

Topic 01 technical recap

1

Confidentiality objective

Leak test score stays below the approved threshold

2

Integrity objective

Approved and executed argument hashes match

3

Authenticity objective

Replay and wrong-audience tests are rejected

4

Availability objective

Error budget and queue latency remain inside SLO

5

Auditability objective

Random sample replays reproduce the ordered action chain

6

Kill-switch baseline

P95 containment time meets the defined target

TOPIC 02

Agent Security Policies, Governance and Compliance

Policy-as-code · evidence · permissions · assurance · communication

LO2 · K3, K4, A3, A4



LO2 and assessment criteria

1

K3

Best practices in information security policies

2

K4

Best practices and emerging technologies in security control

3

A3

Manage compliance with information-security guidelines and classification or permission rules

4

A4

Lead communication of security goals and objectives to the organisation

5

LO2

LO2: Communicate effective agent-security policies and practices, and manage compliance with best practices and technological advancements.

Agent Security Policies, Governance and Compliance: operating model



Every mechanism in this topic is taught through all five layers.

Policy hierarchy: architecture and control boundary



Policy hierarchy: runtime decision trace



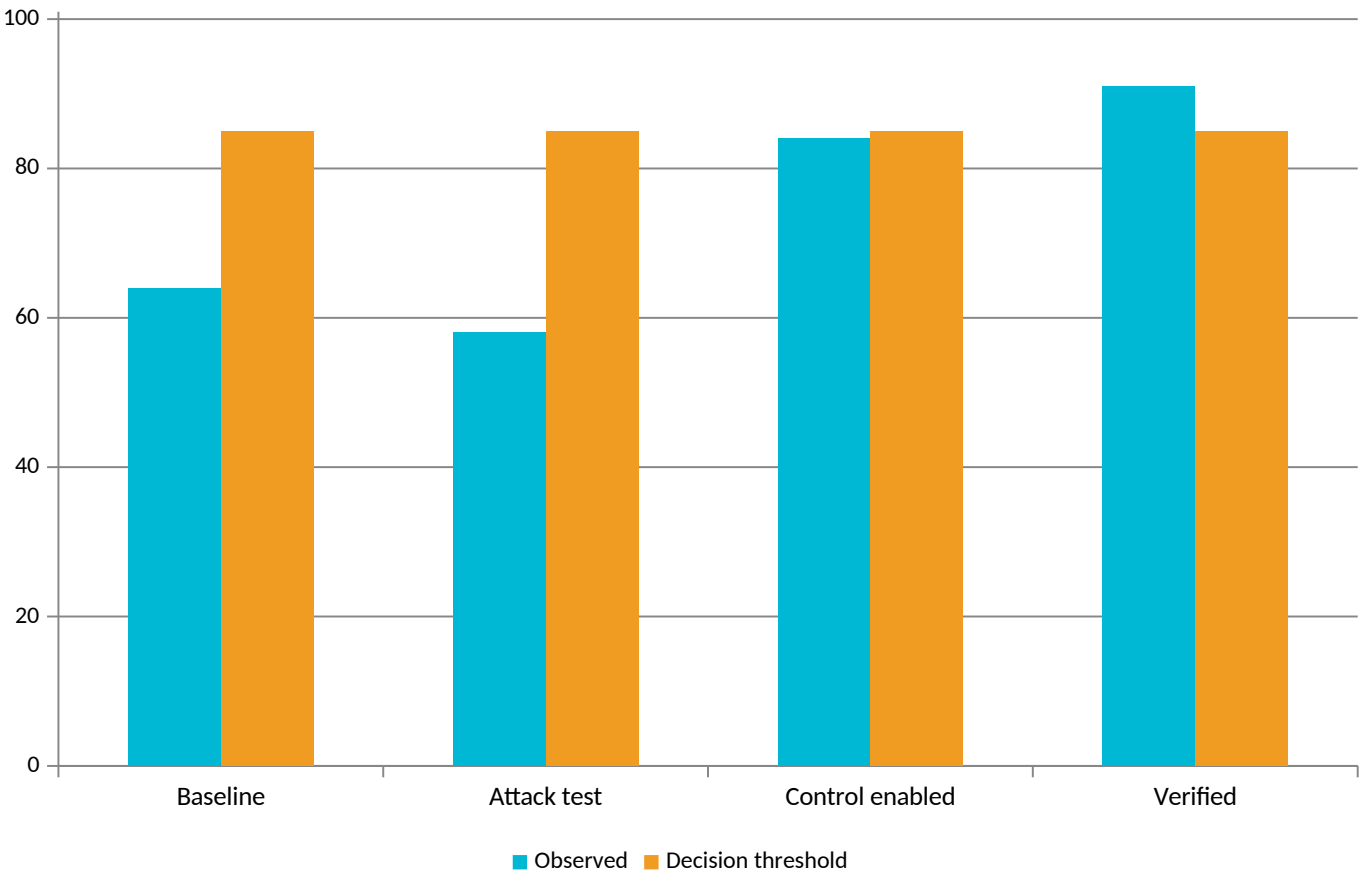
Failure to detect: Runtime rules contradict enterprise requirements

Policy hierarchy: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	policy_id	Required in decision and audit evidence
field_2	parent_id	Required in decision and audit evidence
field_3	scope	Required in decision and audit evidence
field_4	owner	Required in decision and audit evidence
field_5	version	Required in decision and audit evidence
field_6	effective_date	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Policy hierarchy: evidence and acceptance decision



DECISION RULE

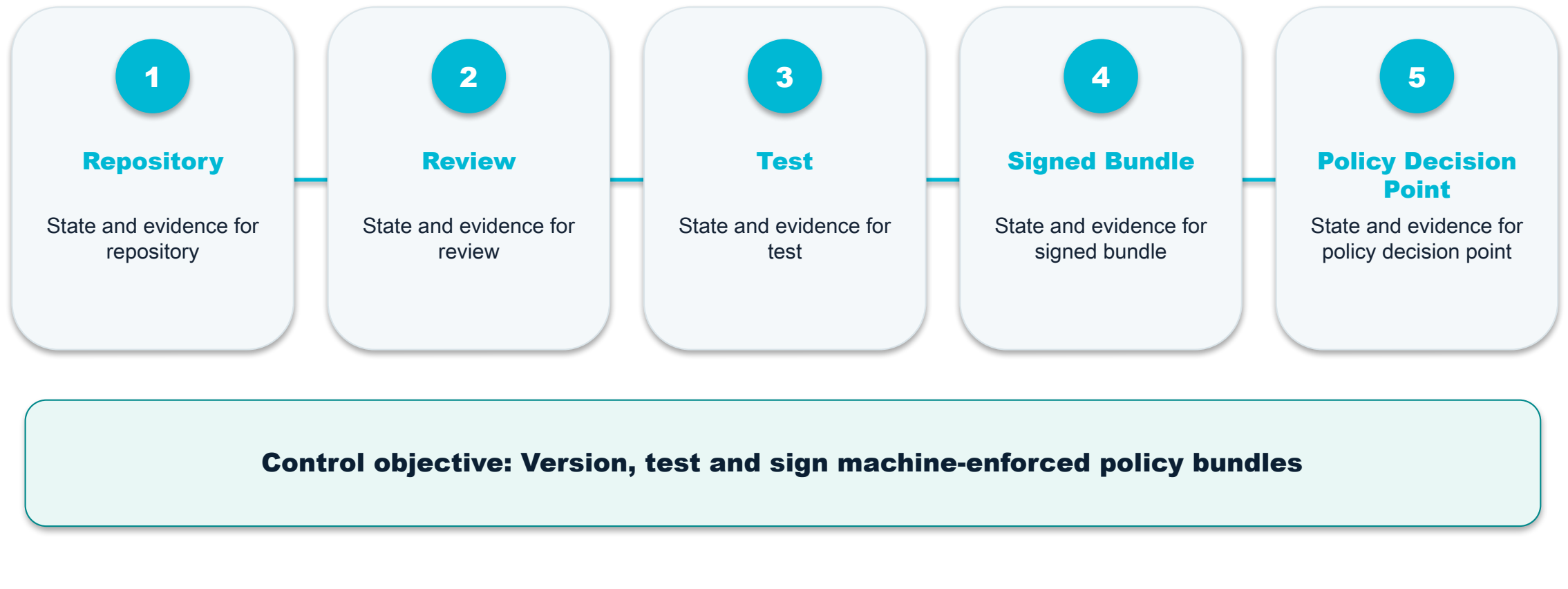
Accept only when No active rule has an orphaned parent_id. Otherwise record a gap, owner and due date.

EVIDENCE STATUS



Synthetic training evidence; values are illustrative, not external performance claims

Policy-as-code: architecture and control boundary



Policy-as-code: runtime decision trace



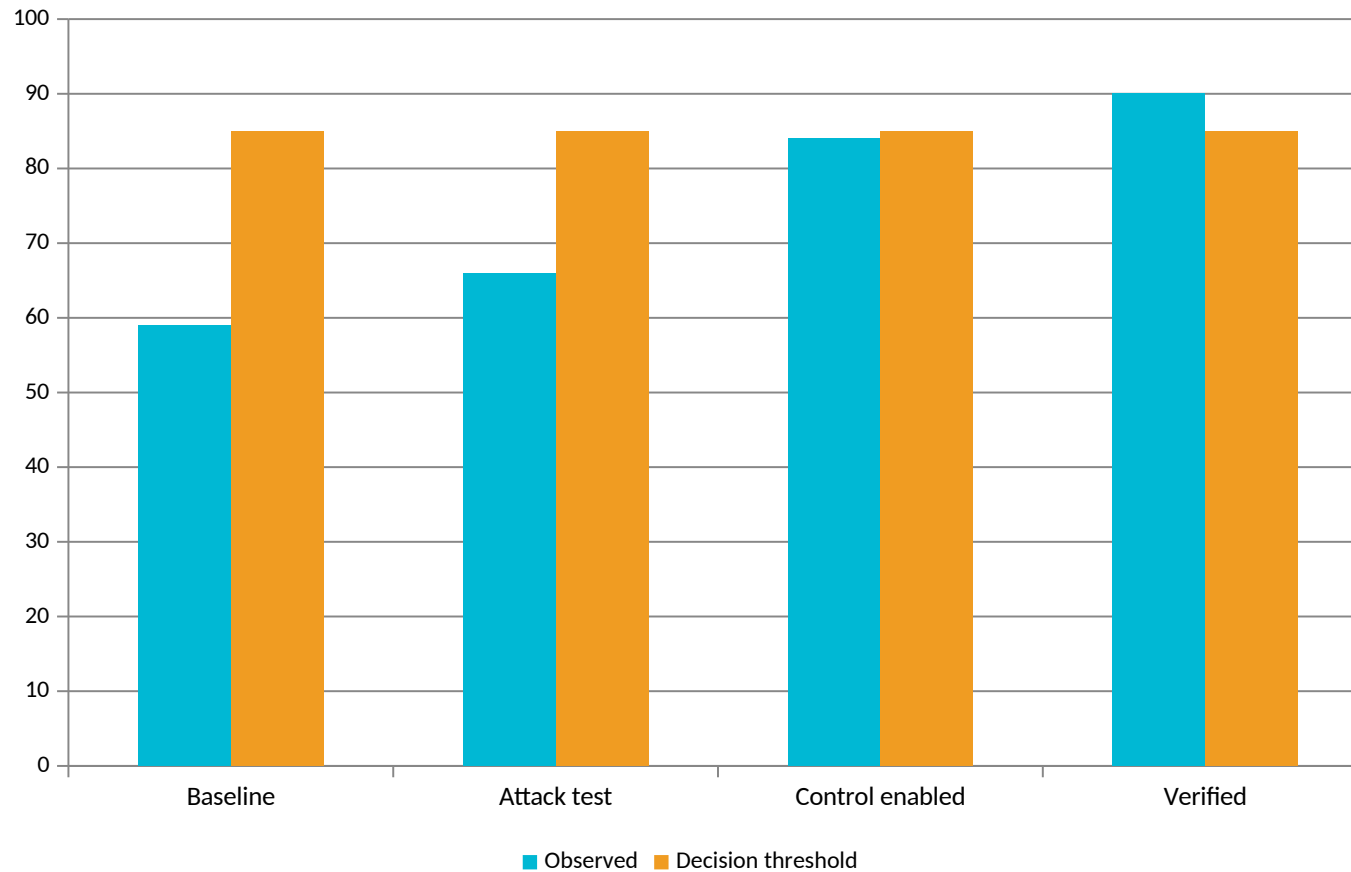
Failure to detect: Manual configuration drifts from approved policy

Policy-as-code: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	rule_id	Required in decision and audit evidence
field_2	input_schema	Required in decision and audit evidence
field_3	effect	Required in decision and audit evidence
field_4	conditions[]	Required in decision and audit evidence
field_5	tests[]	Required in decision and audit evidence
field_6	bundle_hash	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Policy-as-code: evidence and acceptance decision



DECISION RULE

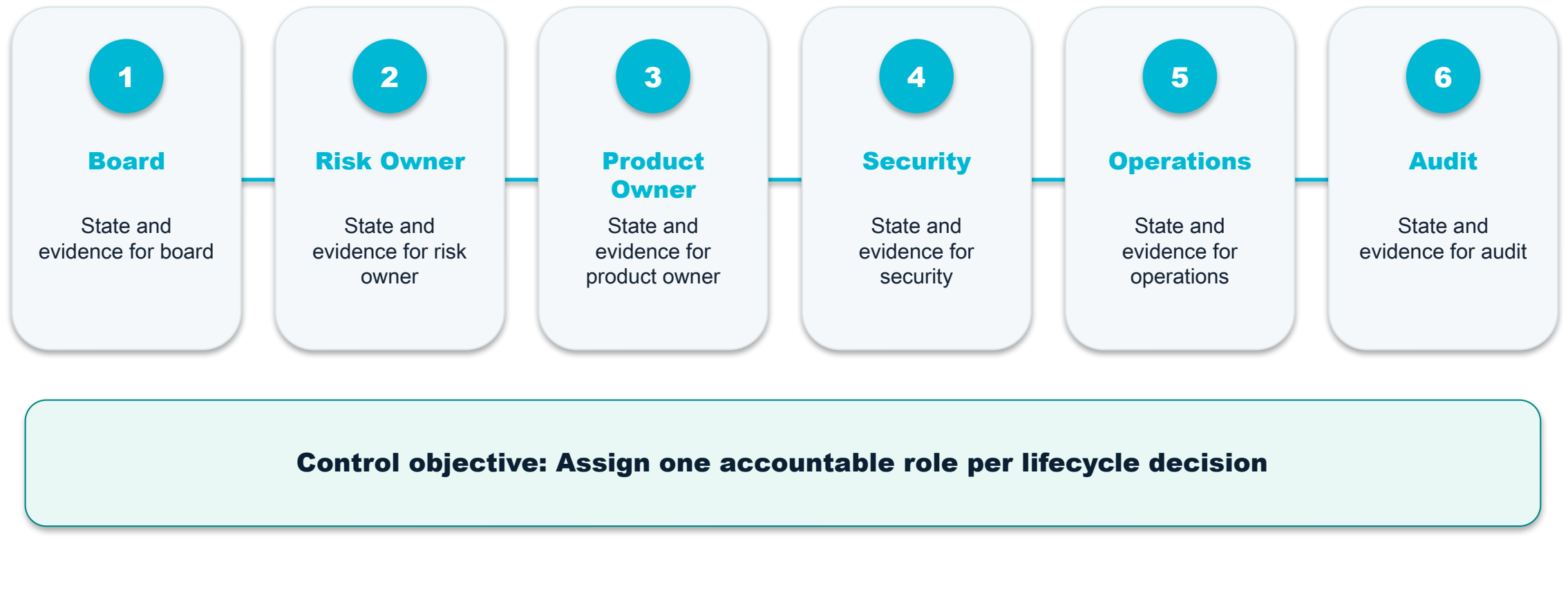
Accept only when Policy tests pass and deployed hash matches release. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Governance RACI: architecture and control boundary



Governance RACI: runtime decision trace



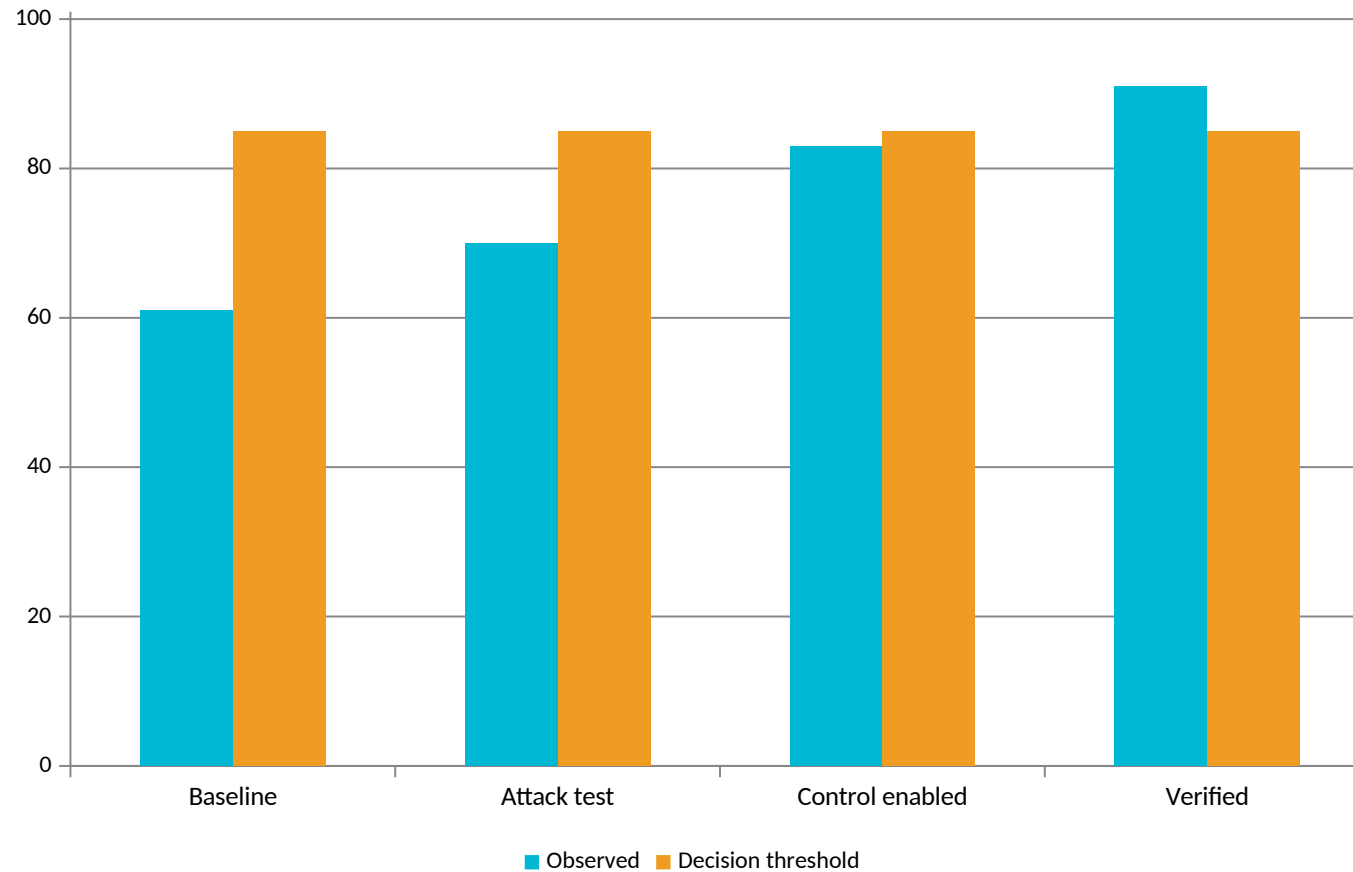
Failure to detect: Two teams assume the other owns a risky agent

Governance RACI: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	decision	Required in decision and audit evidence
field_2	accountable	Required in decision and audit evidence
field_3	responsible	Required in decision and audit evidence
field_4	consulted[]	Required in decision and audit evidence
field_5	informed[]	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Governance RACI: evidence and acceptance decision



DECISION RULE

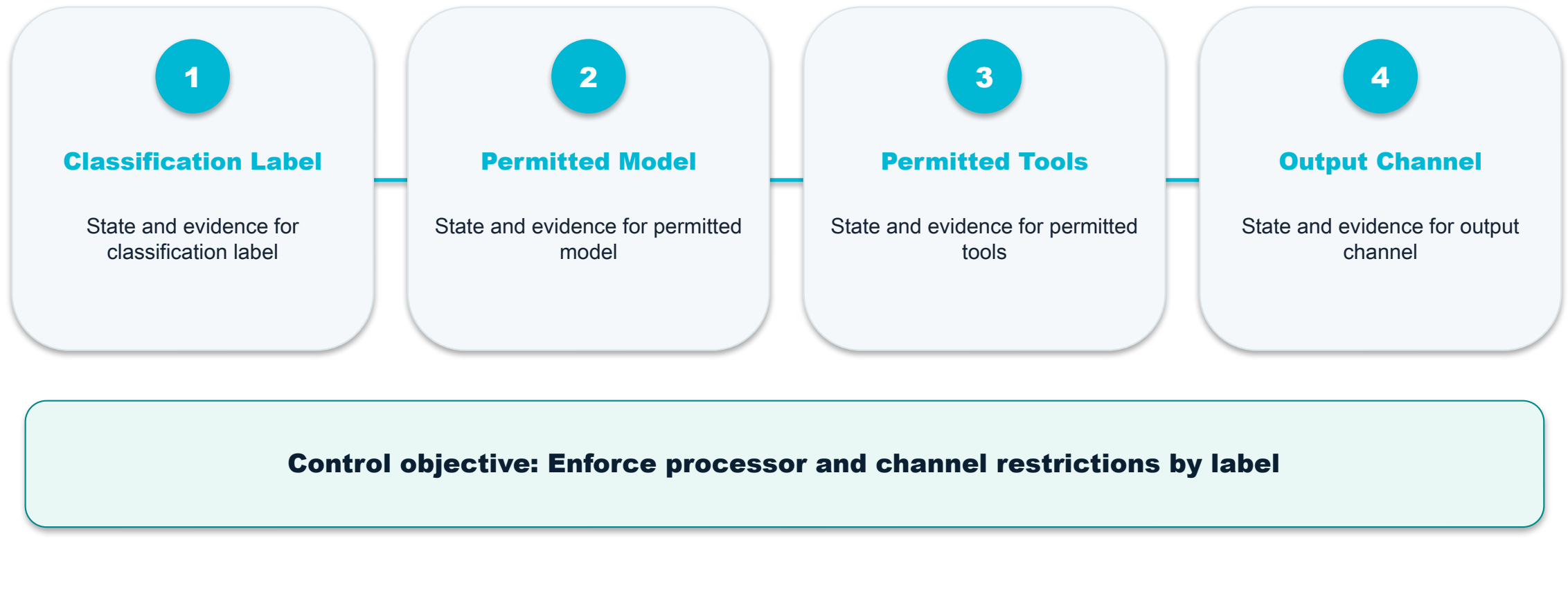
Accept only when Every critical decision has exactly one accountable owner. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

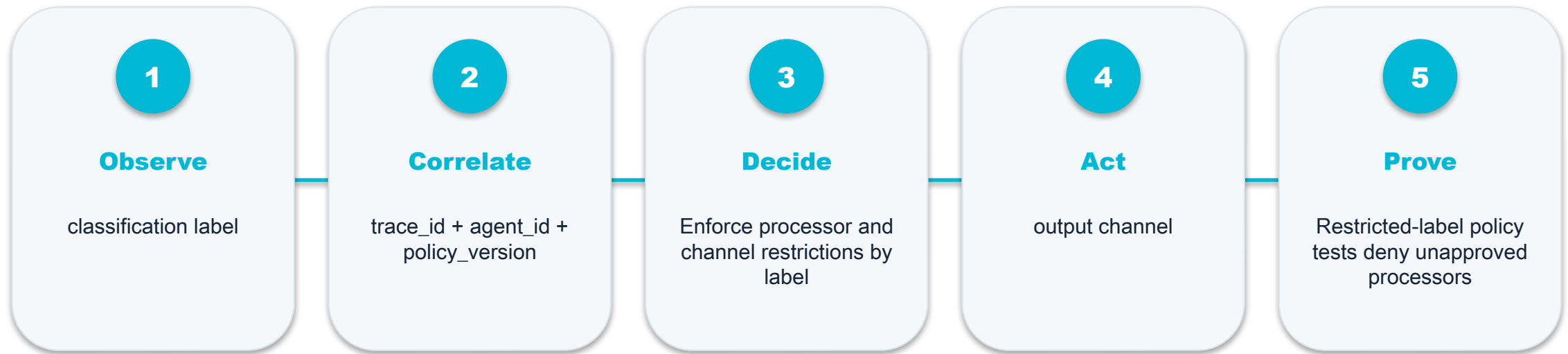
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Information handling rules: architecture and control boundary



Information handling rules: runtime decision trace



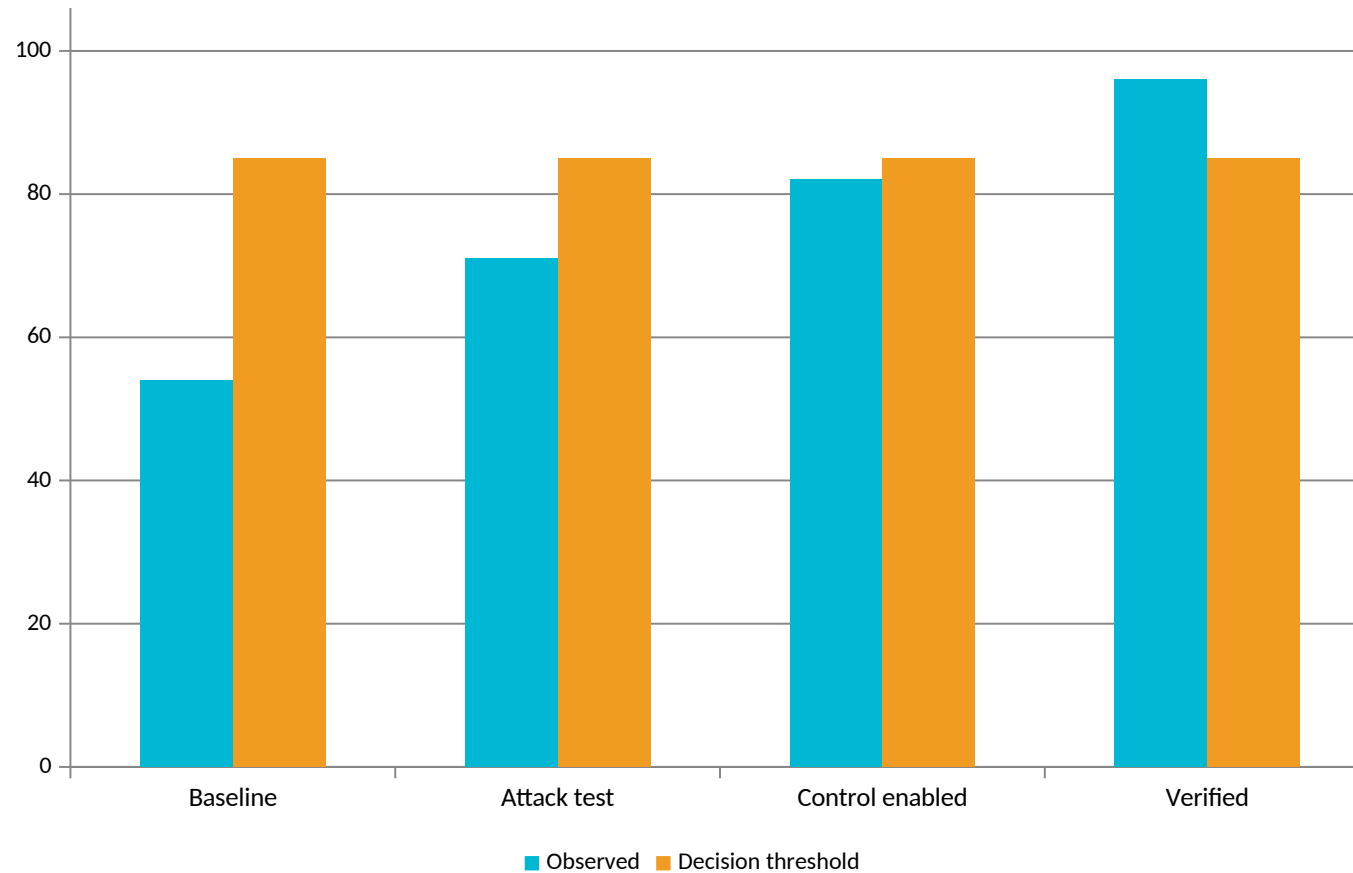
Failure to detect: Restricted records flow to a consumer-grade model endpoint

Information handling rules: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	class	Required in decision and audit evidence
field_2	allowed_processors[]	Required in decision and audit evidence
field_3	allowed_actions[]	Required in decision and audit evidence
field_4	retention_days	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Information handling rules: evidence and acceptance decision



DECISION RULE

Accept only when Restricted-label policy tests deny unapproved processors. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

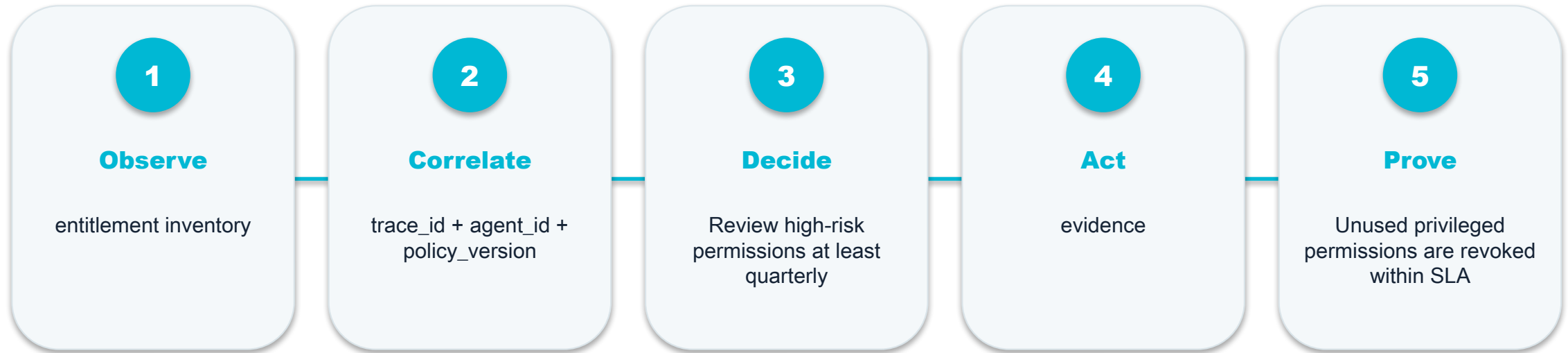
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Permission recertification: architecture and control boundary



Permission recertification: runtime decision trace



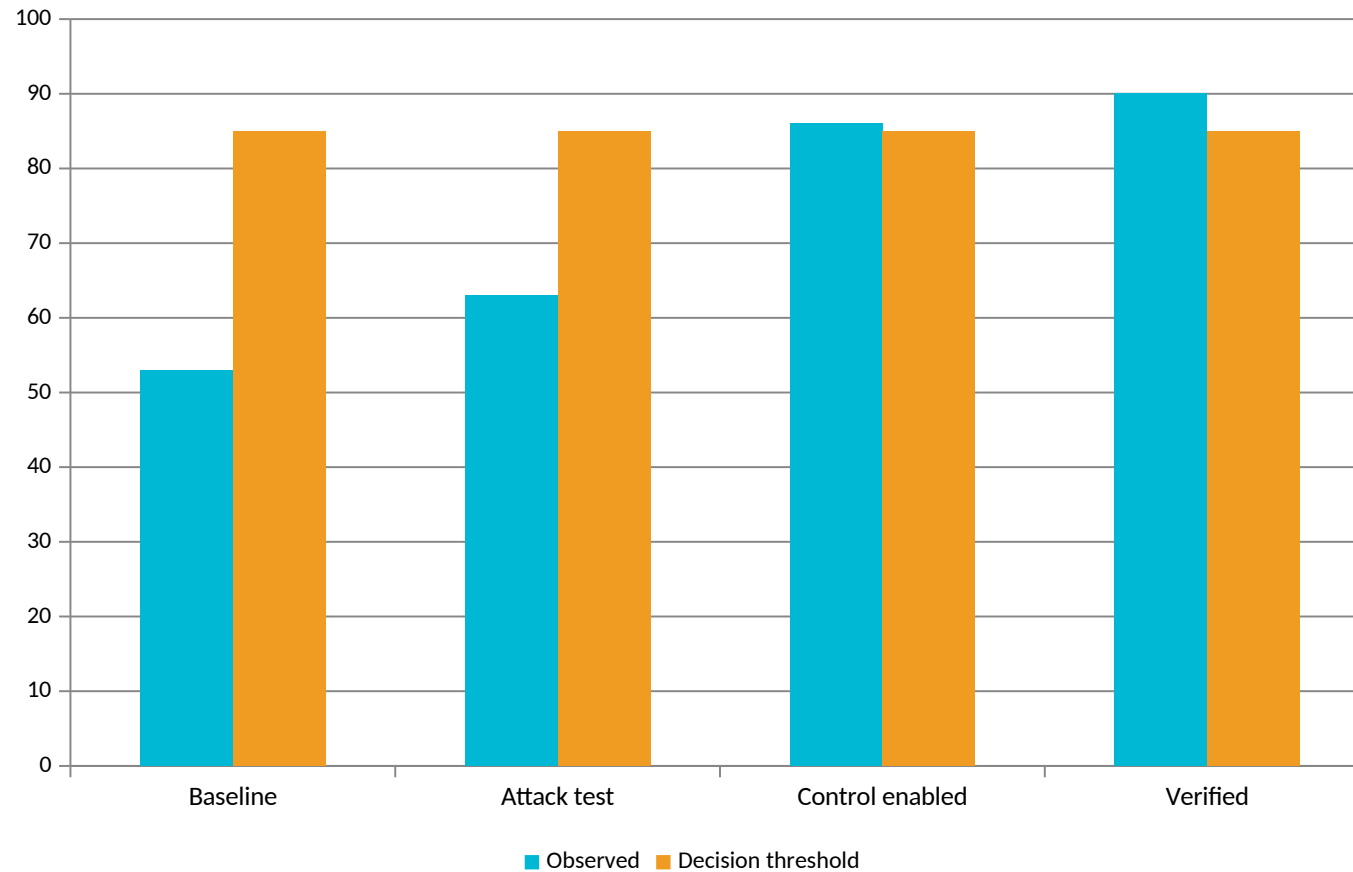
Failure to detect: Dormant high-privilege tool scopes remain active

Permission recertification: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	principal	Required in decision and audit evidence
field_2	permission	Required in decision and audit evidence
field_3	resource	Required in decision and audit evidence
field_4	last_used	Required in decision and audit evidence
field_5	reviewer	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Permission recertification: evidence and acceptance decision



DECISION RULE

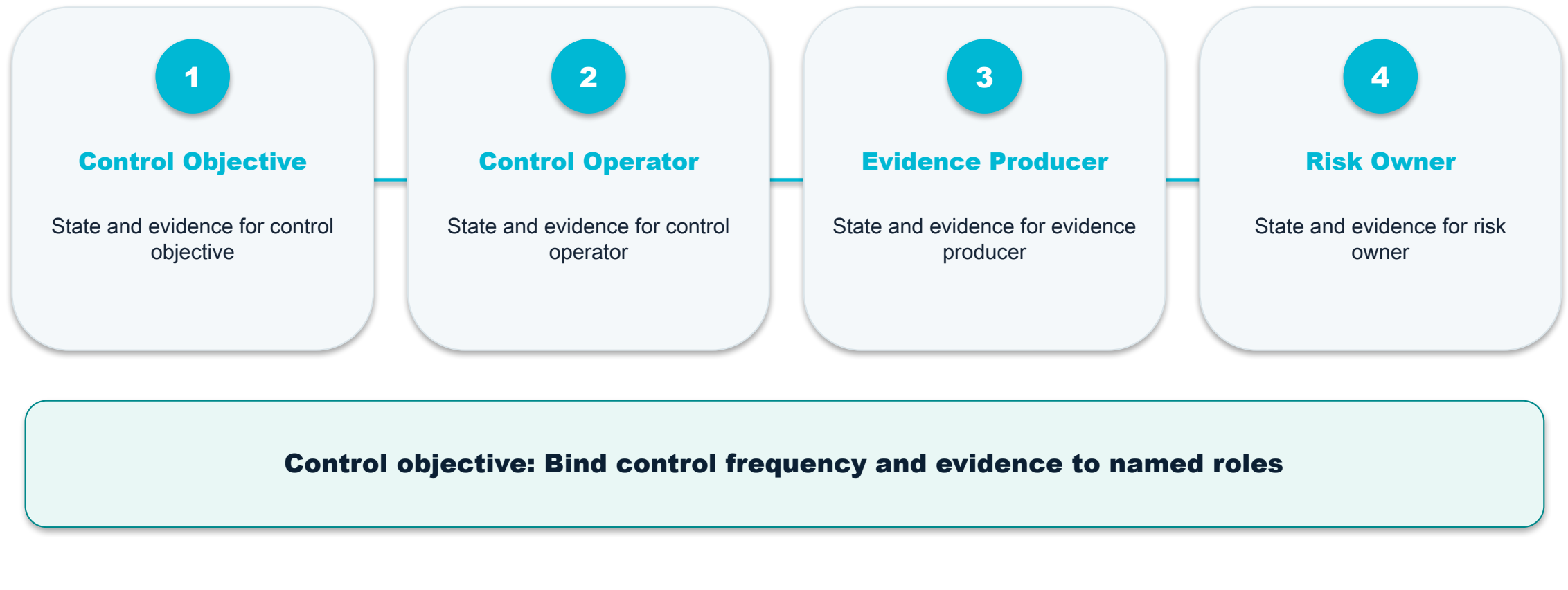
Accept only when Unused privileged permissions are revoked within SLA. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

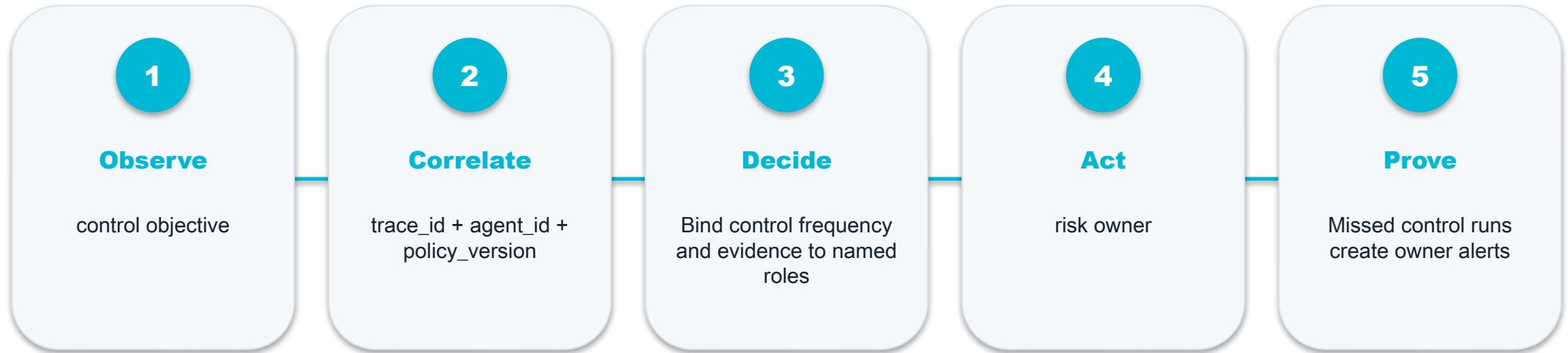
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Control ownership: architecture and control boundary



Control ownership: runtime decision trace



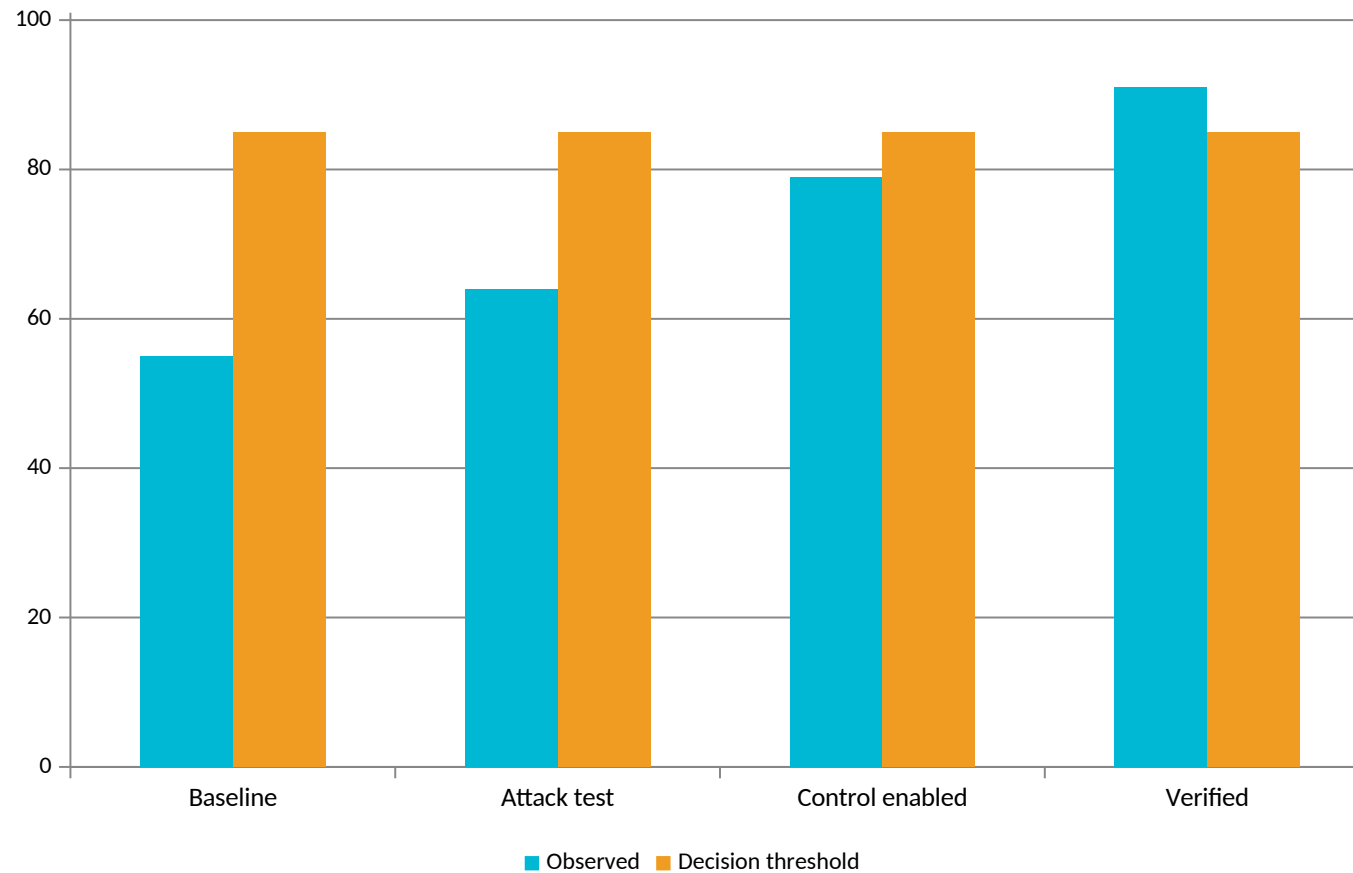
Failure to detect: A control exists on paper but nobody operates it

Control ownership: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	control_id	Required in decision and audit evidence
field_2	operator	Required in decision and audit evidence
field_3	frequency	Required in decision and audit evidence
field_4	evidence	Required in decision and audit evidence
field_5	exception_route	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Control ownership: evidence and acceptance decision



DECISION RULE

**Accept only when
Missed control runs
create owner alerts.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

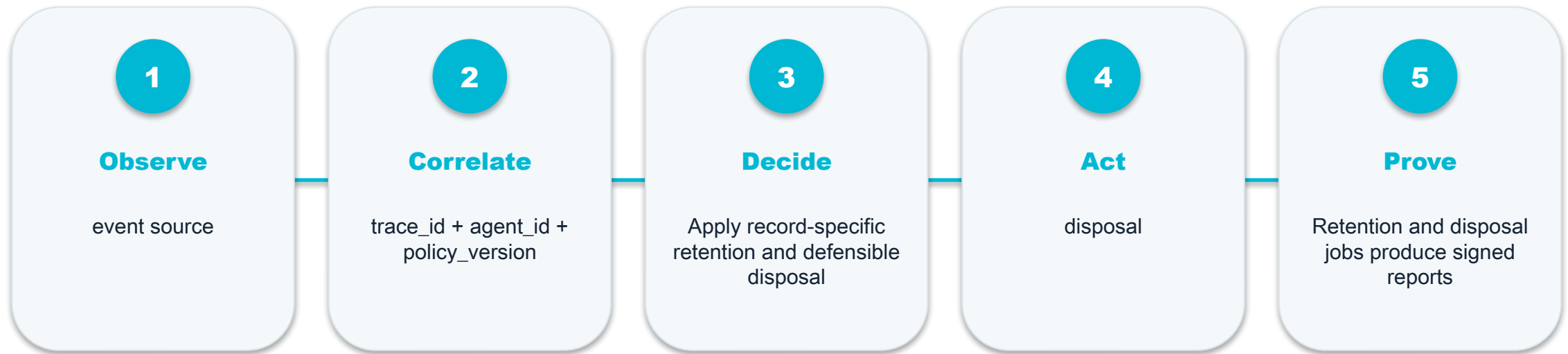
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Evidence retention: architecture and control boundary



Evidence retention: runtime decision trace



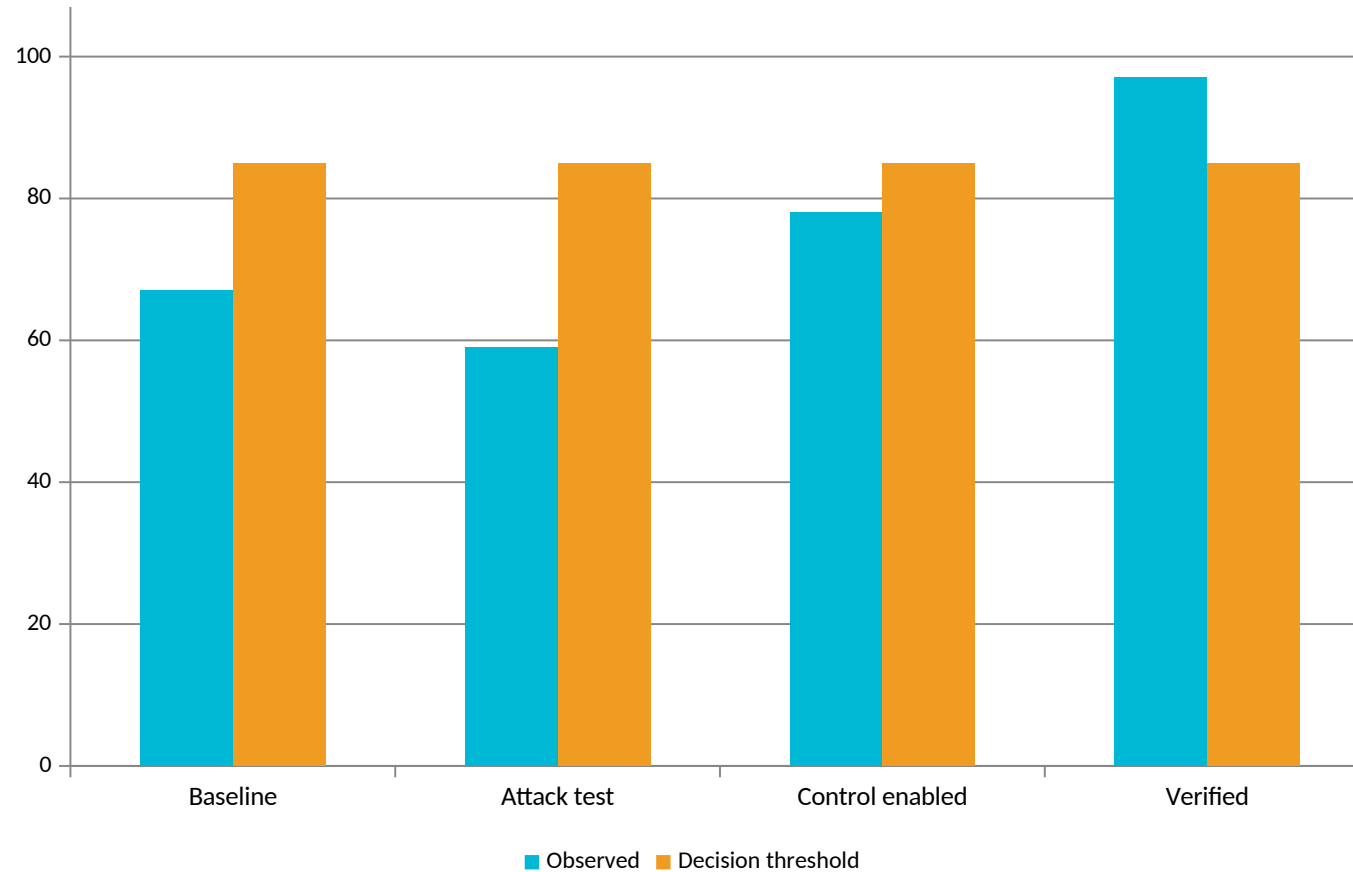
Failure to detect: Evidence is deleted before an investigation or kept indefinitely

Evidence retention: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	record_type	Required in decision and audit evidence
field_2	retention_days	Required in decision and audit evidence
field_3	legal_hold	Required in decision and audit evidence
field_4	hash	Required in decision and audit evidence
field_5	disposal_date	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Evidence retention: evidence and acceptance decision



DECISION RULE

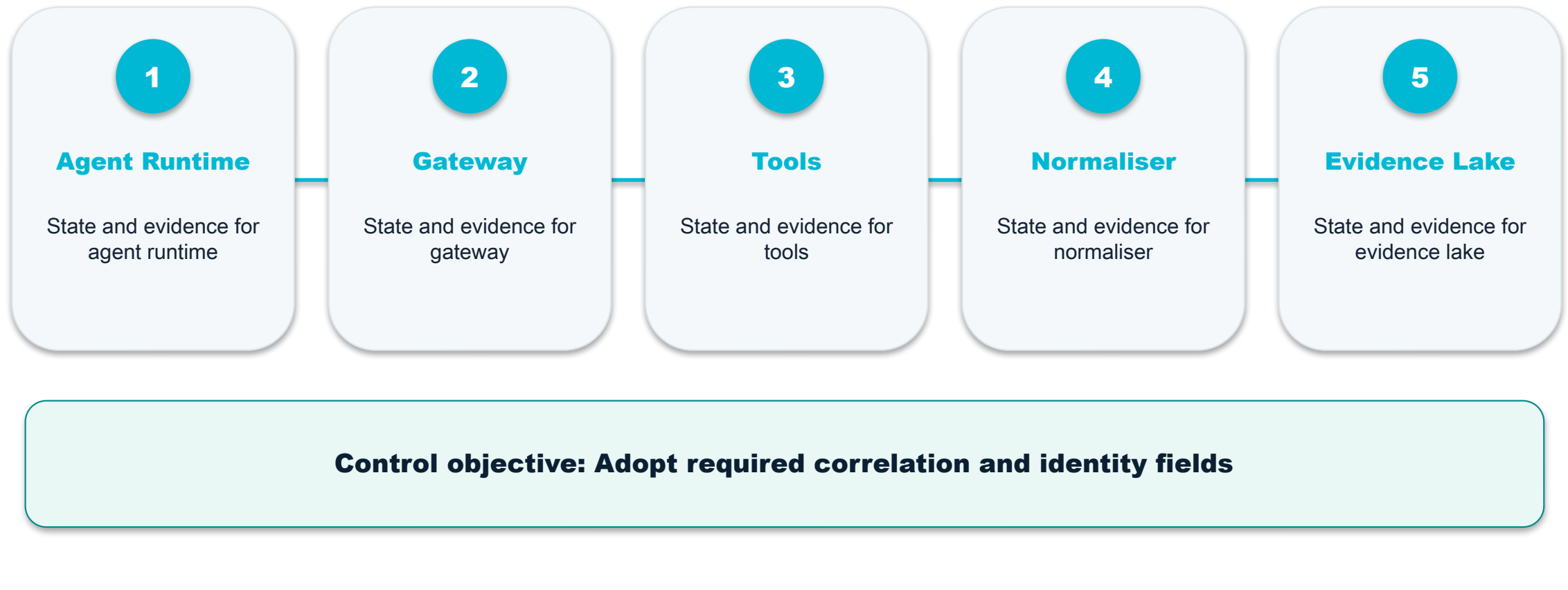
Accept only when Retention and disposal jobs produce signed reports. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

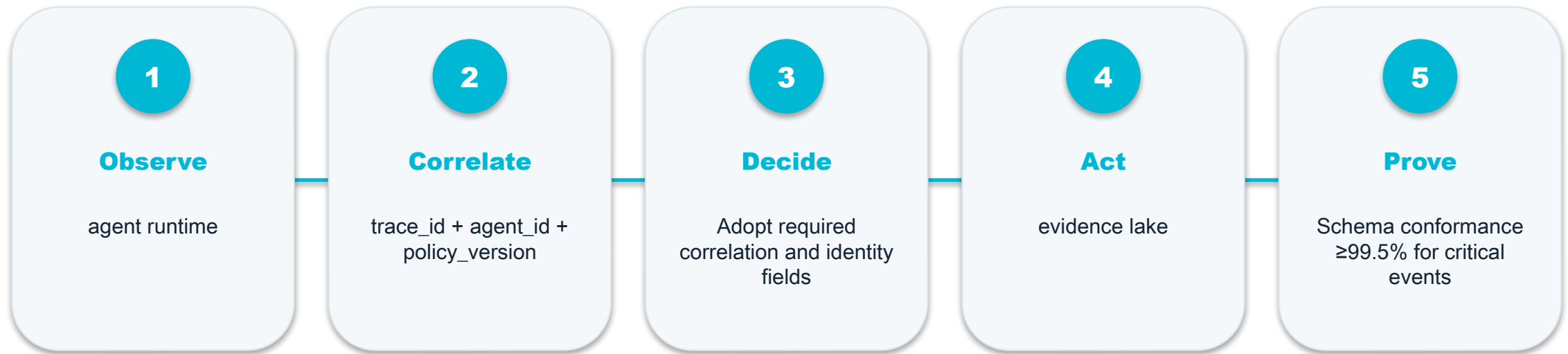
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Audit event schema: architecture and control boundary



Audit event schema: runtime decision trace



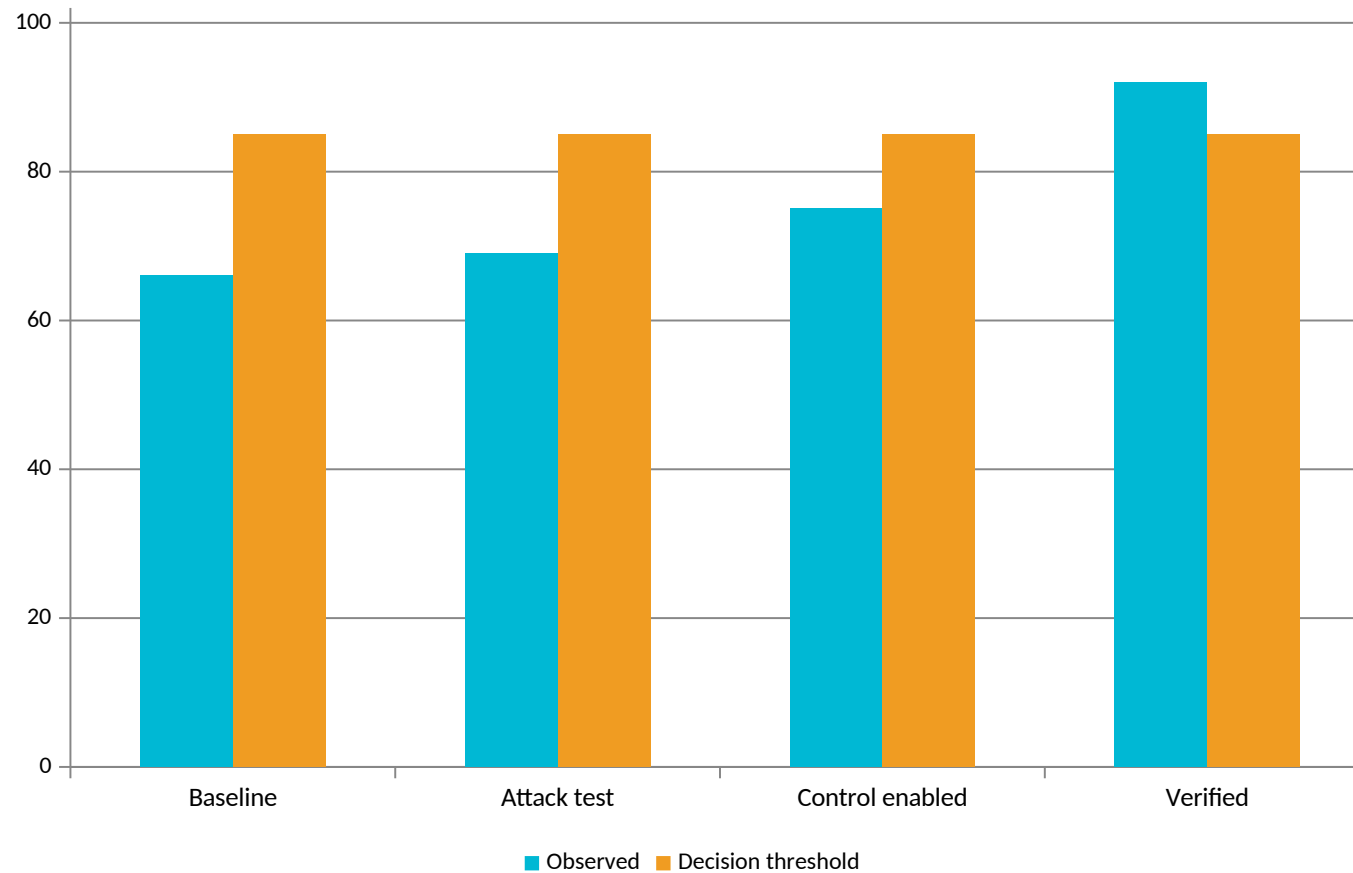
Failure to detect: Logs cannot be joined across components

Audit event schema: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	event_time	Required in decision and audit evidence
field_2	trace_id	Required in decision and audit evidence
field_3	agent_id	Required in decision and audit evidence
field_4	tool_id	Required in decision and audit evidence
field_5	action	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Audit event schema: evidence and acceptance decision



DECISION RULE

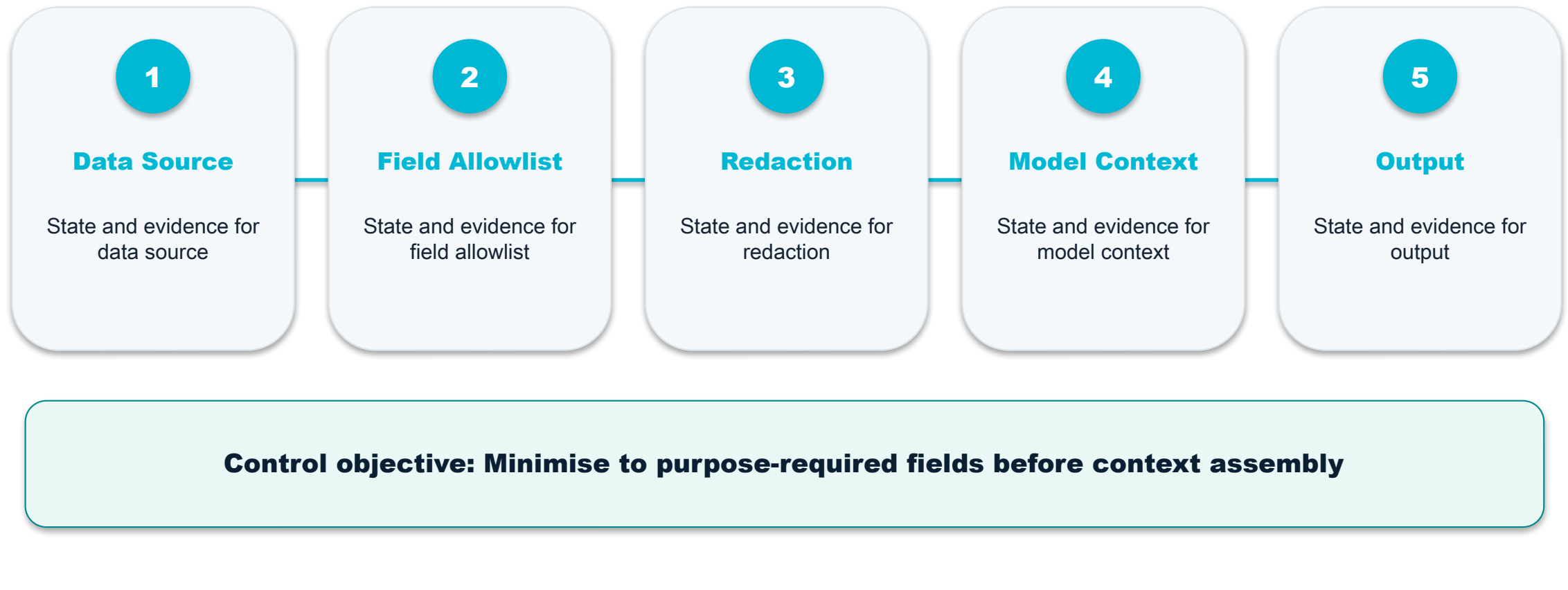
Accept only when Schema conformance $\geq 99.5\%$ for critical events. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

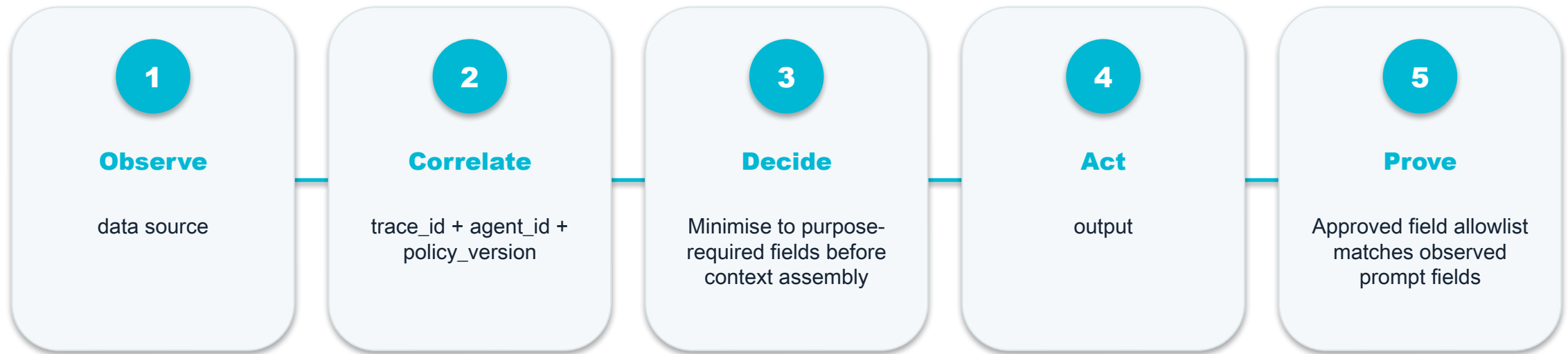
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Privacy minimisation: architecture and control boundary



Privacy minimisation: runtime decision trace



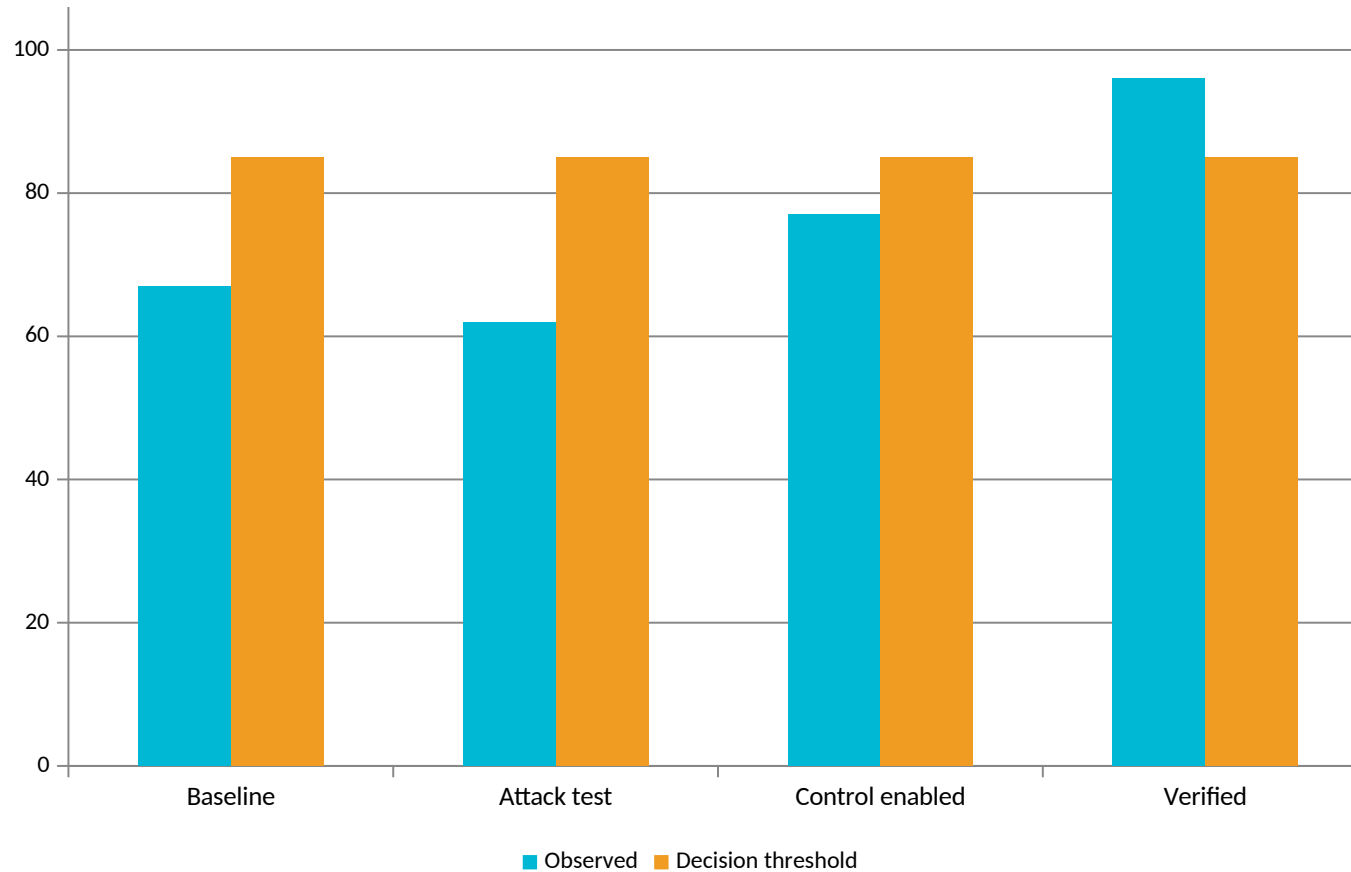
Failure to detect: Entire customer record is sent for a narrow task

Privacy minimisation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	field	Required in decision and audit evidence
field_2	purpose	Required in decision and audit evidence
field_3	necessity	Required in decision and audit evidence
field_4	transformation	Required in decision and audit evidence
field_5	retention	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Privacy minimisation: evidence and acceptance decision



DECISION RULE

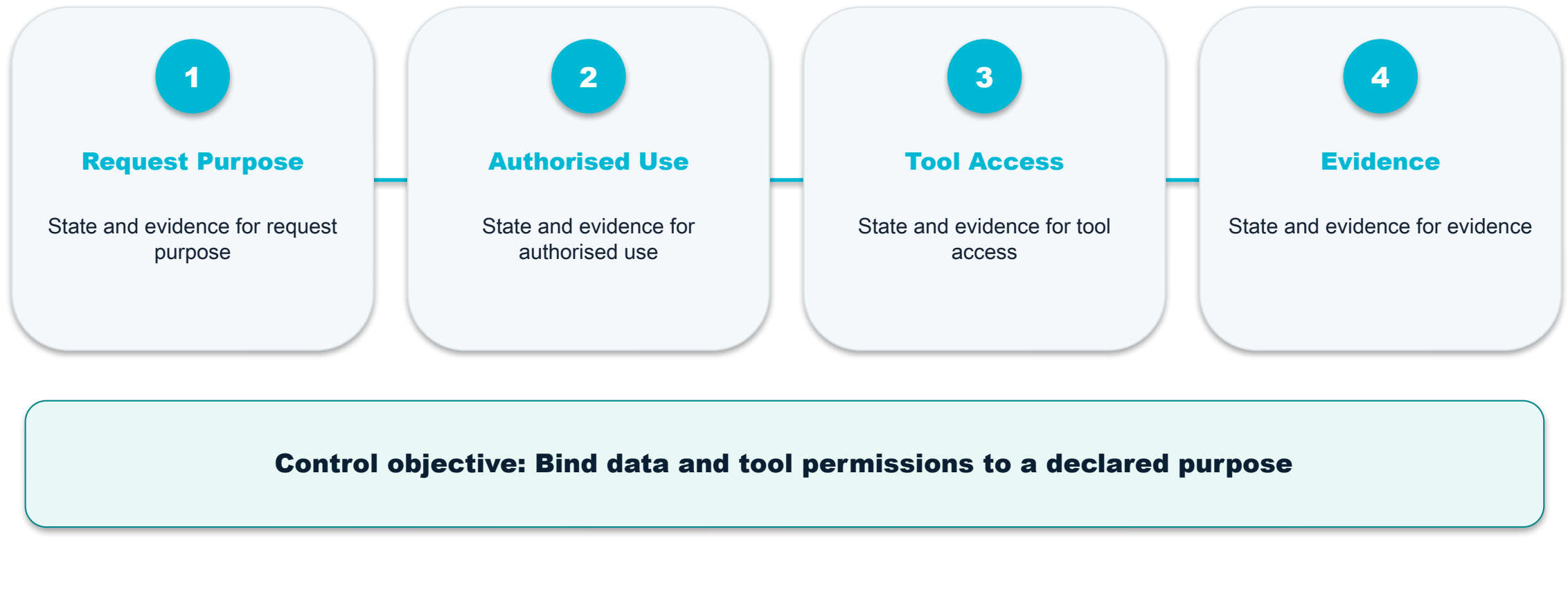
**Accept only when
Approved field allowlist
matches observed
prompt fields.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

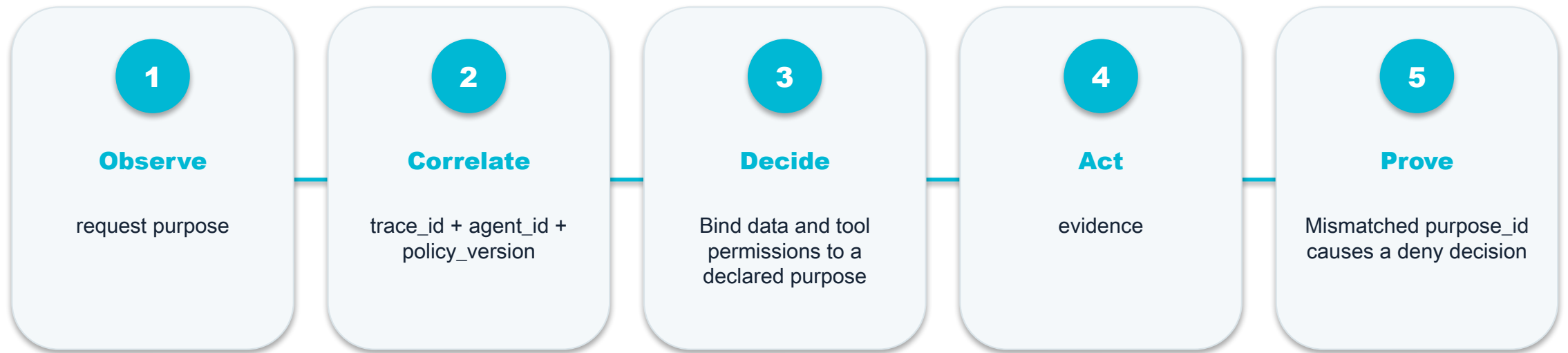
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Purpose limitation: architecture and control boundary



Purpose limitation: runtime decision trace



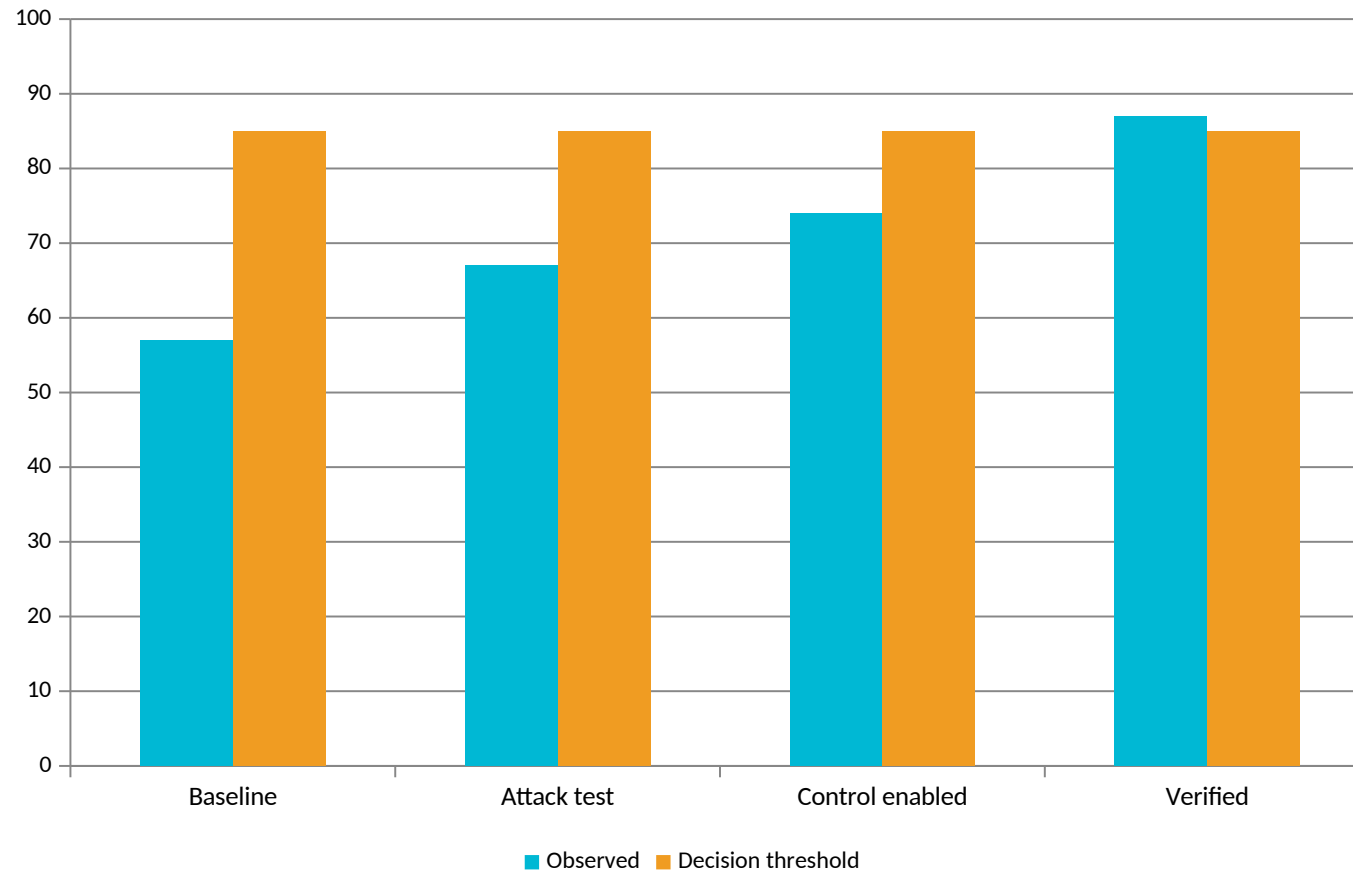
Failure to detect: Agent reuses data for an unrelated task

Purpose limitation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	purpose_id	Required in decision and audit evidence
field_2	lawful_basis	Required in decision and audit evidence
field_3	allowed_data[]	Required in decision and audit evidence
field_4	allowed_actions[]	Required in decision and audit evidence
field_5	expiry	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Purpose limitation: evidence and acceptance decision



DECISION RULE

Accept only when Mismatched purpose_id causes a deny decision. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Change management: architecture and control boundary



Control objective: Treat prompts, policies and tools as controlled changes

Change management: runtime decision trace



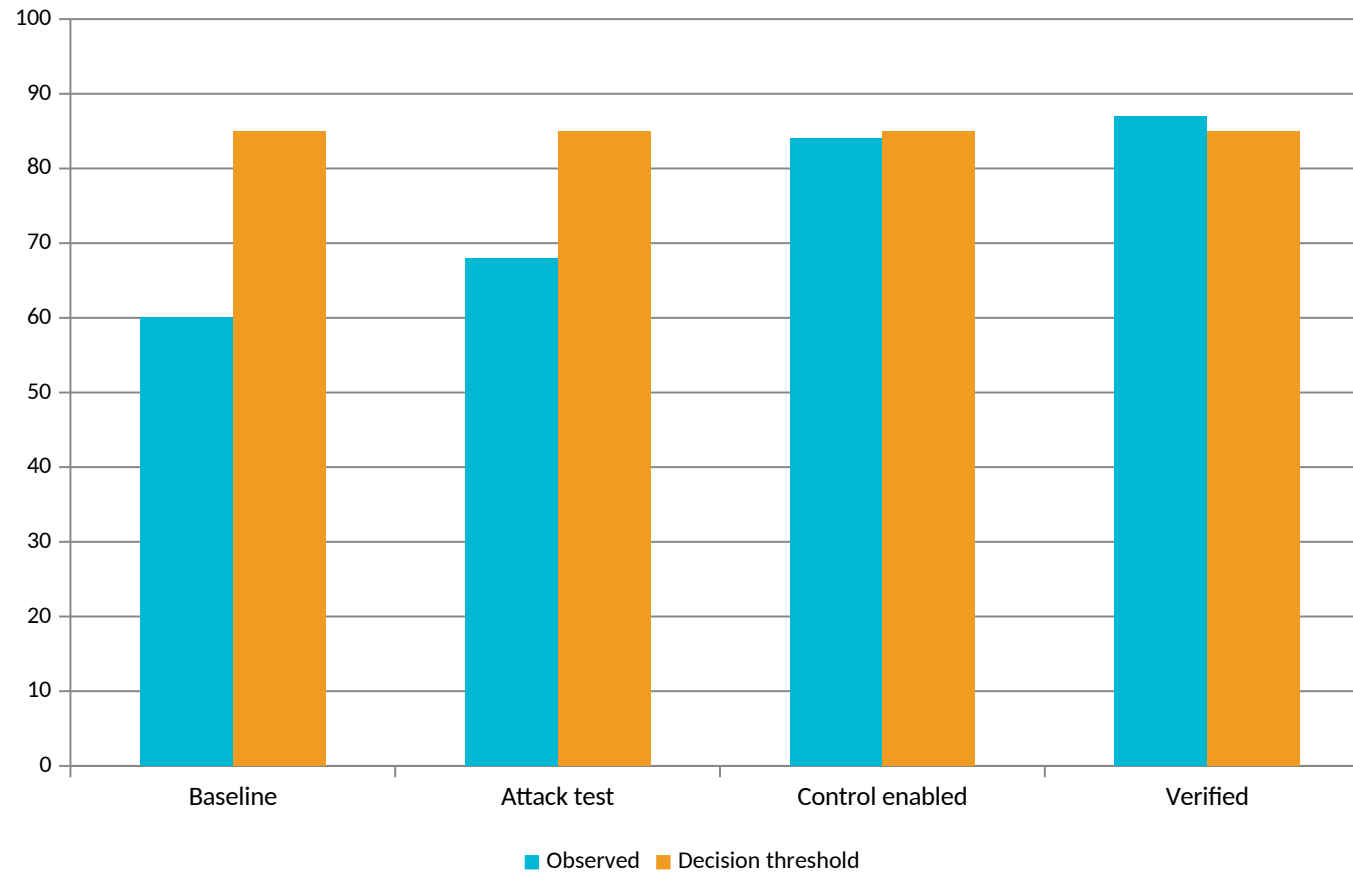
Failure to detect: Prompt or tool change silently expands agency

Change management: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	change_id	Required in decision and audit evidence
field_2	components[]	Required in decision and audit evidence
field_3	risk	Required in decision and audit evidence
field_4	tests[]	Required in decision and audit evidence
field_5	approver	Required in decision and audit evidence
field_6	rollback_ref	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Change management: evidence and acceptance decision



DECISION RULE

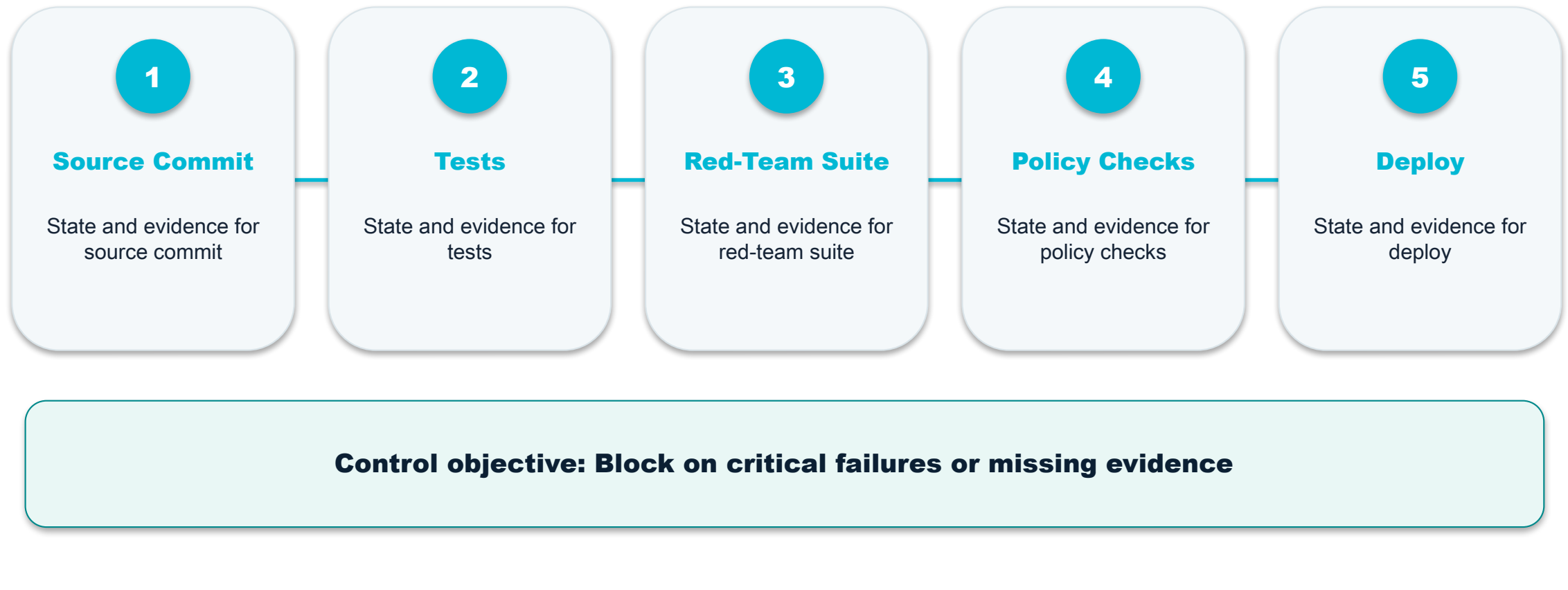
Accept only when Every production diff maps to a change_id and test run. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

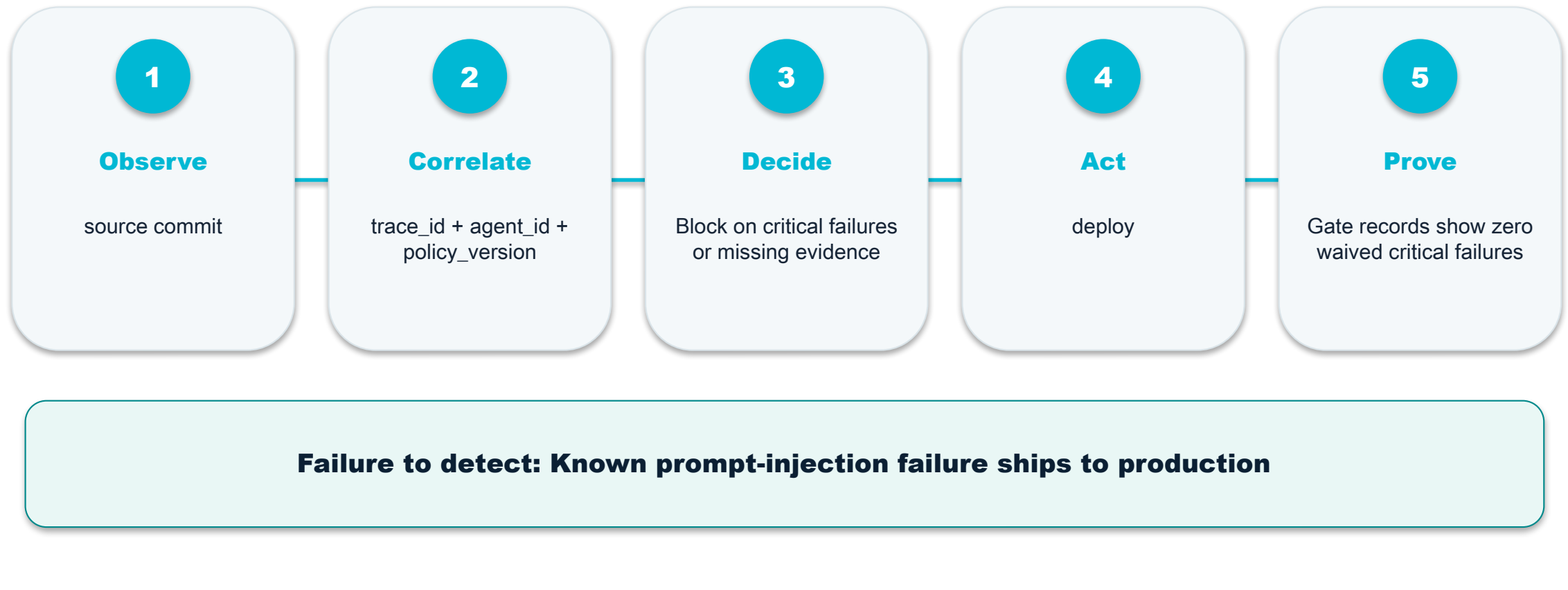
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Release gate: architecture and control boundary



Release gate: runtime decision trace

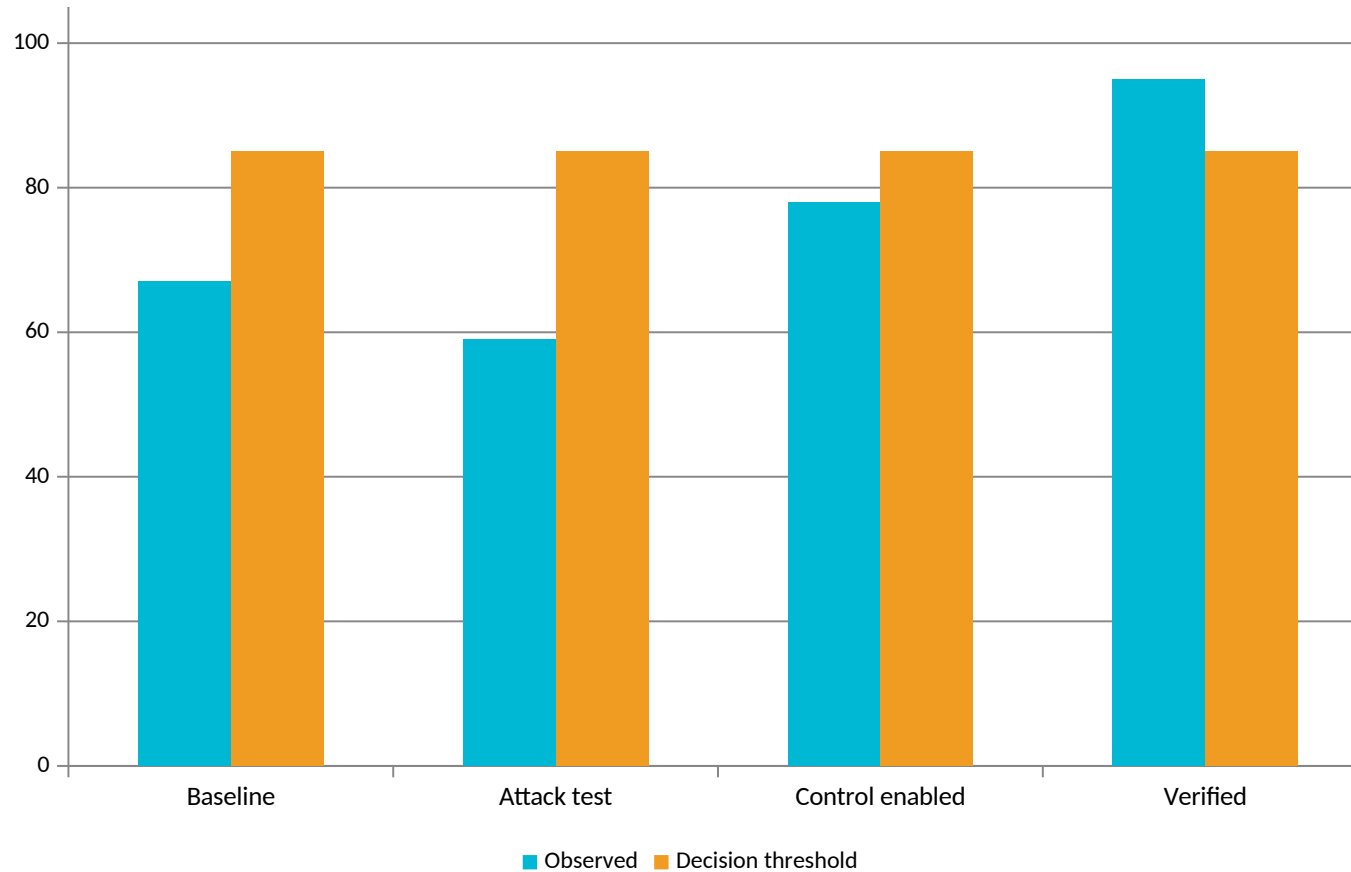


Release gate: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	commit	Required in decision and audit evidence
field_2	test_pass_rate	Required in decision and audit evidence
field_3	critical_failures	Required in decision and audit evidence
field_4	signer	Required in decision and audit evidence
field_5	release_decision	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Release gate: evidence and acceptance decision



DECISION RULE

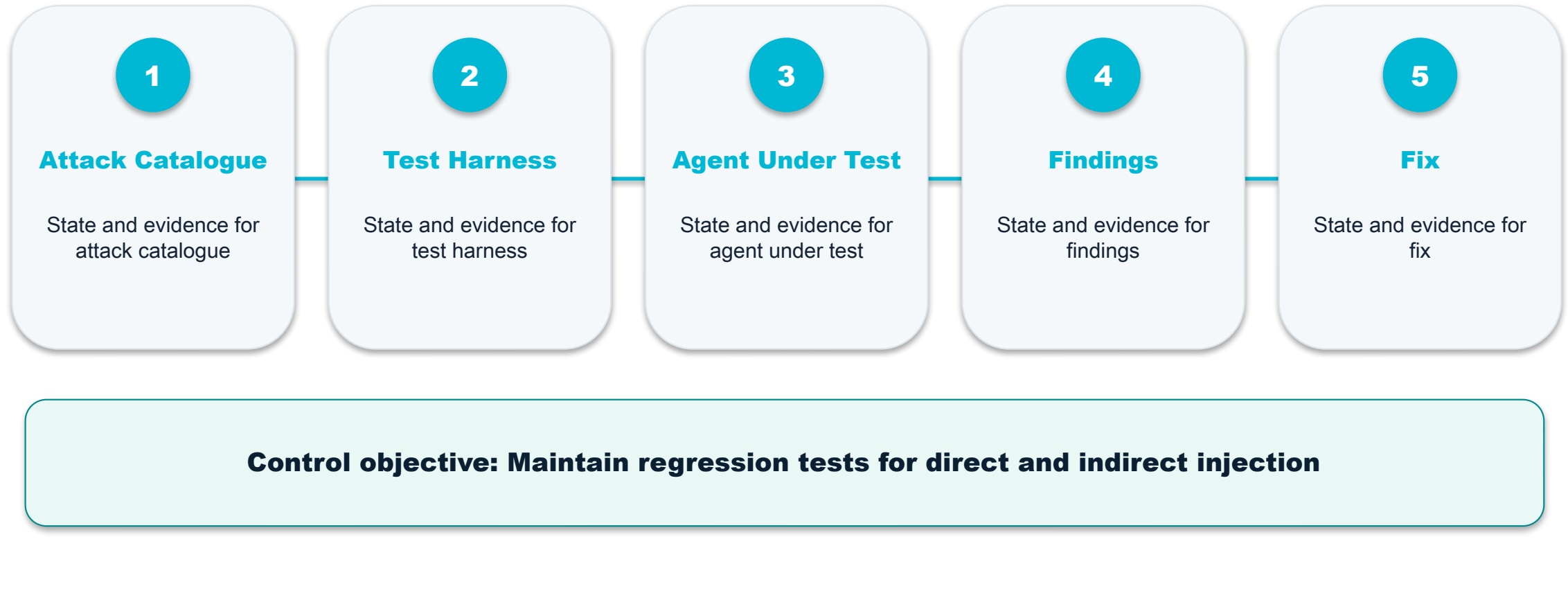
Accept only when Gate records show zero waived critical failures. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Red-team acceptance: architecture and control boundary



Red-team acceptance: runtime decision trace



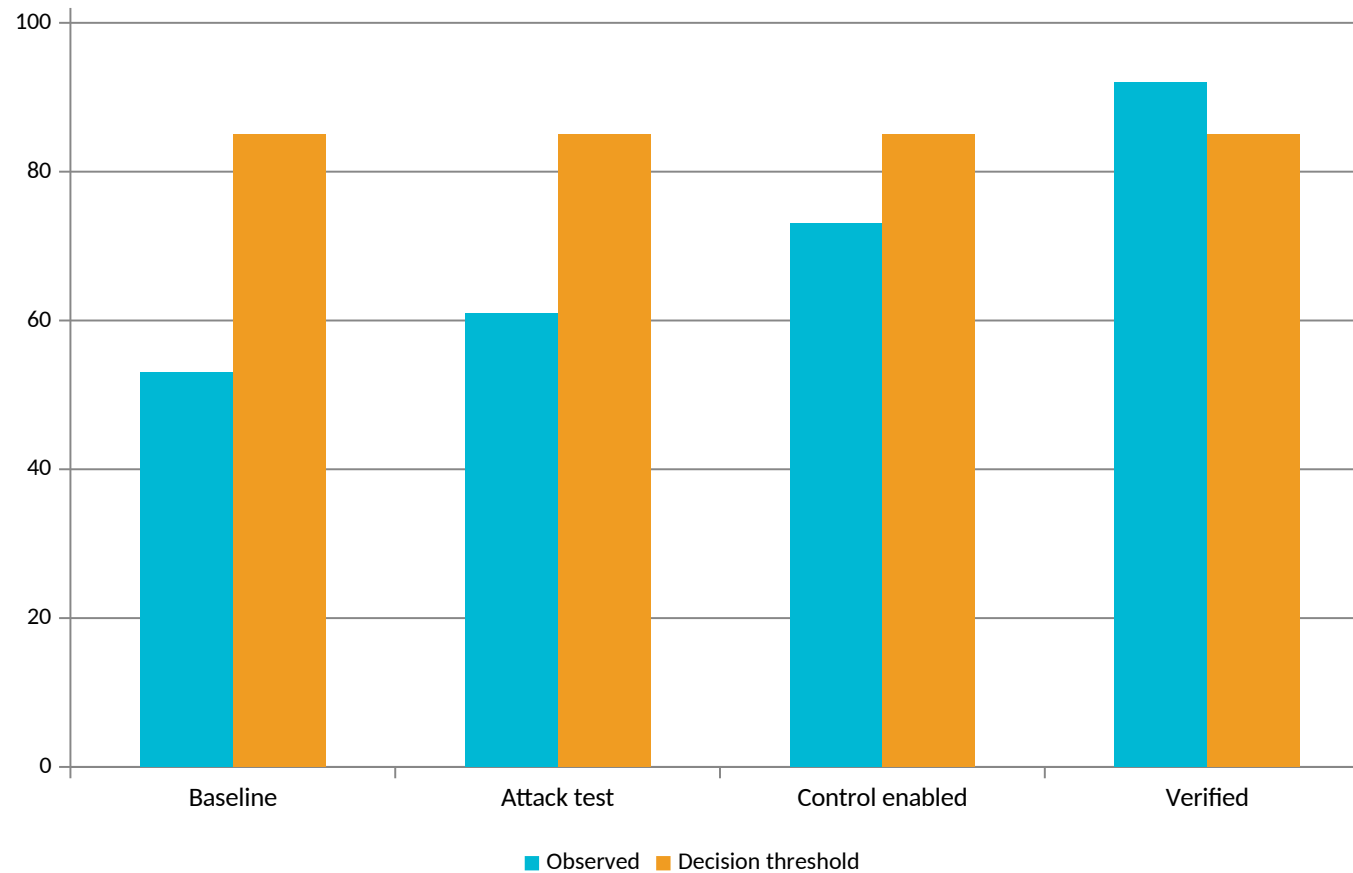
Failure to detect: Security claims are not challenged with adversarial inputs

Red-team acceptance: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	test_id	Required in decision and audit evidence
field_2	attack_class	Required in decision and audit evidence
field_3	expected_control	Required in decision and audit evidence
field_4	observed_result	Required in decision and audit evidence
field_5	severity	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Red-team acceptance: evidence and acceptance decision



DECISION RULE

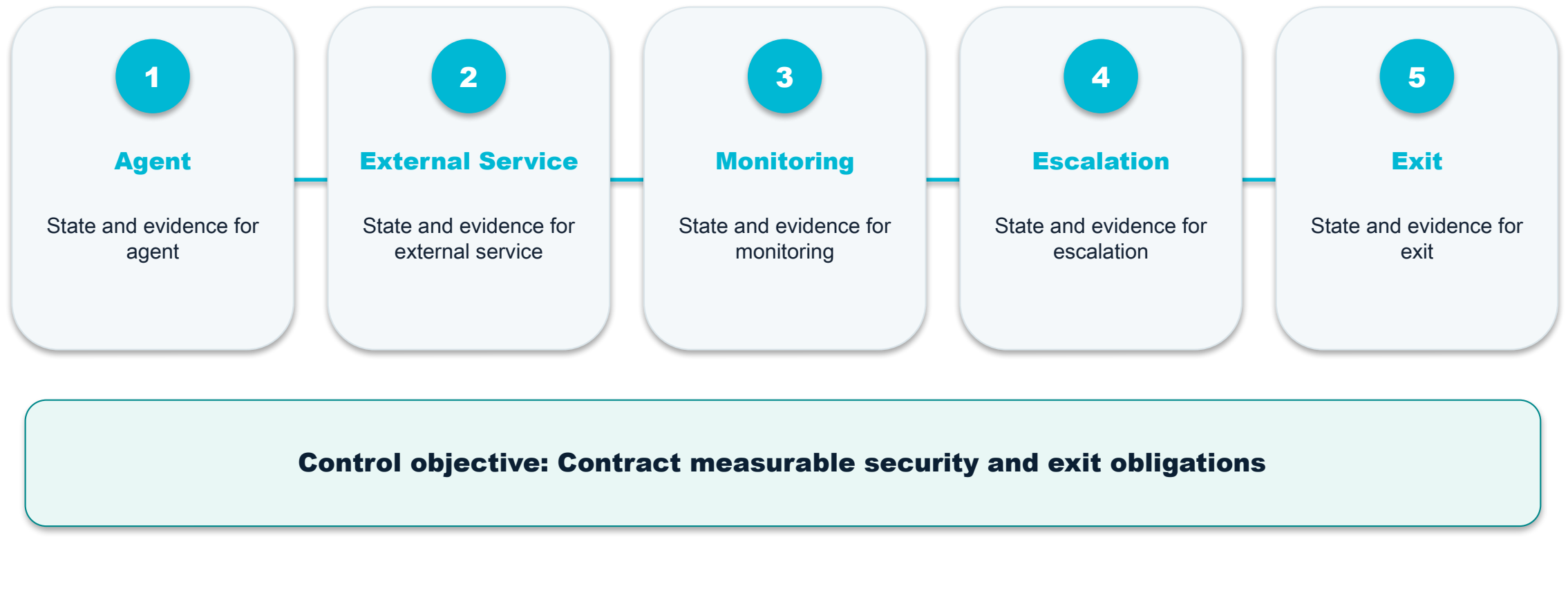
Accept only when Critical attack success rate equals zero at release. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Third-party service level: architecture and control boundary



Third-party service level: runtime decision trace



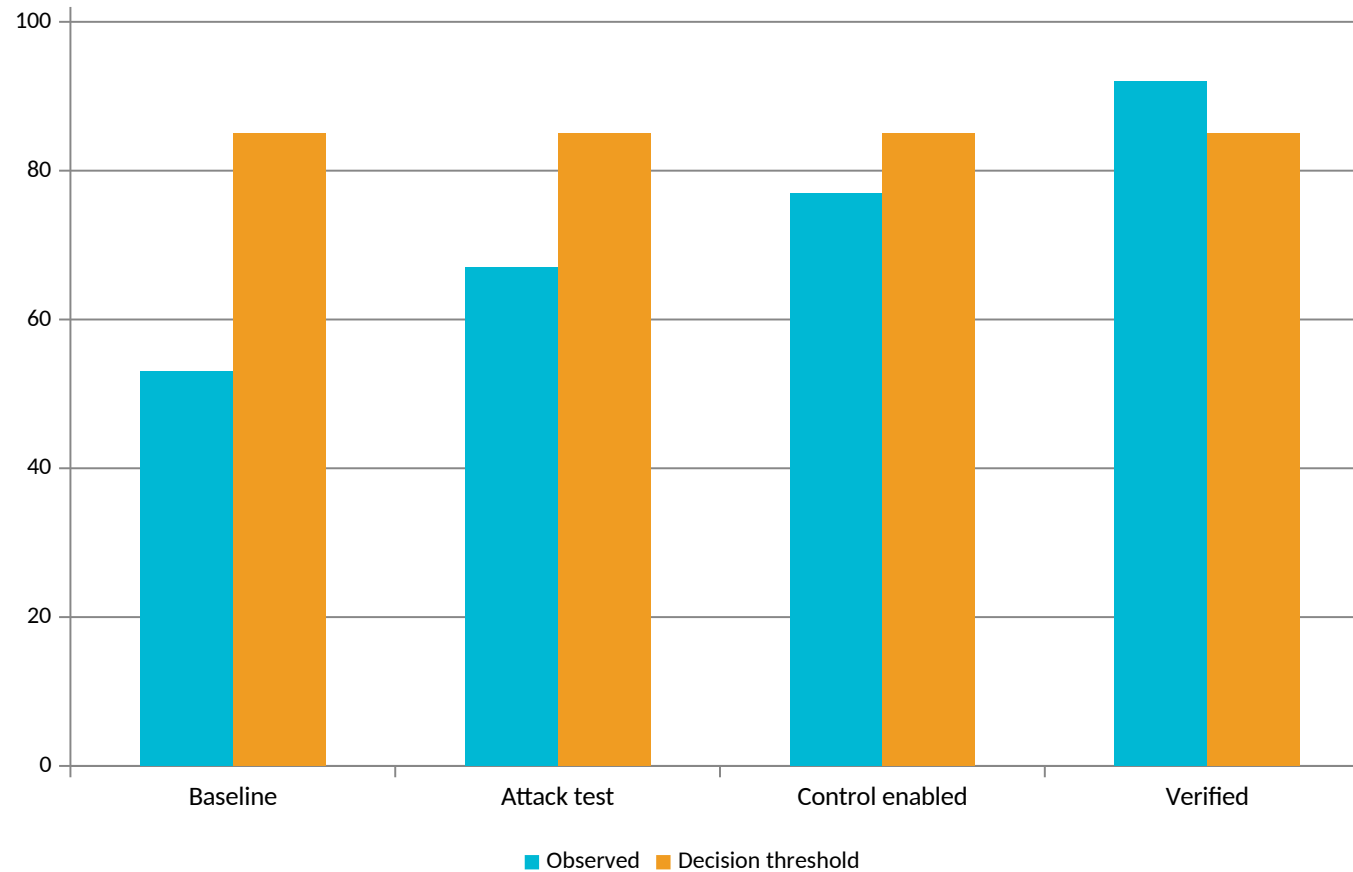
Failure to detect: Provider outage or breach cannot be investigated

Third-party service level: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	availability_slo	Required in decision and audit evidence
field_2	log_access_sla	Required in decision and audit evidence
field_3	breach_notice_hours	Required in decision and audit evidence
field_4	RTO	Required in decision and audit evidence
field_5	export_format	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Third-party service level: evidence and acceptance decision



DECISION RULE

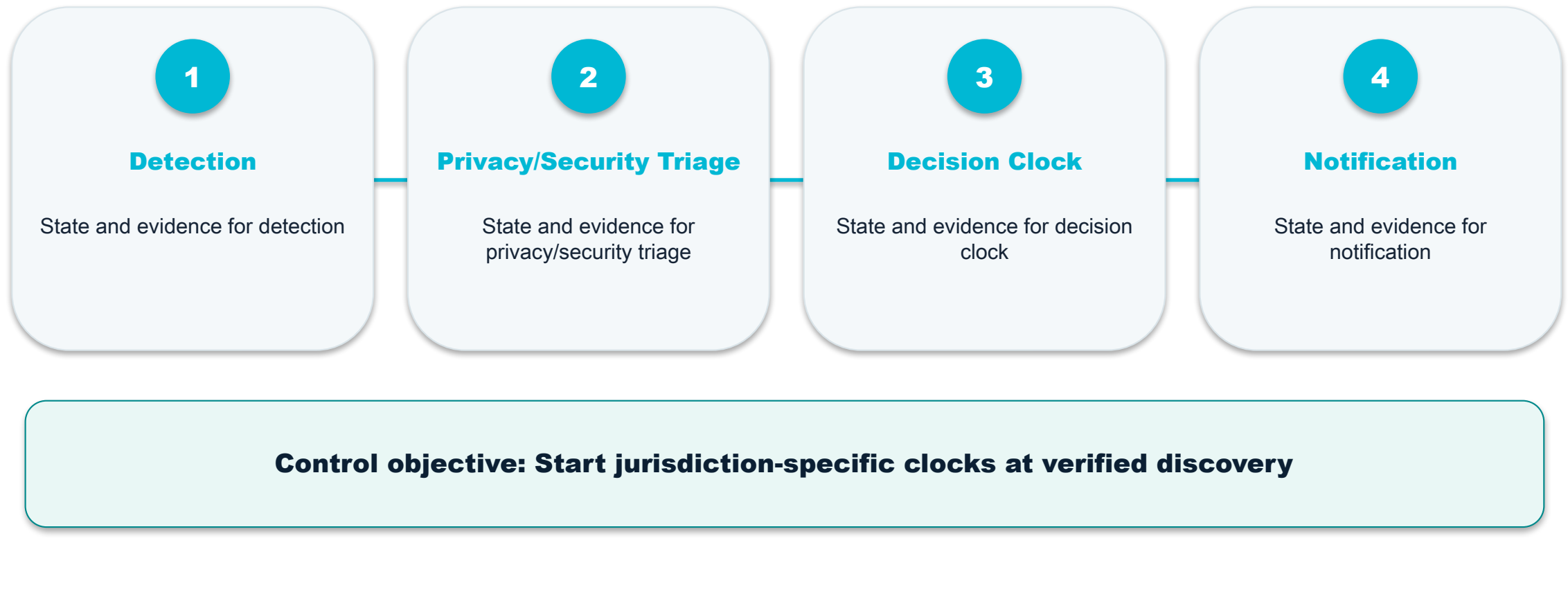
Accept only when Provider evidence meets each contracted SLA. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Incident notification: architecture and control boundary



Incident notification: runtime decision trace



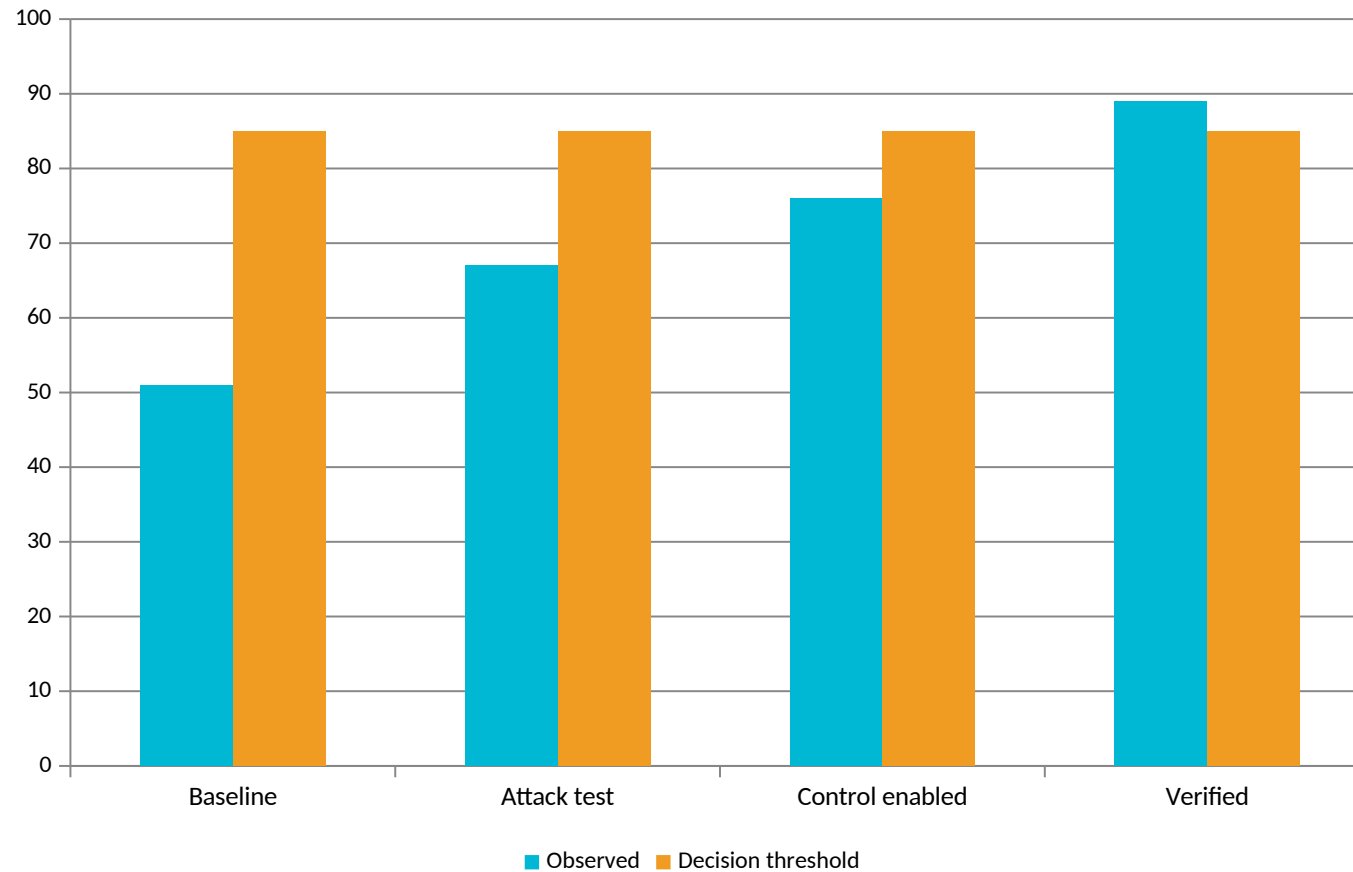
Failure to detect: Notification deadline is missed because ownership is unclear

Incident notification: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	incident_id	Required in decision and audit evidence
field_2	discovered_at	Required in decision and audit evidence
field_3	jurisdictions[]	Required in decision and audit evidence
field_4	impact	Required in decision and audit evidence
field_5	deadline	Required in decision and audit evidence
field_6	owner	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Incident notification: evidence and acceptance decision



DECISION RULE

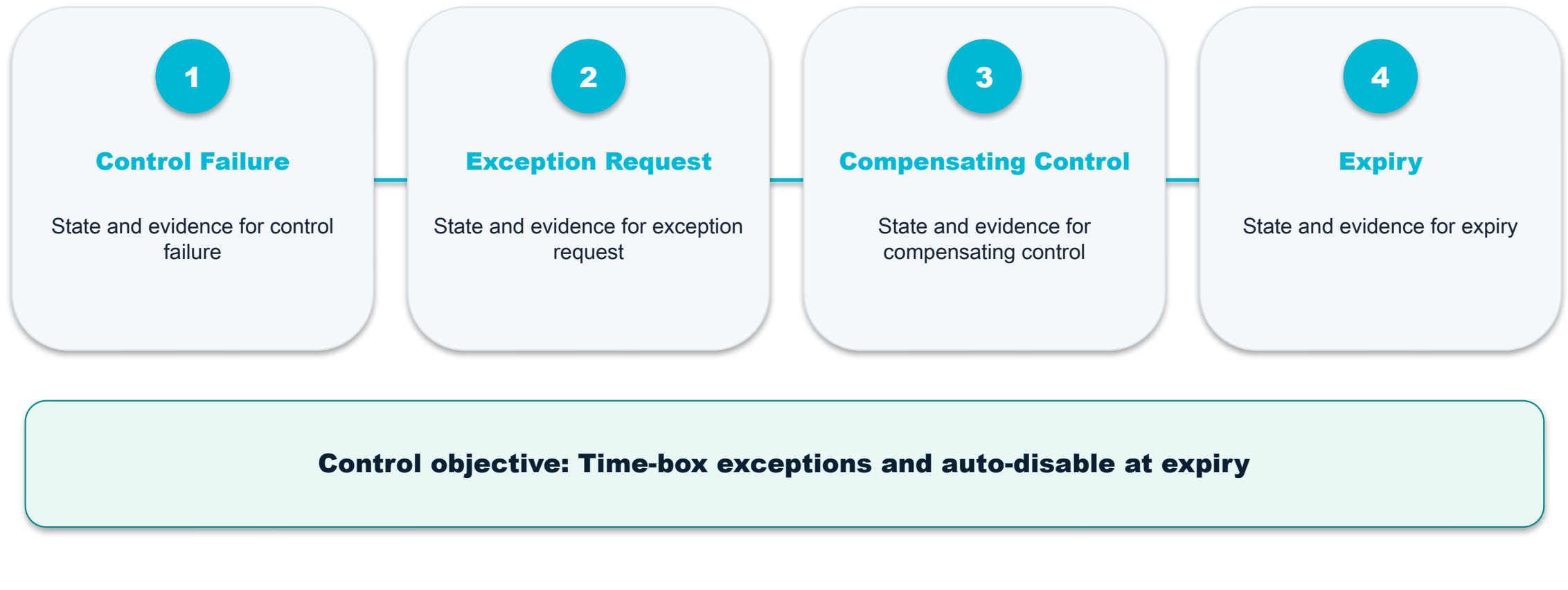
**Accept only when
Decision and
notification timestamps
are auditable.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

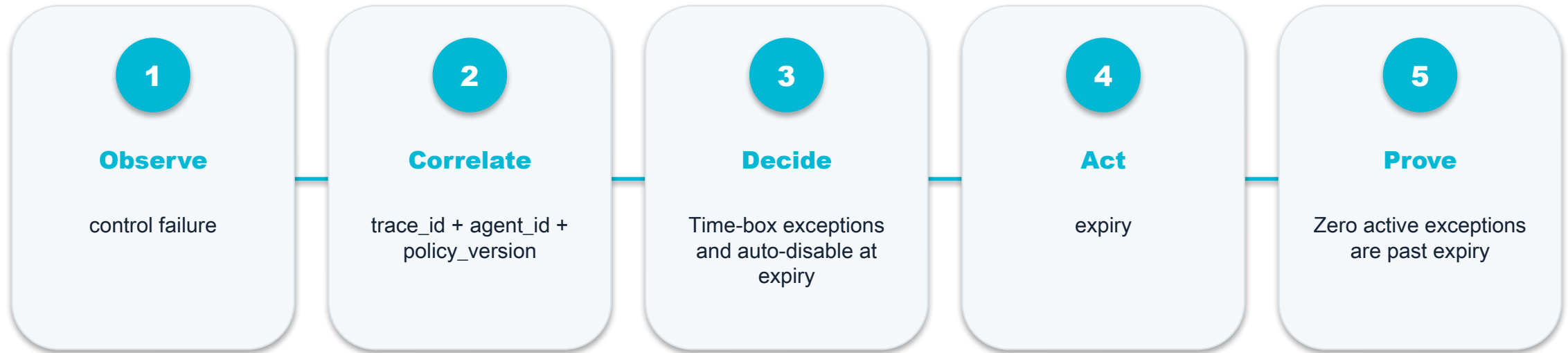
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Exception workflow: architecture and control boundary



Exception workflow: runtime decision trace



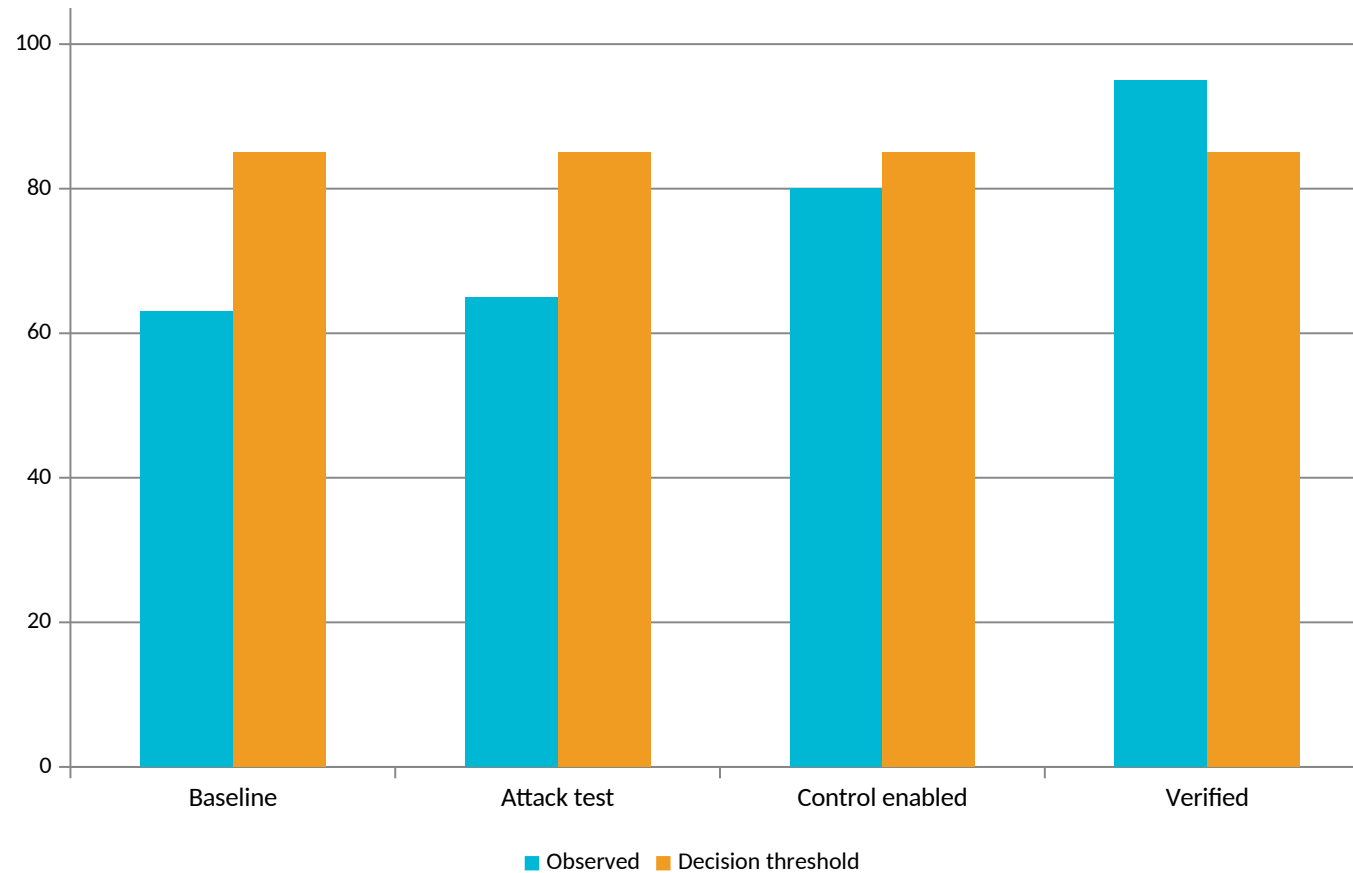
Failure to detect: Temporary bypass becomes permanent shadow policy

Exception workflow: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	exception_id	Required in decision and audit evidence
field_2	control_id	Required in decision and audit evidence
field_3	rationale	Required in decision and audit evidence
field_4	risk_owner	Required in decision and audit evidence
field_5	expiry	Required in decision and audit evidence
field_6	compensation	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Exception workflow: evidence and acceptance decision



DECISION RULE

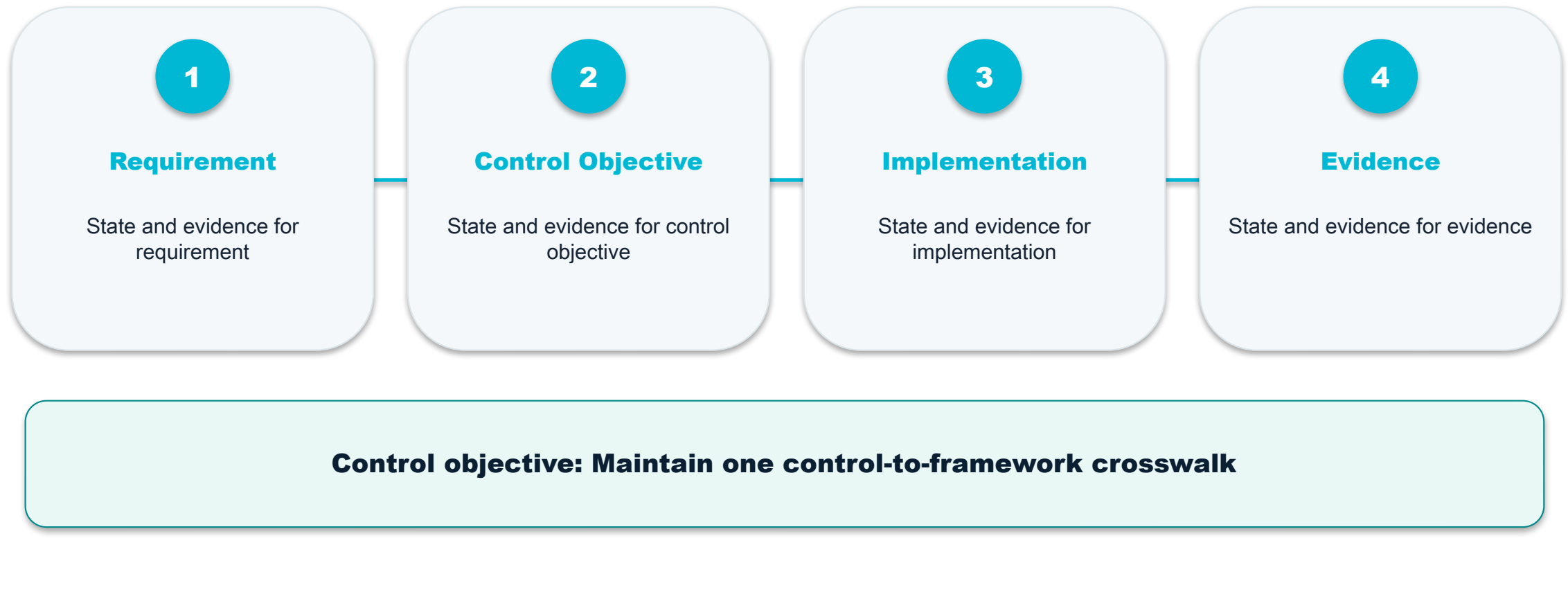
Accept only when Zero active exceptions are past expiry. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

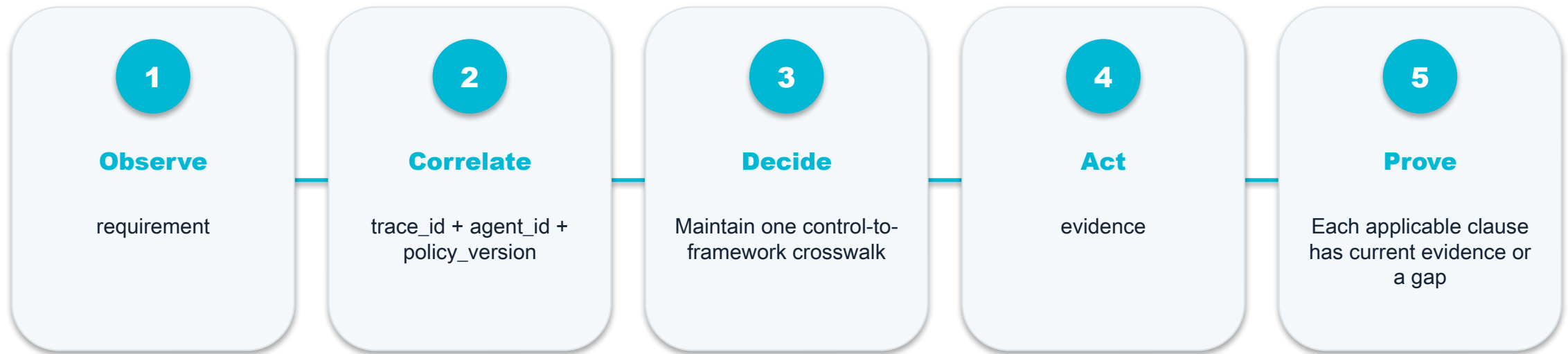
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Framework crosswalk: architecture and control boundary



Framework crosswalk: runtime decision trace



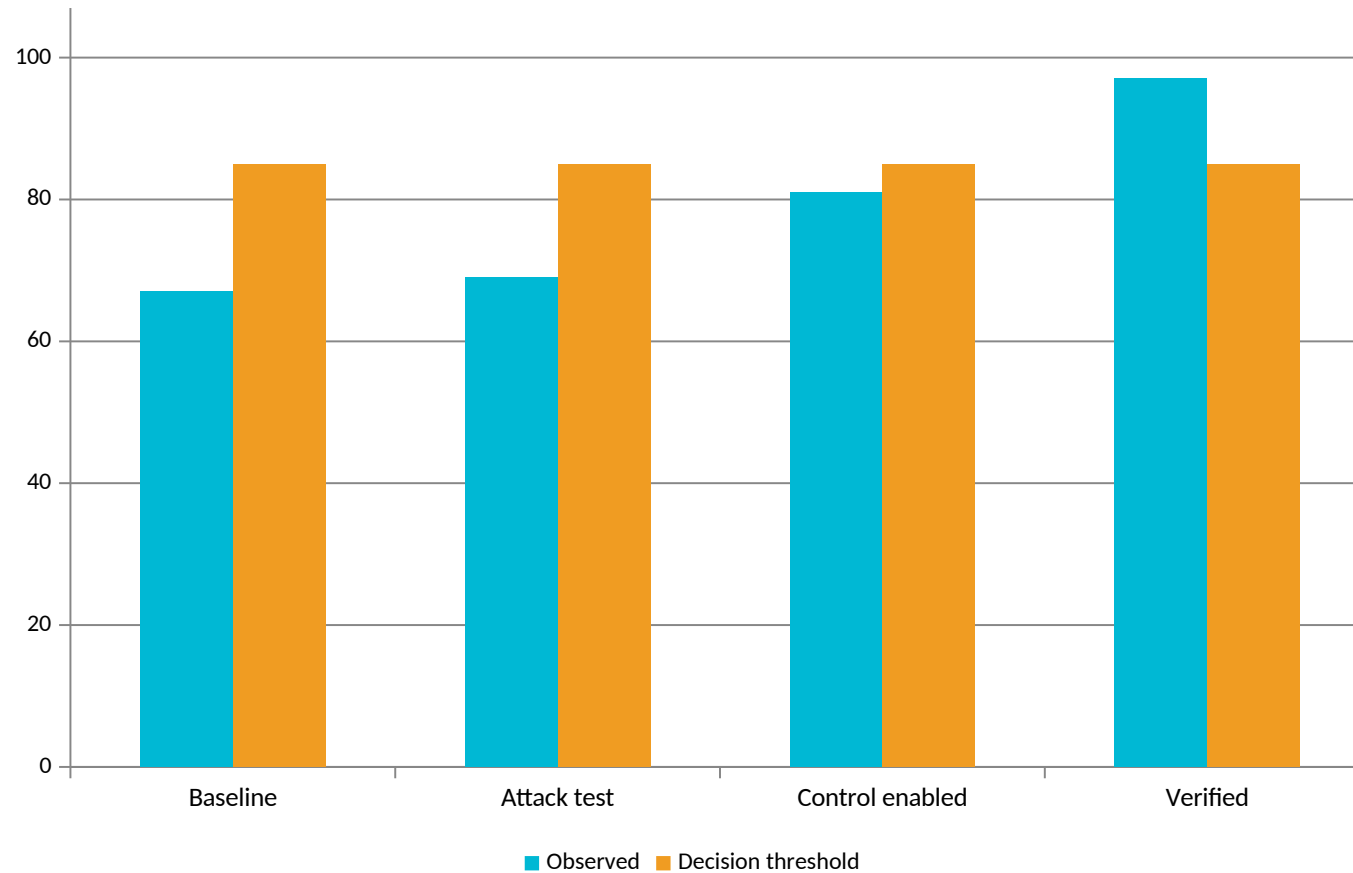
Failure to detect: Multiple audits ask for duplicate evidence with inconsistent answers

Framework crosswalk: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	framework	Required in decision and audit evidence
field_2	clause	Required in decision and audit evidence
field_3	control_id	Required in decision and audit evidence
field_4	evidence_id	Required in decision and audit evidence
field_5	coverage_status	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Framework crosswalk: evidence and acceptance decision



DECISION RULE

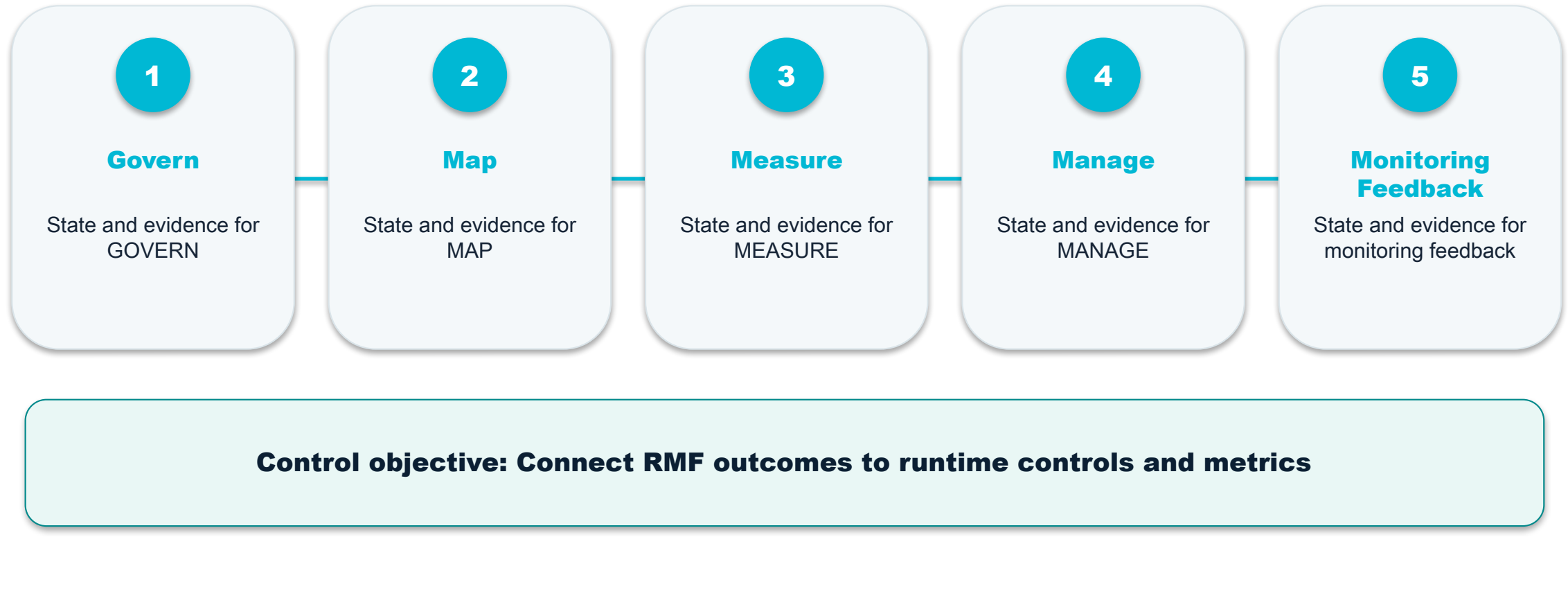
Accept only when Each applicable clause has current evidence or a gap. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

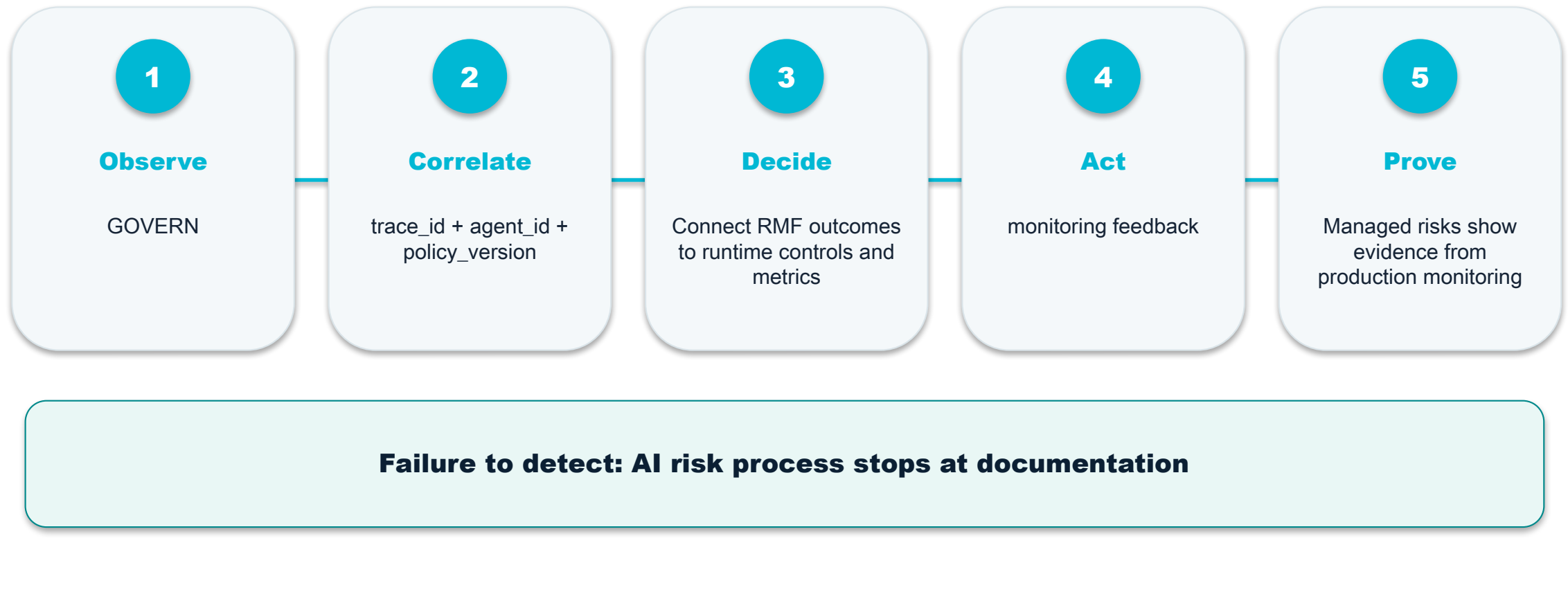
PASS

Synthetic training evidence; values are illustrative, not external performance claims

NIST AI RMF operations: architecture and control boundary



NIST AI RMF operations: runtime decision trace

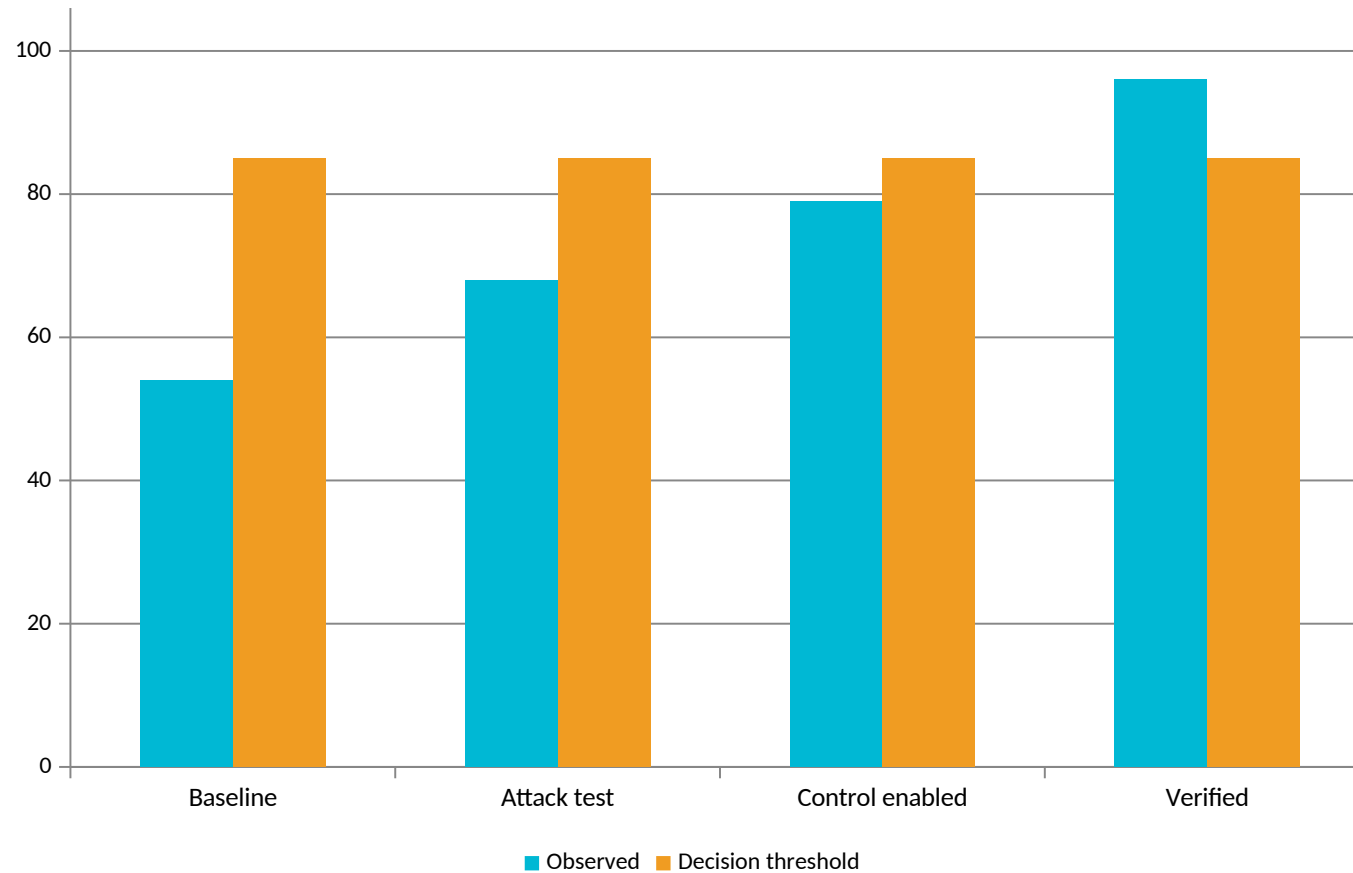


NIST AI RMF operations: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	function	Required in decision and audit evidence
field_2	category	Required in decision and audit evidence
field_3	risk_scenario	Required in decision and audit evidence
field_4	metric	Required in decision and audit evidence
field_5	response	Required in decision and audit evidence
field_6	owner	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

NIST AI RMF operations: evidence and acceptance decision



DECISION RULE

**Accept only when
Managed risks show
evidence from
production monitoring.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

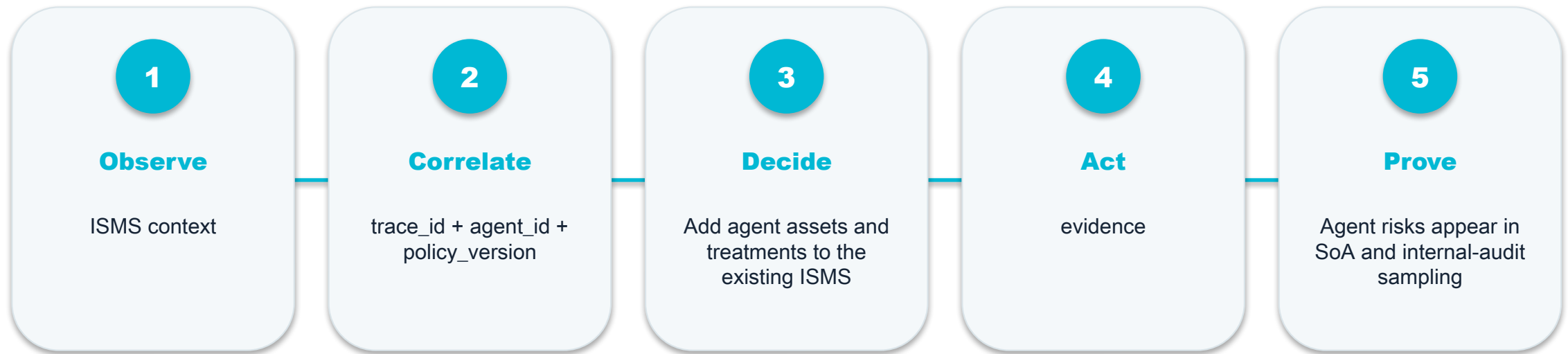
PASS

Synthetic training evidence; values are illustrative, not external performance claims

ISO 27001 integration: architecture and control boundary



ISO 27001 integration: runtime decision trace



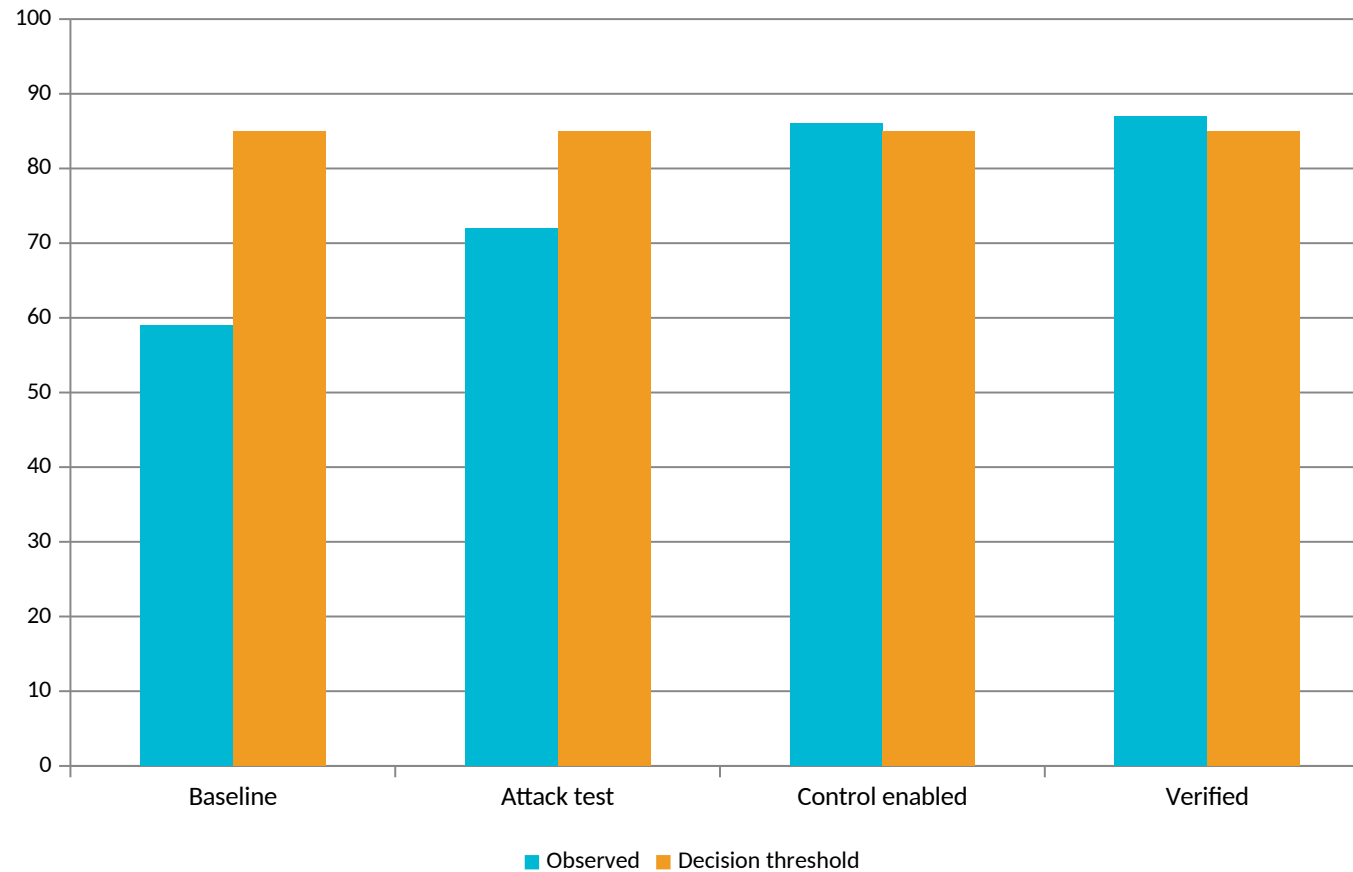
Failure to detect: Agent systems sit outside the established ISMS

ISO 27001 integration: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	asset	Required in decision and audit evidence
field_2	risk_id	Required in decision and audit evidence
field_3	control_ref	Required in decision and audit evidence
field_4	statement_of_applicability	Required in decision and audit evidence
field_5	evidence	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

ISO 27001 integration: evidence and acceptance decision



DECISION RULE

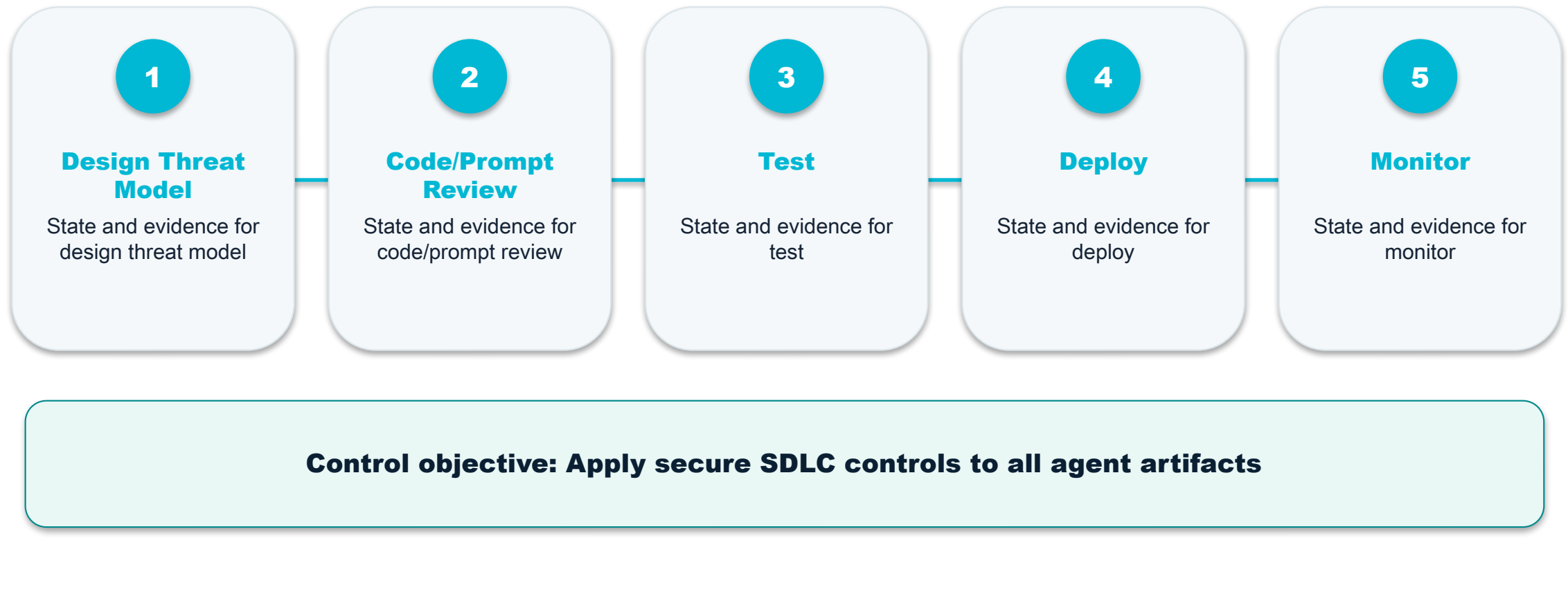
Accept only when Agent risks appear in SoA and internal-audit sampling. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Secure development lifecycle: architecture and control boundary



Secure development lifecycle: runtime decision trace



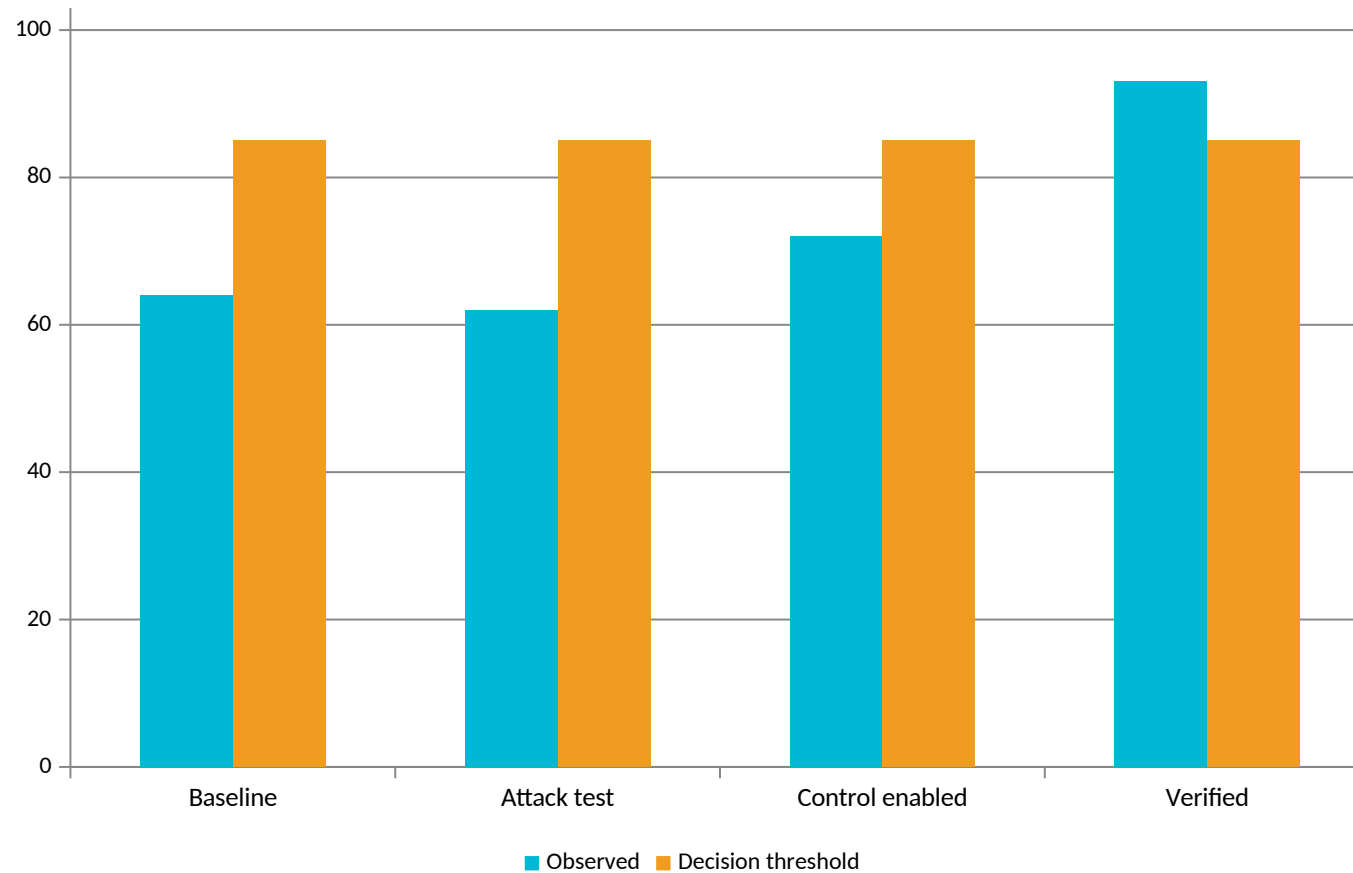
Failure to detect: Prompt/config changes bypass application security review

Secure development lifecycle: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	artifact_type	Required in decision and audit evidence
field_2	reviewer	Required in decision and audit evidence
field_3	tests	Required in decision and audit evidence
field_4	vulnerabilities	Required in decision and audit evidence
field_5	approval	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Secure development lifecycle: evidence and acceptance decision



DECISION RULE

Accept only when No production artifact lacks review and test evidence. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

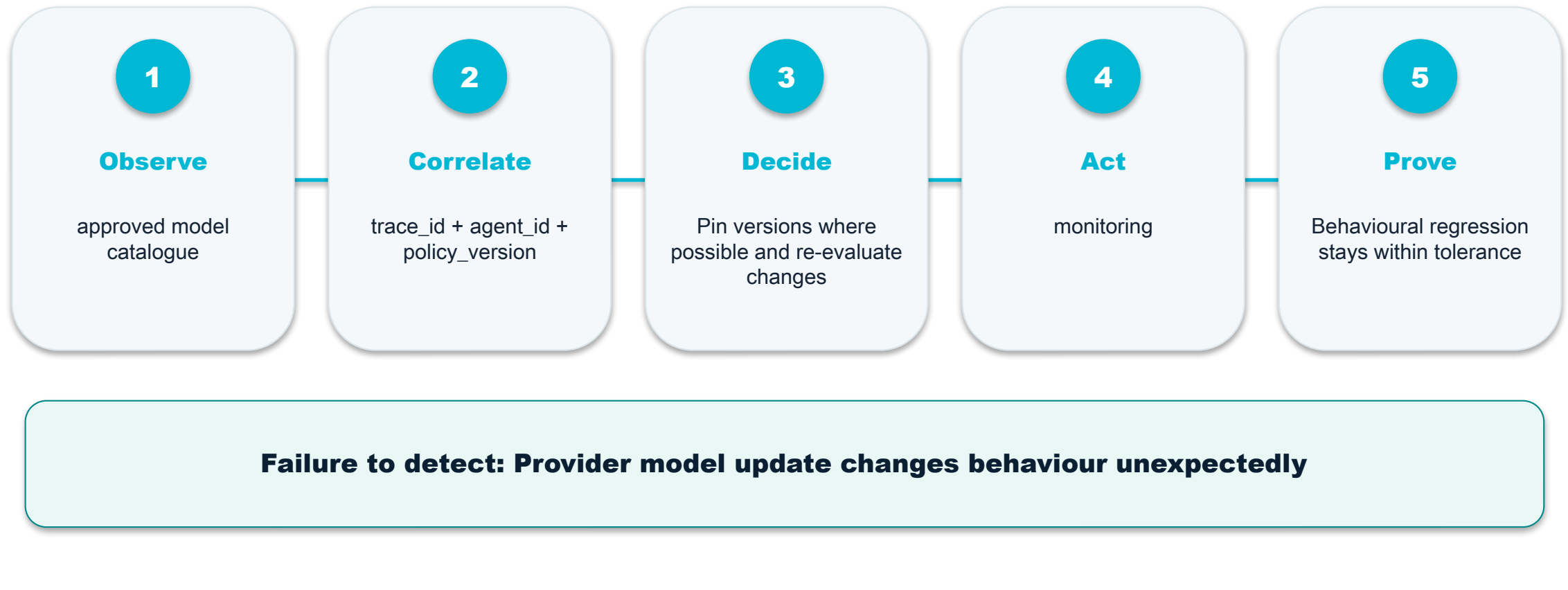
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Model/version governance: architecture and control boundary



Model/version governance: runtime decision trace

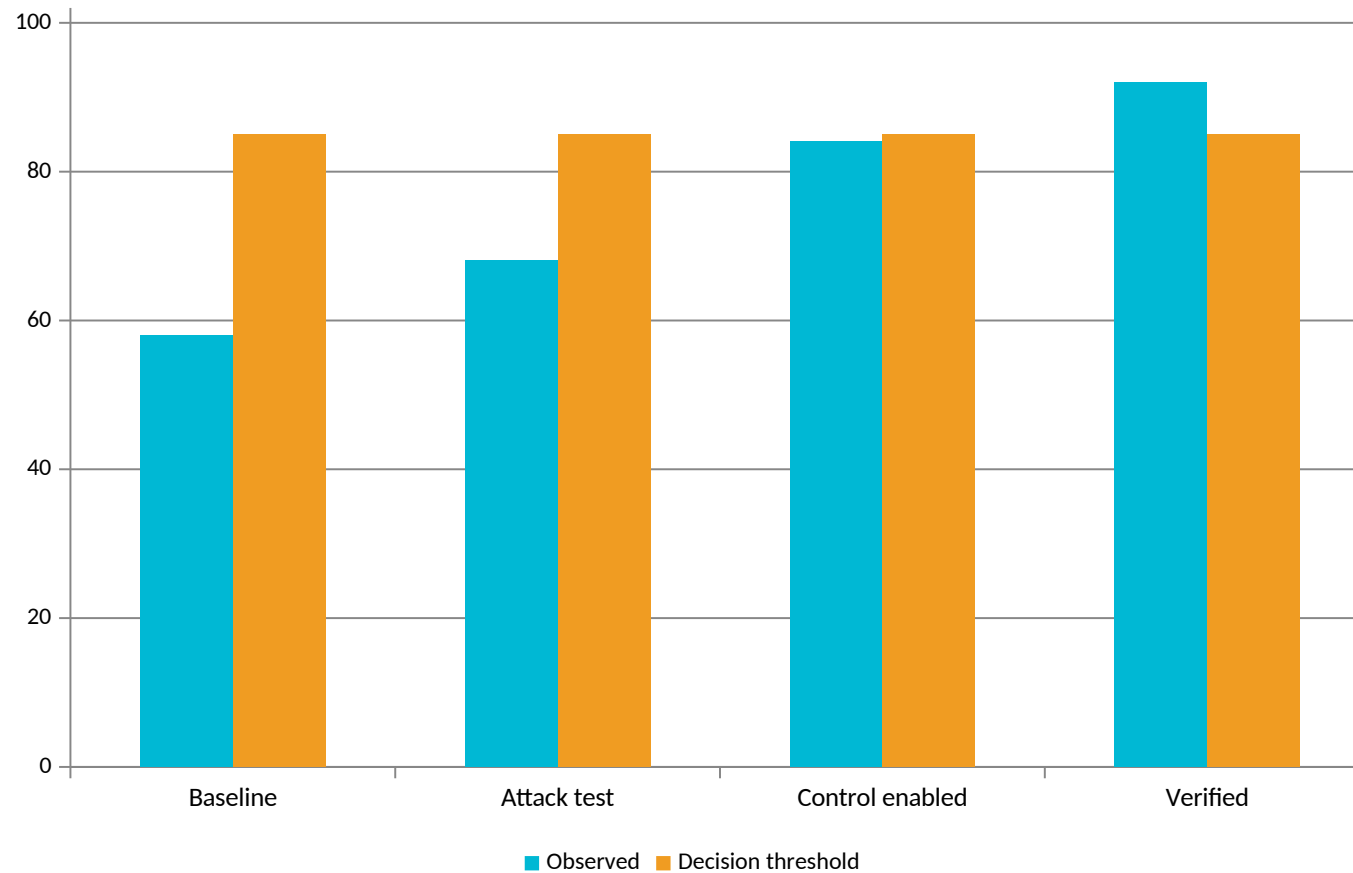


Model/version governance: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	provider	Required in decision and audit evidence
field_2	model_id	Required in decision and audit evidence
field_3	version	Required in decision and audit evidence
field_4	eval_set	Required in decision and audit evidence
field_5	score	Required in decision and audit evidence
field_6	retirement_date	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Model/version governance: evidence and acceptance decision



DECISION RULE

Accept only when Behavioural regression stays within tolerance. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Security communication cadence: architecture and control boundary



Security communication cadence: runtime decision trace



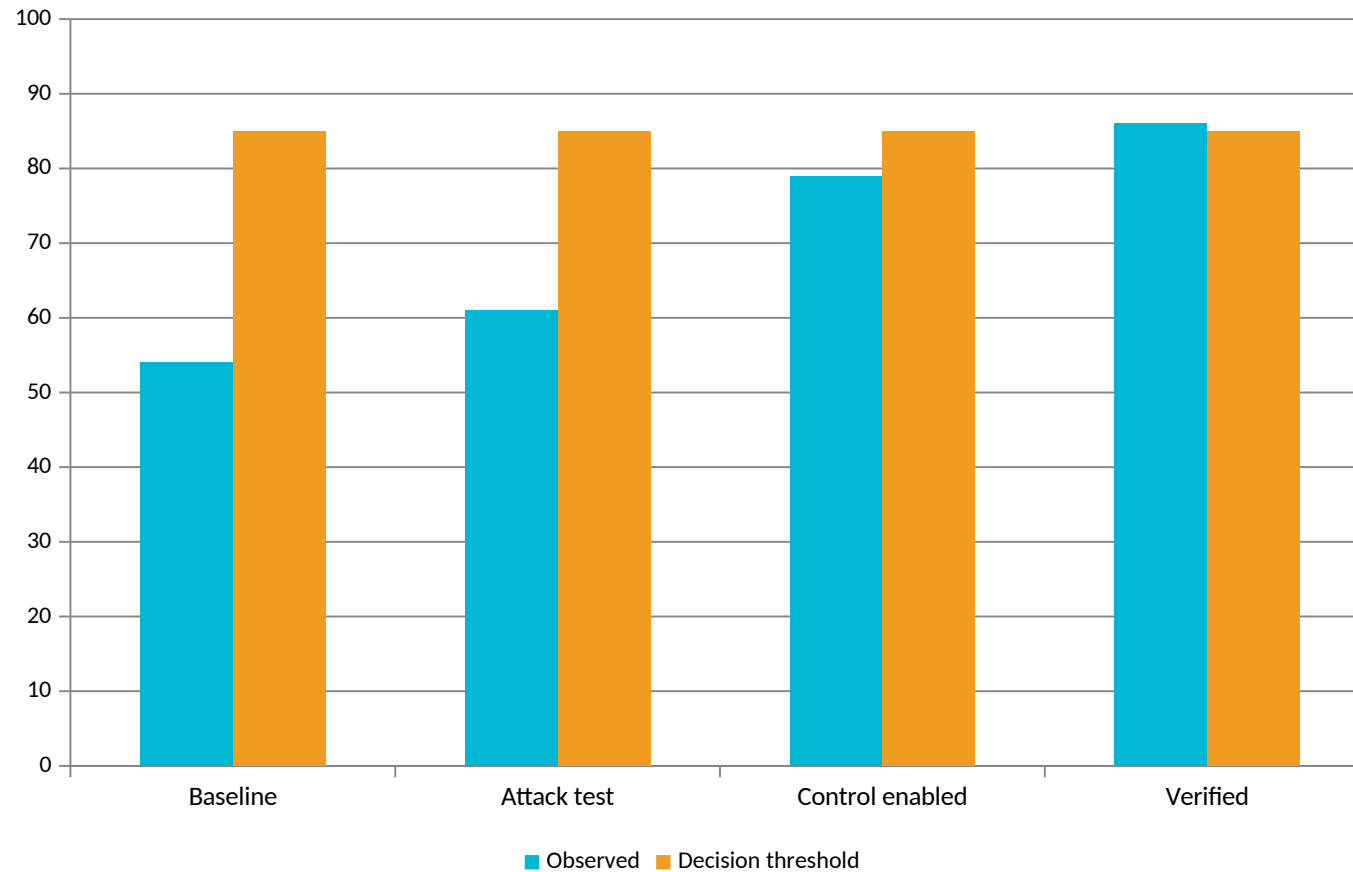
Failure to detect: Technical alerts never become business decisions

Security communication cadence: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	audience	Required in decision and audit evidence
field_2	frequency	Required in decision and audit evidence
field_3	metrics[]	Required in decision and audit evidence
field_4	decisions[]	Required in decision and audit evidence
field_5	action_owner	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Security communication cadence: evidence and acceptance decision



DECISION RULE

Accept only when Actions have owners and due dates after every review. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Assurance dashboard: architecture and control boundary



Assurance dashboard: runtime decision trace



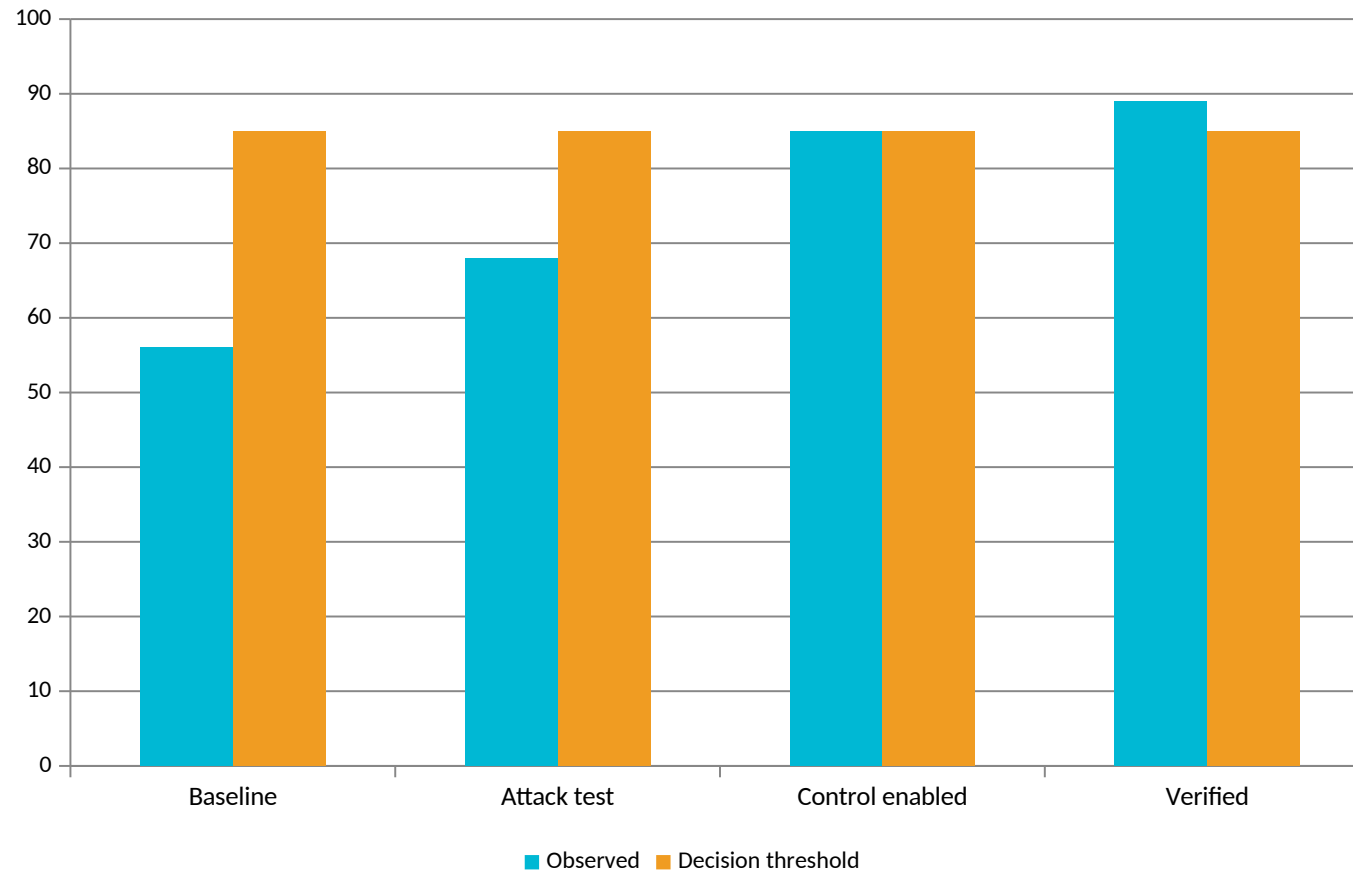
Failure to detect: Vanity metrics hide deteriorating controls

Assurance dashboard: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	metric_id	Required in decision and audit evidence
field_2	numerator	Required in decision and audit evidence
field_3	denominator	Required in decision and audit evidence
field_4	threshold	Required in decision and audit evidence
field_5	trend	Required in decision and audit evidence
field_6	owner	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Assurance dashboard: evidence and acceptance decision



DECISION RULE

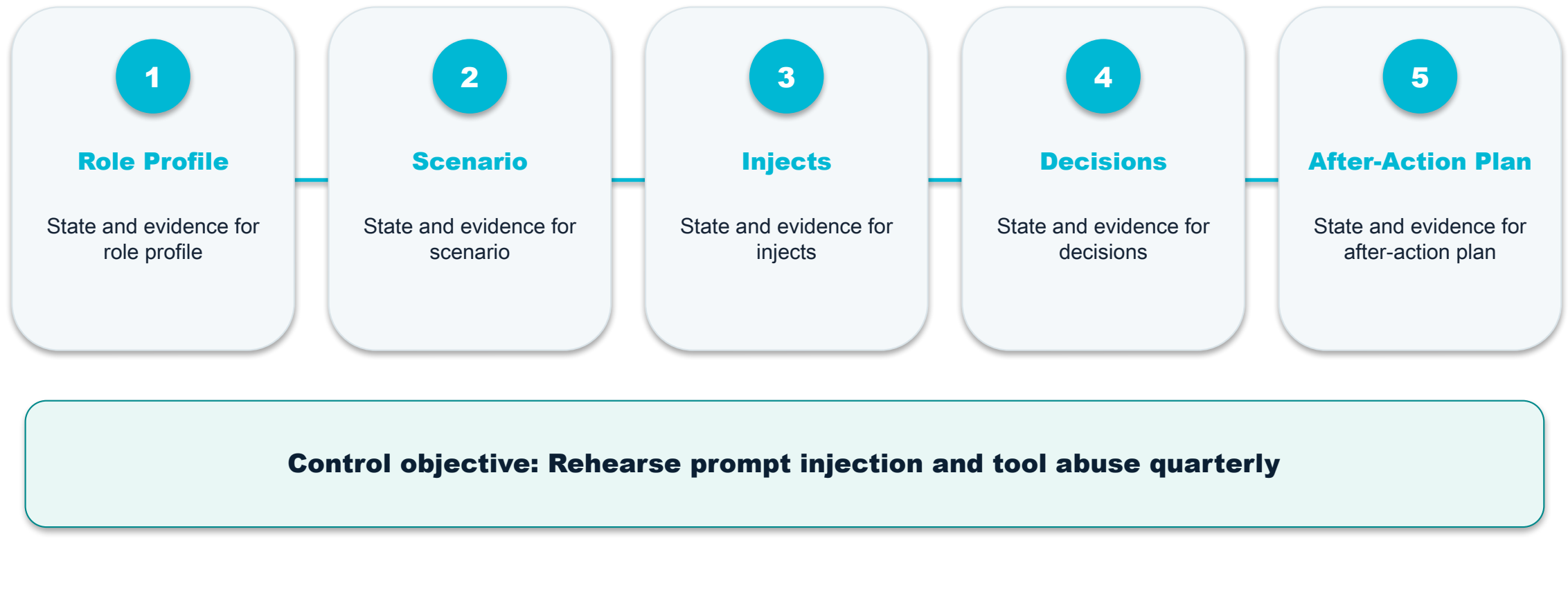
Accept only when Threshold breach triggers a documented action. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

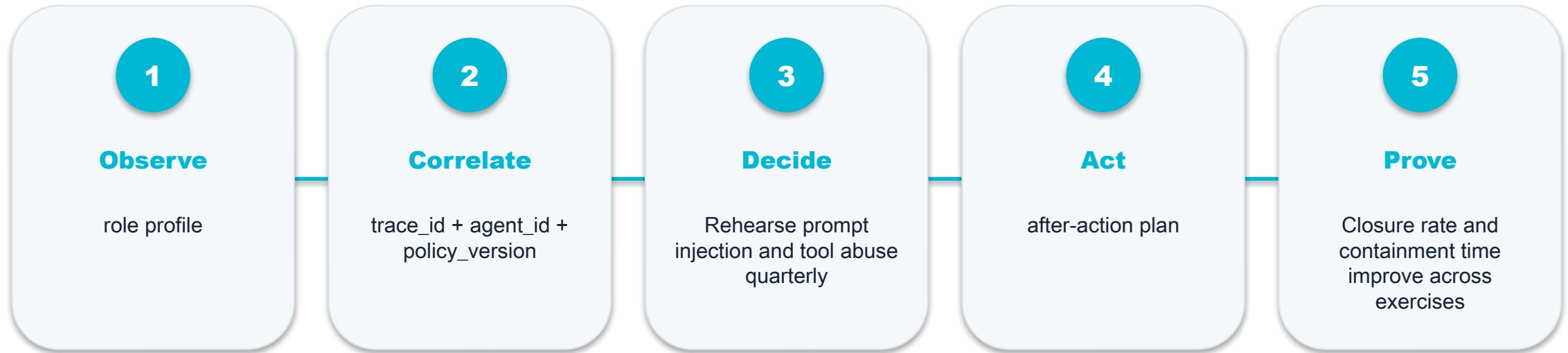
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Training and tabletop: architecture and control boundary



Training and tabletop: runtime decision trace



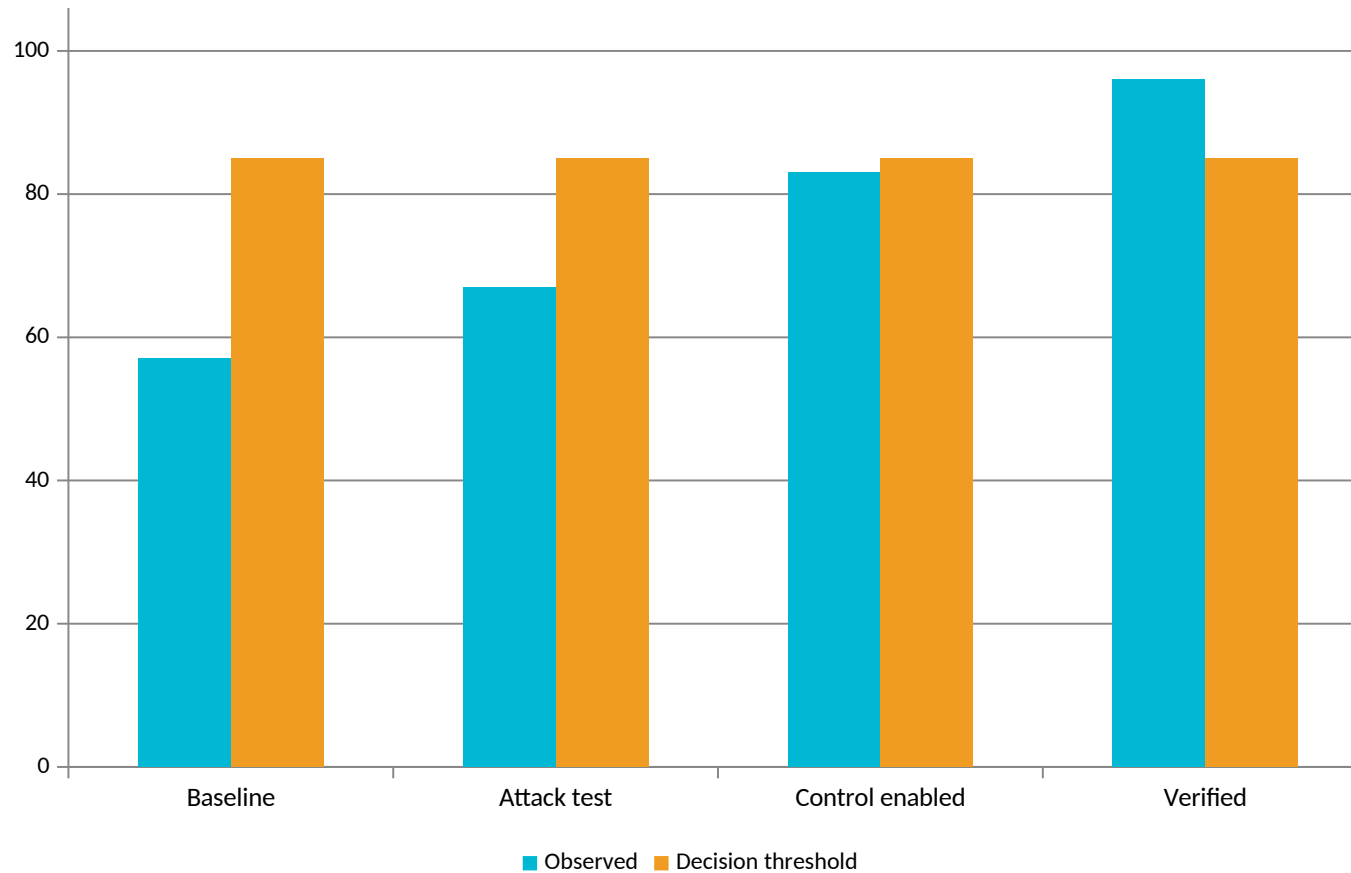
Failure to detect: Teams know policy but cannot execute containment

Training and tabletop: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	exercise_id	Required in decision and audit evidence
field_2	roles[]	Required in decision and audit evidence
field_3	inject_time	Required in decision and audit evidence
field_4	decision	Required in decision and audit evidence
field_5	gap	Required in decision and audit evidence
field_6	due_date	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Training and tabletop: evidence and acceptance decision



DECISION RULE

Accept only when Closure rate and containment time improve across exercises. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Activity 04 — Assemble a Compliance Evidence Pack

1 Scenario

Use synthetic data in activity-04-compliance-evidence to make a security-operations decision.

2 Input contract

mock_evidence.json

3 Runnable evidence

activity.py

4 Output artifact

compliance_report.md

5 Acceptance

Each required control maps to current evidence, an owner and a status; gaps are explicit.

6 Safety boundary

Offline simulation only; no live credentials, systems or external services.

Activity 04: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Activity 04: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_evidence.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	compliance_report.md	Generated locally and reproducible
Criterion	A3	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete compliance_report.md only	Retain script and mock data

Aligned activity files in labs/

Activity 05 — Communicate Agent Security Posture

1 Scenario

Use synthetic data in activity-05-security-communication to make a security-operations decision.

2 Input contract

mock_metrics.csv

3 Runnable evidence

activity.py

4 Output artifact

security_brief.html

5 Acceptance

The generated brief shows thresholds, current values, decisions, owners and due dates for two audiences.

6 Safety boundary

Offline simulation only; no live credentials, systems or external services.

Activity 05: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Activity 05: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_metrics.csv	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	security_brief.html	Generated locally and reproducible
Criterion	A4	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete security_brief.html only	Retain script and mock data

Aligned activity files in labs/

Topic 02 technical recap

1

ISO 27001 integration

Agent risks appear in SoA and internal-audit sampling

2

Secure development lifecycle

No production artifact lacks review and test evidence

3

Model/version governance

Behavioural regression stays within tolerance

4

Security communication cadence

Actions have owners and due dates after every review

5

Assurance dashboard

Threshold breach triggers a documented action

6

Training and tabletop

Closure rate and containment time improve across exercises

TOPIC 03

AI Agent Risk Assessment and Security Control Evaluation

Threat paths · risk quantification · control effectiveness · remediation

LO3 · K5, K6, A5, A6



LO3 and assessment criteria

1

K5

Knowledge of security management benchmarks

2

K6

Gap analysis in organisation security

3

A5

Review existing security controls against current and future business costs and risks

4

A6

Develop strategies and plans to resolve security gaps

5

LO3

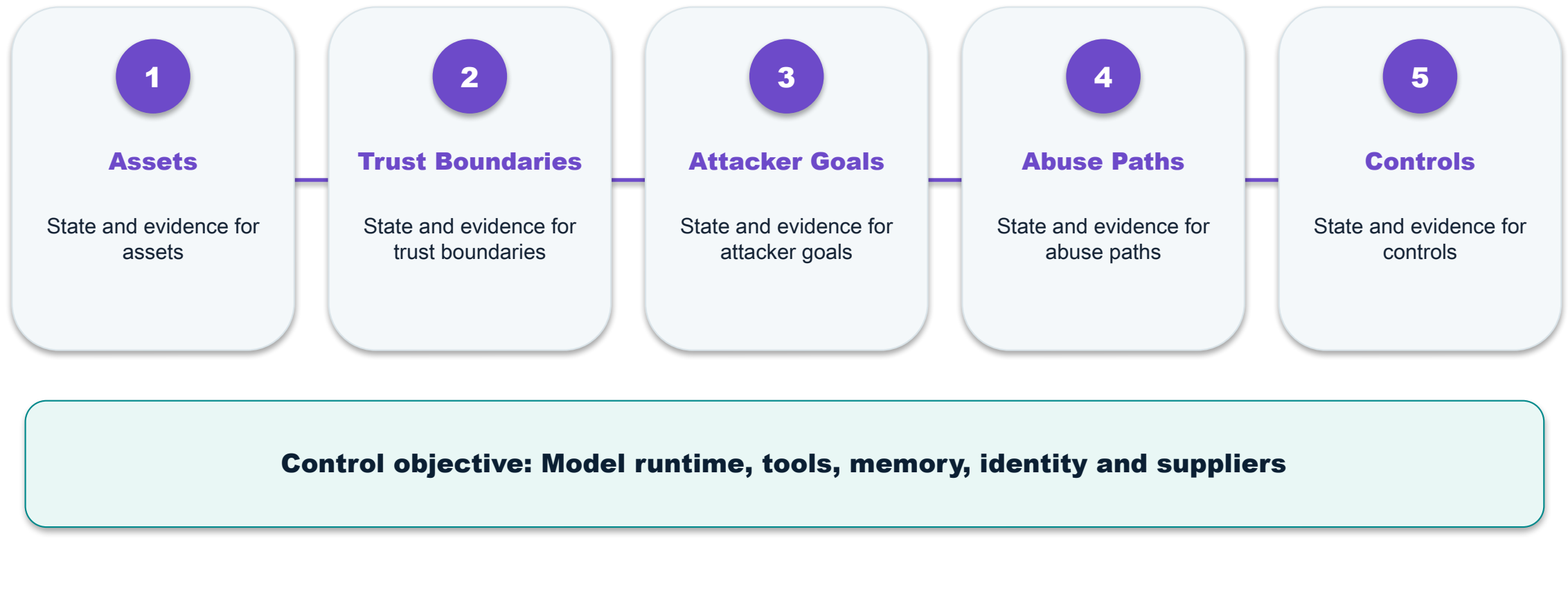
LO3: Evaluate existing agent-security controls against business risks and costs, and develop strategies to resolve identified security gaps.

AI Agent Risk Assessment and Security Control Evaluation: operating model

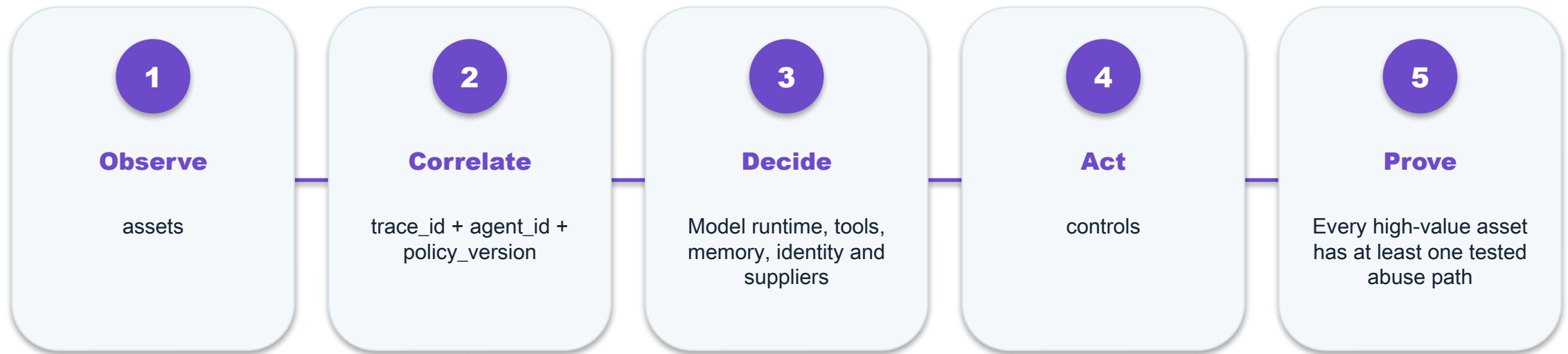


Every mechanism in this topic is taught through all five layers.

Agent threat model: architecture and control boundary



Agent threat model: runtime decision trace



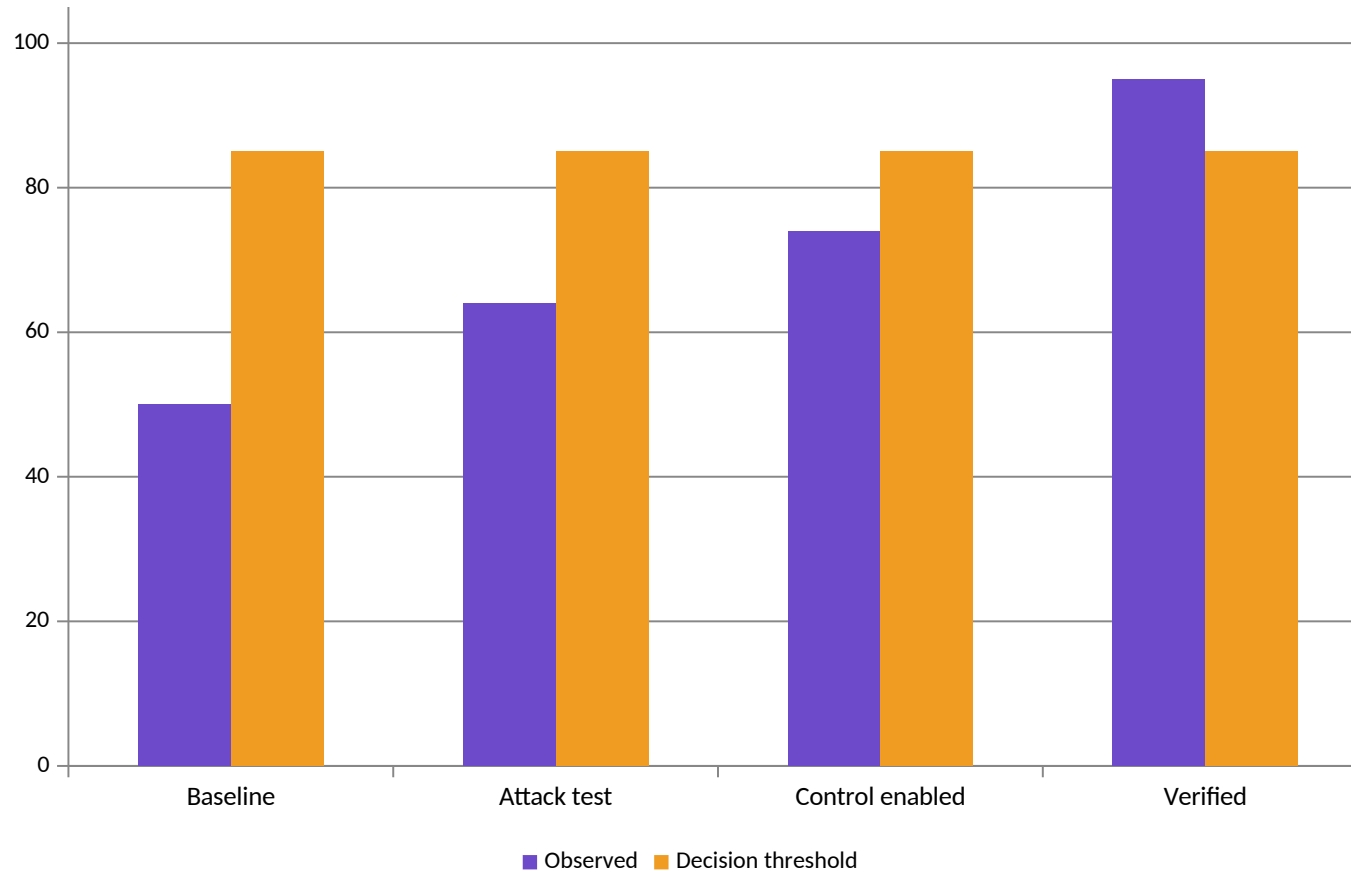
Failure to detect: Threat model treats the LLM as the only attack surface

Agent threat model: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	asset	Required in decision and audit evidence
field_2	entry_point	Required in decision and audit evidence
field_3	precondition	Required in decision and audit evidence
field_4	technique	Required in decision and audit evidence
field_5	impact	Required in decision and audit evidence
field_6	mitigation	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Agent threat model: evidence and acceptance decision



DECISION RULE

Accept only when Every high-value asset has at least one tested abuse path. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

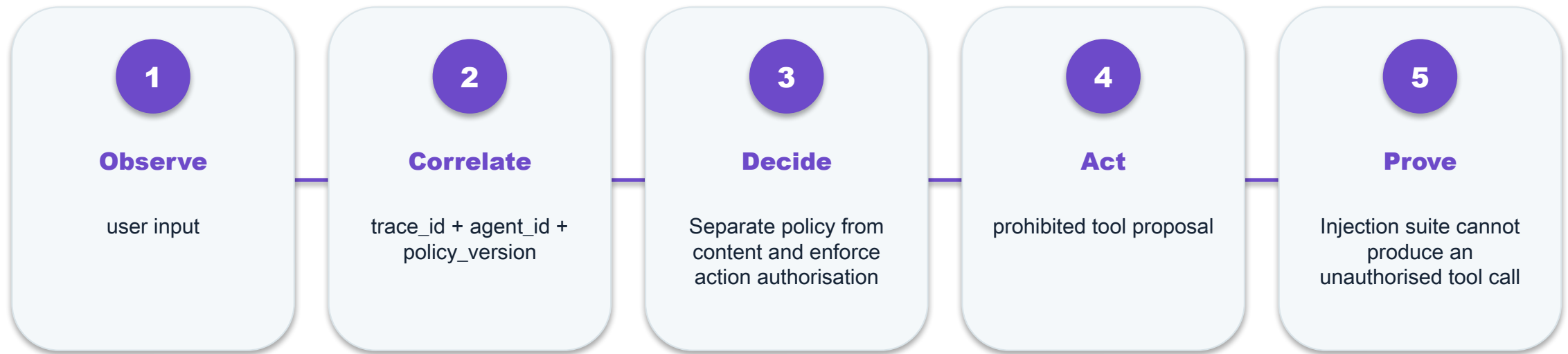
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Direct prompt injection: architecture and control boundary



Direct prompt injection: runtime decision trace



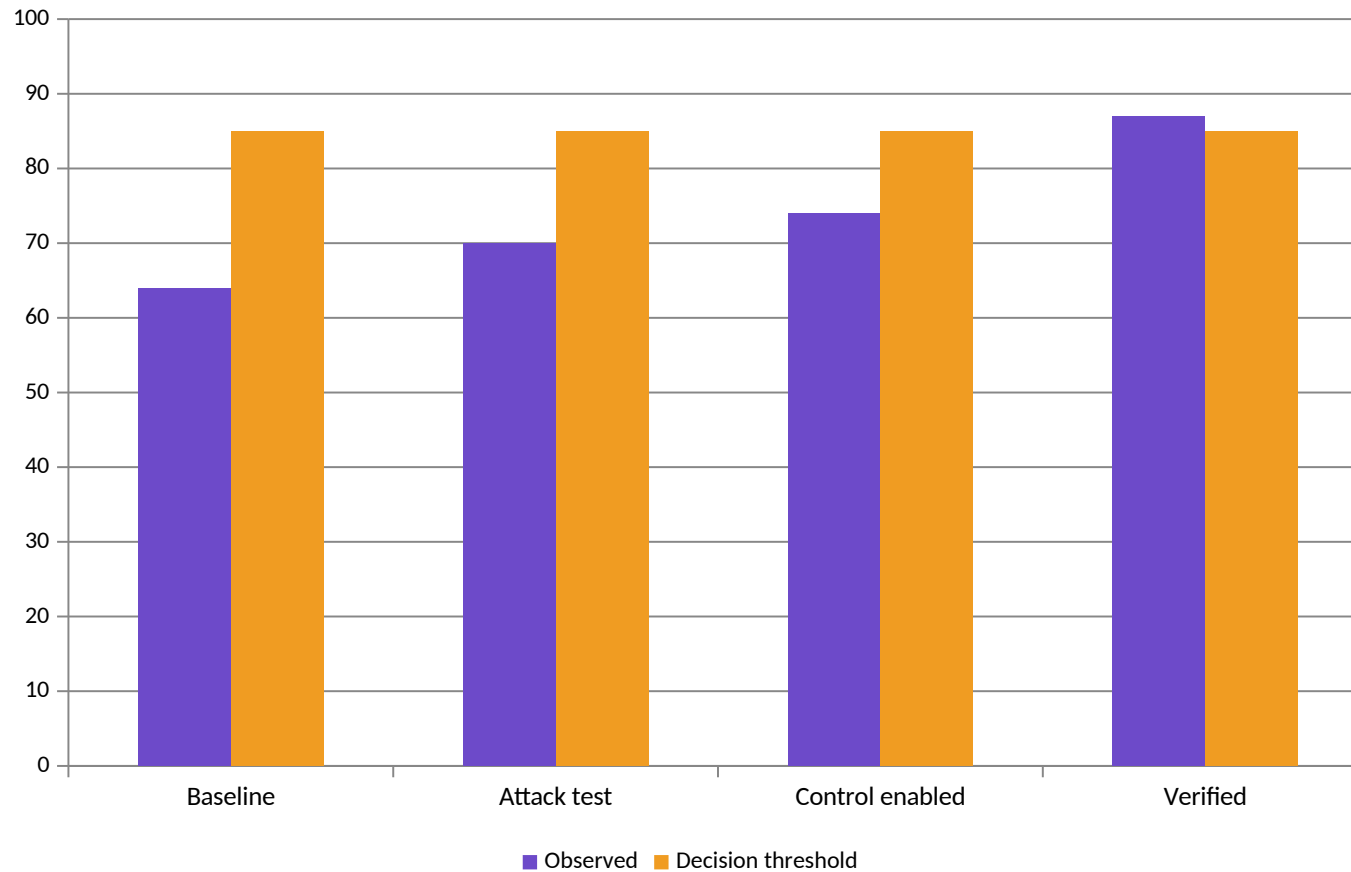
Failure to detect: User instruction overrides policy and triggers a sensitive action

Direct prompt injection: concrete control contract

Field / key	Expected value or type	Enforcement
input_source	user	Required in decision and audit evidence
field_2	injection_score	Required in decision and audit evidence
field_3	target_instruction	Required in decision and audit evidence
field_4	decision	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Direct prompt injection: evidence and acceptance decision



DECISION RULE

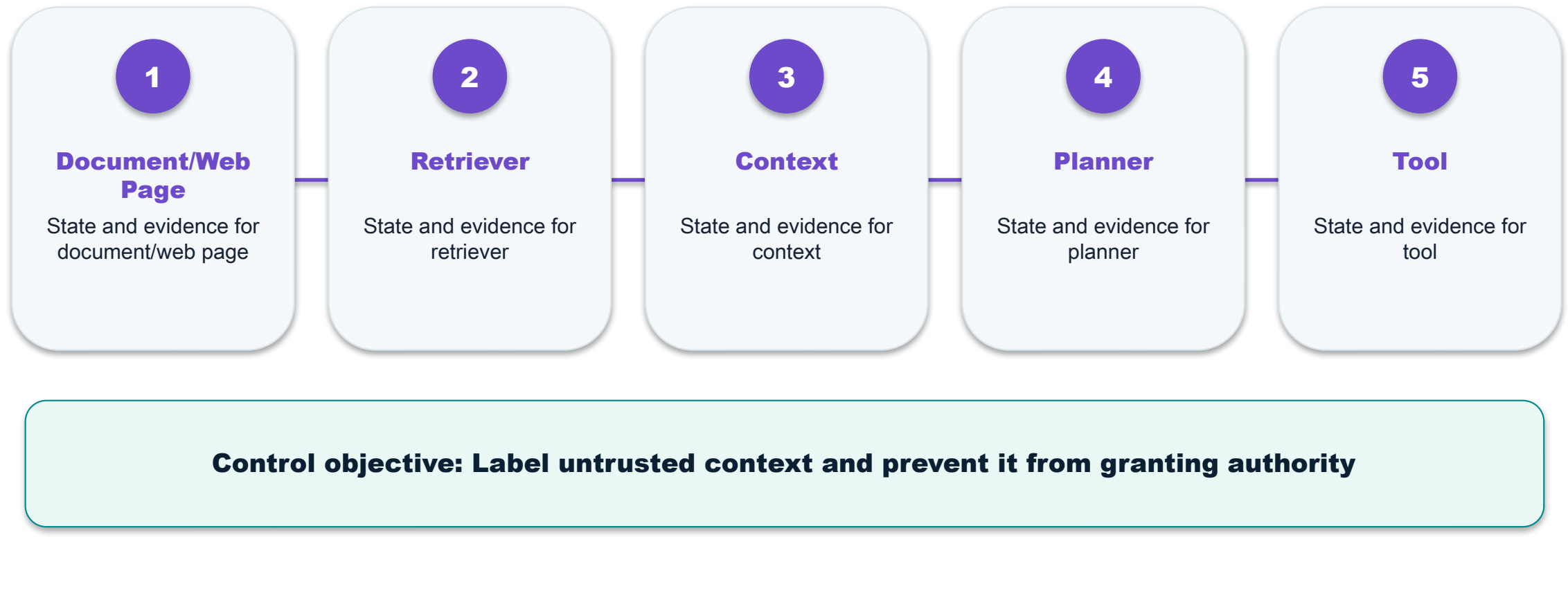
Accept only when Injection suite cannot produce an unauthorised tool call. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

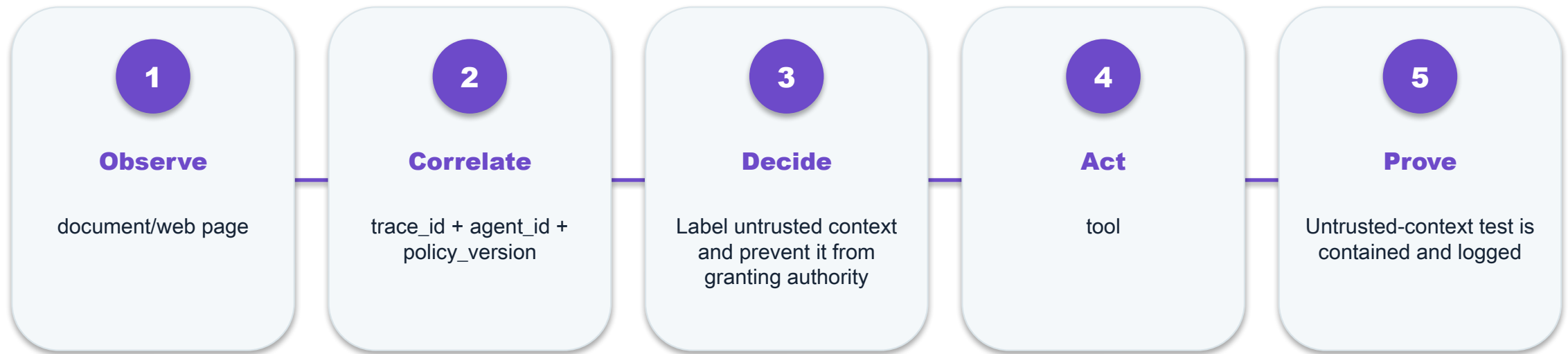
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Indirect prompt injection: architecture and control boundary



Indirect prompt injection: runtime decision trace



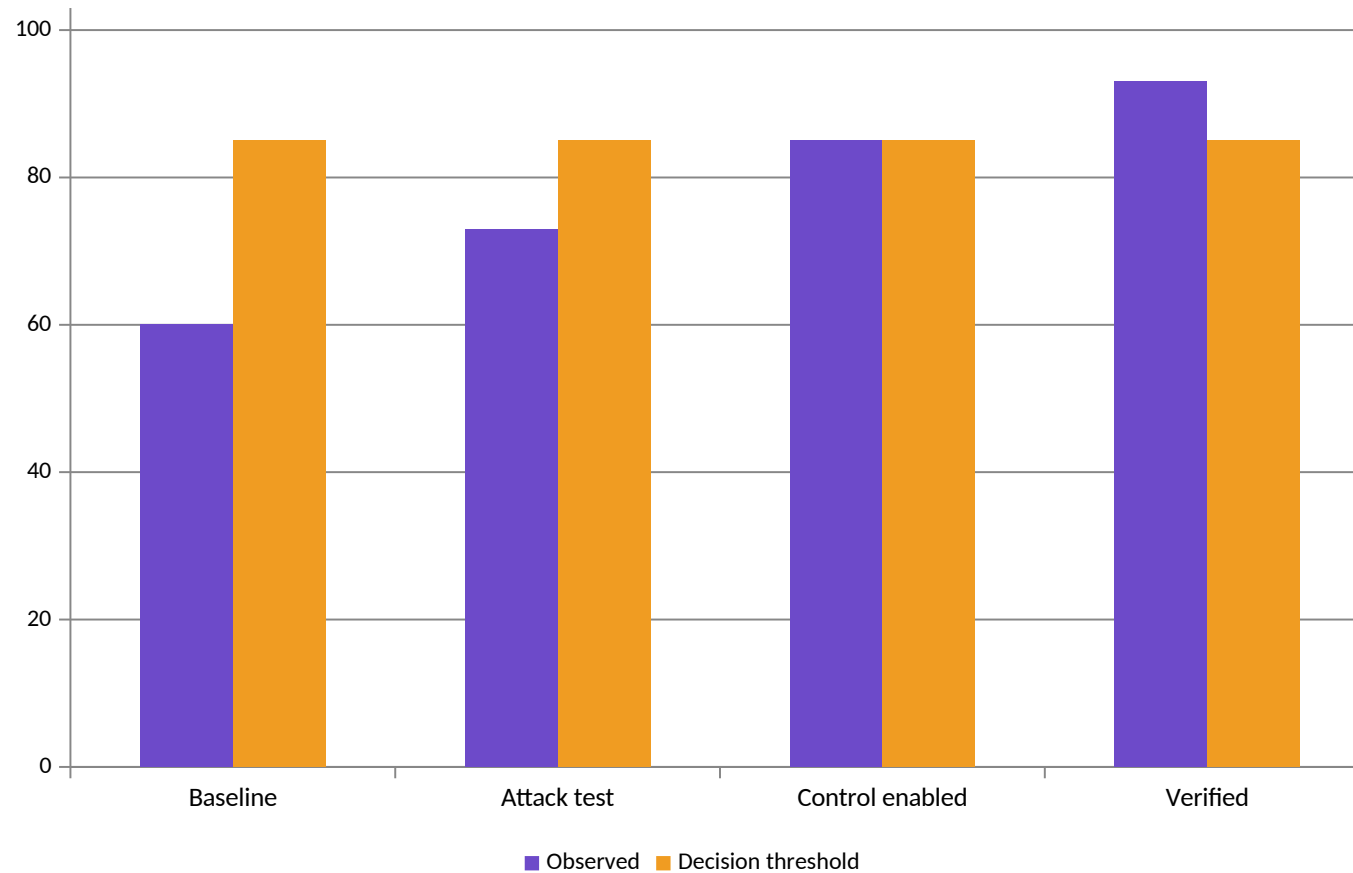
Failure to detect: Retrieved content tells the agent to exfiltrate data

Indirect prompt injection: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	source_uri	Required in decision and audit evidence
field_2	trust_label	Required in decision and audit evidence
field_3	content_hash	Required in decision and audit evidence
field_4	instruction_markers	Required in decision and audit evidence
field_5	action	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Indirect prompt injection: evidence and acceptance decision



DECISION RULE

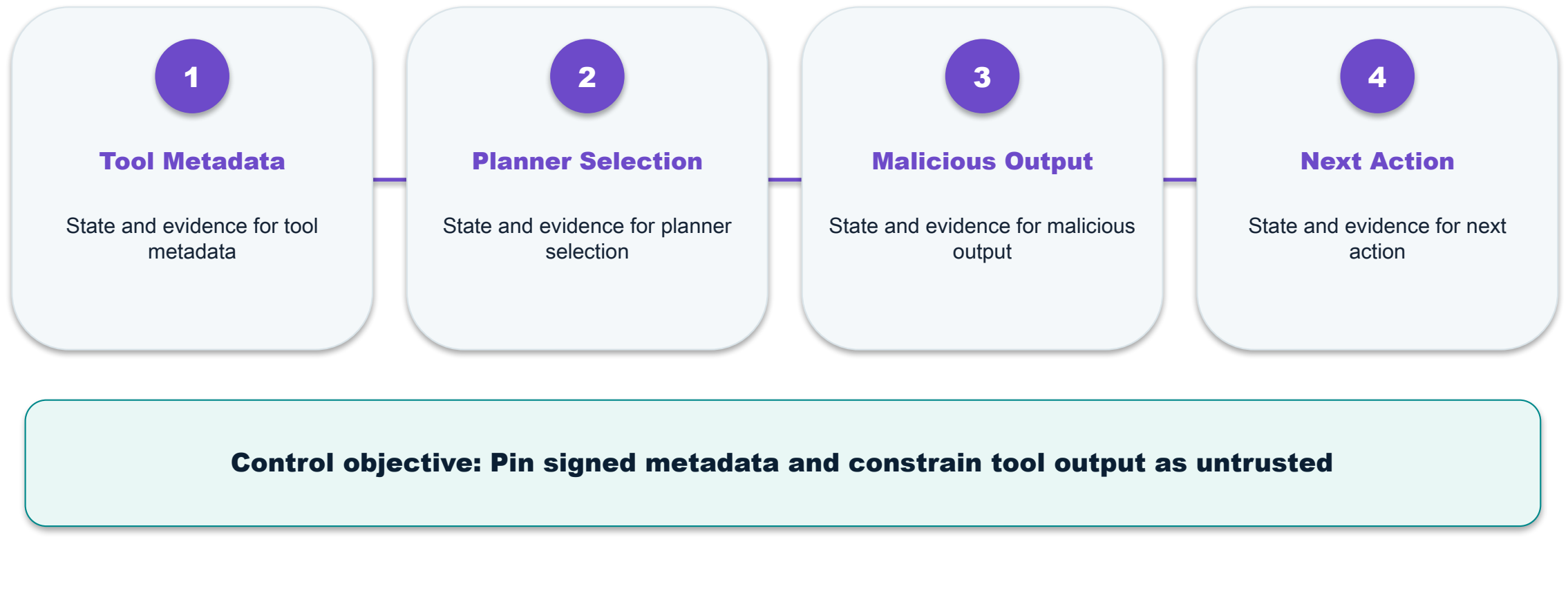
Accept only when Untrusted-context test is contained and logged. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

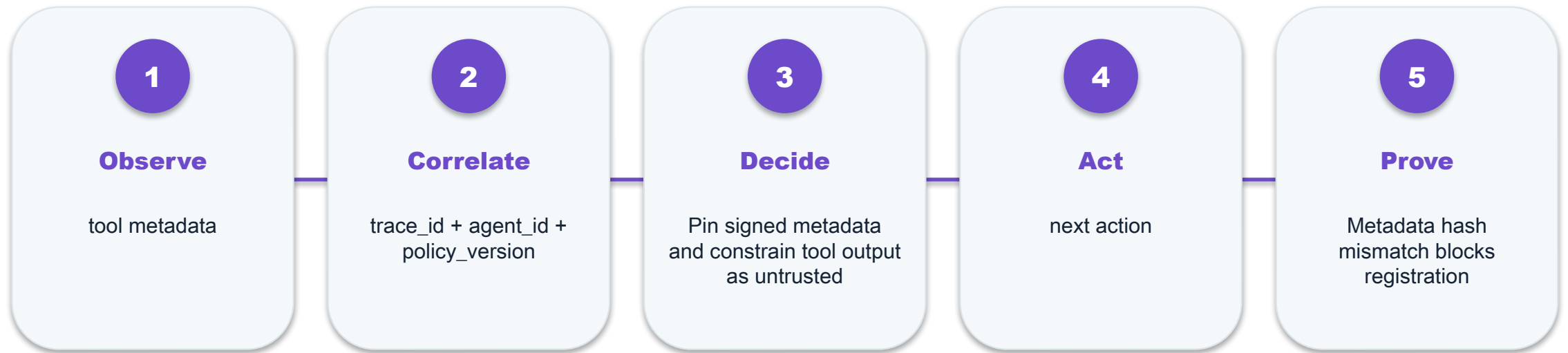
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Tool poisoning: architecture and control boundary



Tool poisoning: runtime decision trace



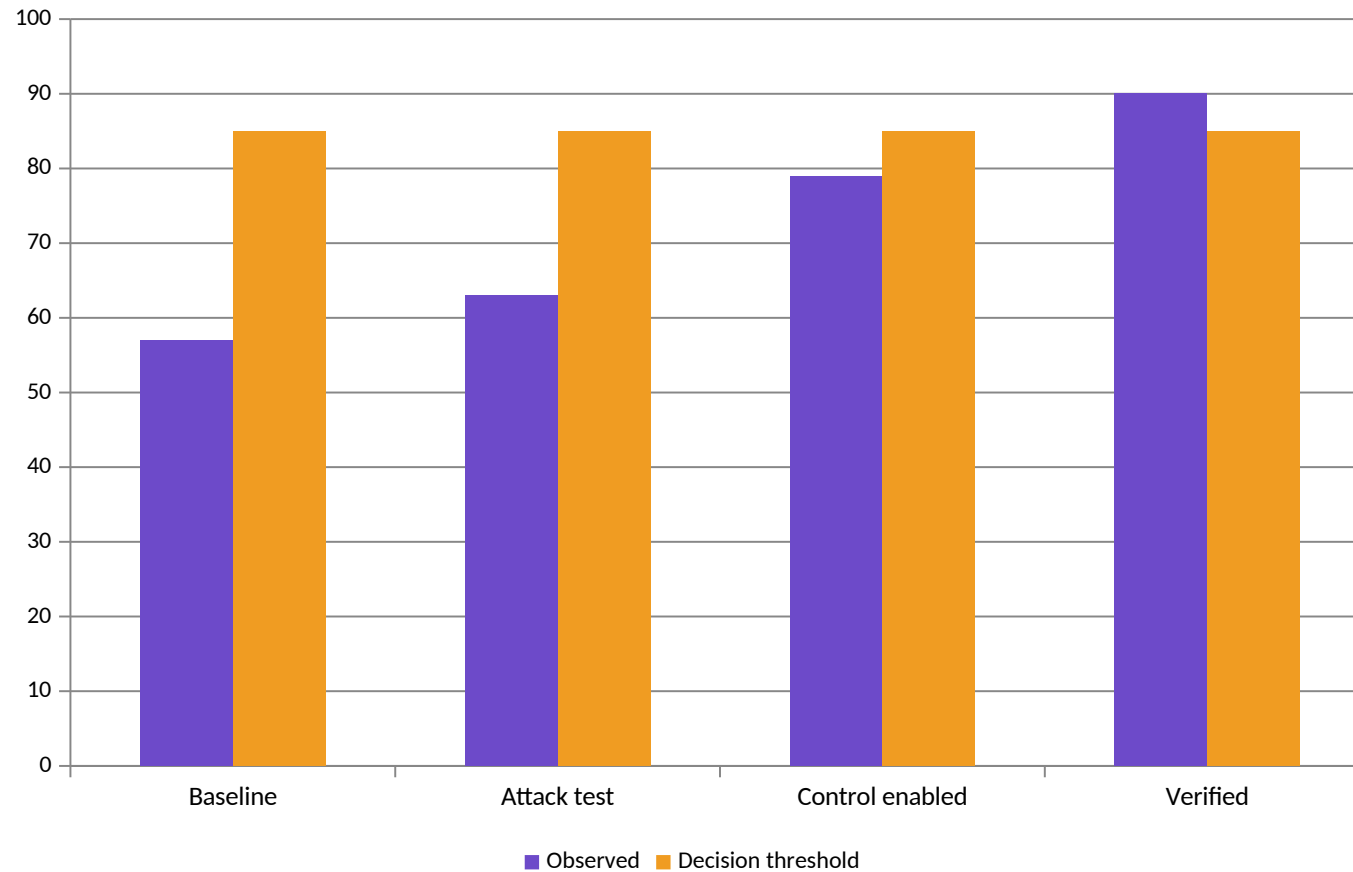
Failure to detect: Poisoned tool description manipulates planning

Tool poisoning: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	tool_id	Required in decision and audit evidence
field_2	description_hash	Required in decision and audit evidence
field_3	publisher	Required in decision and audit evidence
field_4	output_trust	Required in decision and audit evidence
field_5	version	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Tool poisoning: evidence and acceptance decision



DECISION RULE

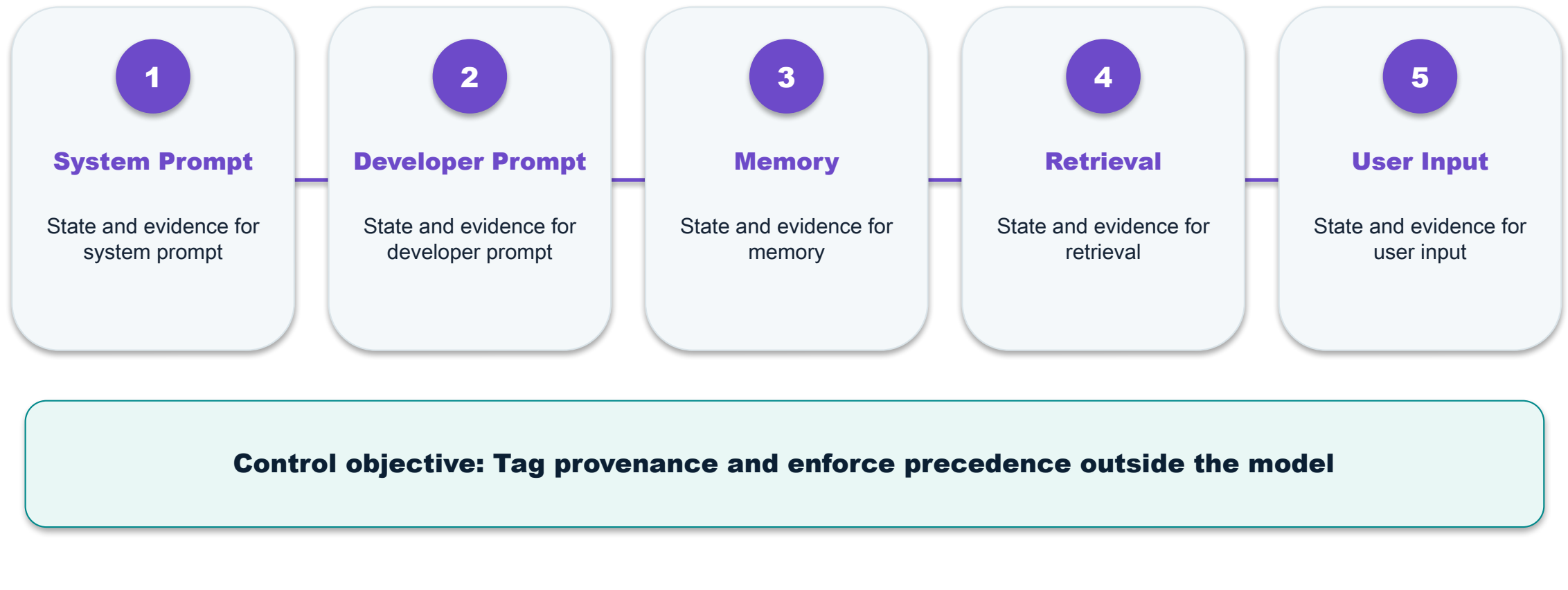
Accept only when Metadata hash mismatch blocks registration. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Context poisoning: architecture and control boundary



Context poisoning: runtime decision trace



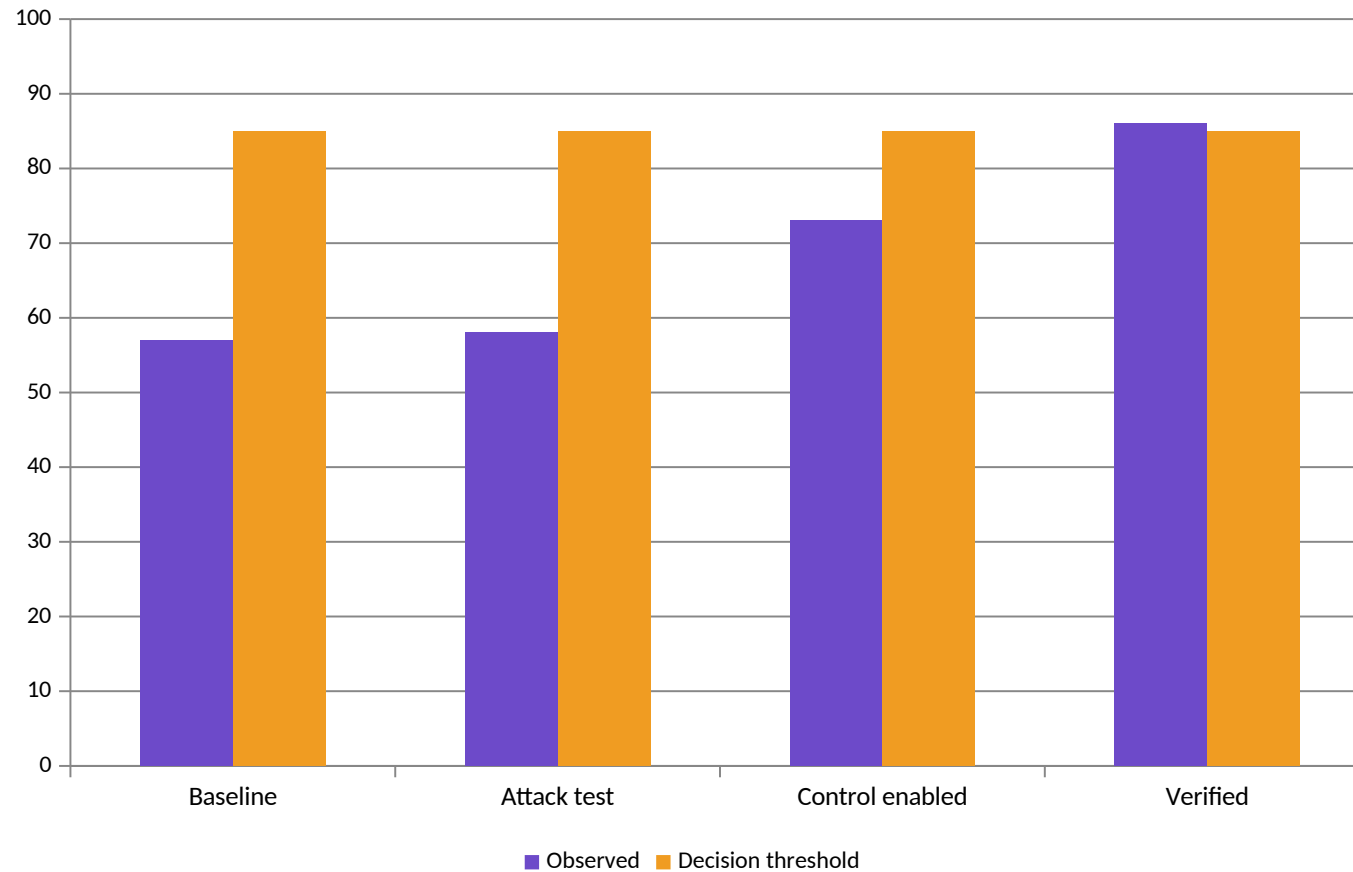
Failure to detect: Low-trust context persists as high-trust instruction

Context poisoning: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	segment_type	Required in decision and audit evidence
field_2	source	Required in decision and audit evidence
field_3	precedence	Required in decision and audit evidence
field_4	integrity_hash	Required in decision and audit evidence
field_5	trust_level	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Context poisoning: evidence and acceptance decision



DECISION RULE

Accept only when Context assembly report shows trust labels and hashes. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Memory poisoning: architecture and control boundary



Memory poisoning: runtime decision trace



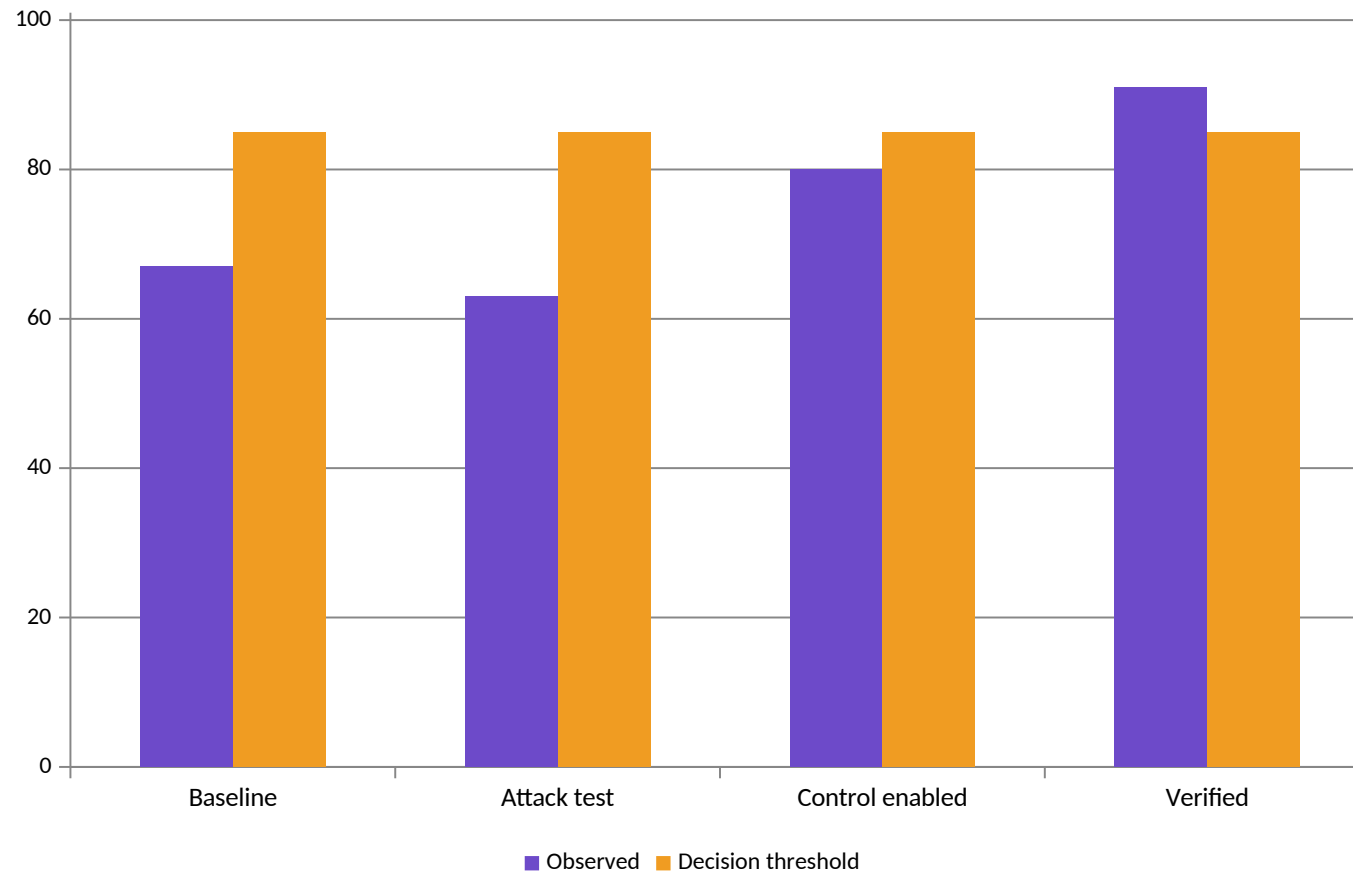
Failure to detect: Adversarial fact persists and steers future sessions

Memory poisoning: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	memory_id	Required in decision and audit evidence
field_2	author	Required in decision and audit evidence
field_3	tenant	Required in decision and audit evidence
field_4	confidence	Required in decision and audit evidence
field_5	ttl	Required in decision and audit evidence
field_6	approved_for_reuse	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Memory poisoning: evidence and acceptance decision



DECISION RULE

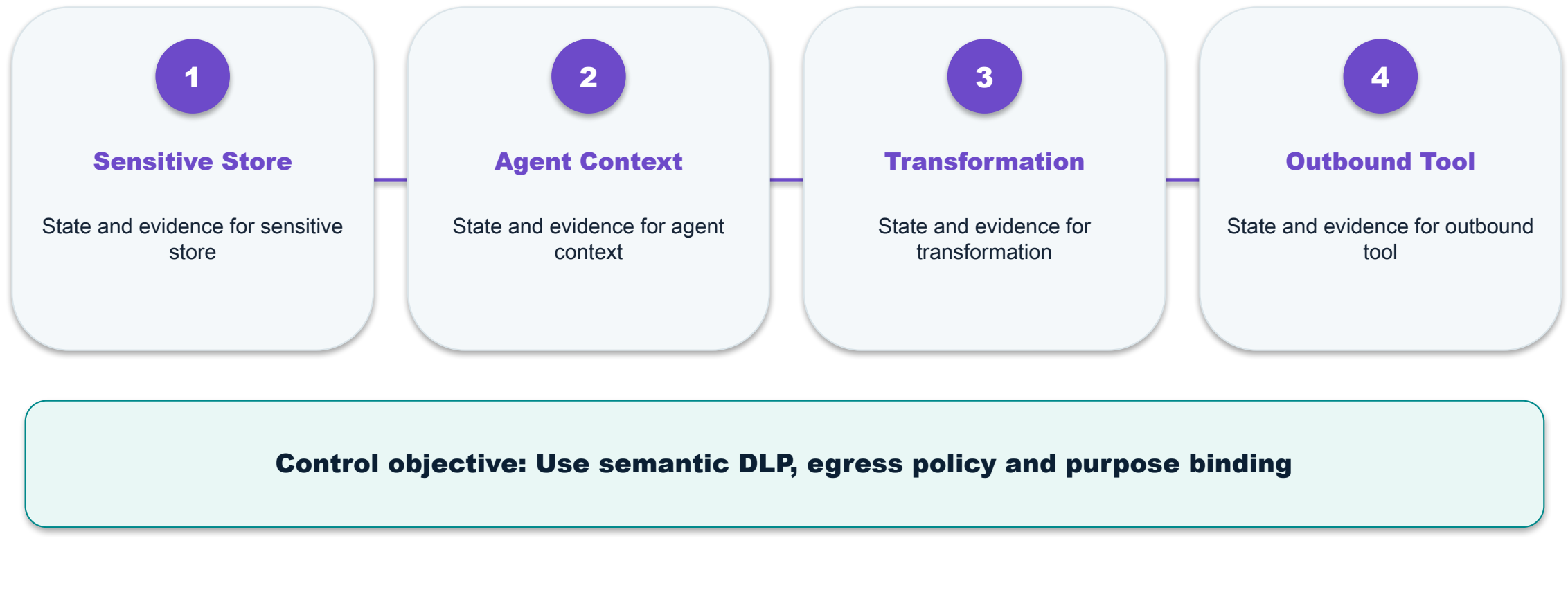
**Accept only when
Poison seed is absent
from unrelated session
retrieval. Otherwise
record a gap, owner and
due date.**

EVIDENCE STATUS

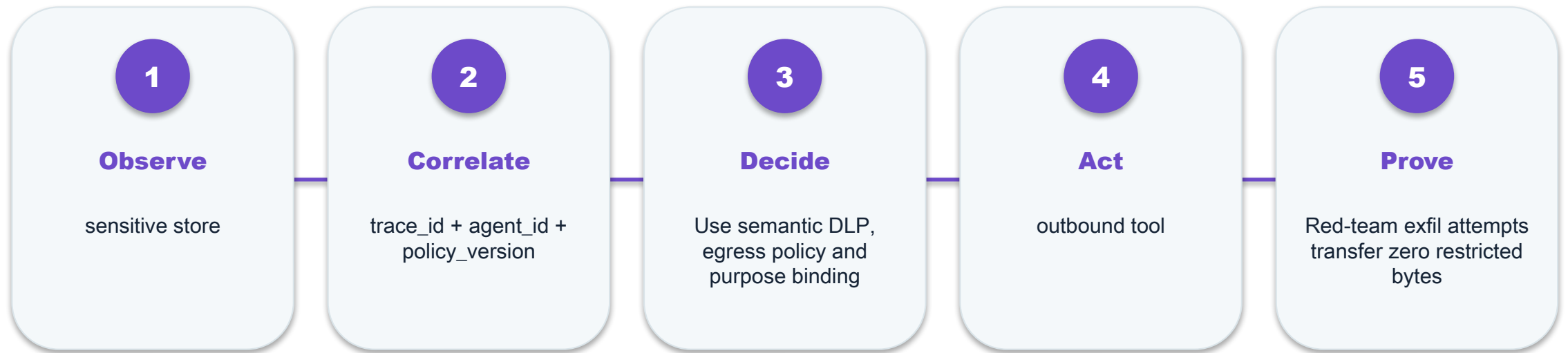
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Data exfiltration: architecture and control boundary



Data exfiltration: runtime decision trace



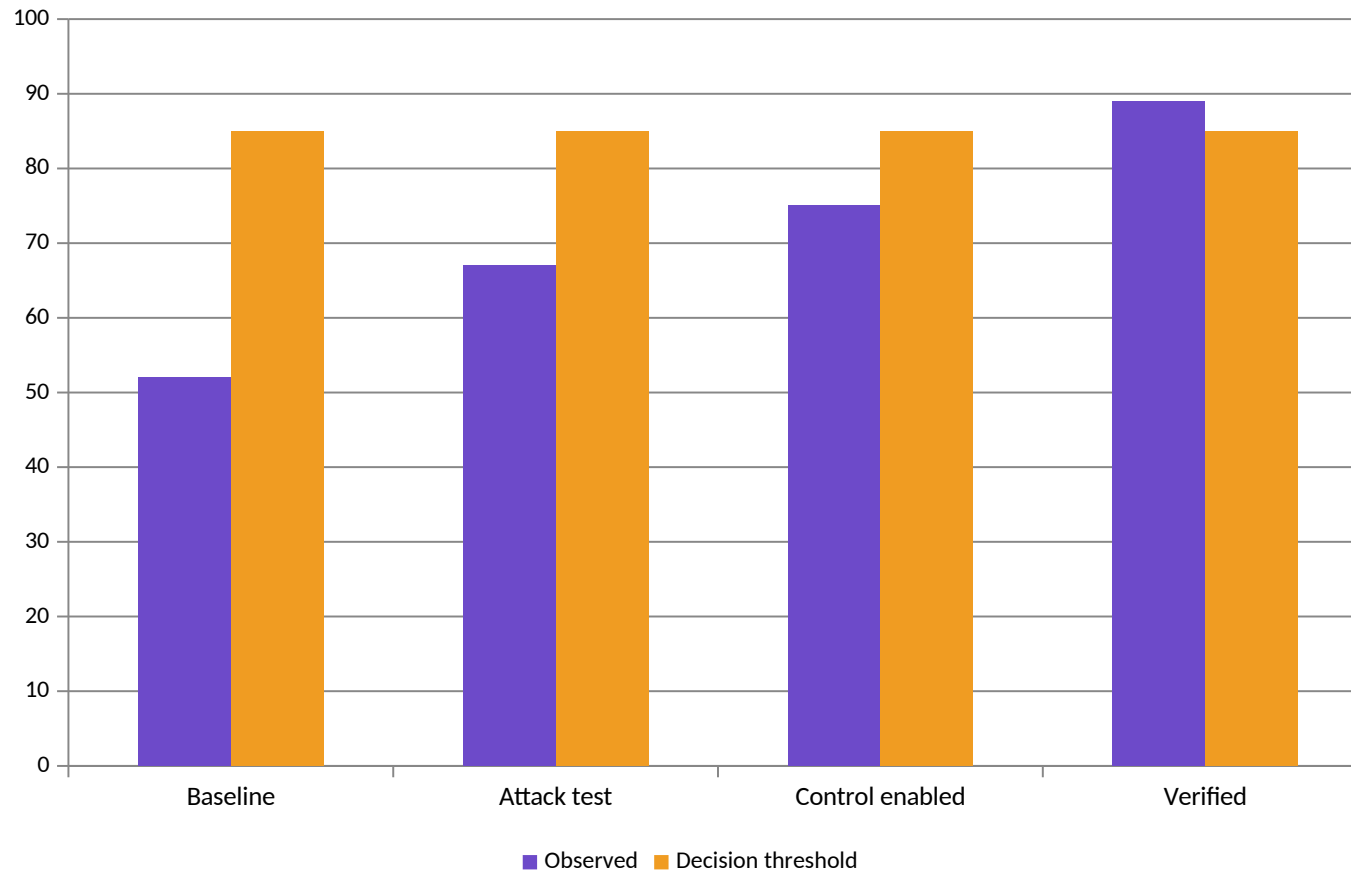
Failure to detect: Agent encodes restricted data to evade keyword DLP

Data exfiltration: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	data_class	Required in decision and audit evidence
field_2	bytes_read	Required in decision and audit evidence
field_3	destination	Required in decision and audit evidence
field_4	bytes_out	Required in decision and audit evidence
field_5	leak_score	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Data exfiltration: evidence and acceptance decision



DECISION RULE

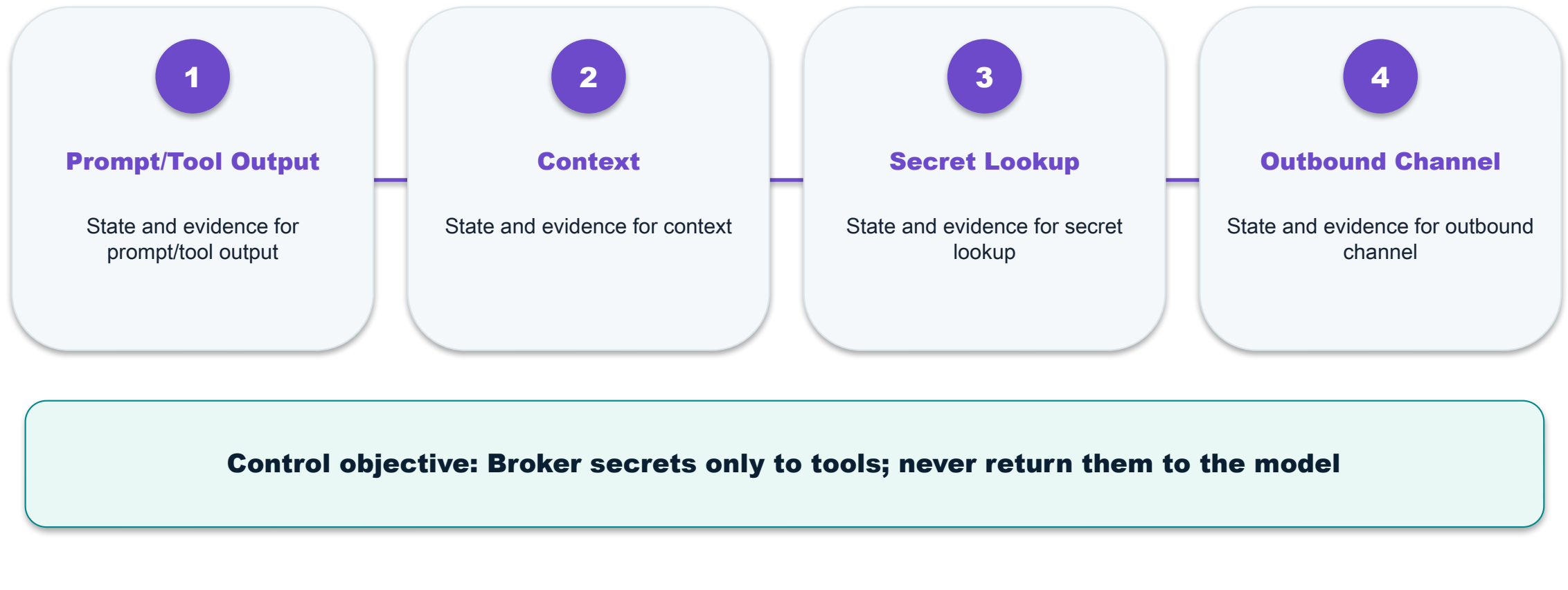
Accept only when Red-team exfil attempts transfer zero restricted bytes. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

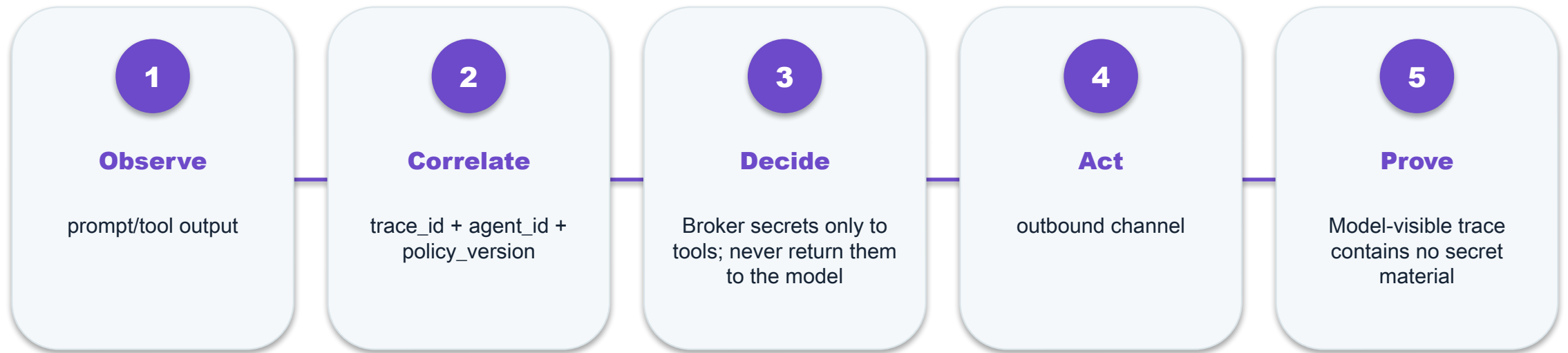
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Credential harvesting: architecture and control boundary



Credential harvesting: runtime decision trace



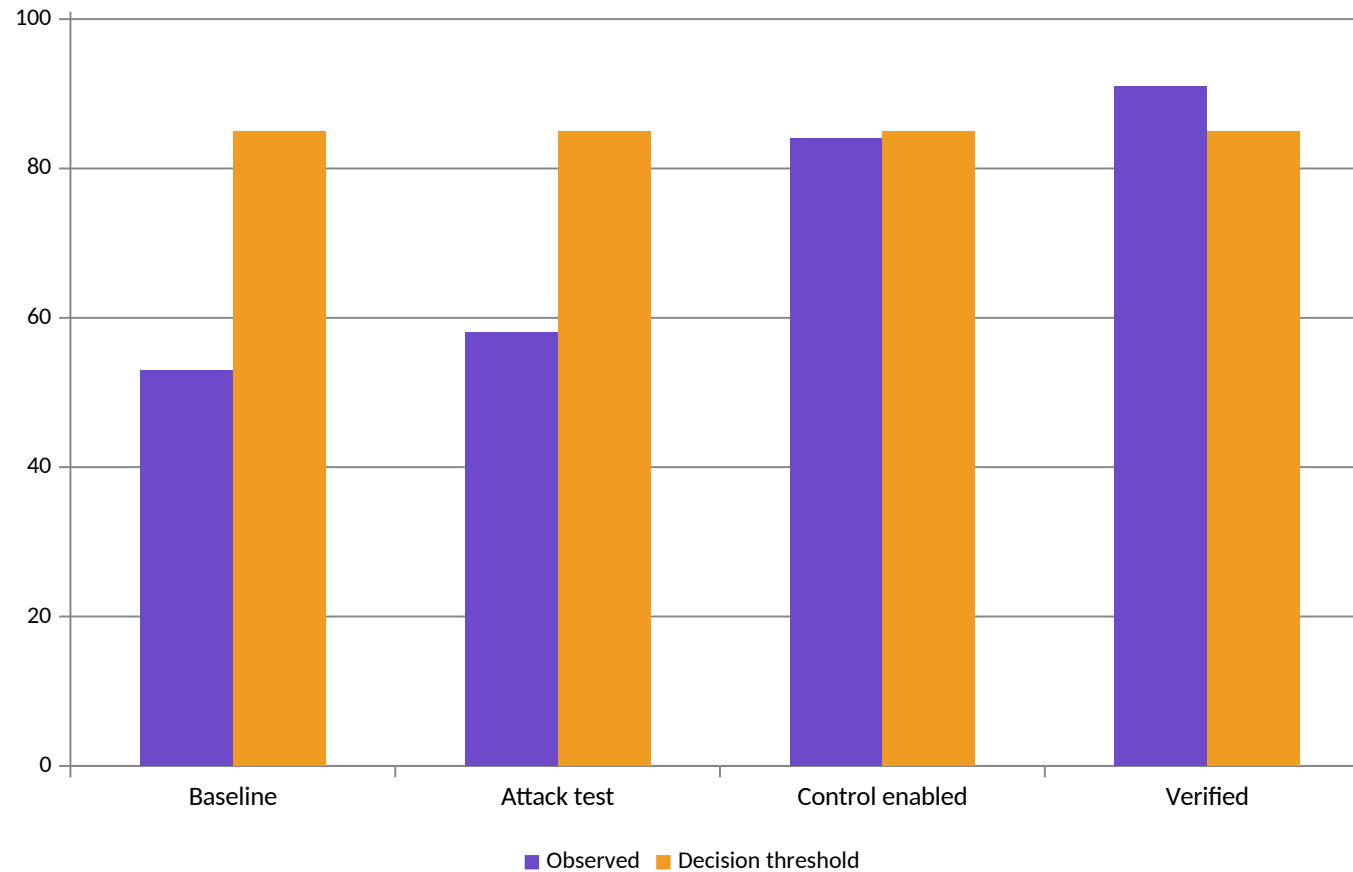
Failure to detect: Agent is tricked into retrieving and revealing a token

Credential harvesting: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	secret_ref	Required in decision and audit evidence
field_2	access_reason	Required in decision and audit evidence
field_3	requester	Required in decision and audit evidence
field_4	destination	Required in decision and audit evidence
field_5	decision	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Credential harvesting: evidence and acceptance decision



DECISION RULE

Accept only when Model-visible trace contains no secret material. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

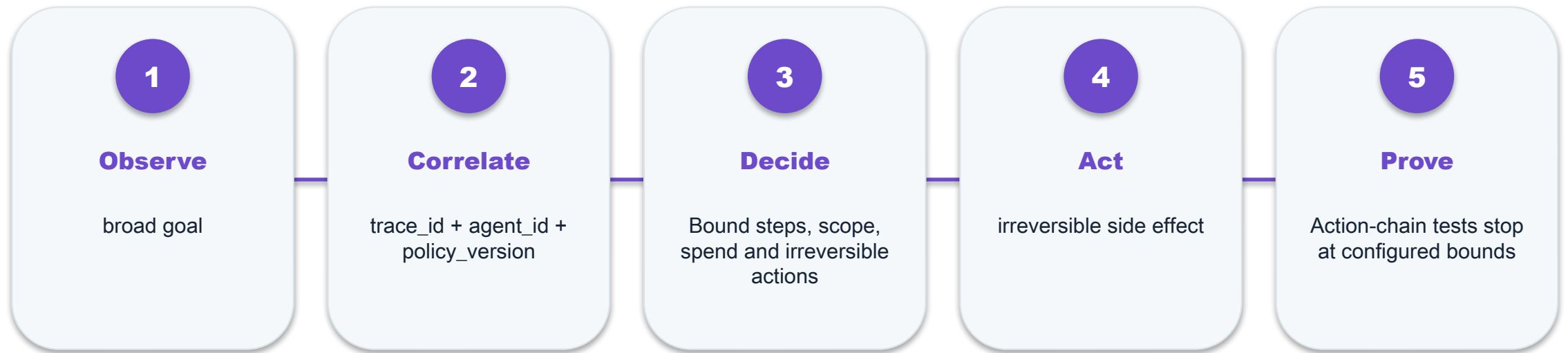
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Excessive agency: architecture and control boundary



Excessive agency: runtime decision trace



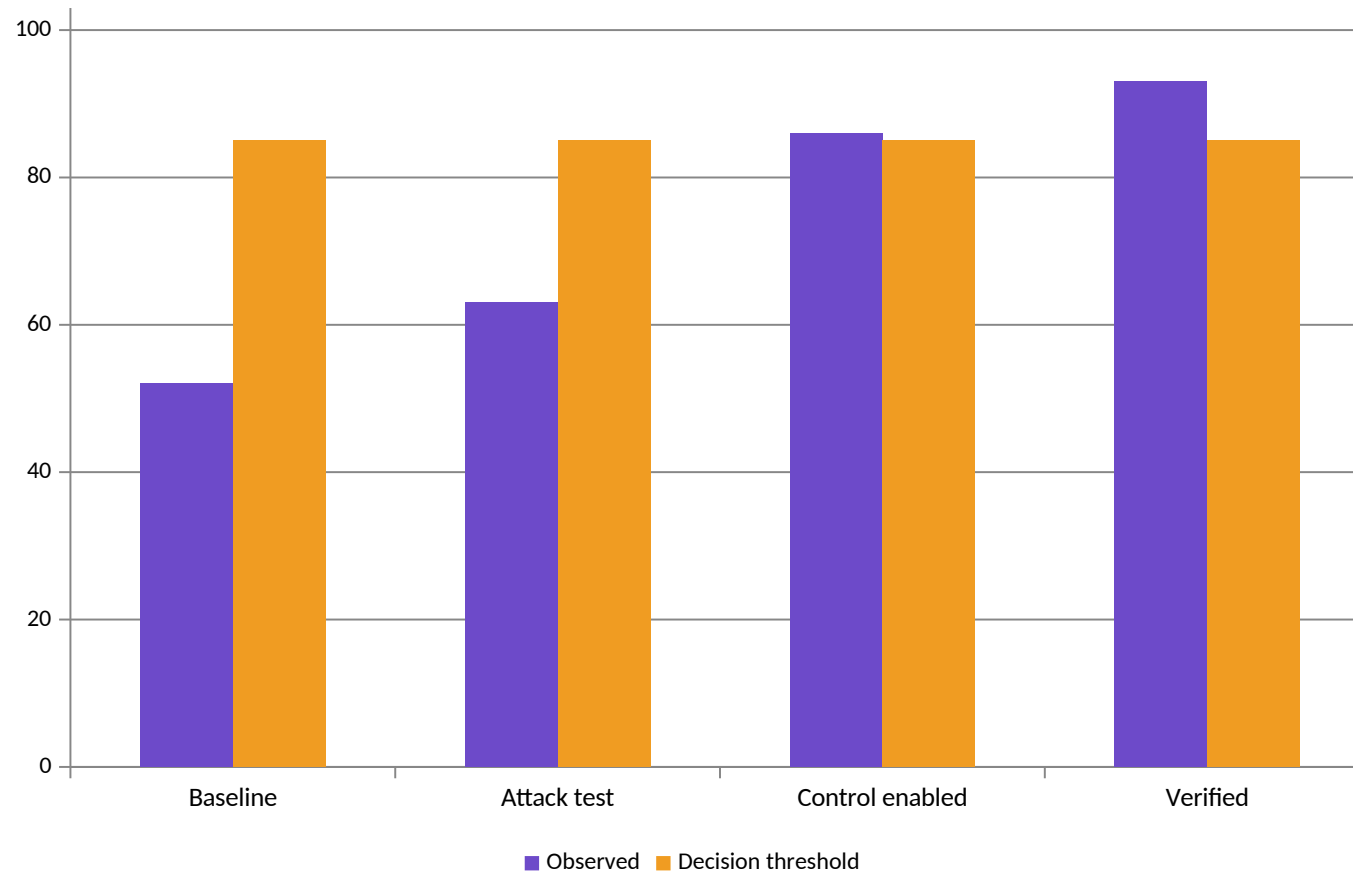
Failure to detect: Agent performs destructive actions beyond user intent

Excessive agency: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	max_steps	Required in decision and audit evidence
field_2	allowed_tools	Required in decision and audit evidence
field_3	spend_limit	Required in decision and audit evidence
field_4	approval_tier	Required in decision and audit evidence
field_5	stop_reason	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Excessive agency: evidence and acceptance decision



DECISION RULE

**Accept only when
Action-chain tests stop
at configured bounds.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Privilege escalation: architecture and control boundary



Privilege escalation: runtime decision trace



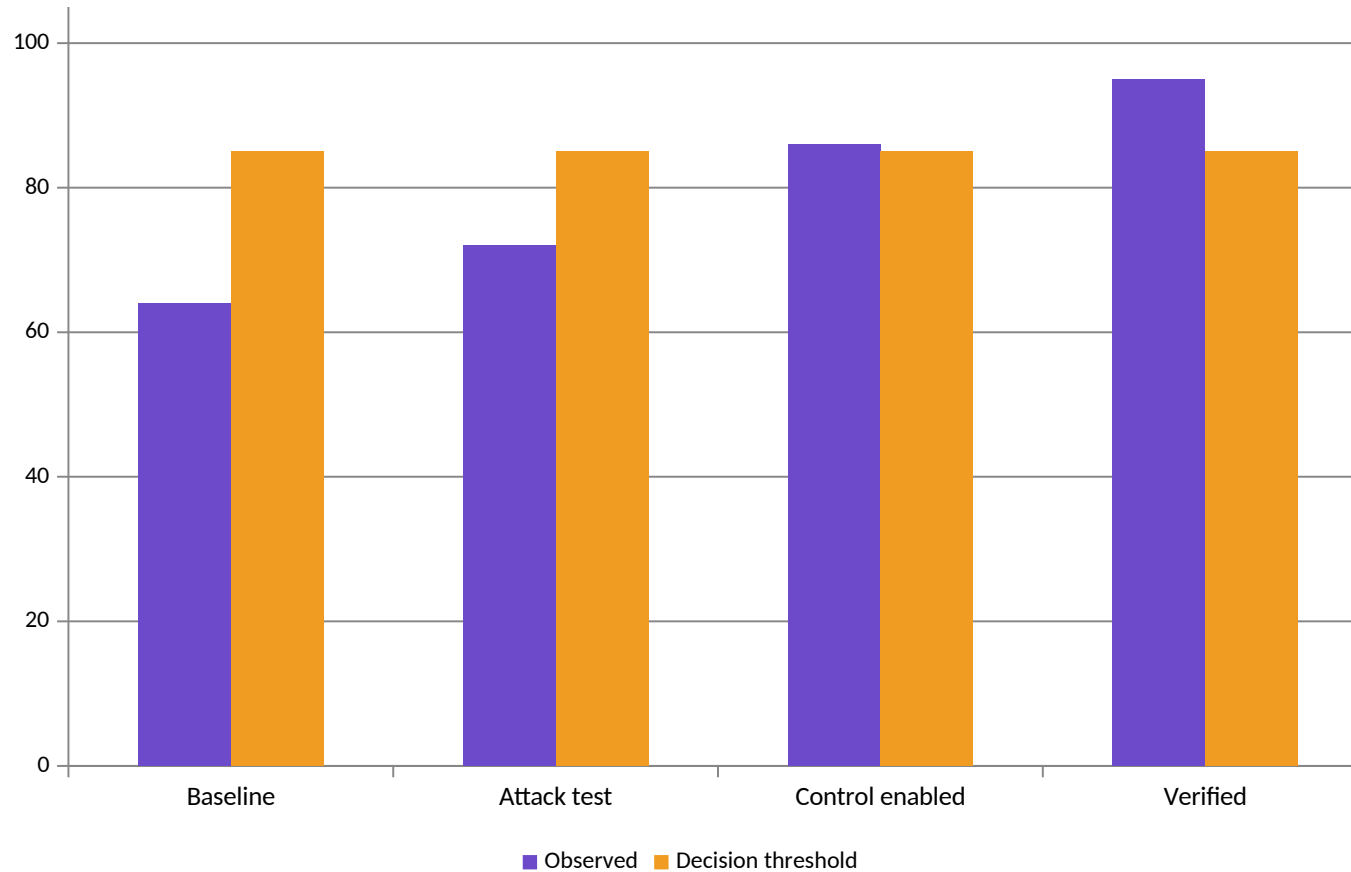
Failure to detect: Wildcard permission makes admin action reachable

Privilege escalation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	principal	Required in decision and audit evidence
field_2	role	Required in decision and audit evidence
field_3	scope	Required in decision and audit evidence
field_4	resource	Required in decision and audit evidence
field_5	requested_action	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Privilege escalation: evidence and acceptance decision



DECISION RULE

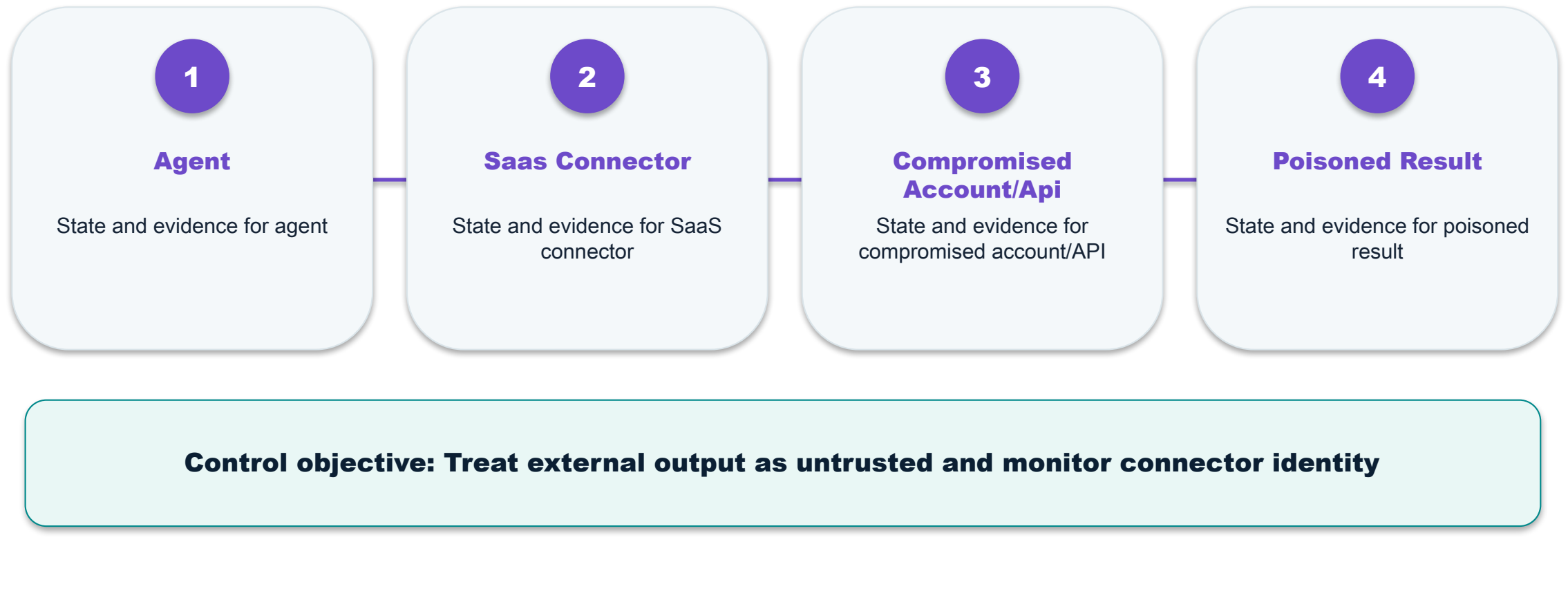
**Accept only when
Privilege-path graph has
no unintended route.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

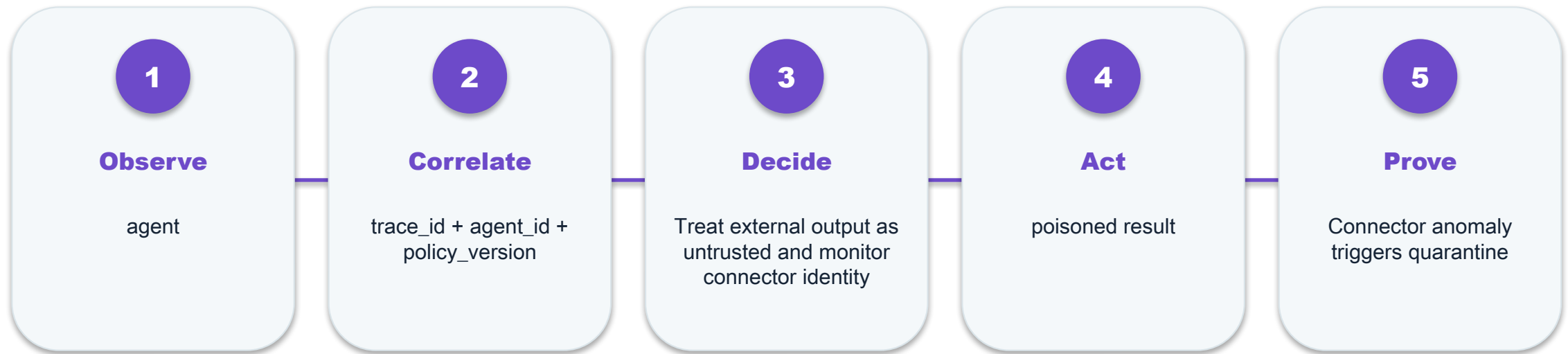
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Compromised integration: architecture and control boundary



Compromised integration: runtime decision trace



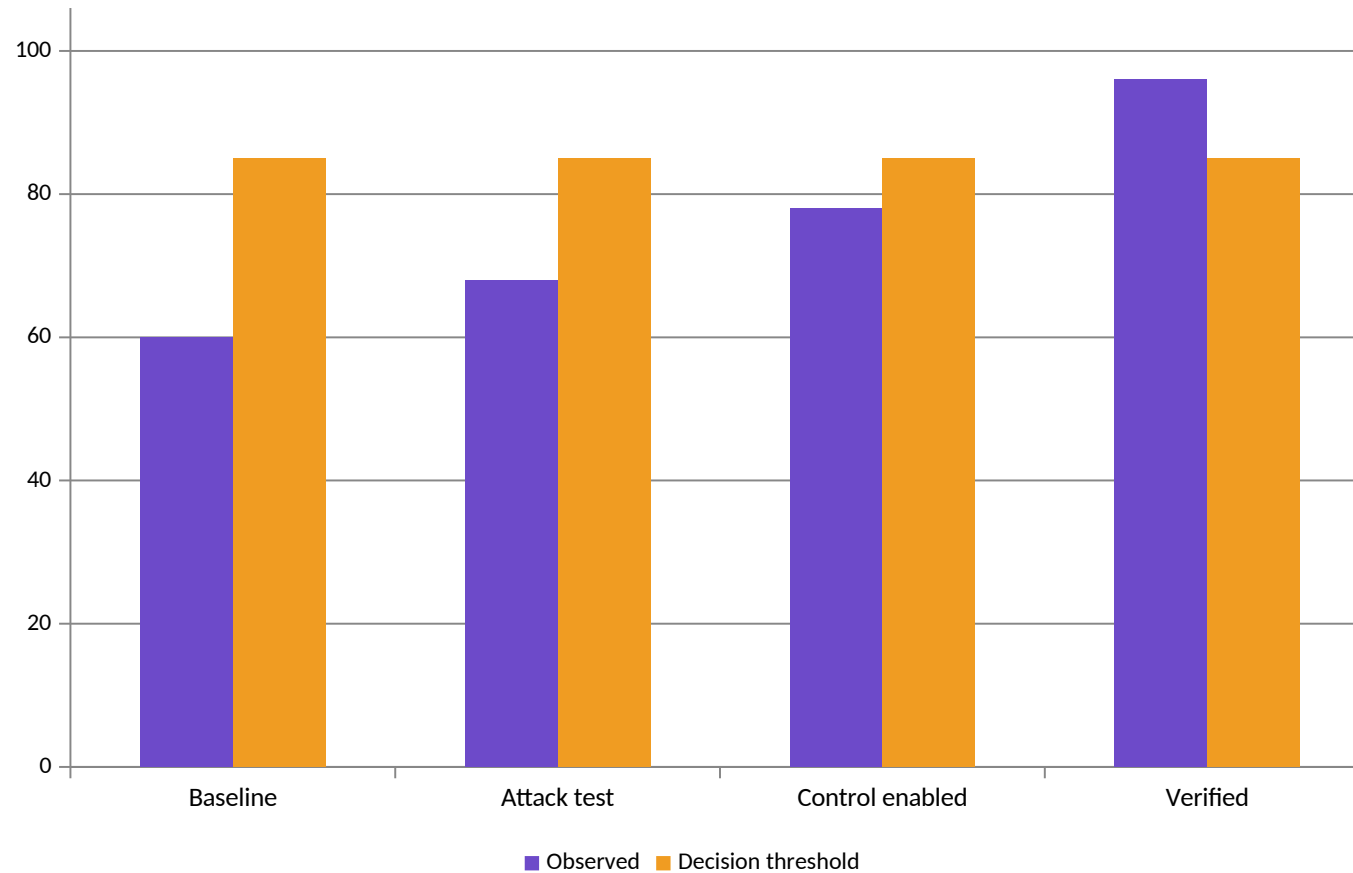
Failure to detect: Trusted integration returns malicious instructions or data

Compromised integration: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	connector_id	Required in decision and audit evidence
field_2	auth_subject	Required in decision and audit evidence
field_3	endpoint	Required in decision and audit evidence
field_4	tls_peer	Required in decision and audit evidence
field_5	response_hash	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Compromised integration: evidence and acceptance decision



DECISION RULE

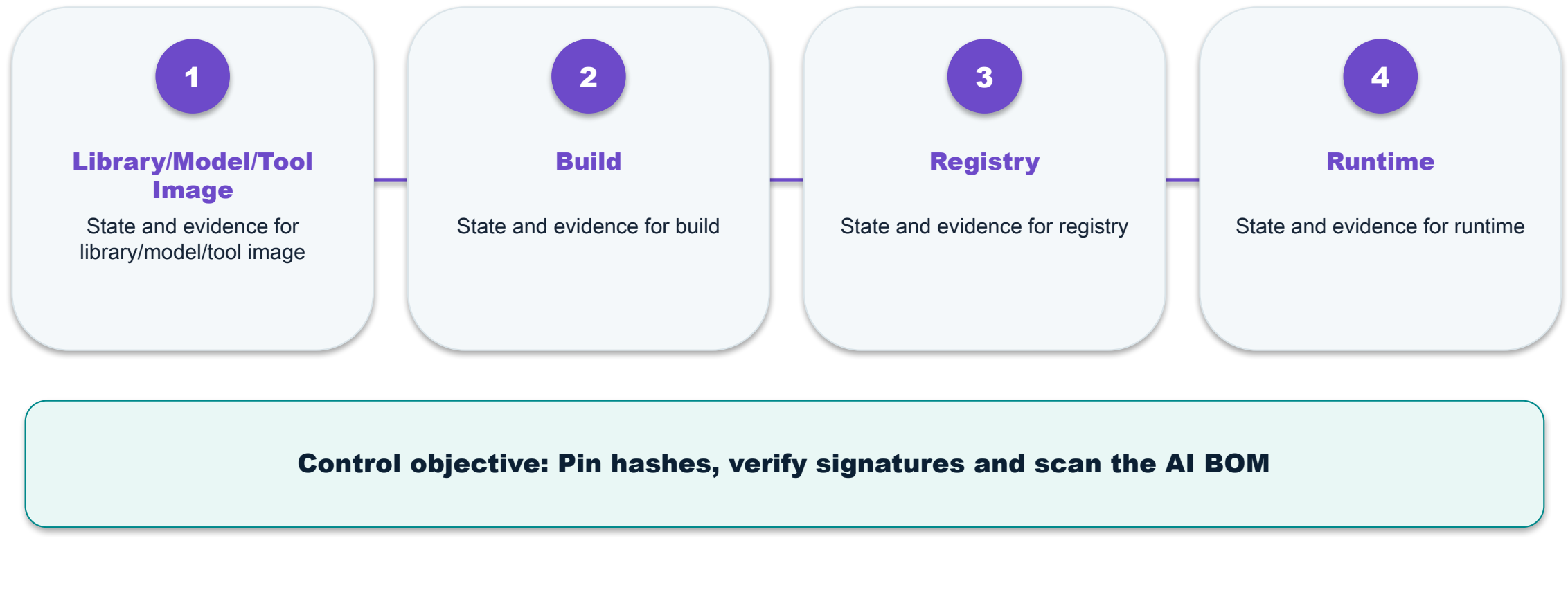
Accept only when Connector anomaly triggers quarantine. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

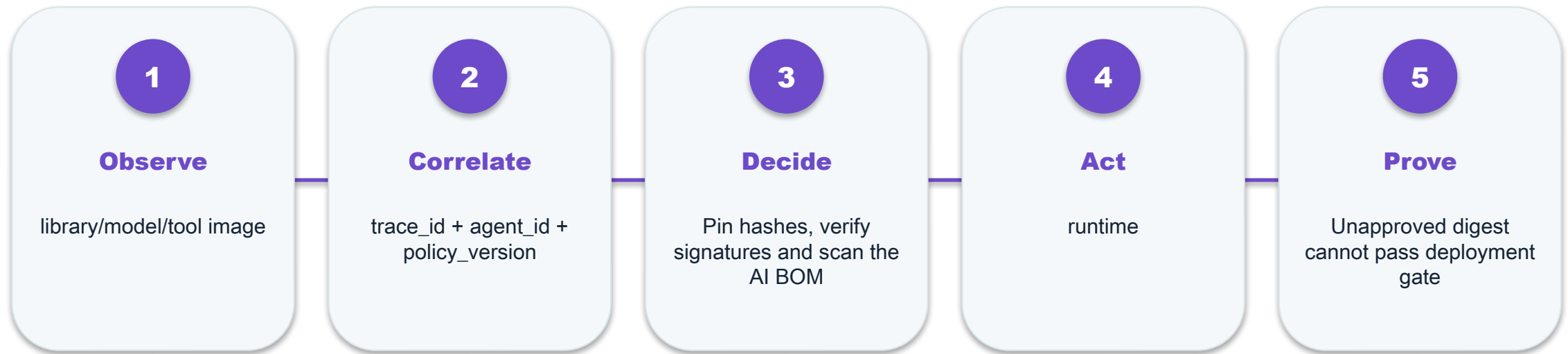
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Supply-chain compromise: architecture and control boundary



Supply-chain compromise: runtime decision trace



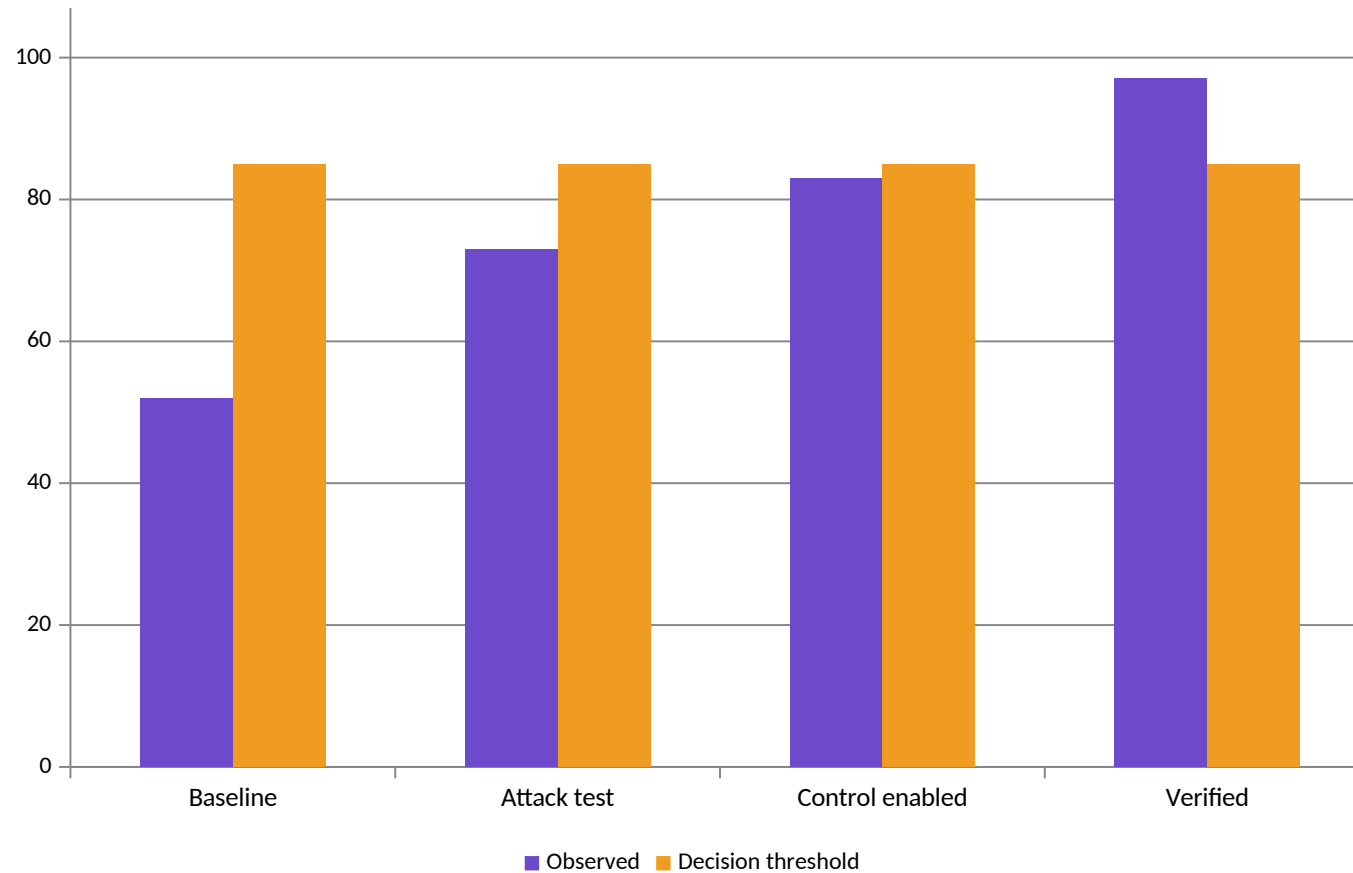
Failure to detect: Typosquatted dependency executes inside the agent runtime

Supply-chain compromise: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	package	Required in decision and audit evidence
field_2	version	Required in decision and audit evidence
field_3	digest	Required in decision and audit evidence
field_4	signer	Required in decision and audit evidence
field_5	SBOM_ref	Required in decision and audit evidence
field_6	vuln_status	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Supply-chain compromise: evidence and acceptance decision



DECISION RULE

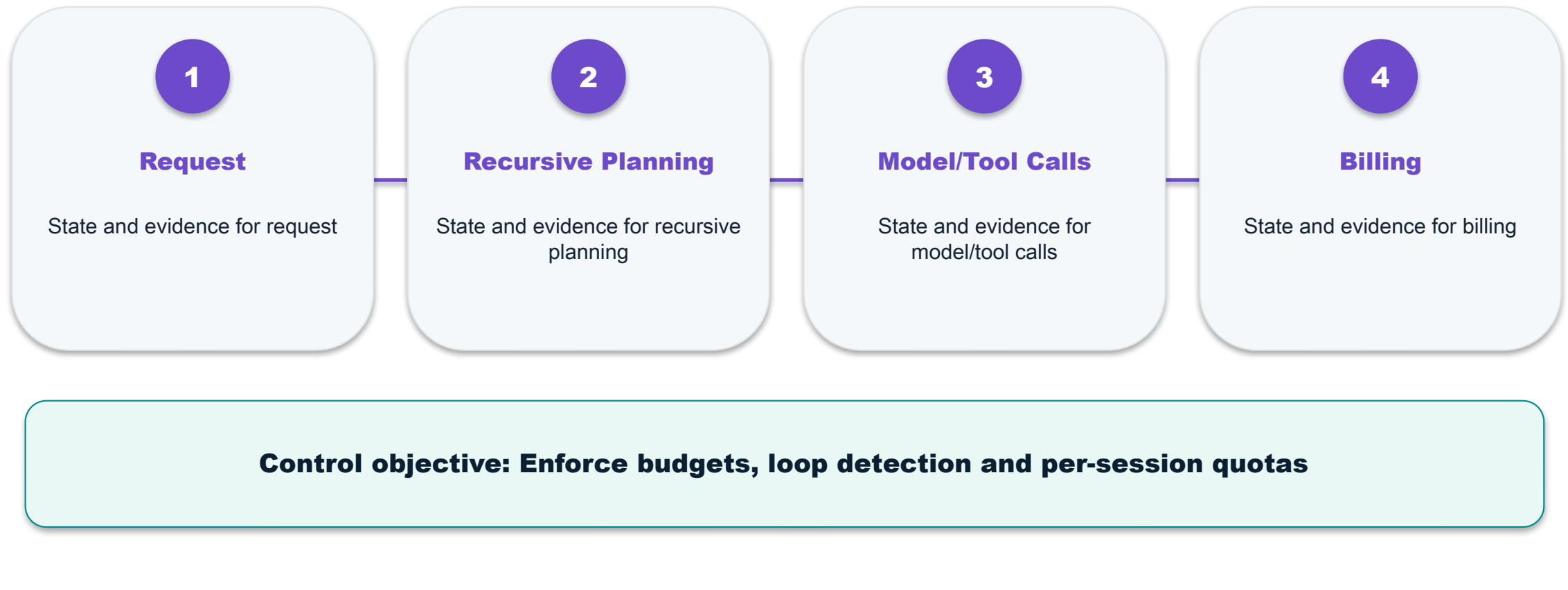
Accept only when Unapproved digest cannot pass deployment gate. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

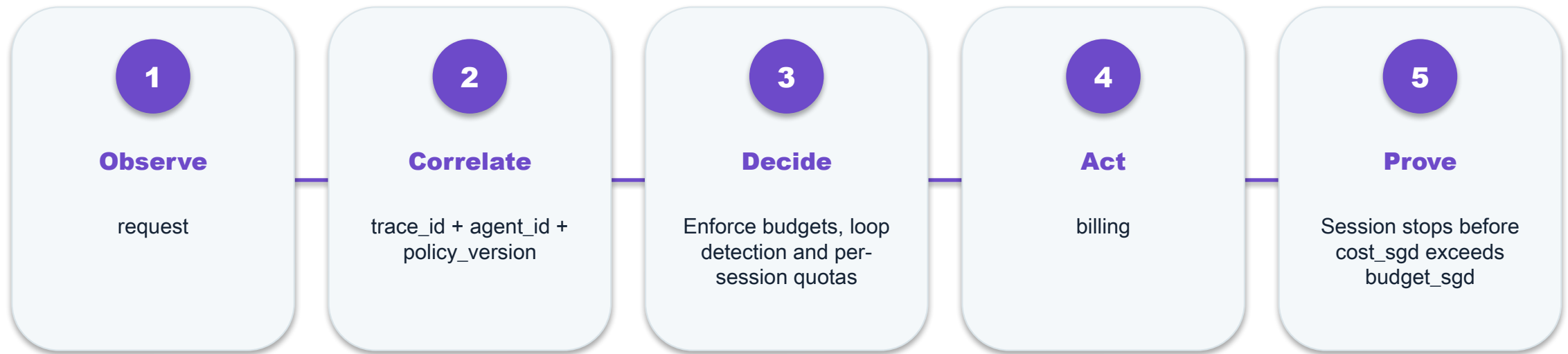
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Unbounded consumption: architecture and control boundary



Unbounded consumption: runtime decision trace



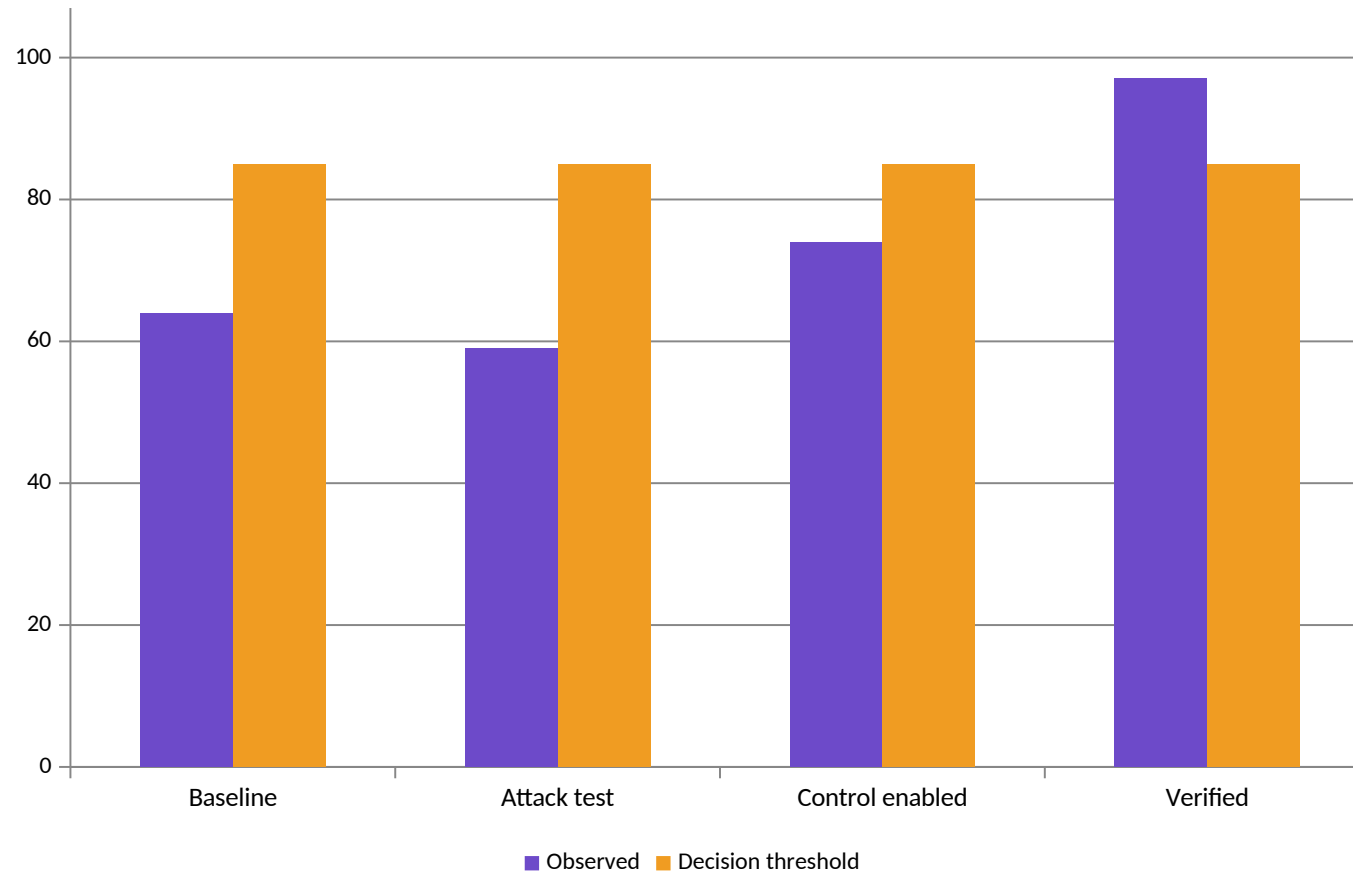
Failure to detect: Loop consumes quota and creates unexpected cost

Unbounded consumption: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	call_count	Required in decision and audit evidence
field_2	token_count	Required in decision and audit evidence
field_3	elapsed_ms	Required in decision and audit evidence
field_4	cost_sgd	Required in decision and audit evidence
field_5	budget_sgd	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Unbounded consumption: evidence and acceptance decision



DECISION RULE

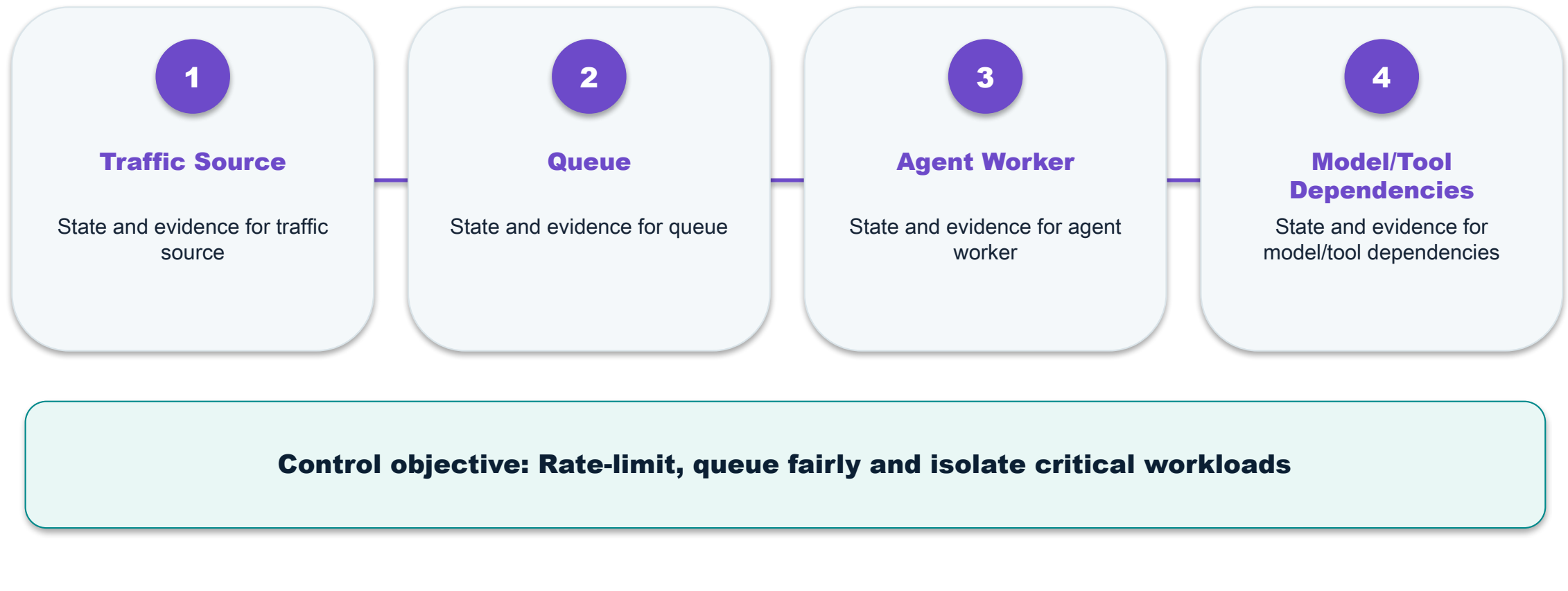
Accept only when Session stops before `cost_sgd` exceeds `budget_sgd`. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

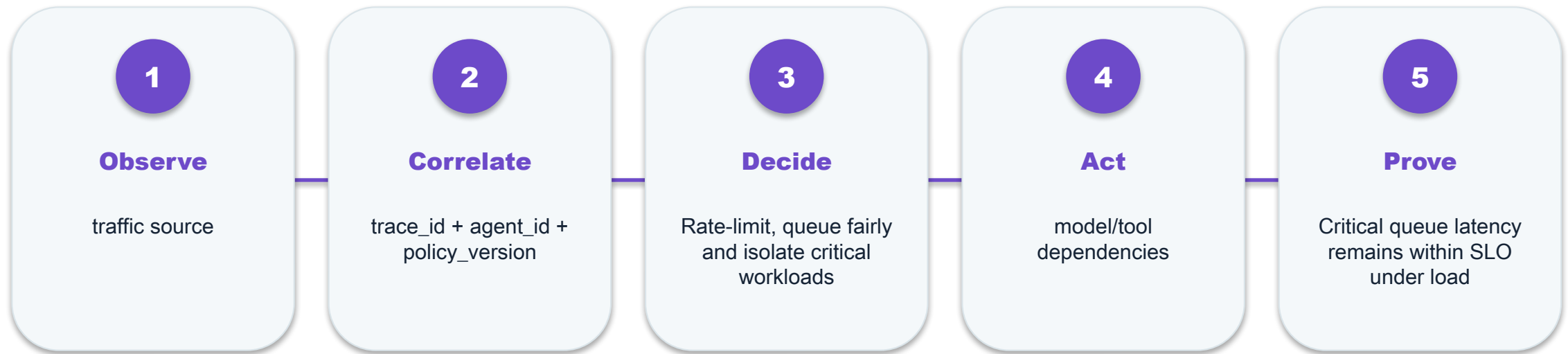
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Service denial: architecture and control boundary



Service denial: runtime decision trace



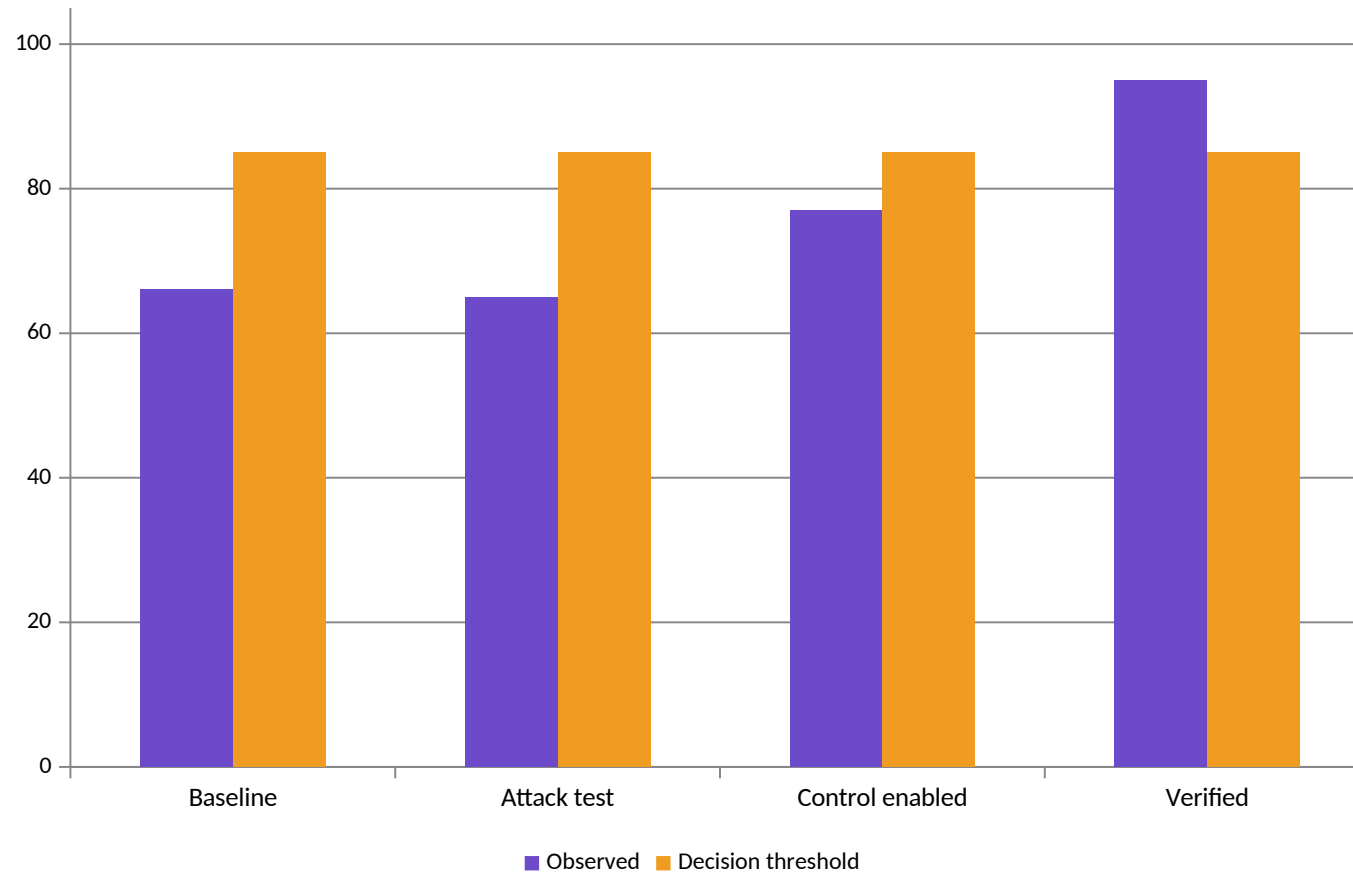
Failure to detect: Prompt flood saturates limited agent workers

Service denial: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	request_rate	Required in decision and audit evidence
field_2	queue_depth	Required in decision and audit evidence
field_3	timeout_rate	Required in decision and audit evidence
field_4	dependency_health	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Service denial: evidence and acceptance decision



DECISION RULE

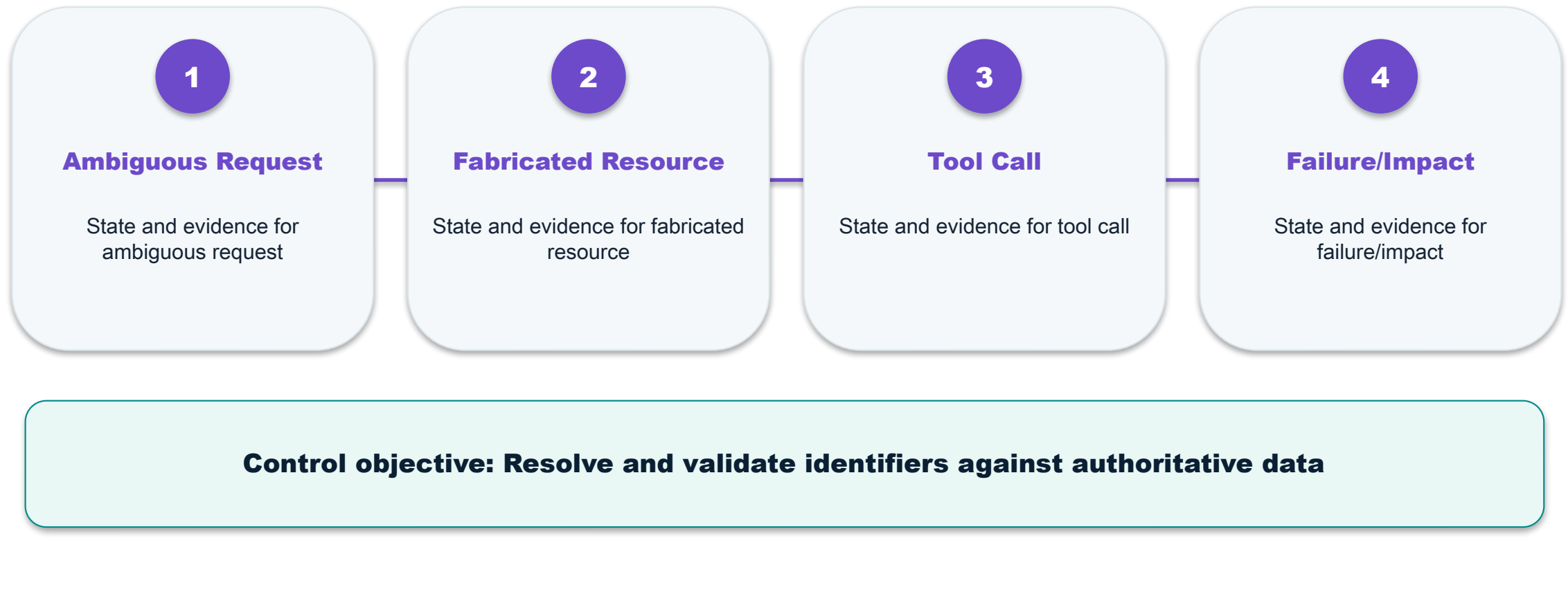
Accept only when Critical queue latency remains within SLO under load. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

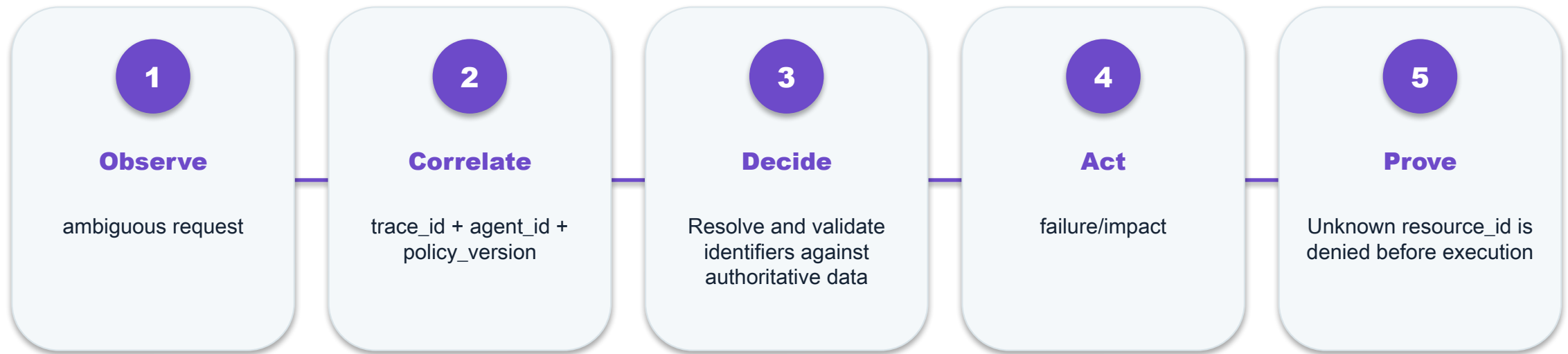
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Hallucinated action: architecture and control boundary



Hallucinated action: runtime decision trace



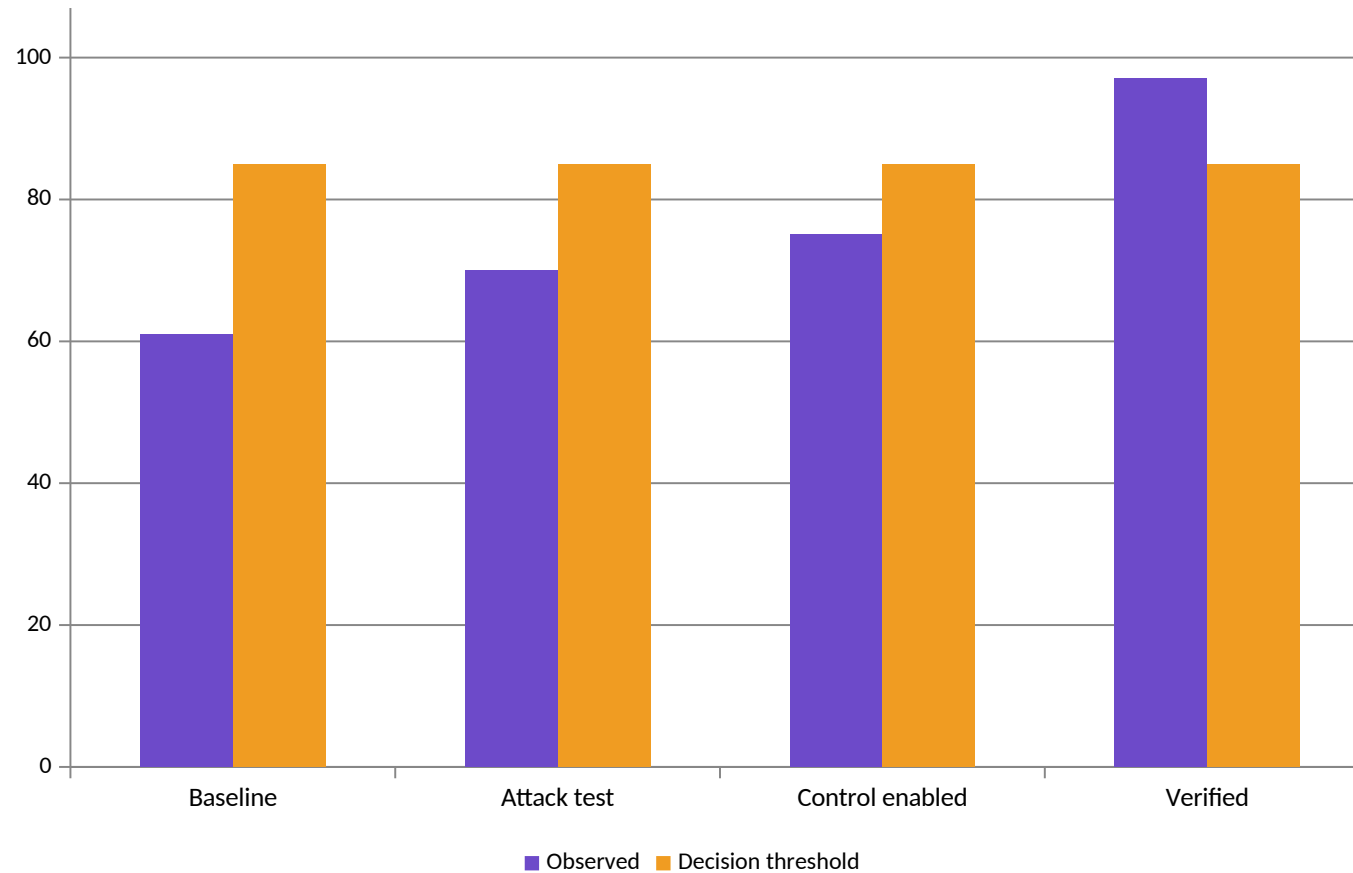
Failure to detect: Agent invents an account and changes the wrong record

Hallucinated action: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	resource_id	Required in decision and audit evidence
field_2	source_of_truth	Required in decision and audit evidence
field_3	confidence	Required in decision and audit evidence
field_4	validation_result	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Hallucinated action: evidence and acceptance decision



DECISION RULE

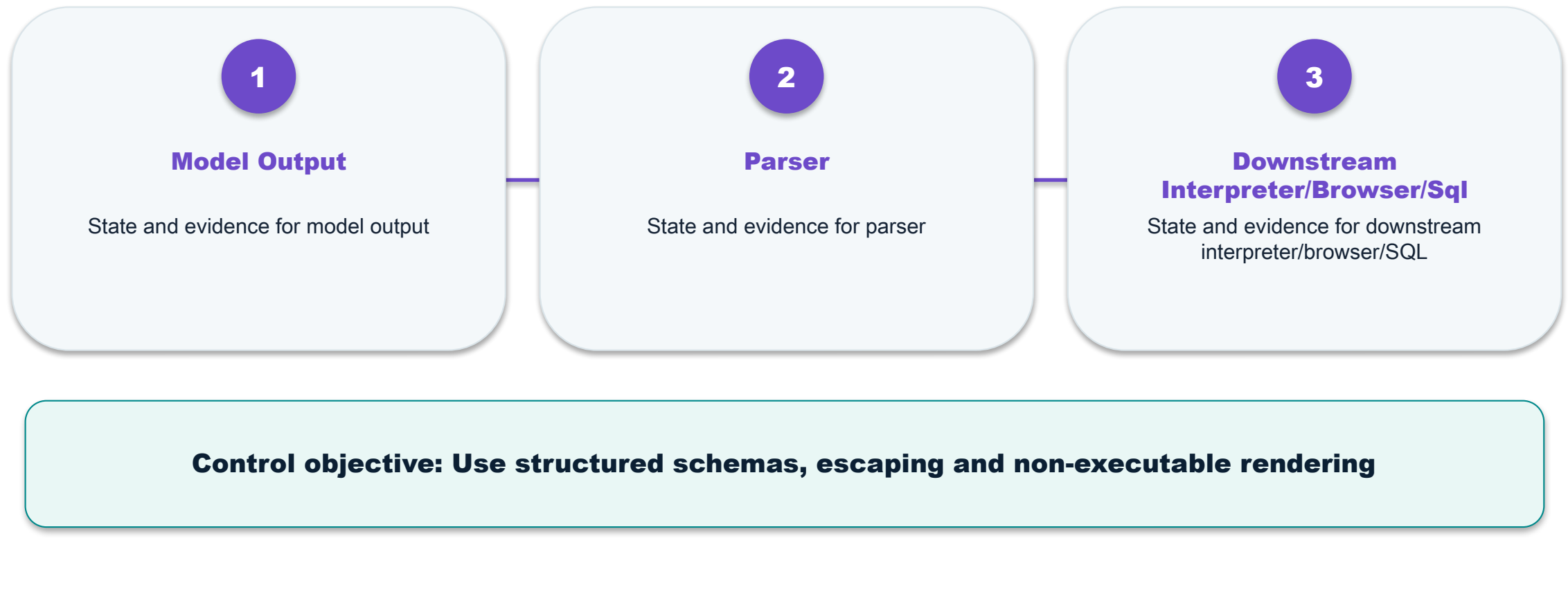
Accept only when Unknown resource_id is denied before execution. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

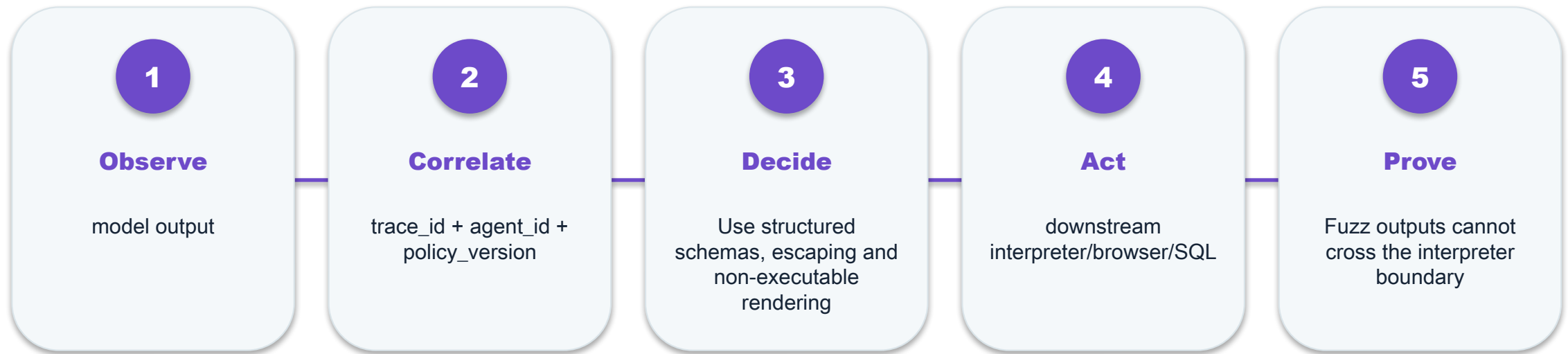
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Insecure output handling: architecture and control boundary



Insecure output handling: runtime decision trace



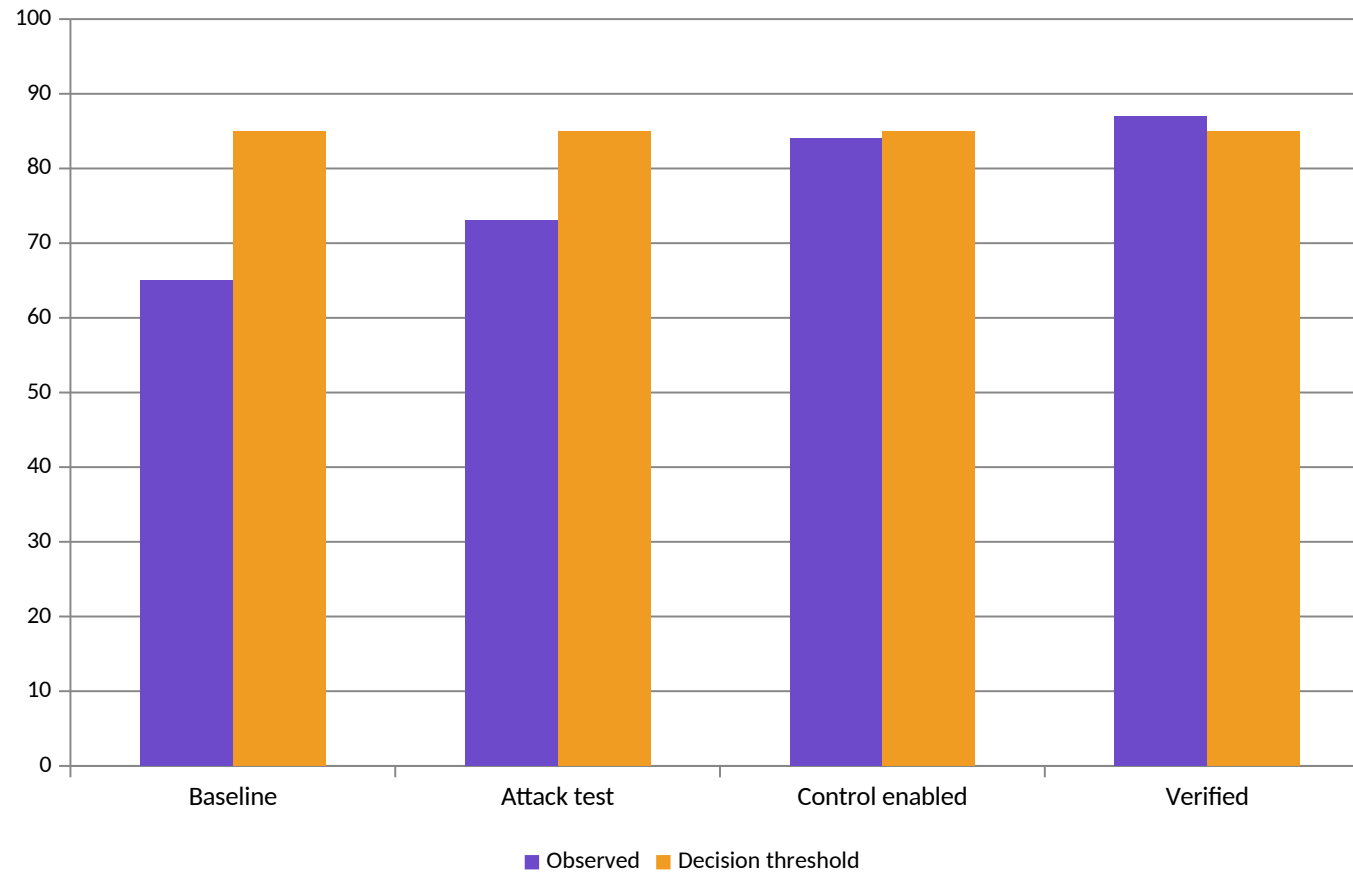
Failure to detect: Generated command or markup executes as code

Insecure output handling: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	output_type	Required in decision and audit evidence
field_2	schema_valid	Required in decision and audit evidence
field_3	encoding	Required in decision and audit evidence
field_4	destination	Required in decision and audit evidence
field_5	sanitised	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Insecure output handling: evidence and acceptance decision



DECISION RULE

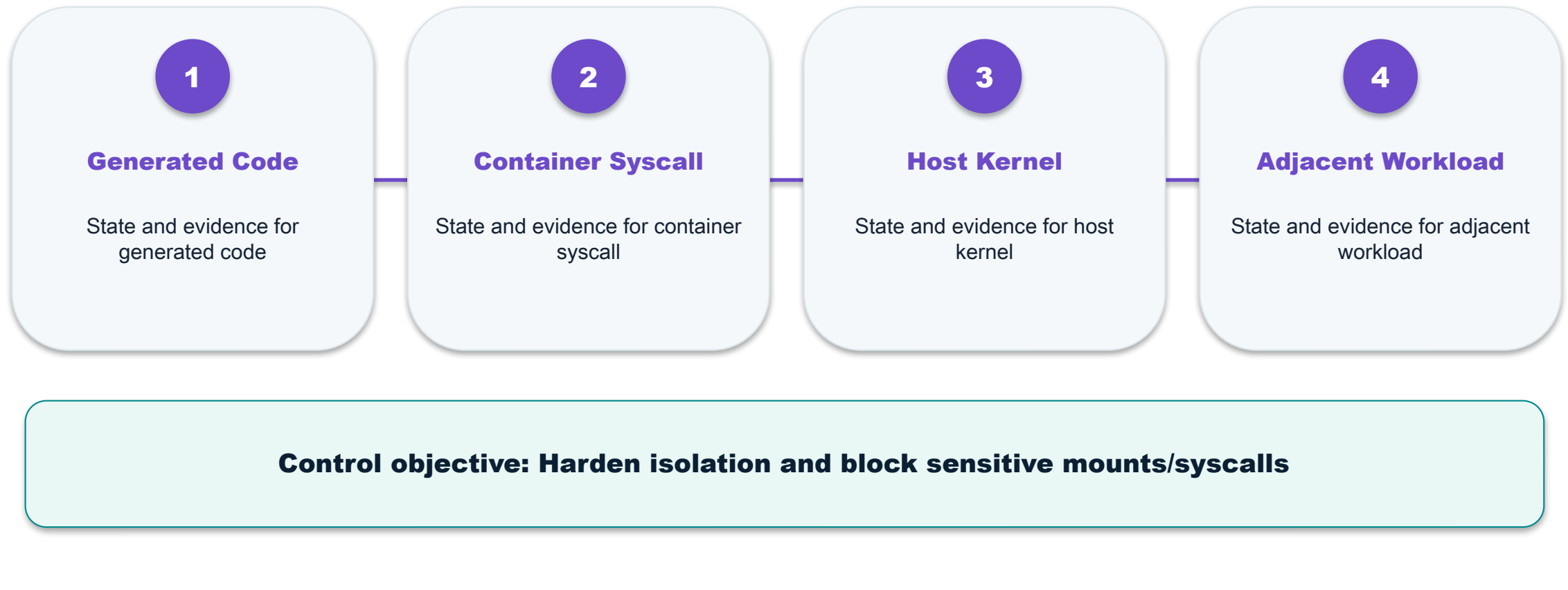
Accept only when Fuzz outputs cannot cross the interpreter boundary. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

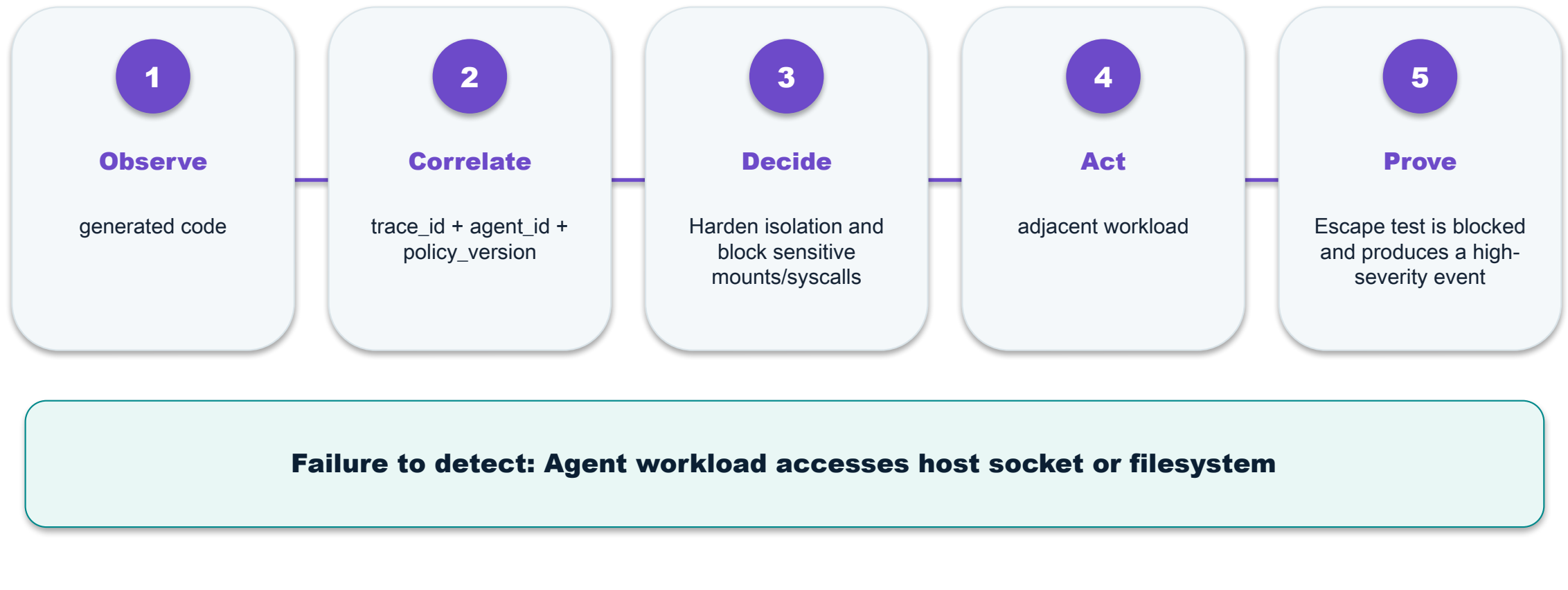
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Sandbox escape: architecture and control boundary



Sandbox escape: runtime decision trace

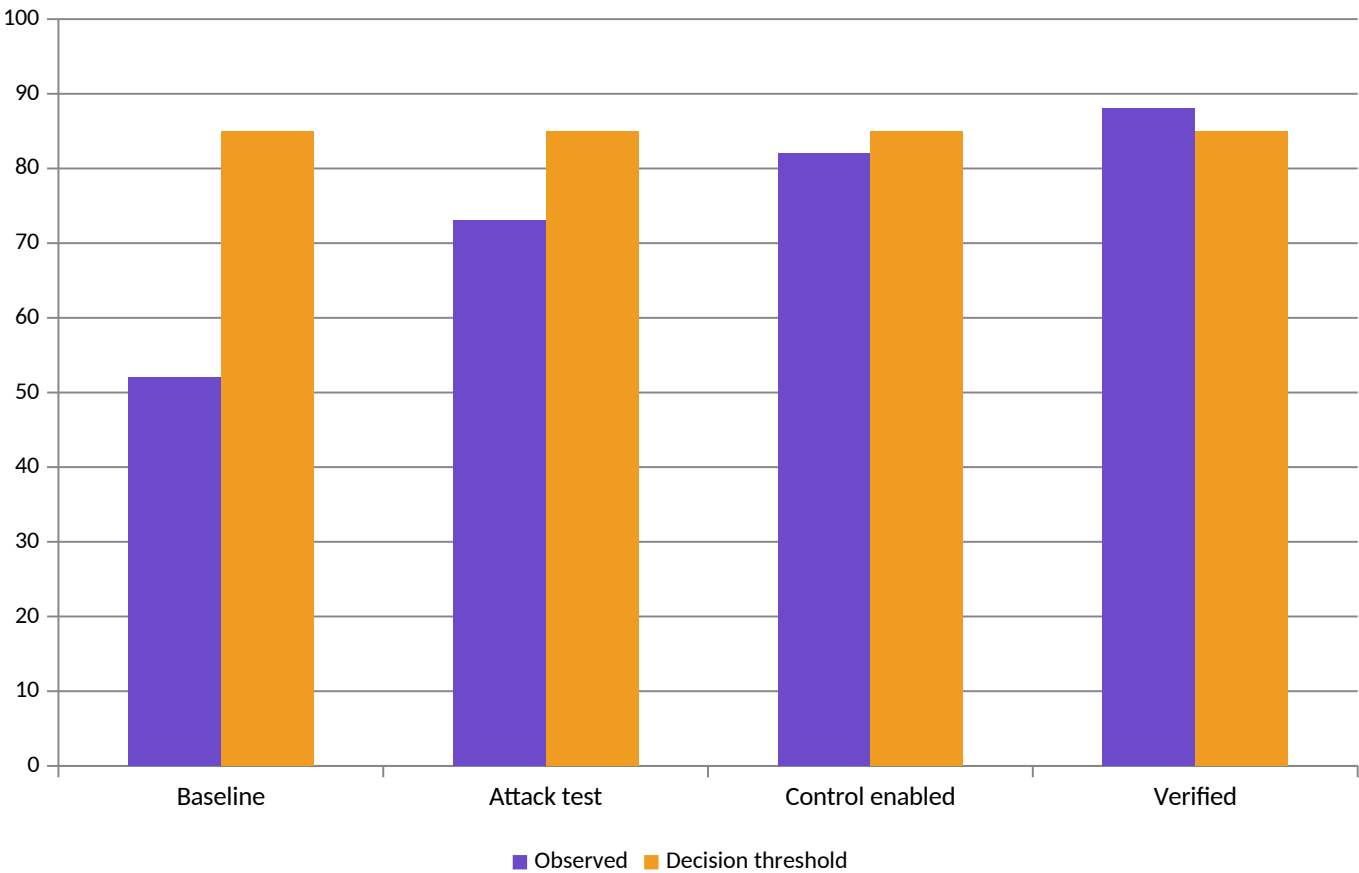


Sandbox escape: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	syscall	Required in decision and audit evidence
field_2	seccomp_action	Required in decision and audit evidence
field_3	uid	Required in decision and audit evidence
field_4	mount	Required in decision and audit evidence
field_5	network_namespace	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Sandbox escape: evidence and acceptance decision



DECISION RULE

**Accept only when
Escape test is blocked
and produces a high-
severity event.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS



Synthetic training evidence; values are illustrative, not external performance claims

Human override failure: architecture and control boundary



Control objective: Show exact diff, impact and evidence; support reject and edit

Human override failure: runtime decision trace



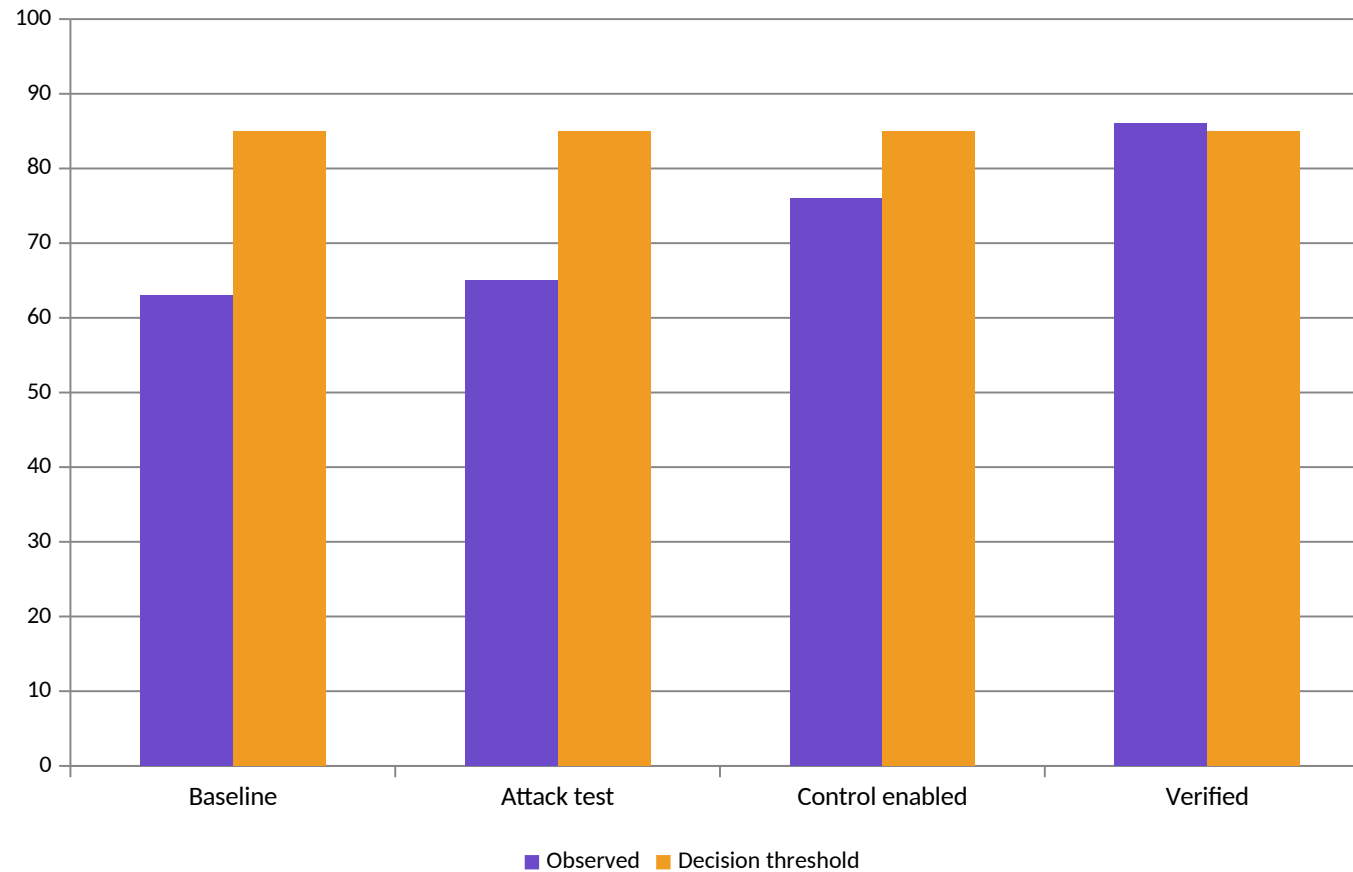
Failure to detect: Reviewer rubber-stamps an opaque or misleading proposal

Human override failure: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	approval_id	Required in decision and audit evidence
field_2	action_summary	Required in decision and audit evidence
field_3	risk	Required in decision and audit evidence
field_4	evidence	Required in decision and audit evidence
field_5	reviewer	Required in decision and audit evidence
field_6	decision	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Human override failure: evidence and acceptance decision



DECISION RULE

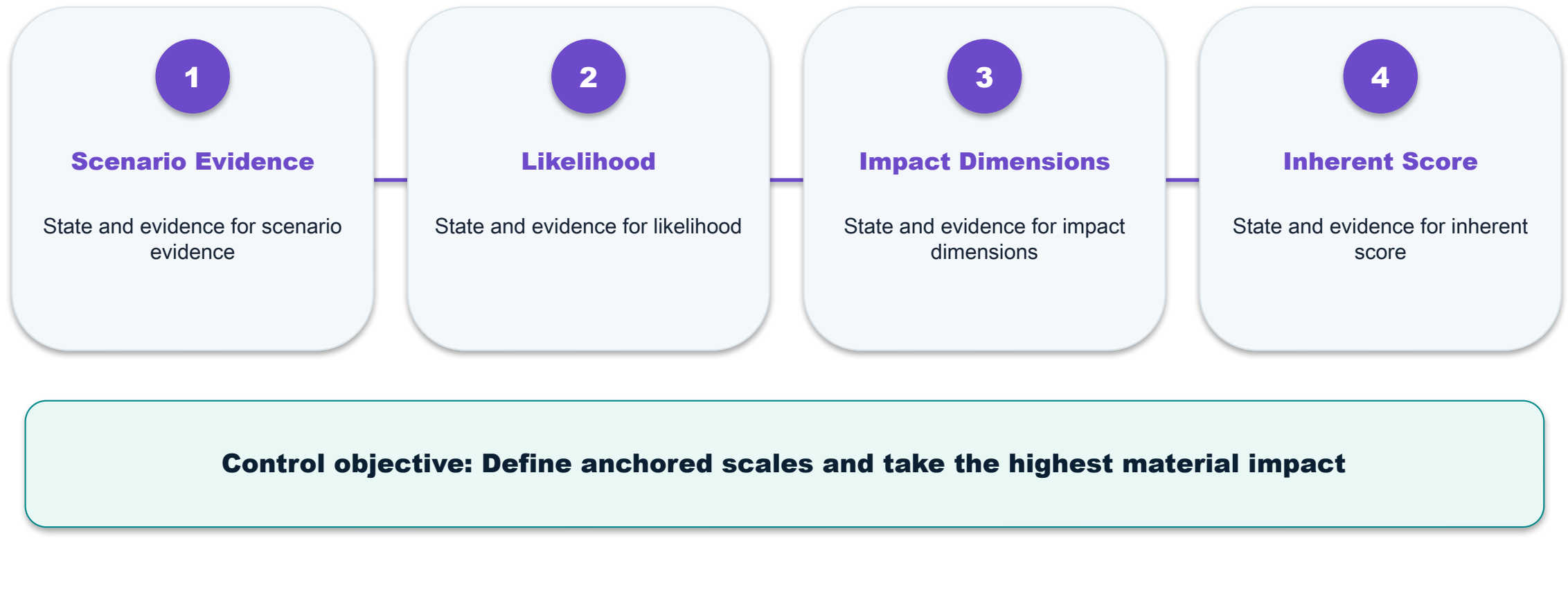
Accept only when Review sampling finds complete decision context. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

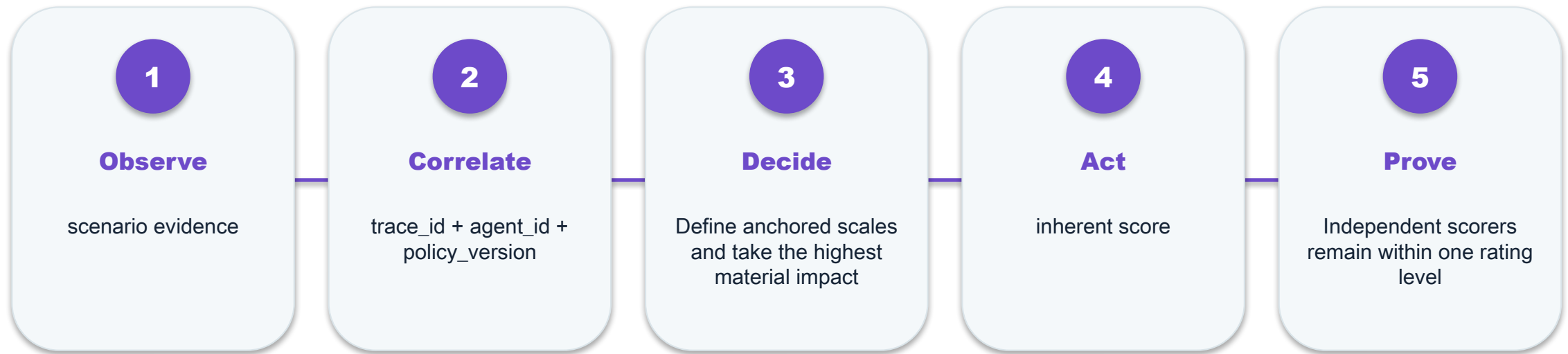
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Likelihood-impact scoring: architecture and control boundary



Likelihood-impact scoring: runtime decision trace



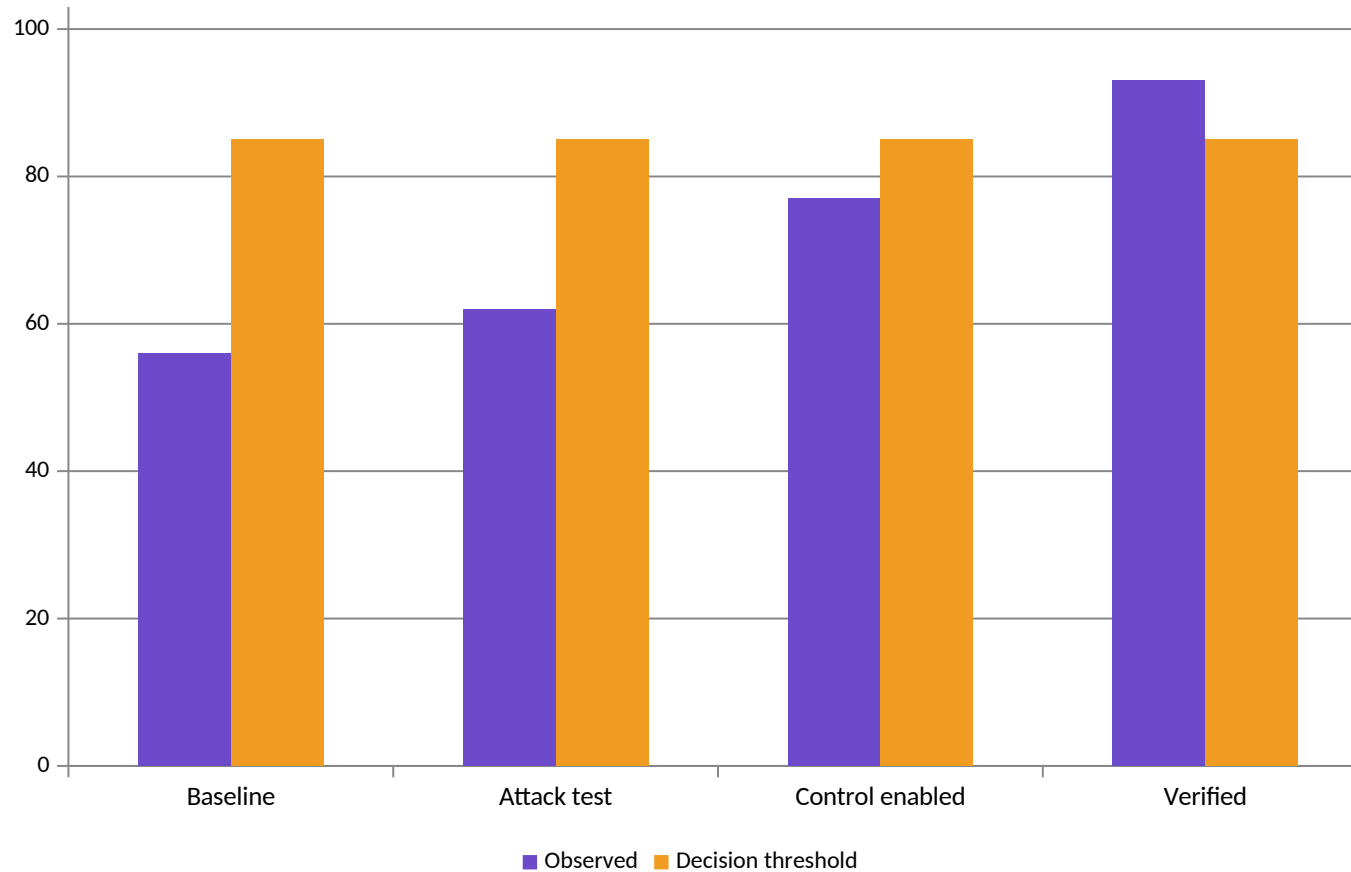
Failure to detect: Teams compare risks using inconsistent scales

Likelihood-impact scoring: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	likelihood_1_5	Required in decision and audit evidence
field_2	financial	Required in decision and audit evidence
field_3	operational	Required in decision and audit evidence
field_4	legal	Required in decision and audit evidence
field_5	safety	Required in decision and audit evidence
field_6	score	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Likelihood-impact scoring: evidence and acceptance decision



DECISION RULE

Accept only when Independent scorers remain within one rating level. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Expected-loss estimate: architecture and control boundary

1

Event Frequency × Loss Magnitude

State and evidence for event frequency × loss magnitude

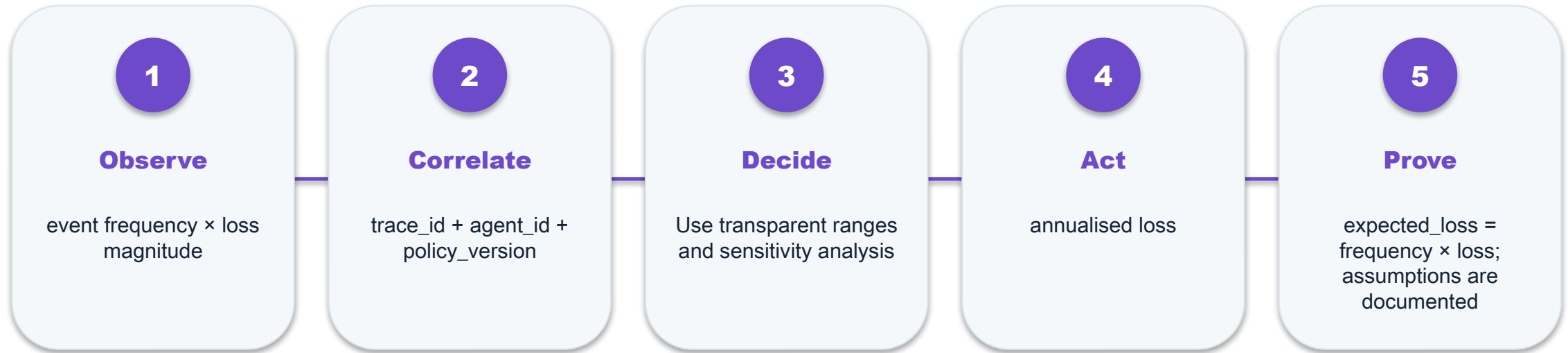
2

Annualised Loss

State and evidence for annualised loss

Control objective: Use transparent ranges and sensitivity analysis

Expected-loss estimate: runtime decision trace



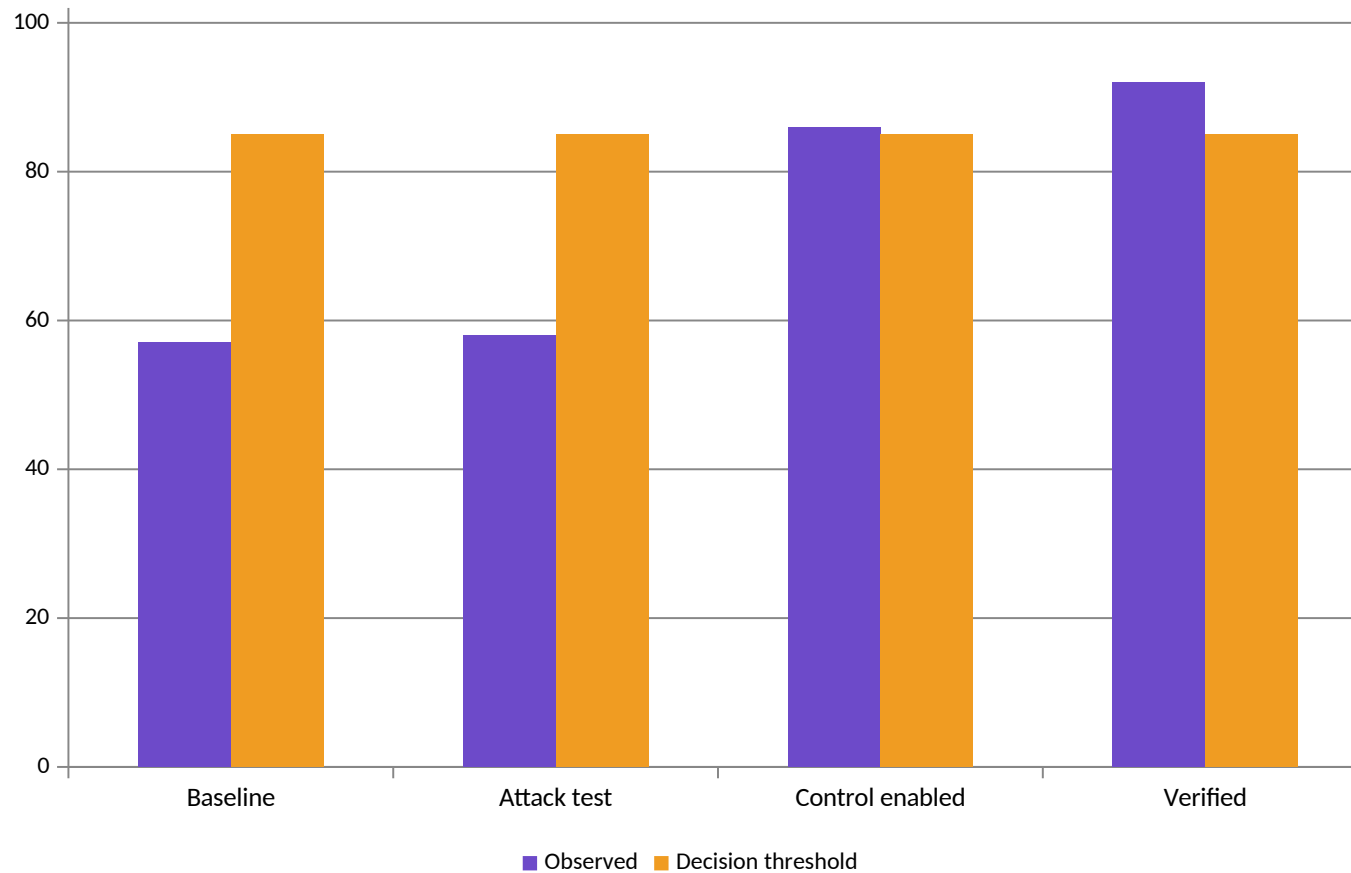
Failure to detect: Control spend is justified only by qualitative fear

Expected-loss estimate: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	frequency_per_year	Required in decision and audit evidence
field_2	loss_sgd	Required in decision and audit evidence
field_3	expected_loss_sgd	Required in decision and audit evidence
field_4	confidence	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Expected-loss estimate: evidence and acceptance decision



DECISION RULE

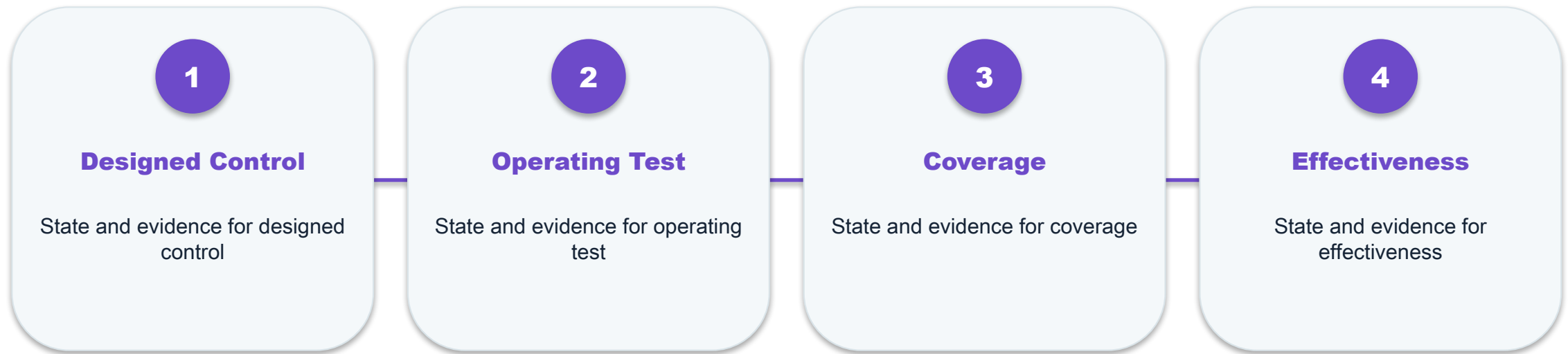
Accept only when $\text{expected_loss} = \text{frequency} \times \text{loss}$; assumptions are documented. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Control effectiveness: architecture and control boundary



Control objective: Test design, operation and coverage separately

Control effectiveness: runtime decision trace



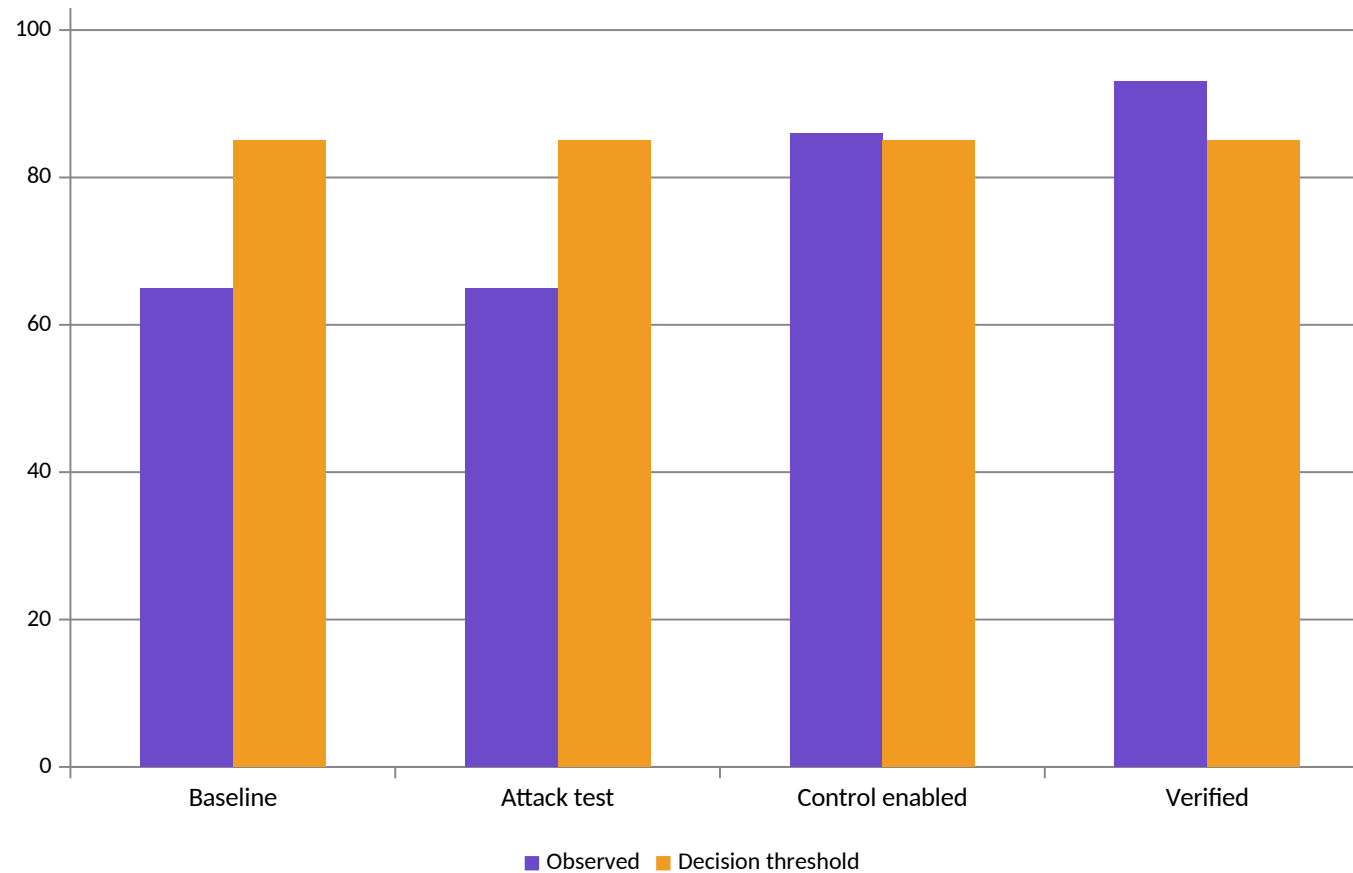
Failure to detect: Control exists but operates on only part of the fleet

Control effectiveness: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	design_score	Required in decision and audit evidence
field_2	operation_score	Required in decision and audit evidence
field_3	coverage_pct	Required in decision and audit evidence
field_4	effectiveness_pct	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Control effectiveness: evidence and acceptance decision



DECISION RULE

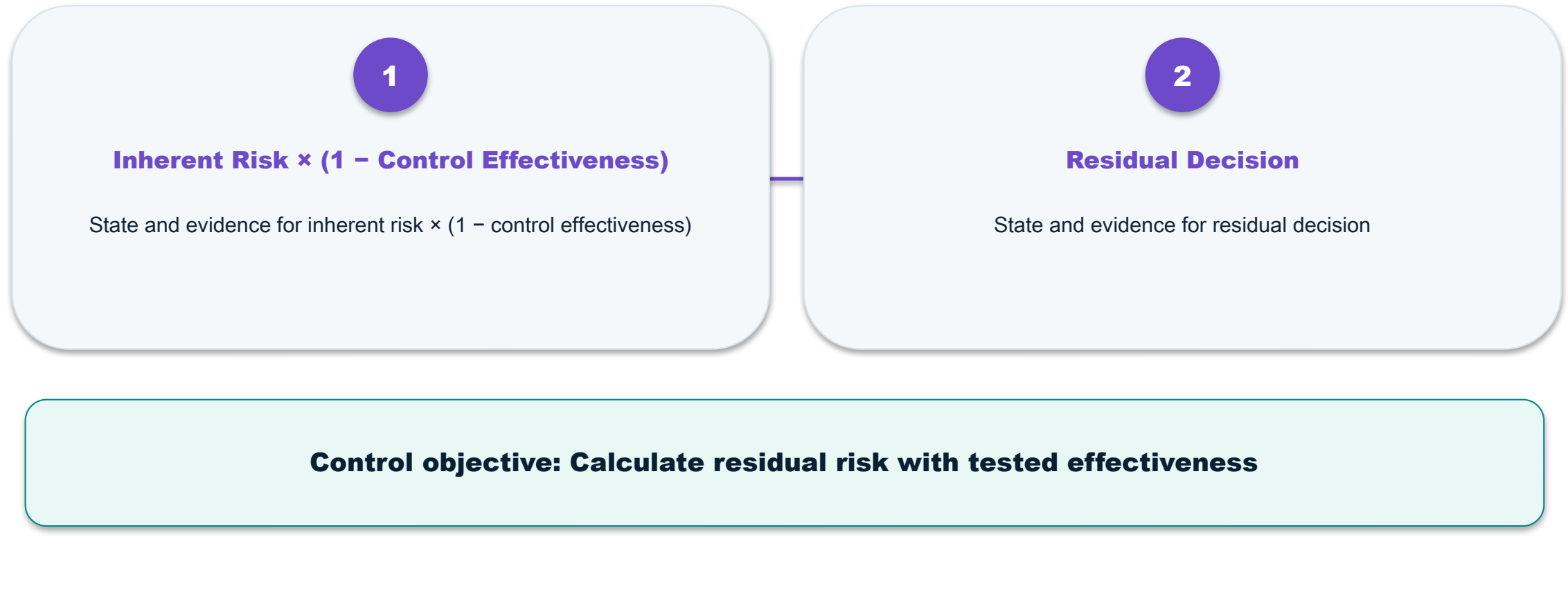
Accept only when effective coverage meets the risk-treatment target. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

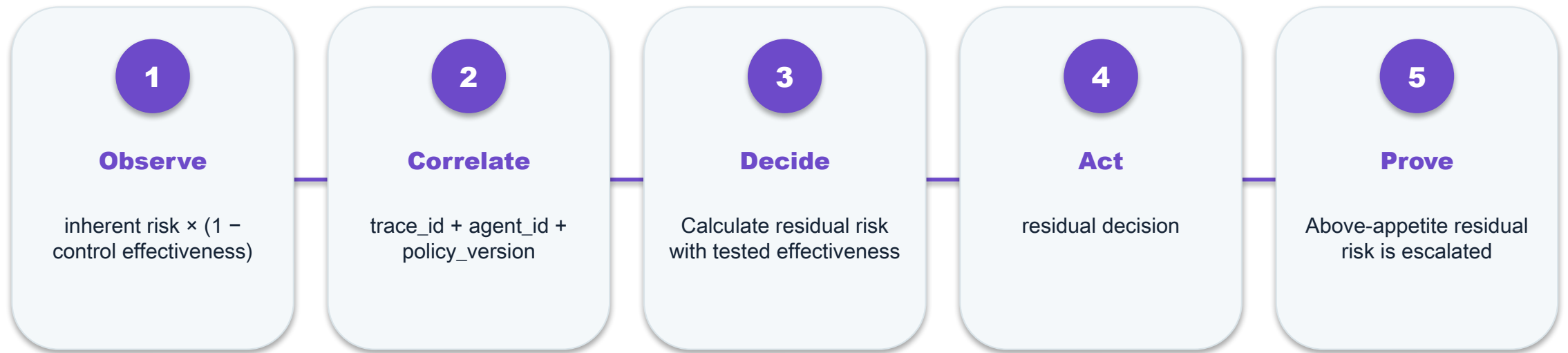
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Residual risk: architecture and control boundary



Residual risk: runtime decision trace



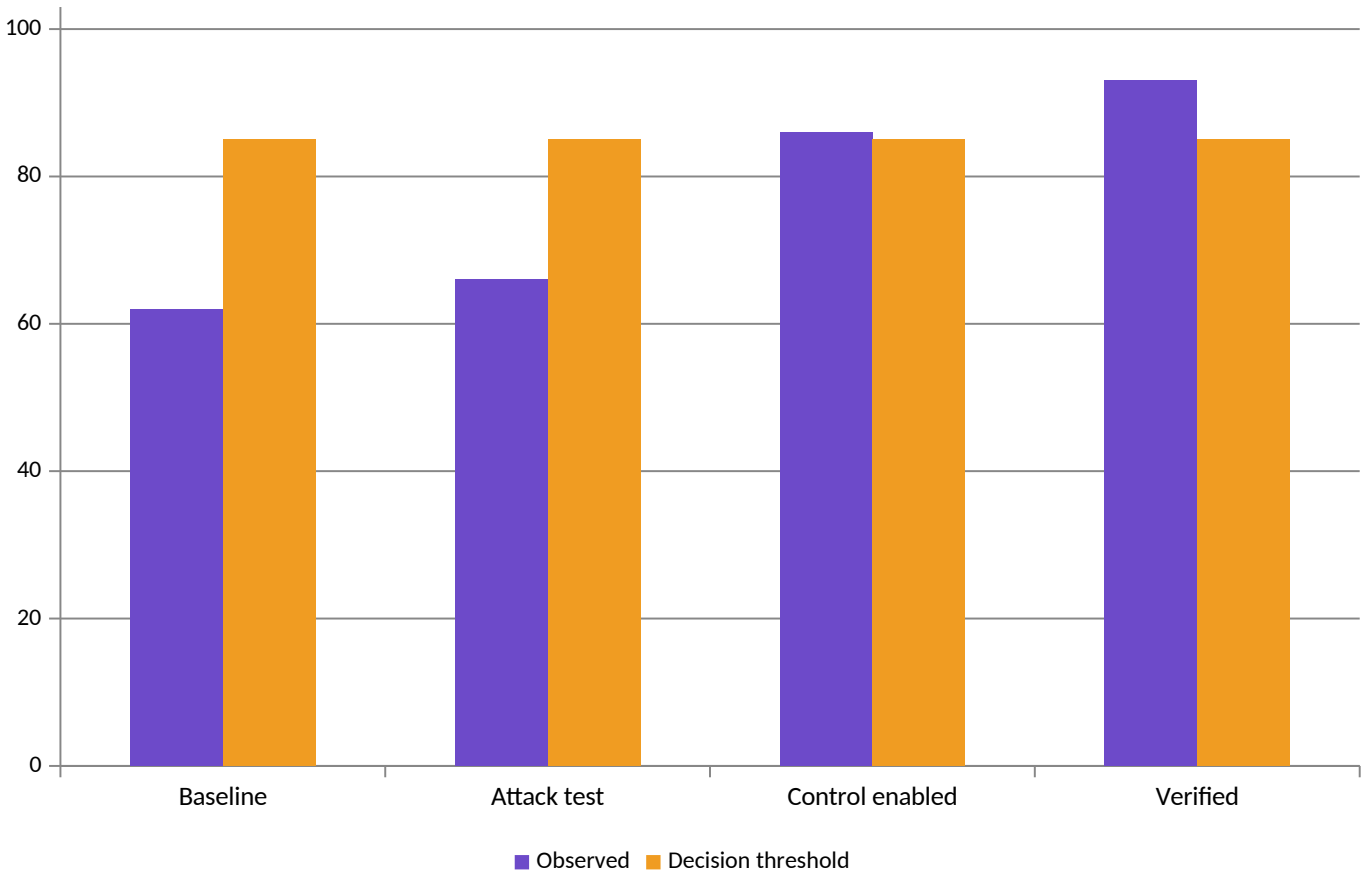
Failure to detect: Reported risk ignores control failure or coverage gaps

Residual risk: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	inherent_score	Required in decision and audit evidence
field_2	effectiveness_pct	Required in decision and audit evidence
field_3	residual_score	Required in decision and audit evidence
field_4	appetite	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Residual risk: evidence and acceptance decision



DECISION RULE

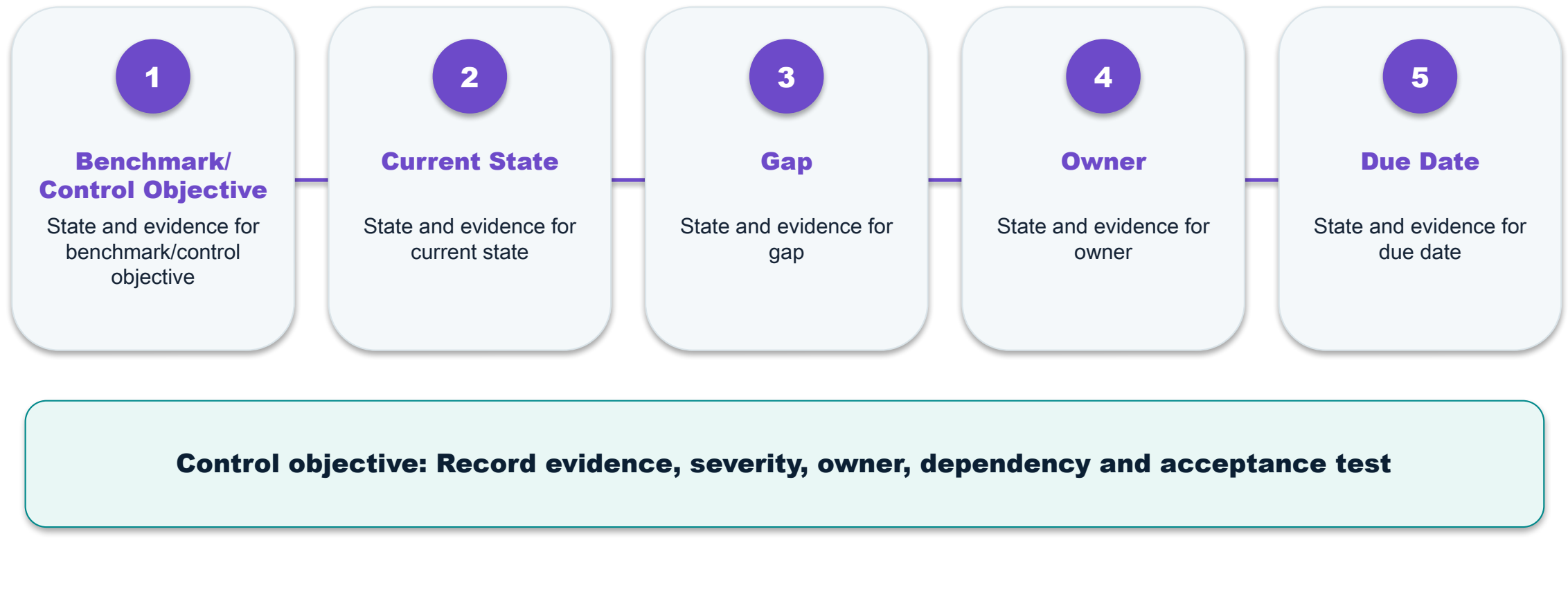
**Accept only when
Above-appetite residual
risk is escalated.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS



Synthetic training evidence; values are illustrative, not external performance claims

Gap register: architecture and control boundary



Gap register: runtime decision trace



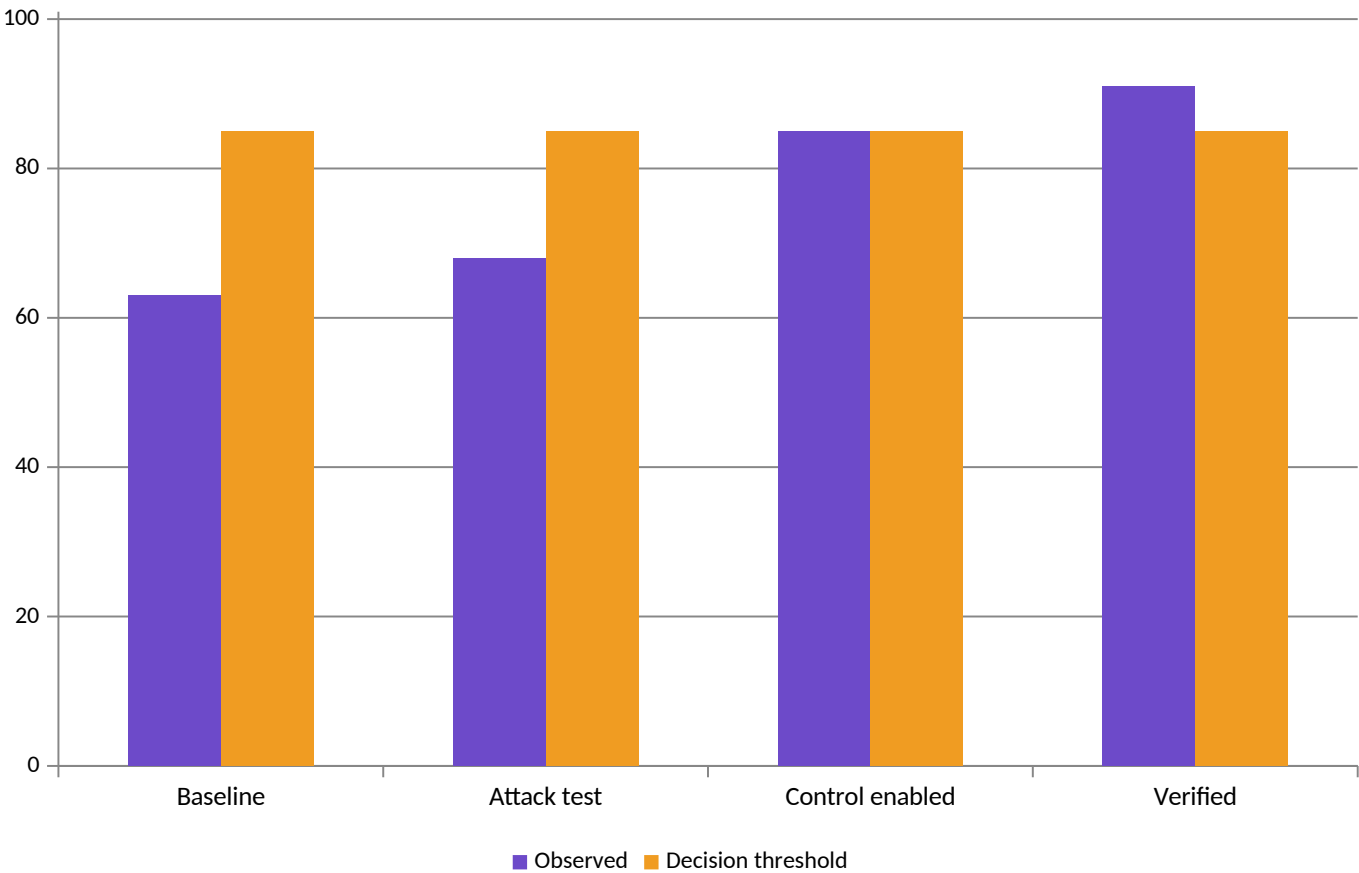
Failure to detect: Findings lack an actionable path to closure

Gap register: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	gap_id	Required in decision and audit evidence
field_2	requirement	Required in decision and audit evidence
field_3	evidence	Required in decision and audit evidence
field_4	severity	Required in decision and audit evidence
field_5	owner	Required in decision and audit evidence
field_6	target_date	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Gap register: evidence and acceptance decision



DECISION RULE

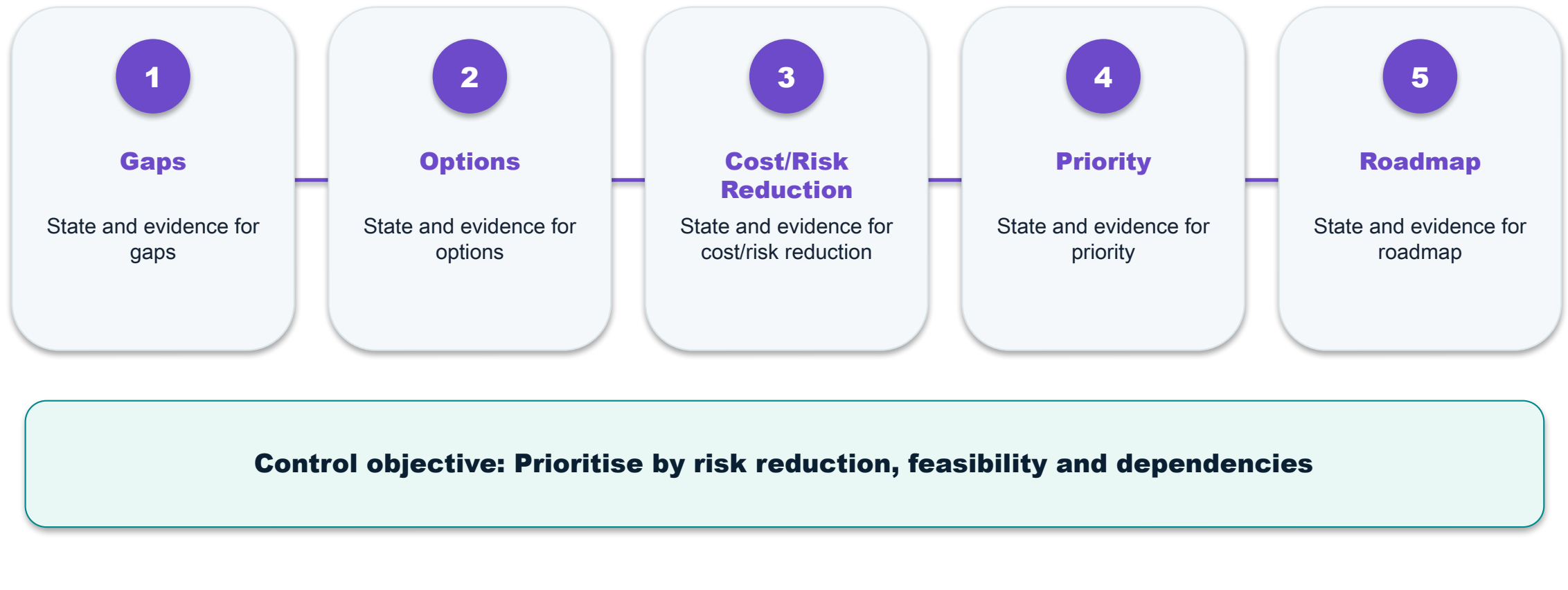
Accept only when No high gap is ownerless or overdue without escalation. Otherwise record a gap, owner and due date.

EVIDENCE STATUS



Synthetic training evidence; values are illustrative, not external performance claims

Remediation portfolio: architecture and control boundary



Remediation portfolio: runtime decision trace



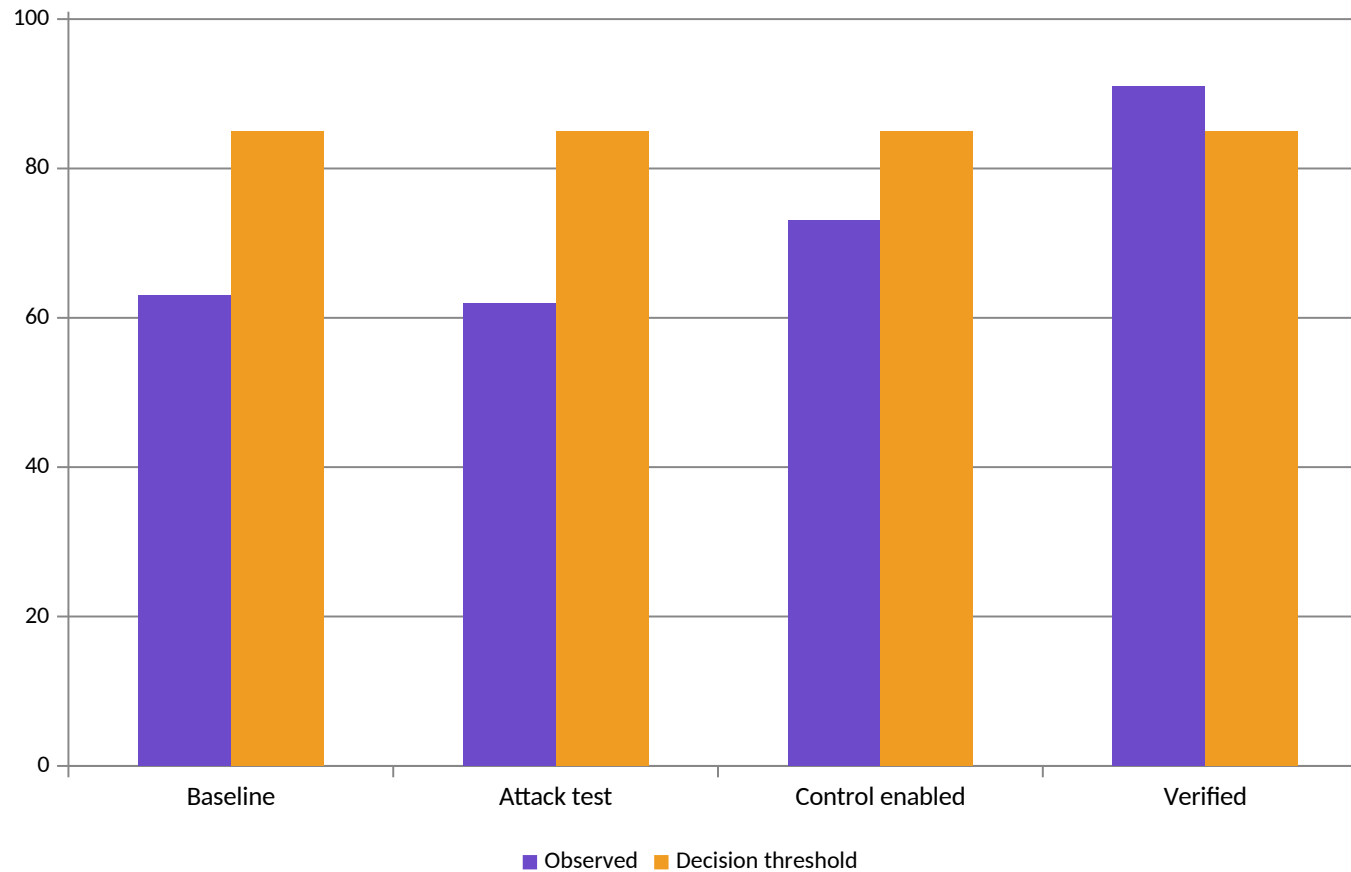
Failure to detect: Teams fund visible controls instead of highest value

Remediation portfolio: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	option	Required in decision and audit evidence
field_2	cost_sgd	Required in decision and audit evidence
field_3	effort_days	Required in decision and audit evidence
field_4	risk_reduction	Required in decision and audit evidence
field_5	dependency	Required in decision and audit evidence
field_6	priority	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Remediation portfolio: evidence and acceptance decision



DECISION RULE

**Accept only when
Selected portfolio fits
budget and maximises
risk reduction.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Activity 06 — Quantify Agent Risks and Control Value

1 Scenario

Use synthetic data in activity-06-risk-control-evaluation to make a security-operations decision.

2 Input contract

mock_risks.csv

3 Runnable evidence

activity.py

4 Output artifact

risk_evaluation.csv

5 Acceptance

Residual risk and expected annual loss calculations reconcile with the input assumptions.

6 Safety boundary

Offline simulation only; no live credentials, systems or external services.

Activity 06: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Activity 06: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_risks.csv	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	risk_evaluation.csv	Generated locally and reproducible
Criterion	A5	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete risk_evaluation.csv only	Retain script and mock data

Aligned activity files in labs/

Activity 07 — Prioritise a Security Gap Remediation Roadmap

1 Scenario

Use synthetic data in activity-07-gap-remediation to make a security-operations decision.

2 Input contract

mock_gaps.csv

3 Runnable evidence

activity.py

4 Output artifact

remediation_roadmap.md

5 Acceptance

The roadmap stays within the mock budget and prioritises overdue and above-appetite gaps.

6 Safety boundary

Offline simulation only; no live credentials, systems or external services.

Activity 07: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Activity 07: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_gaps.csv	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	remediation_roadmap.md	Generated locally and reproducible
Criterion	A6	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete remediation_roadmap.md only	Retain script and mock data

Aligned activity files in labs/

Topic 03 technical recap

1

Likelihood-impact scoring

Independent scorers remain within one rating level

2

Expected-loss estimate

$\text{expected_loss} = \text{frequency} \times \text{loss}$; assumptions are documented

3

Control effectiveness

effective coverage meets the risk-treatment target

4

Residual risk

Above-appetite residual risk is escalated

5

Gap register

No high gap is ownerless or overdue without escalation

6

Remediation portfolio

Selected portfolio fits budget and maximises risk reduction

TOPIC 04

Implementing and Monitoring Organisation-Wide Agent Security Operations

Telemetry · detection · incident response · containment · recovery

LO4 · K7, A7



L04 and assessment criteria

1

K7

Implications and impact of security gaps

2

A7

Drive organisation-wide security initiatives in line with internal and external standards

3

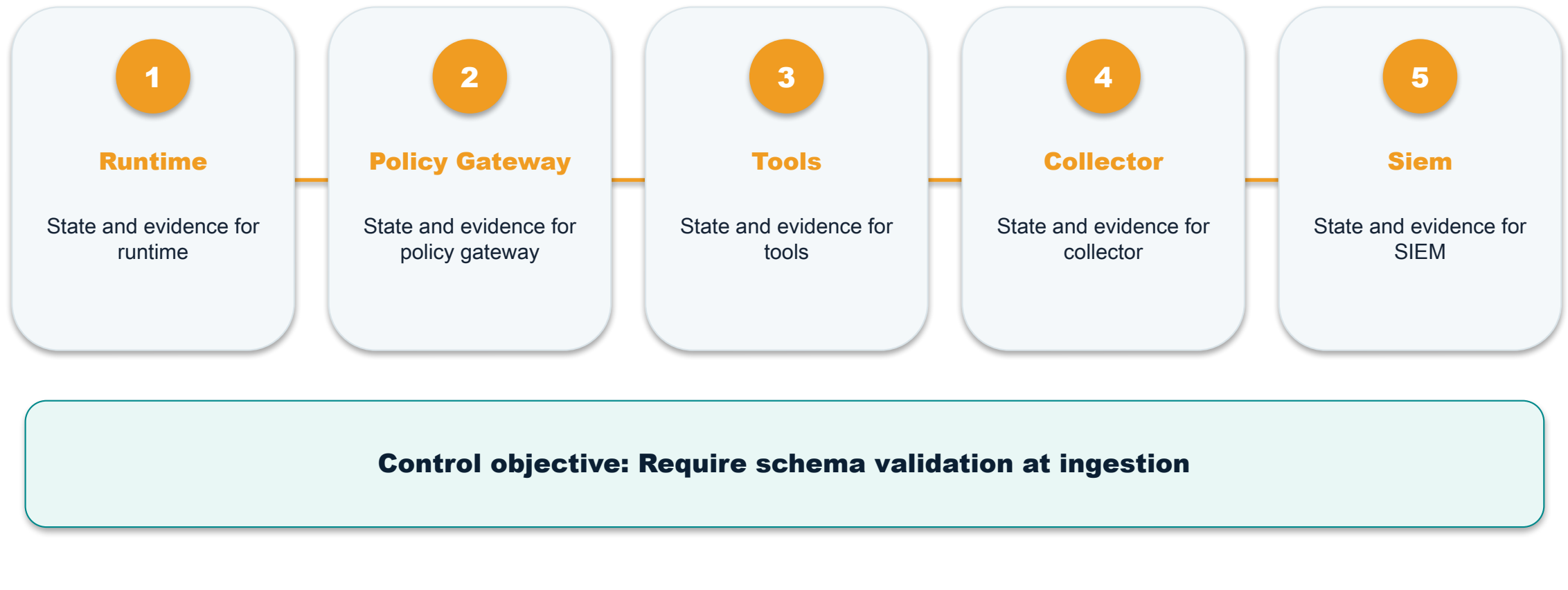
LO4

LO4: Implement organisation-wide agent-security initiatives, assessing and addressing the impact of security gaps.

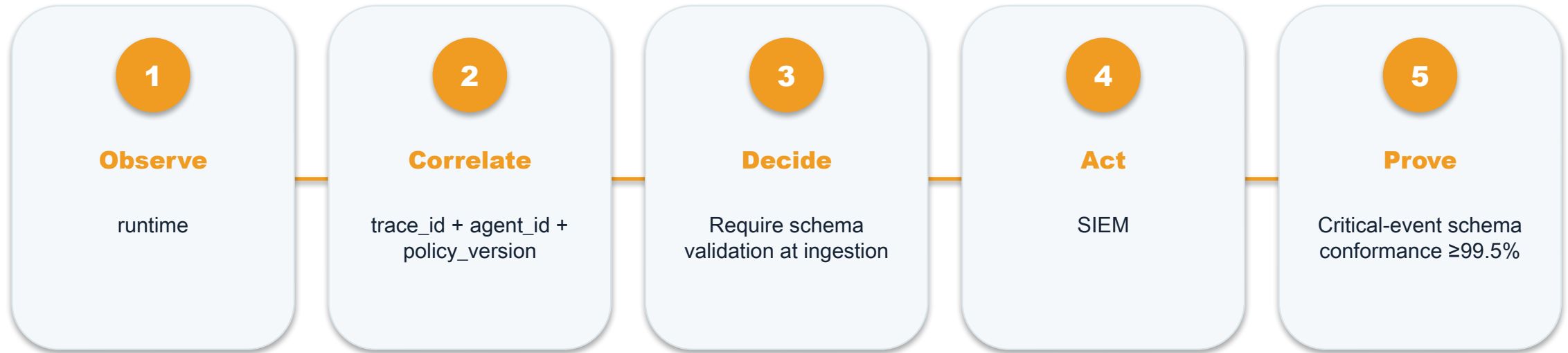
Implementing and Monitoring Organisation-Wide Agent Security Operations: operating model



Telemetry contract: architecture and control boundary



Telemetry contract: runtime decision trace



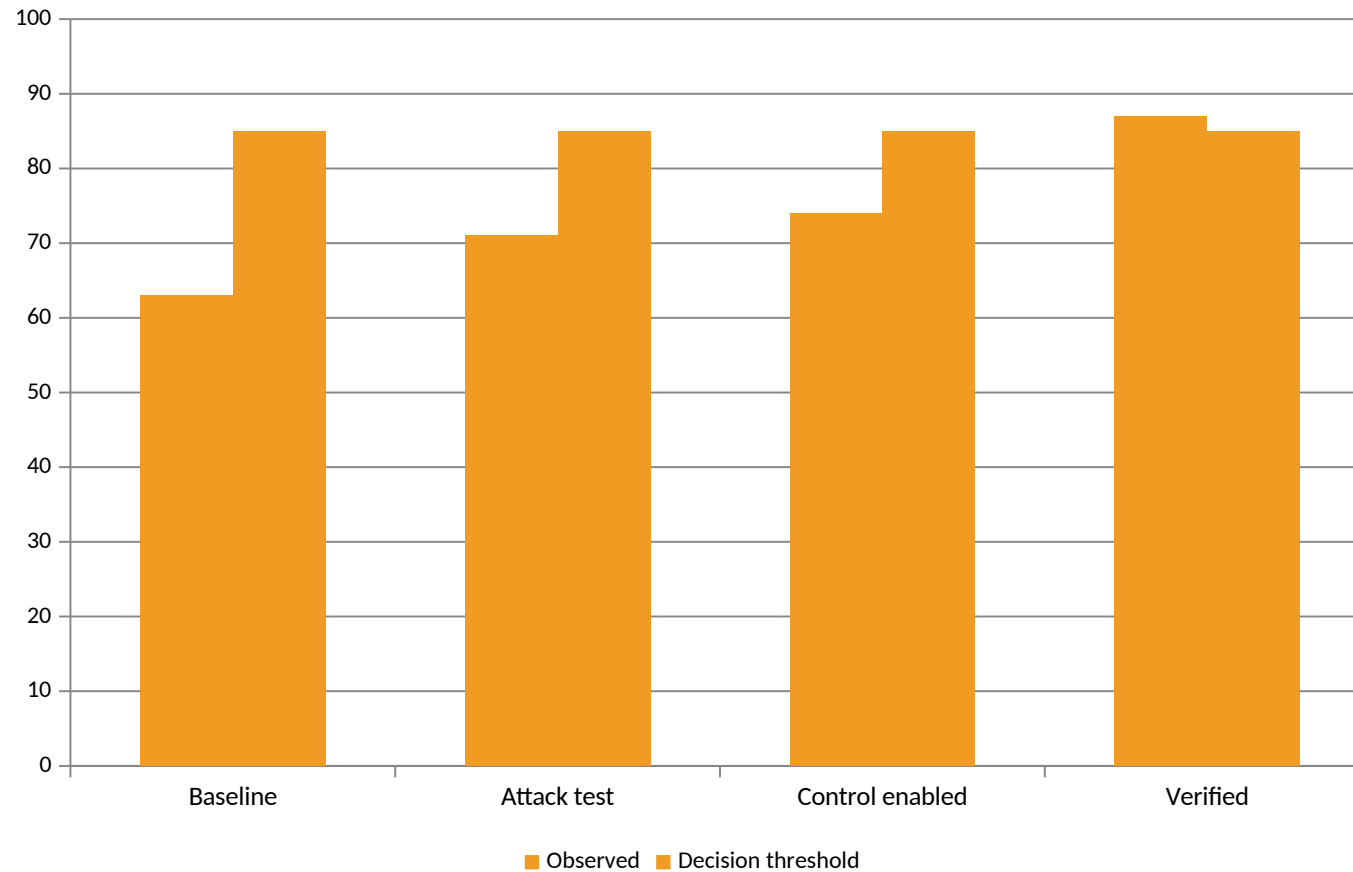
Failure to detect: Missing fields break investigations and detections

Telemetry contract: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	event_time	Required in decision and audit evidence
field_2	trace_id	Required in decision and audit evidence
field_3	span_id	Required in decision and audit evidence
field_4	agent_id	Required in decision and audit evidence
field_5	session_id	Required in decision and audit evidence
field_6	event_type	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Telemetry contract: evidence and acceptance decision



DECISION RULE

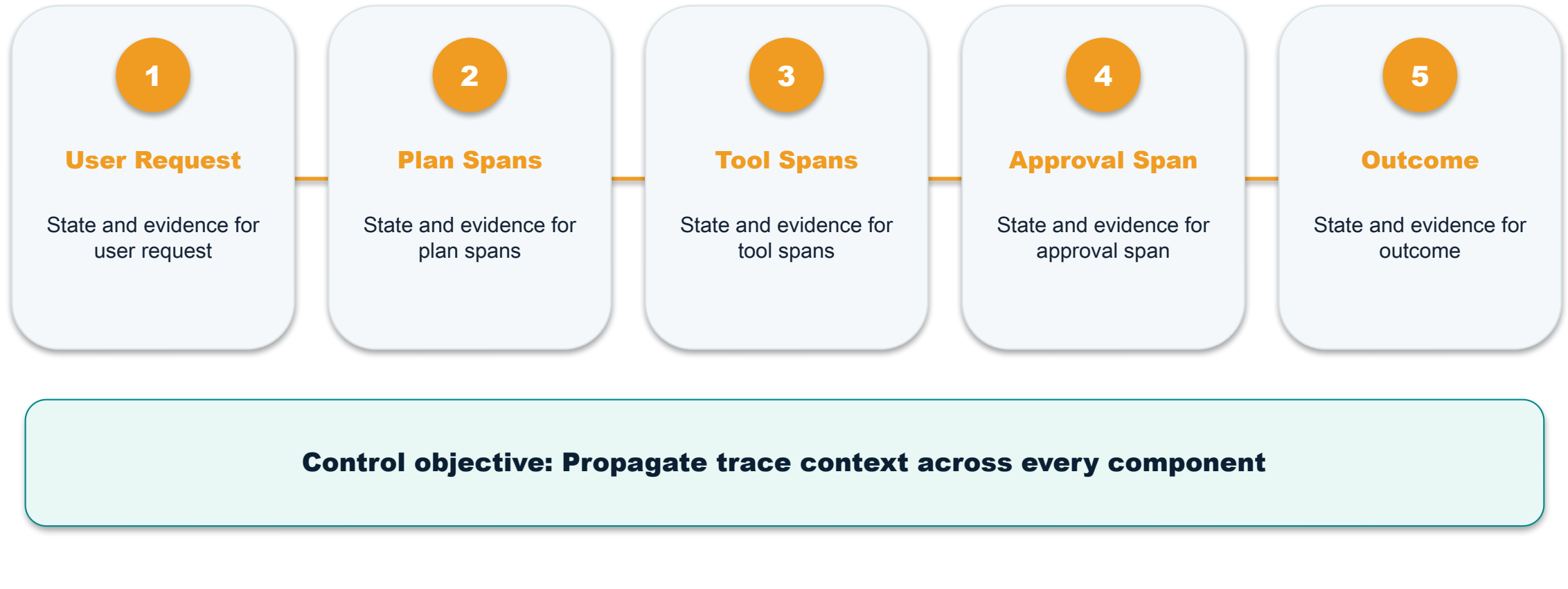
**Accept only when
Critical-event schema
conformance $\geq 99.5\%$.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Trace correlation: architecture and control boundary



Trace correlation: runtime decision trace



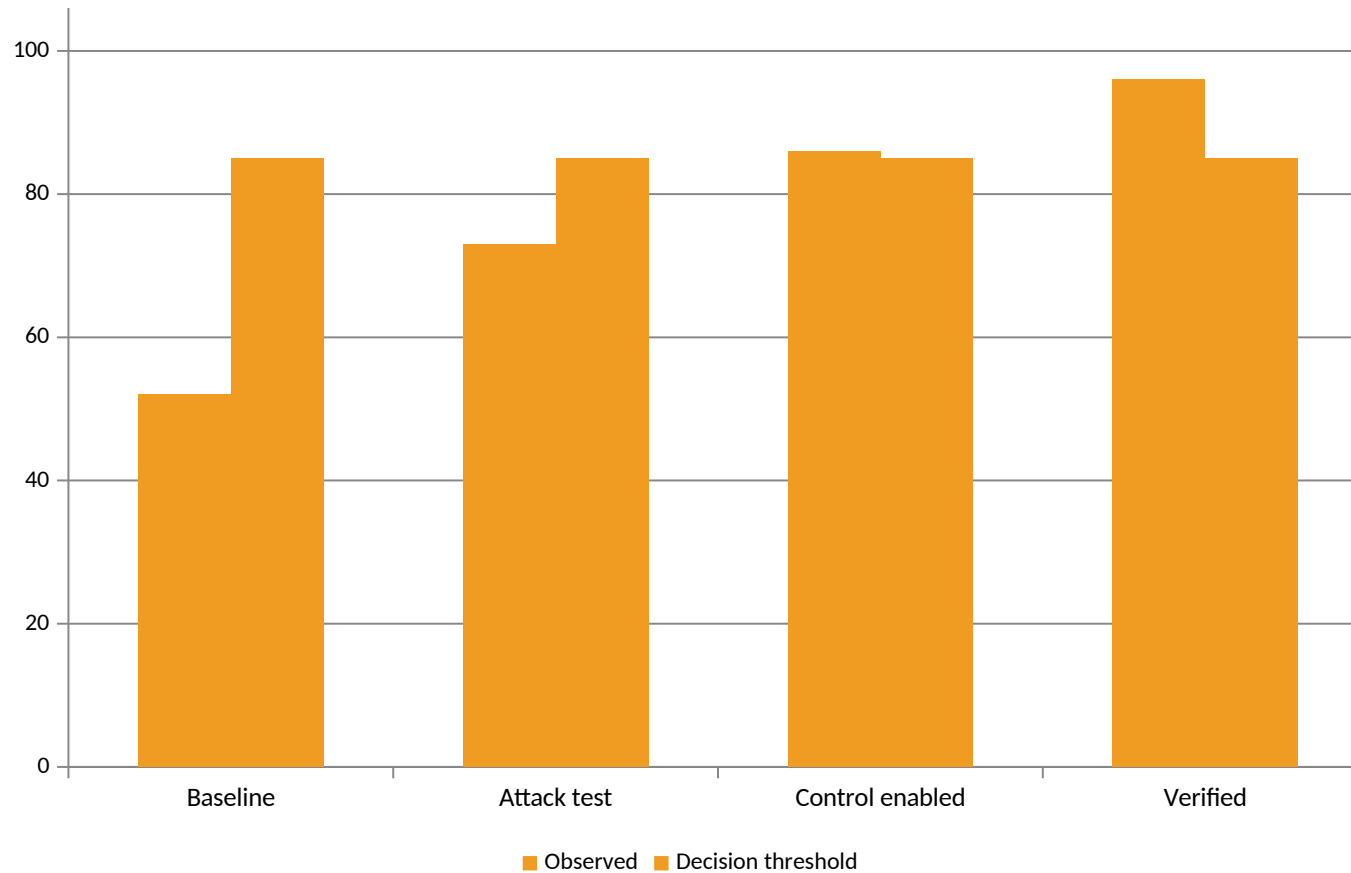
Failure to detect: Events exist but cannot be ordered into one incident

Trace correlation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	trace_id	Required in decision and audit evidence
field_2	parent_span_id	Required in decision and audit evidence
field_3	actor	Required in decision and audit evidence
field_4	action	Required in decision and audit evidence
field_5	resource	Required in decision and audit evidence
field_6	status	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Trace correlation: evidence and acceptance decision



DECISION RULE

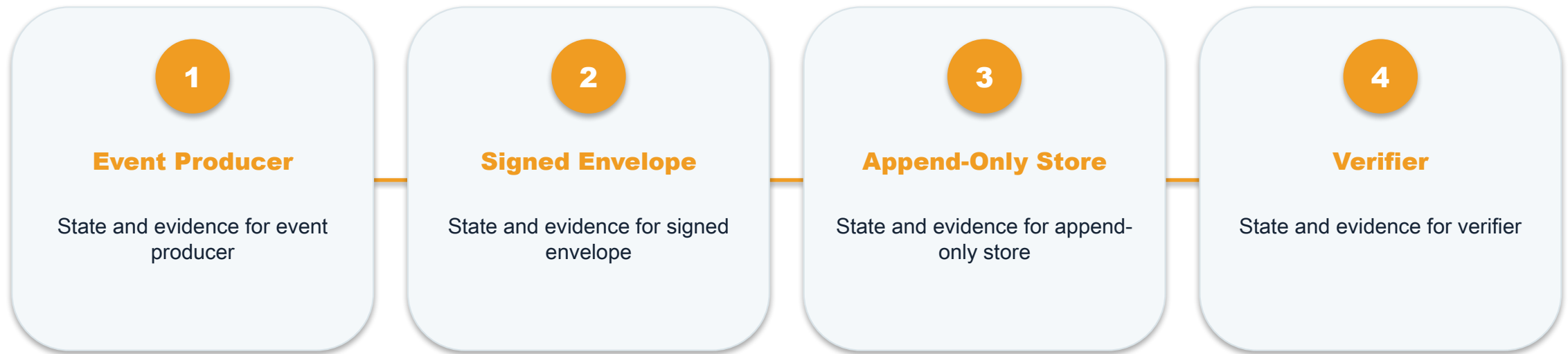
Accept only when A sampled session reconstructs without orphan spans. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Immutable audit log: architecture and control boundary



Control objective: Use append-only storage and hash chaining

Immutable audit log: runtime decision trace



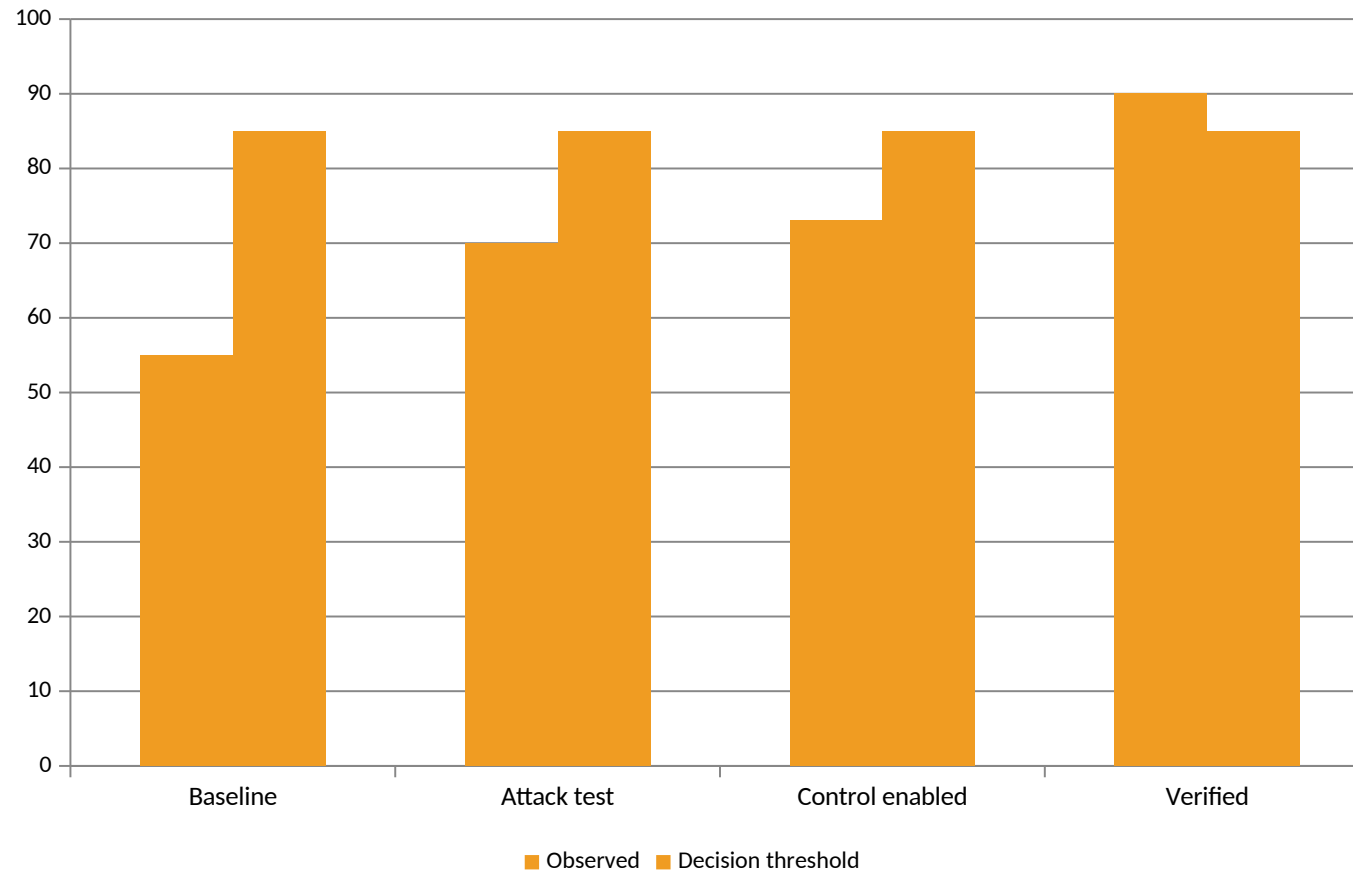
Failure to detect: Attacker deletes or edits incriminating tool events

Immutable audit log: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	sequence	Required in decision and audit evidence
field_2	previous_hash	Required in decision and audit evidence
field_3	event_hash	Required in decision and audit evidence
field_4	signer	Required in decision and audit evidence
field_5	timestamp	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Immutable audit log: evidence and acceptance decision



DECISION RULE

Accept only when Verification detects any missing or altered record. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

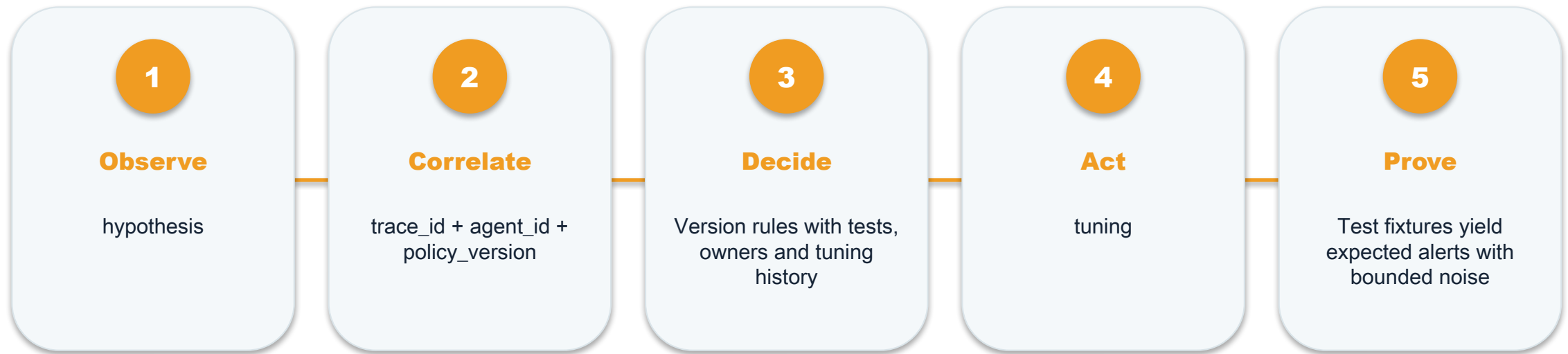
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Detection rule lifecycle: architecture and control boundary



Detection rule lifecycle: runtime decision trace



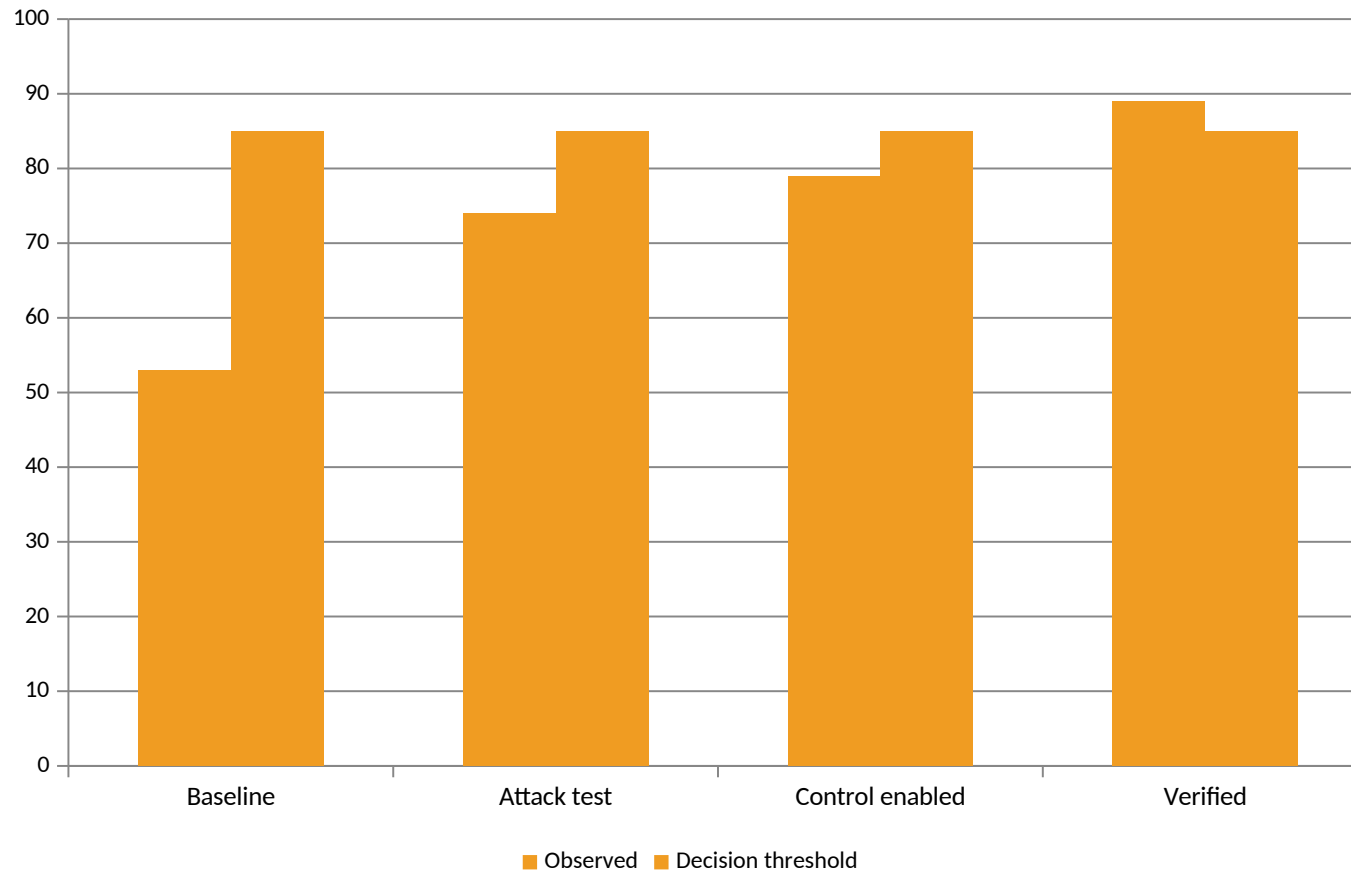
Failure to detect: Unreviewed rule floods analysts or misses known attacks

Detection rule lifecycle: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	rule_id	Required in decision and audit evidence
field_2	version	Required in decision and audit evidence
field_3	query	Required in decision and audit evidence
field_4	severity	Required in decision and audit evidence
field_5	threshold	Required in decision and audit evidence
field_6	owner	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Detection rule lifecycle: evidence and acceptance decision



DECISION RULE

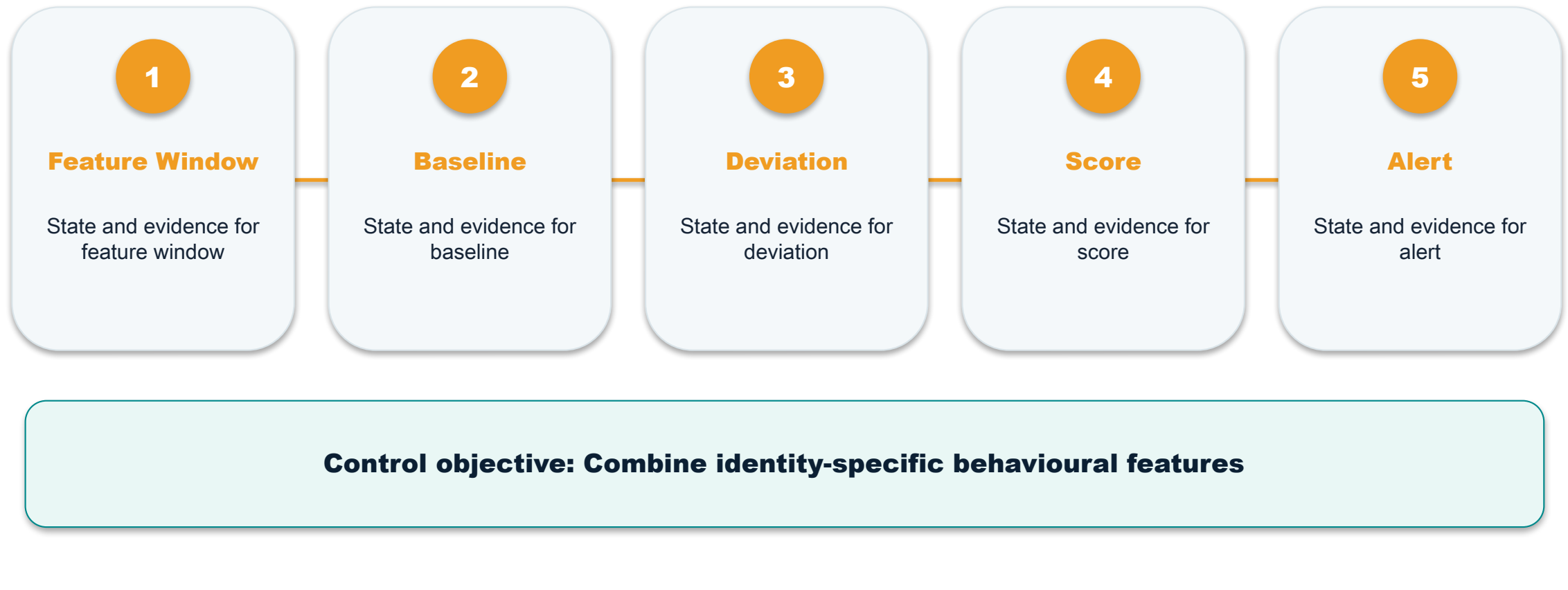
Accept only when Test fixtures yield expected alerts with bounded noise. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

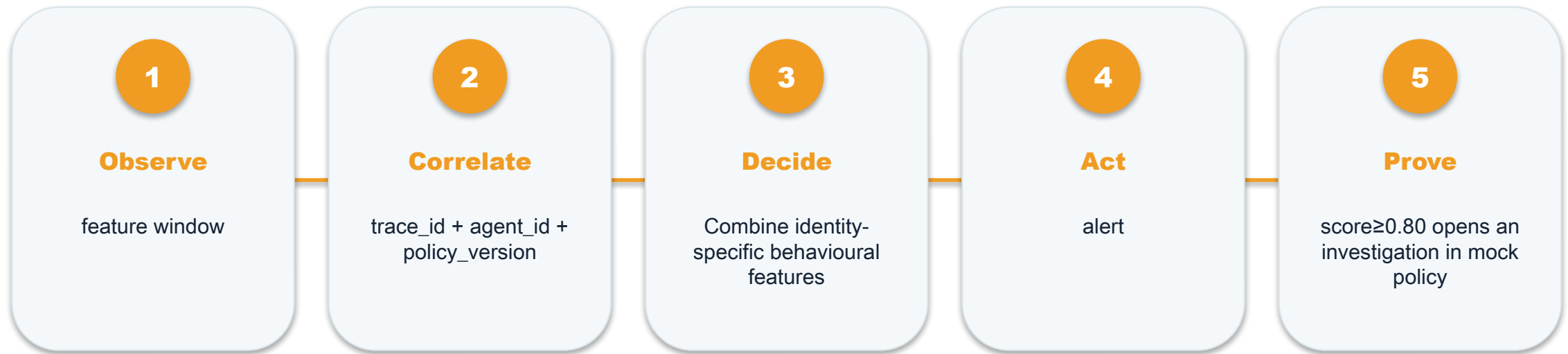
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Behaviour anomaly score: architecture and control boundary



Behaviour anomaly score: runtime decision trace



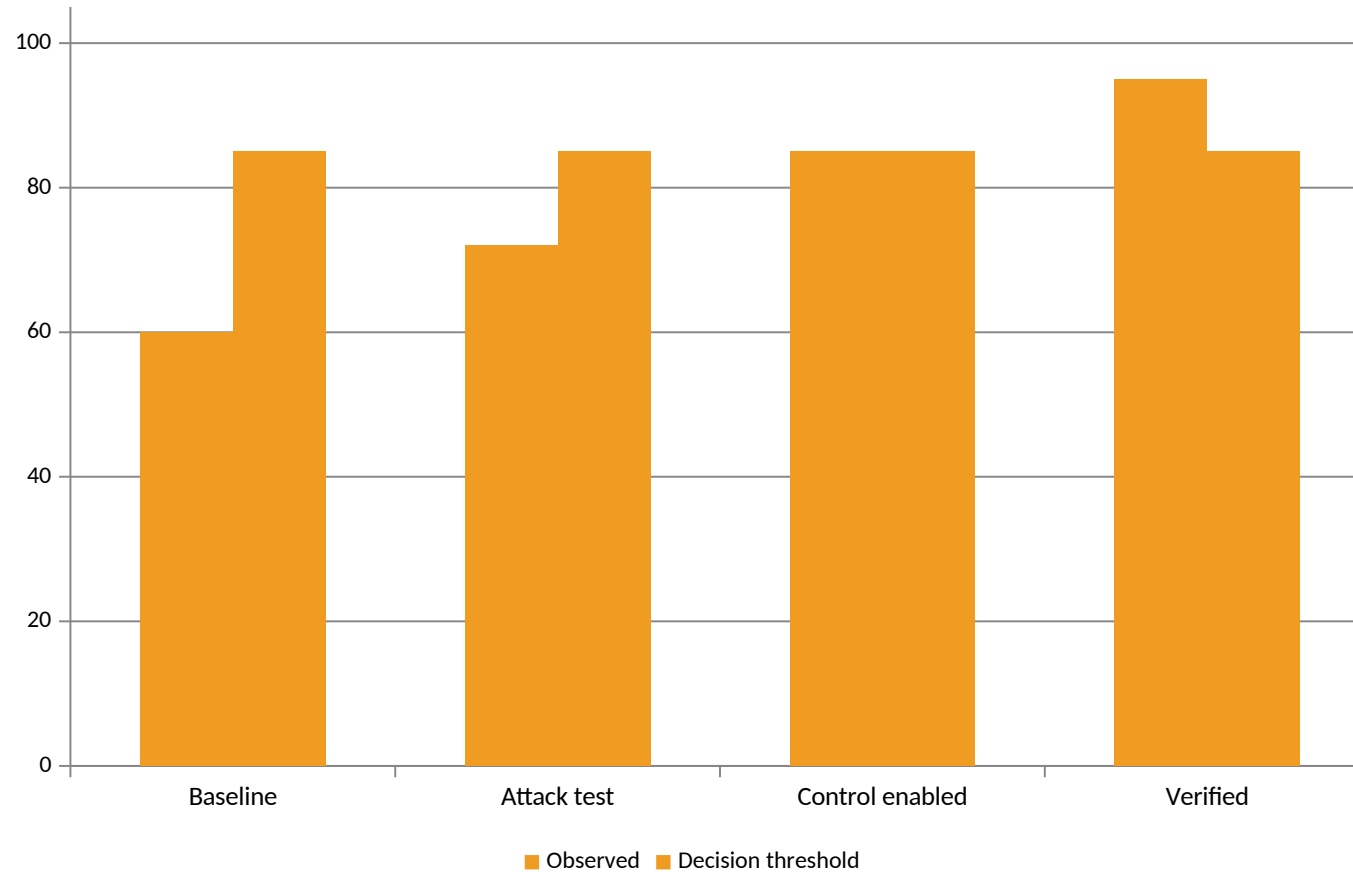
Failure to detect: Compromised agent stays within simple static limits

Behaviour anomaly score: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	tool_rate_z	Required in decision and audit evidence
field_2	new_destination	Required in decision and audit evidence
field_3	denied_ratio	Required in decision and audit evidence
field_4	step_count	Required in decision and audit evidence
field_5	score	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Behaviour anomaly score: evidence and acceptance decision



DECISION RULE

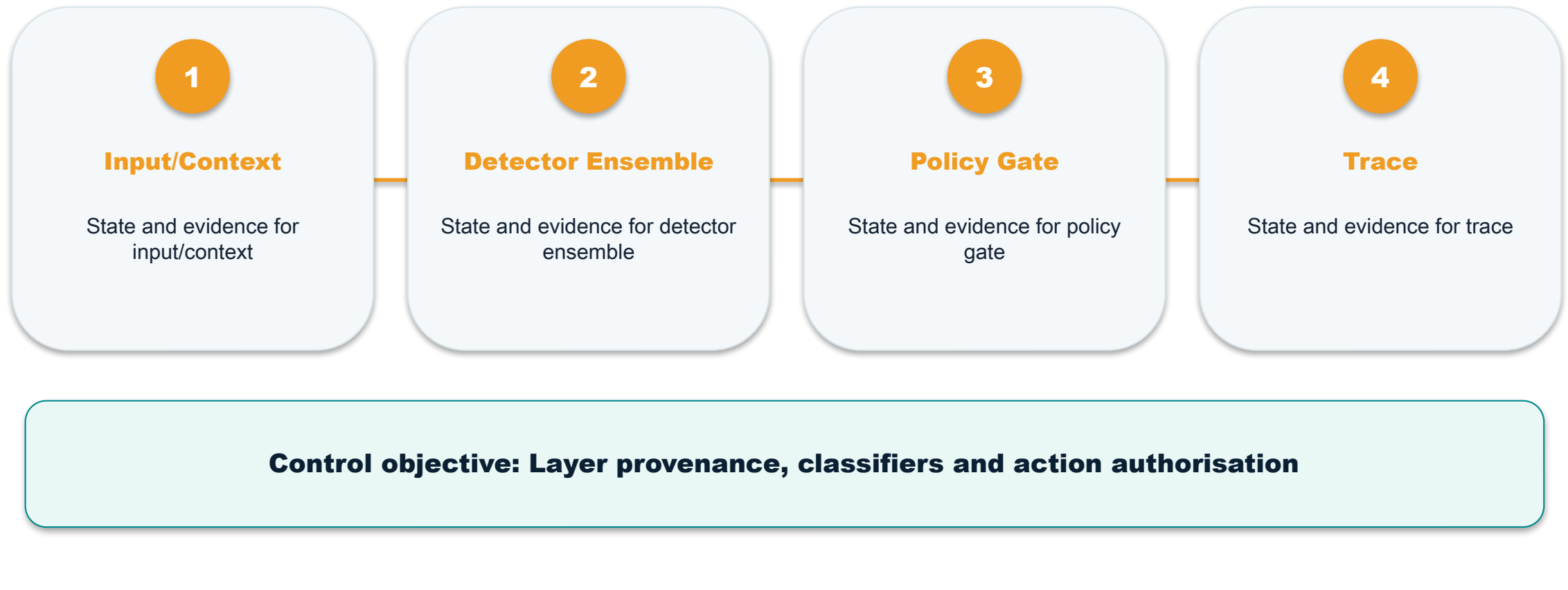
Accept only when score ≥ 0.80 opens an investigation in mock policy. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

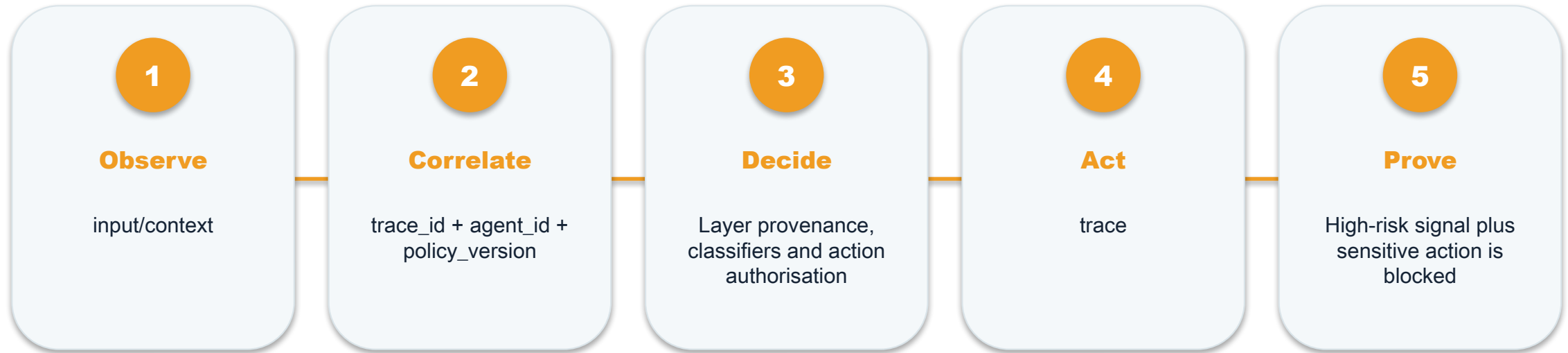
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Prompt-injection signals: architecture and control boundary



Prompt-injection signals: runtime decision trace



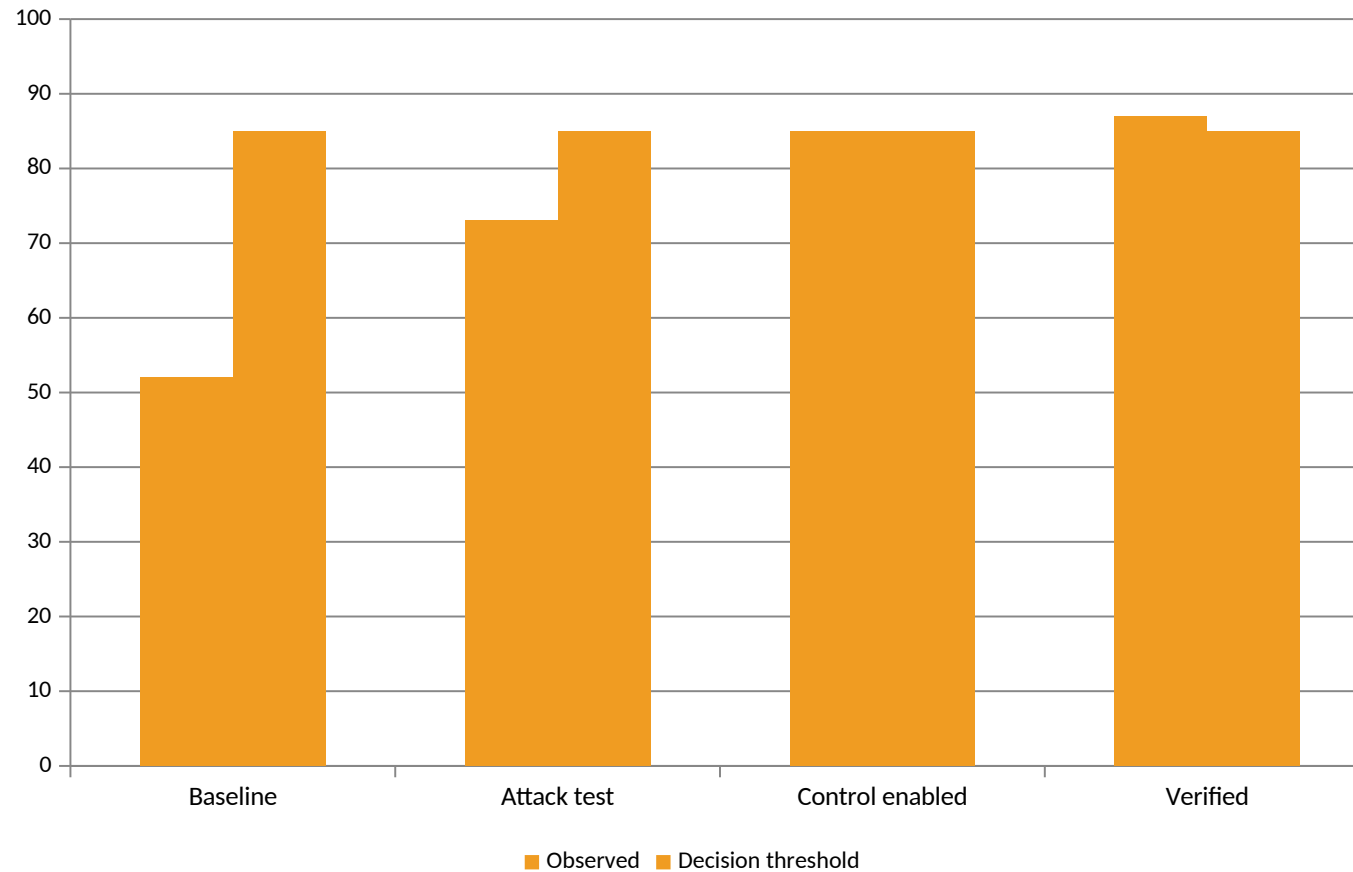
Failure to detect: Obfuscated injection bypasses one keyword filter

Prompt-injection signals: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	instruction_density	Required in decision and audit evidence
field_2	trust_mismatch	Required in decision and audit evidence
field_3	secret_request	Required in decision and audit evidence
field_4	override_phrase	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Prompt-injection signals: evidence and acceptance decision



DECISION RULE

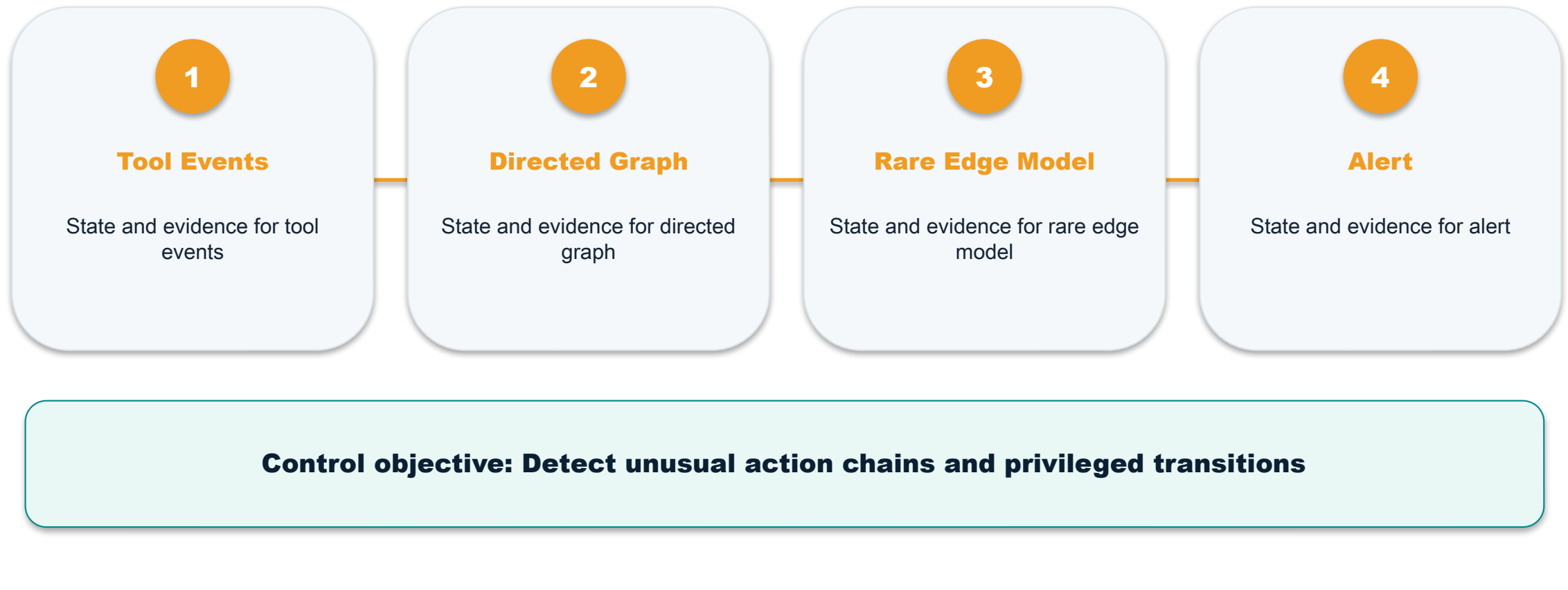
Accept only when High-risk signal plus sensitive action is blocked. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Tool-sequence graph: architecture and control boundary



Tool-sequence graph: runtime decision trace



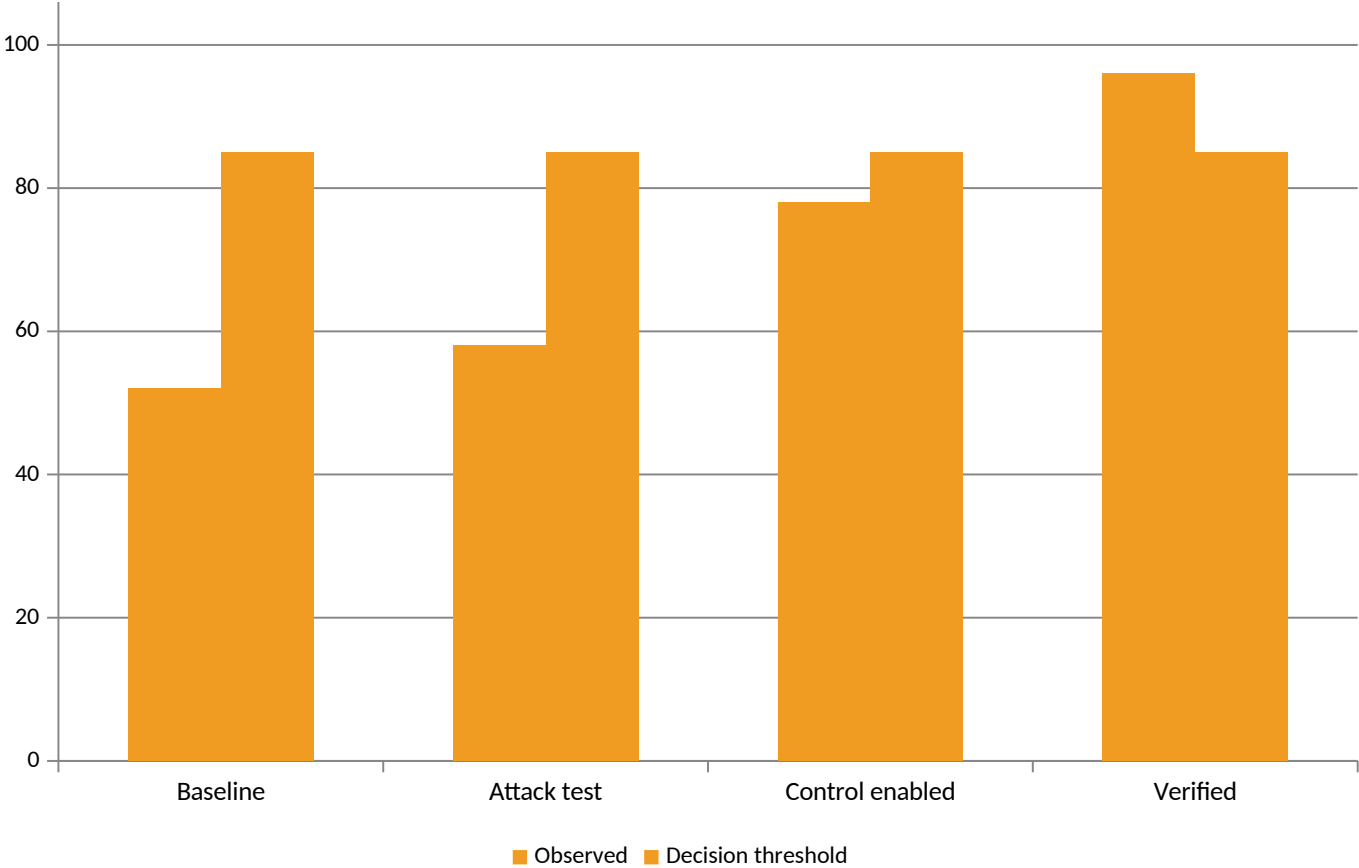
Failure to detect: Each call is allowed but the sequence is malicious

Tool-sequence graph: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	from_tool	Required in decision and audit evidence
field_2	to_tool	Required in decision and audit evidence
field_3	edge_count	Required in decision and audit evidence
field_4	baseline_prob	Required in decision and audit evidence
field_5	risk_weight	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Tool-sequence graph: evidence and acceptance decision



DECISION RULE

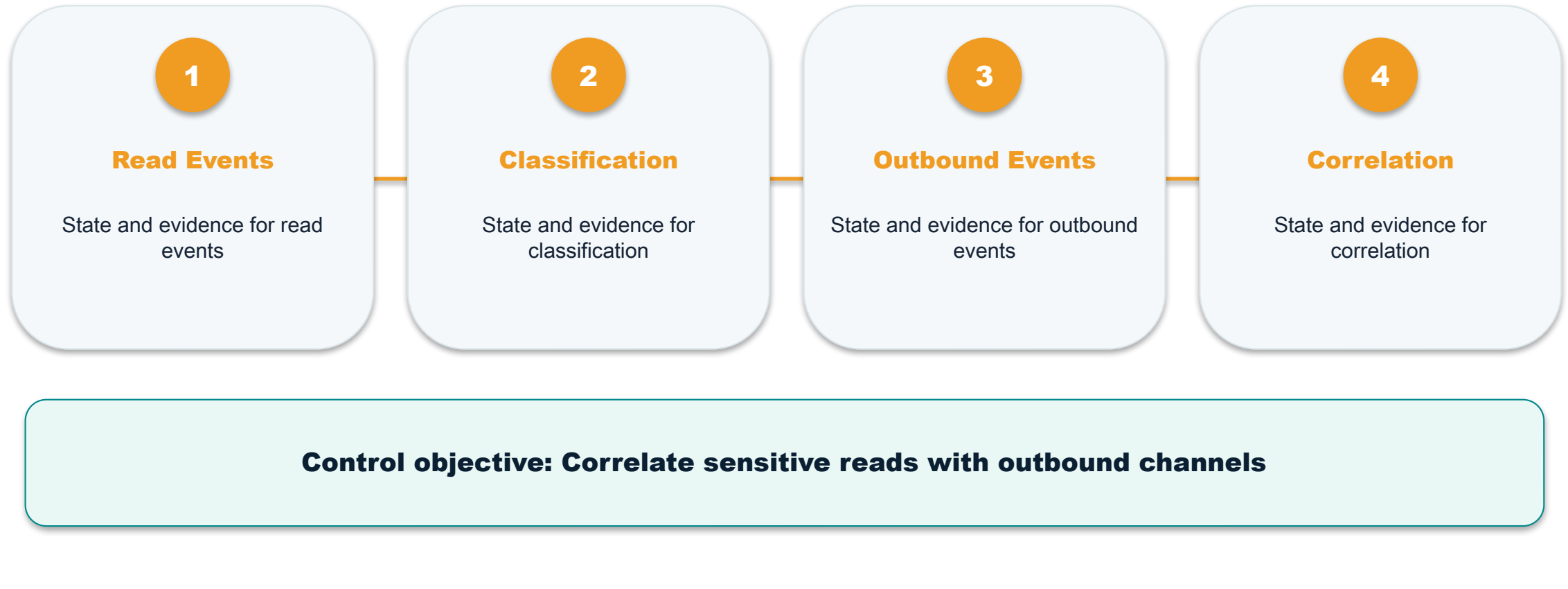
Accept only when Rare high-risk edge produces a correlated alert. Otherwise record a gap, owner and due date.

EVIDENCE STATUS



Synthetic training evidence; values are illustrative, not external performance claims

Data-egress detection: architecture and control boundary



Data-egress detection: runtime decision trace



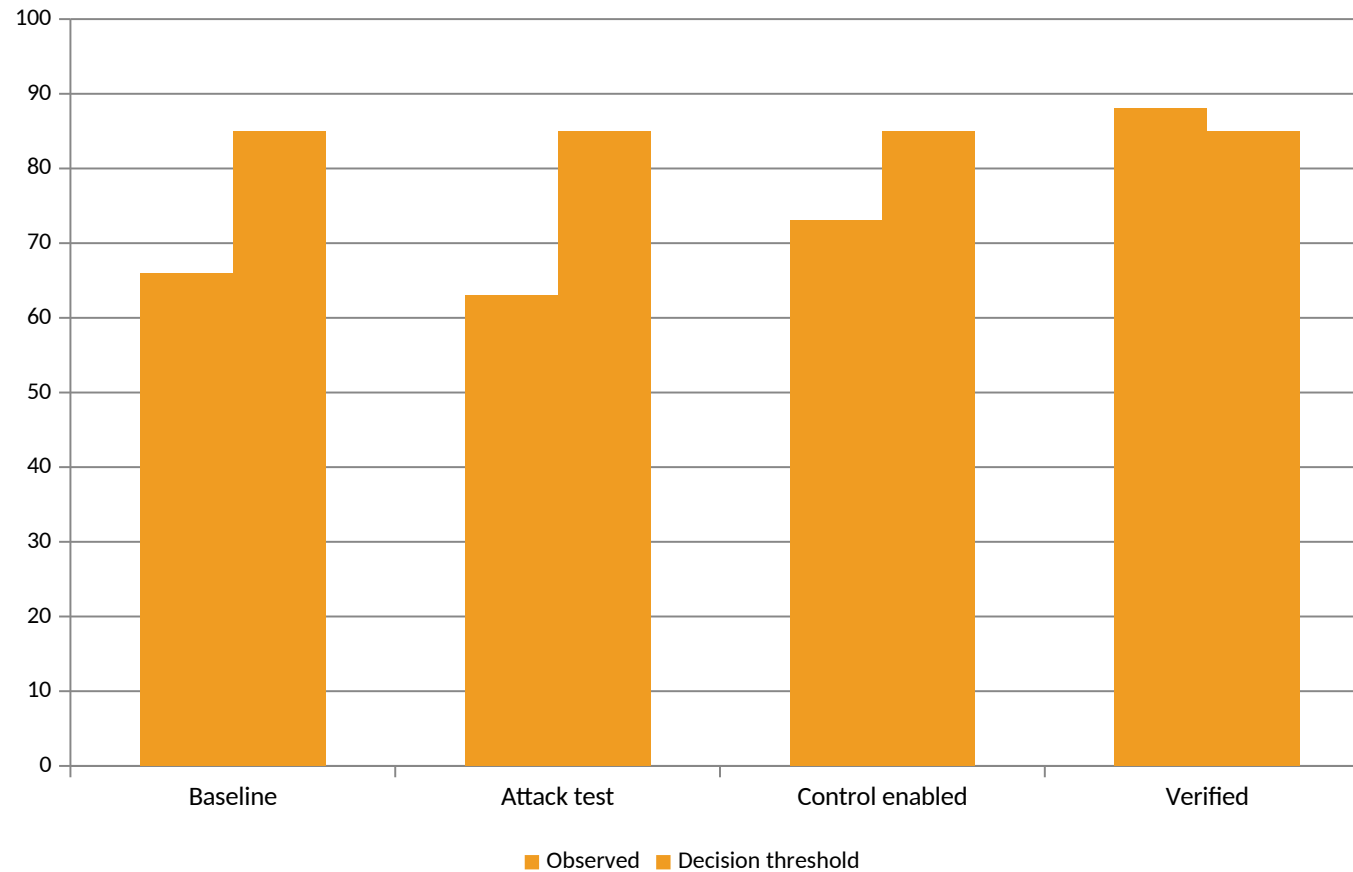
Failure to detect: Agent reads restricted data then sends a small encoded payload

Data-egress detection: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	bytes_read	Required in decision and audit evidence
field_2	data_class	Required in decision and audit evidence
field_3	destination	Required in decision and audit evidence
field_4	bytes_out	Required in decision and audit evidence
field_5	time_delta	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Data-egress detection: evidence and acceptance decision



DECISION RULE

Accept only when Restricted-read + unknown-egress within 5 min alerts. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Secret-access anomaly: architecture and control boundary



Secret-access anomaly: runtime decision trace



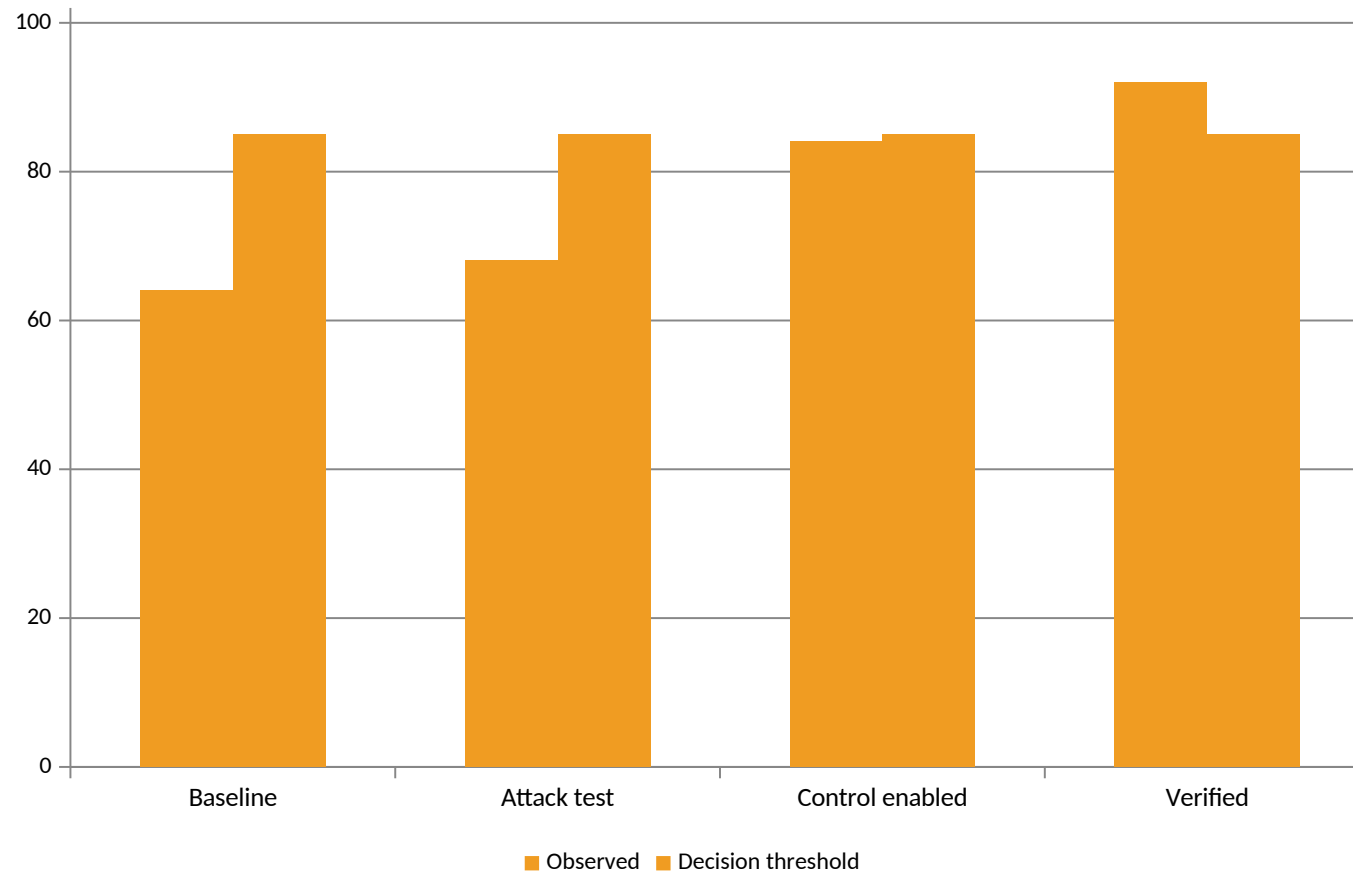
Failure to detect: Agent requests a secret unrelated to its normal tools

Secret-access anomaly: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	secret_ref	Required in decision and audit evidence
field_2	agent_id	Required in decision and audit evidence
field_3	tool_id	Required in decision and audit evidence
field_4	reason	Required in decision and audit evidence
field_5	usual_for_agent	Required in decision and audit evidence
field_6	result	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Secret-access anomaly: evidence and acceptance decision



DECISION RULE

Accept only when Unexpected secret_ref/tool_id pair is denied. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Session drift: runtime decision trace



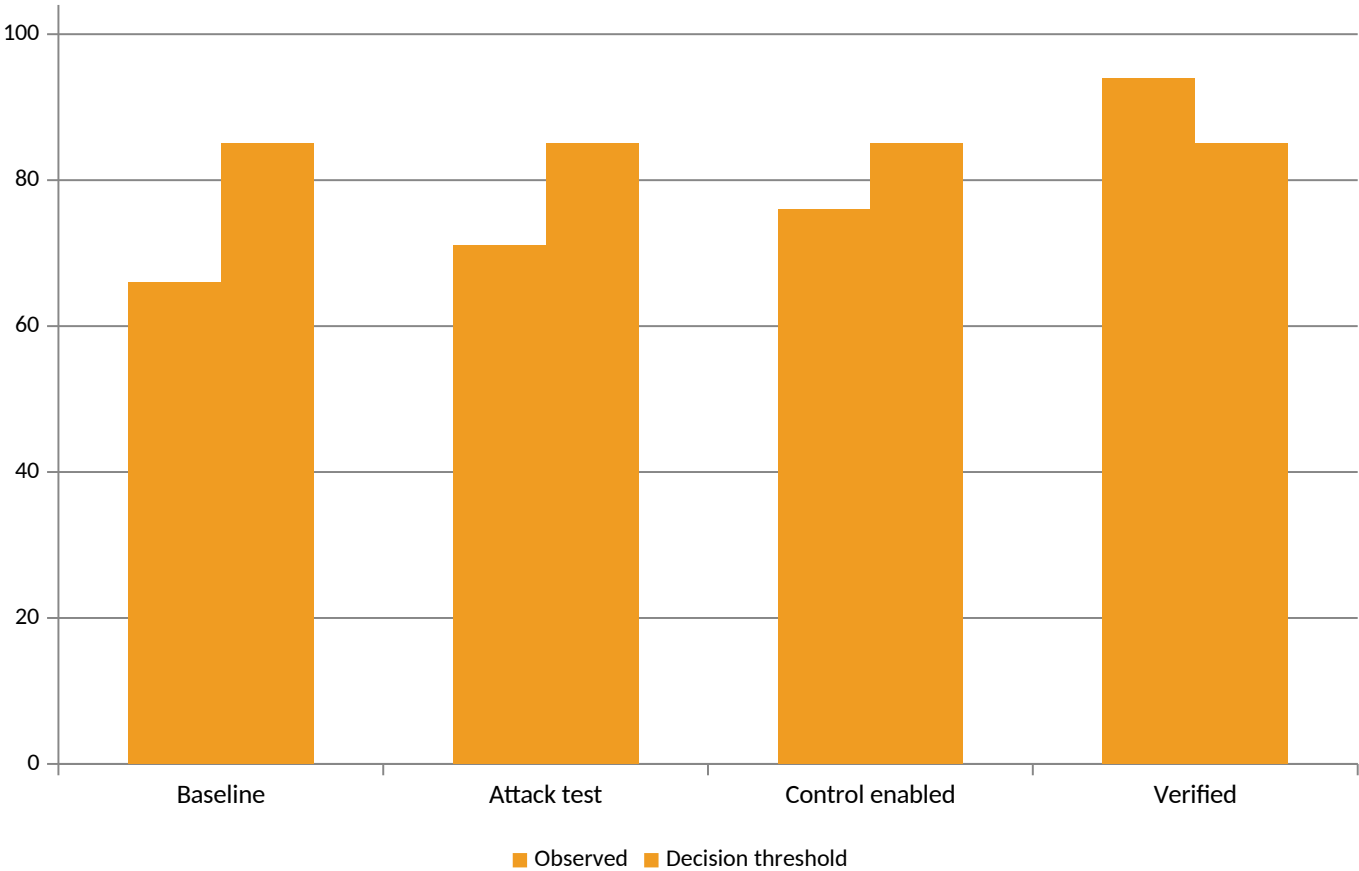
Failure to detect: Long session gradually moves away from approved goal

Session drift: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	goal_hash	Required in decision and audit evidence
field_2	active_tools[]	Required in decision and audit evidence
field_3	destinations[]	Required in decision and audit evidence
field_4	step_count	Required in decision and audit evidence
field_5	drift_score	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Session drift: evidence and acceptance decision



DECISION RULE

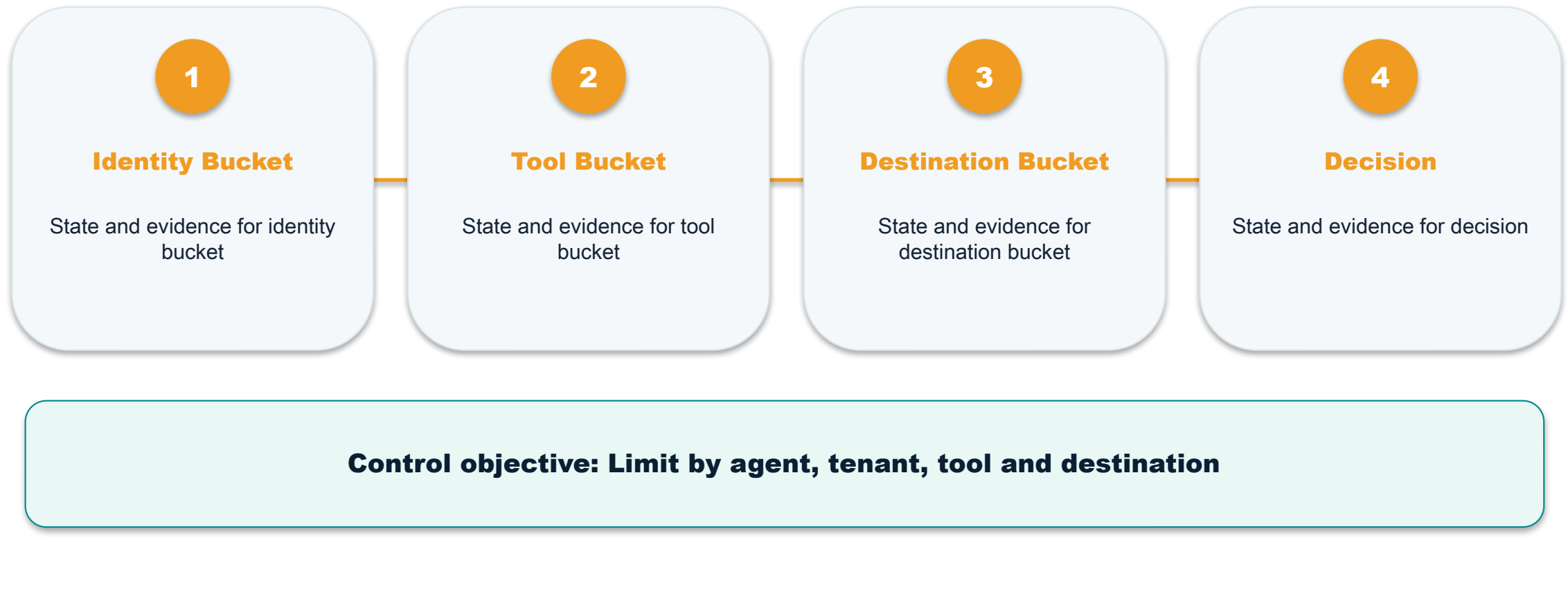
Accept only when $\text{drift_score} \geq 0.70$ pauses before the next action. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

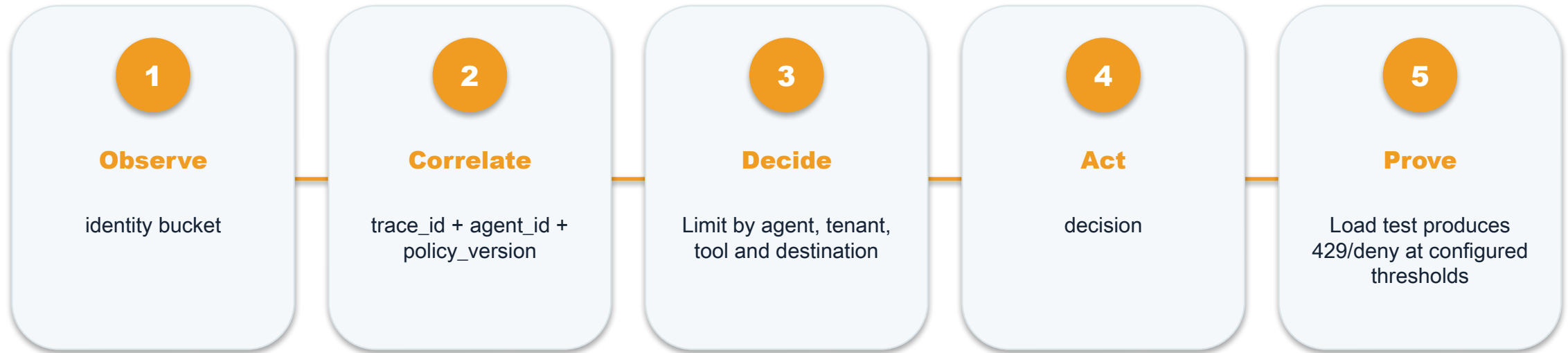


Synthetic training evidence; values are illustrative, not external performance claims

Rate-limit control: architecture and control boundary



Rate-limit control: runtime decision trace



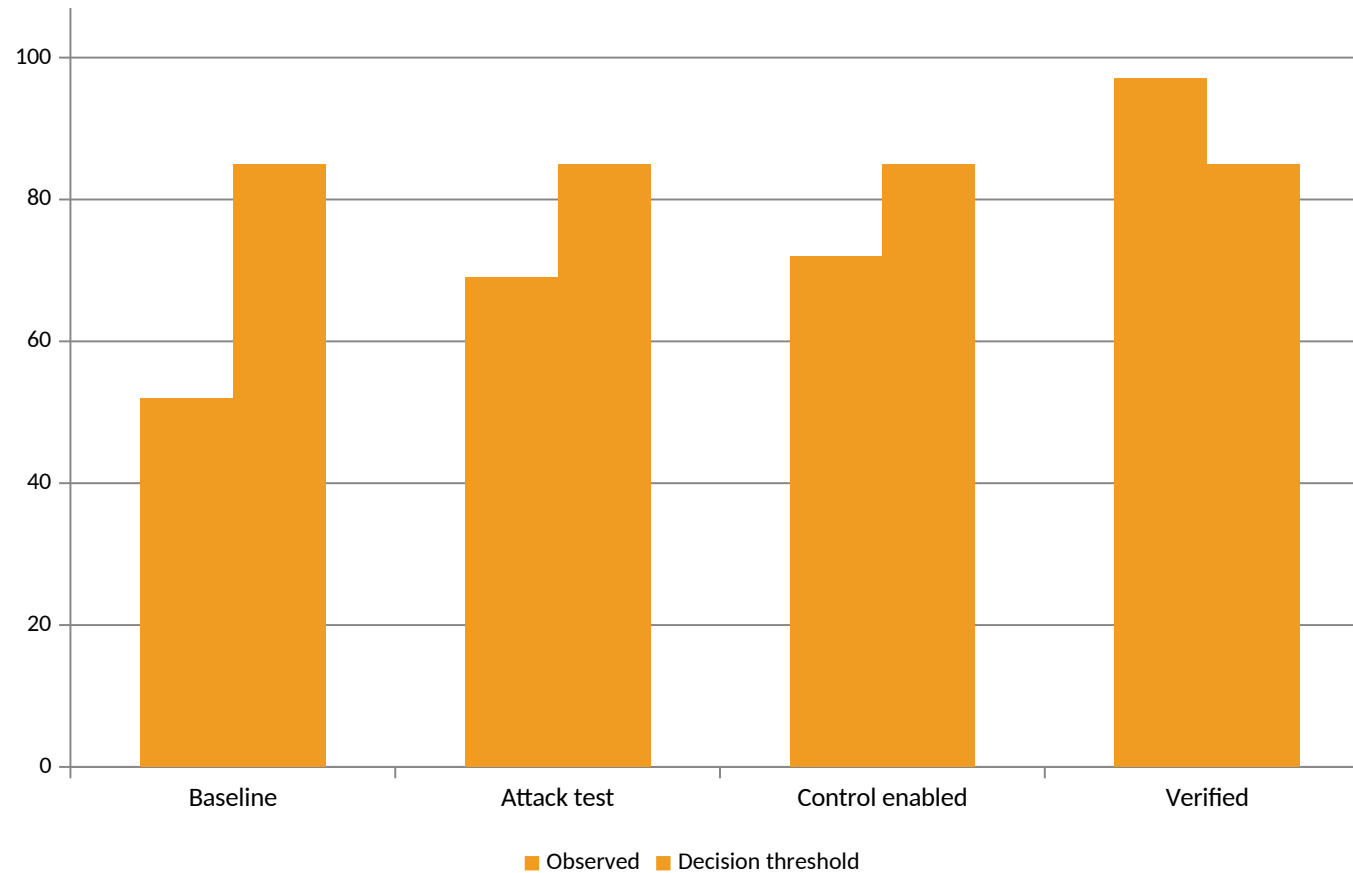
Failure to detect: Distributed calls evade a single global counter

Rate-limit control: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	window_seconds	Required in decision and audit evidence
field_2	max_requests	Required in decision and audit evidence
field_3	current_count	Required in decision and audit evidence
field_4	retry_after	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Rate-limit control: evidence and acceptance decision



DECISION RULE

Accept only when Load test produces 429/deny at configured thresholds. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Budget alert: architecture and control boundary



Budget alert: runtime decision trace



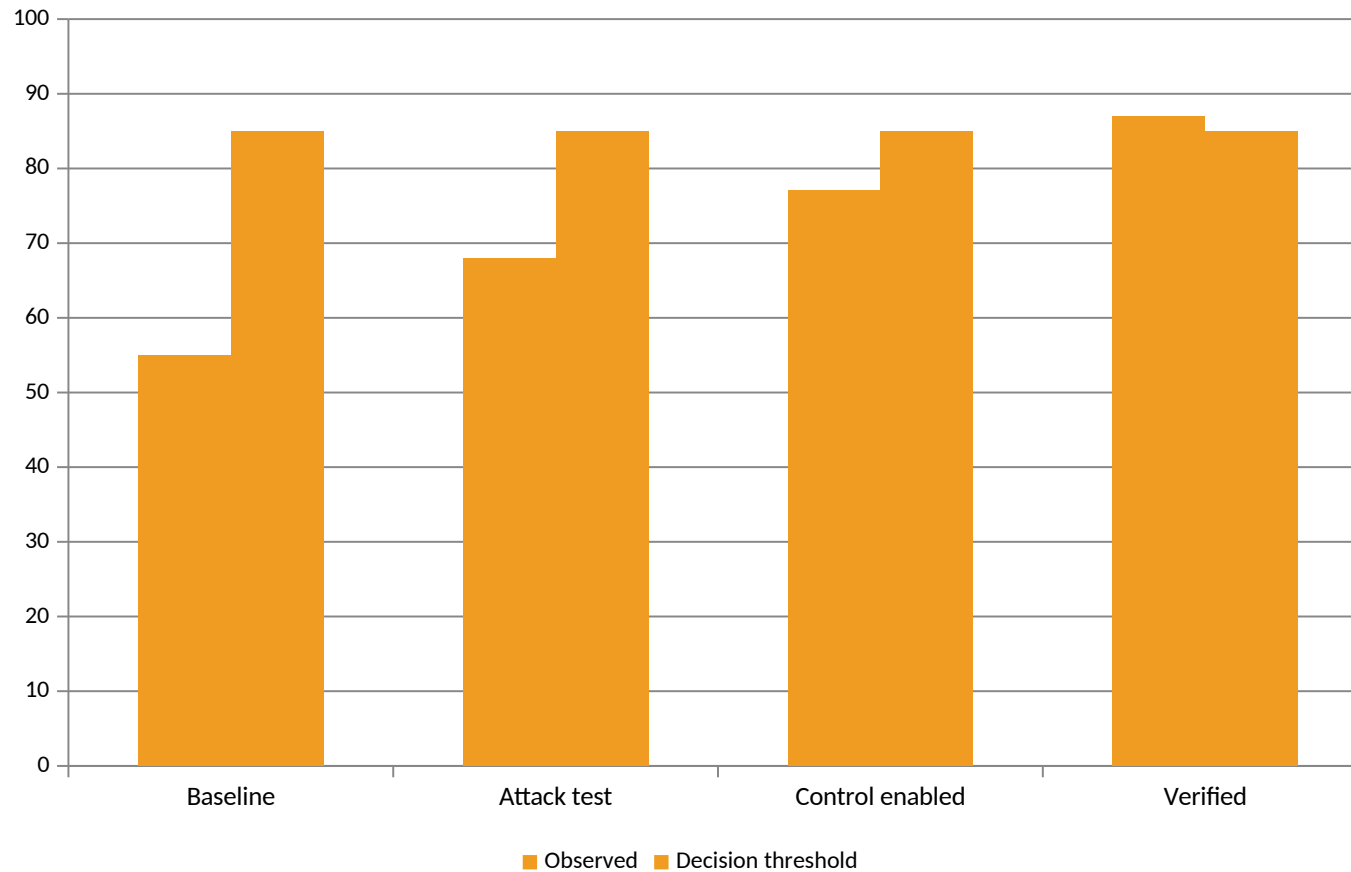
Failure to detect: Runaway planning consumes budget without operational alert

Budget alert: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	session_cost	Required in decision and audit evidence
field_2	daily_cost	Required in decision and audit evidence
field_3	warning_pct	Required in decision and audit evidence
field_4	hard_limit	Required in decision and audit evidence
field_5	action	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Budget alert: evidence and acceptance decision



DECISION RULE

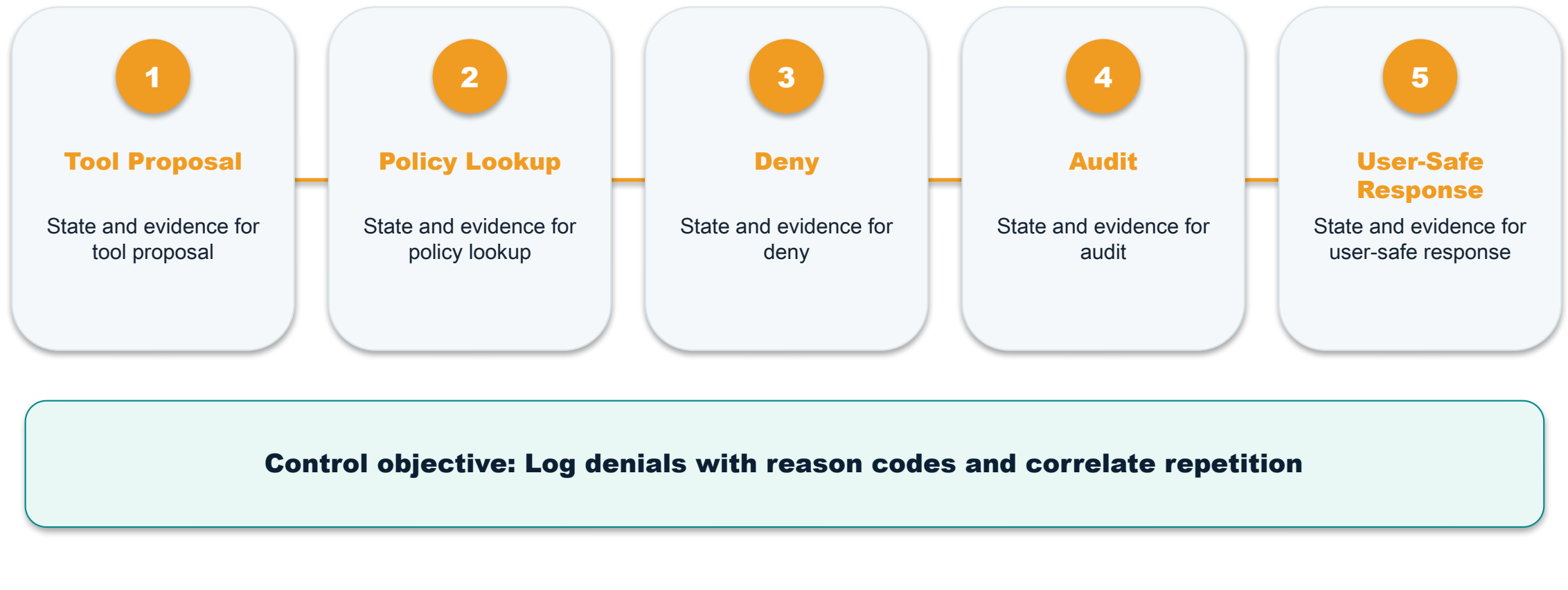
Accept only when Budget breach stops new calls and preserves trace. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Allowlist denial: architecture and control boundary



Allowlist denial: runtime decision trace



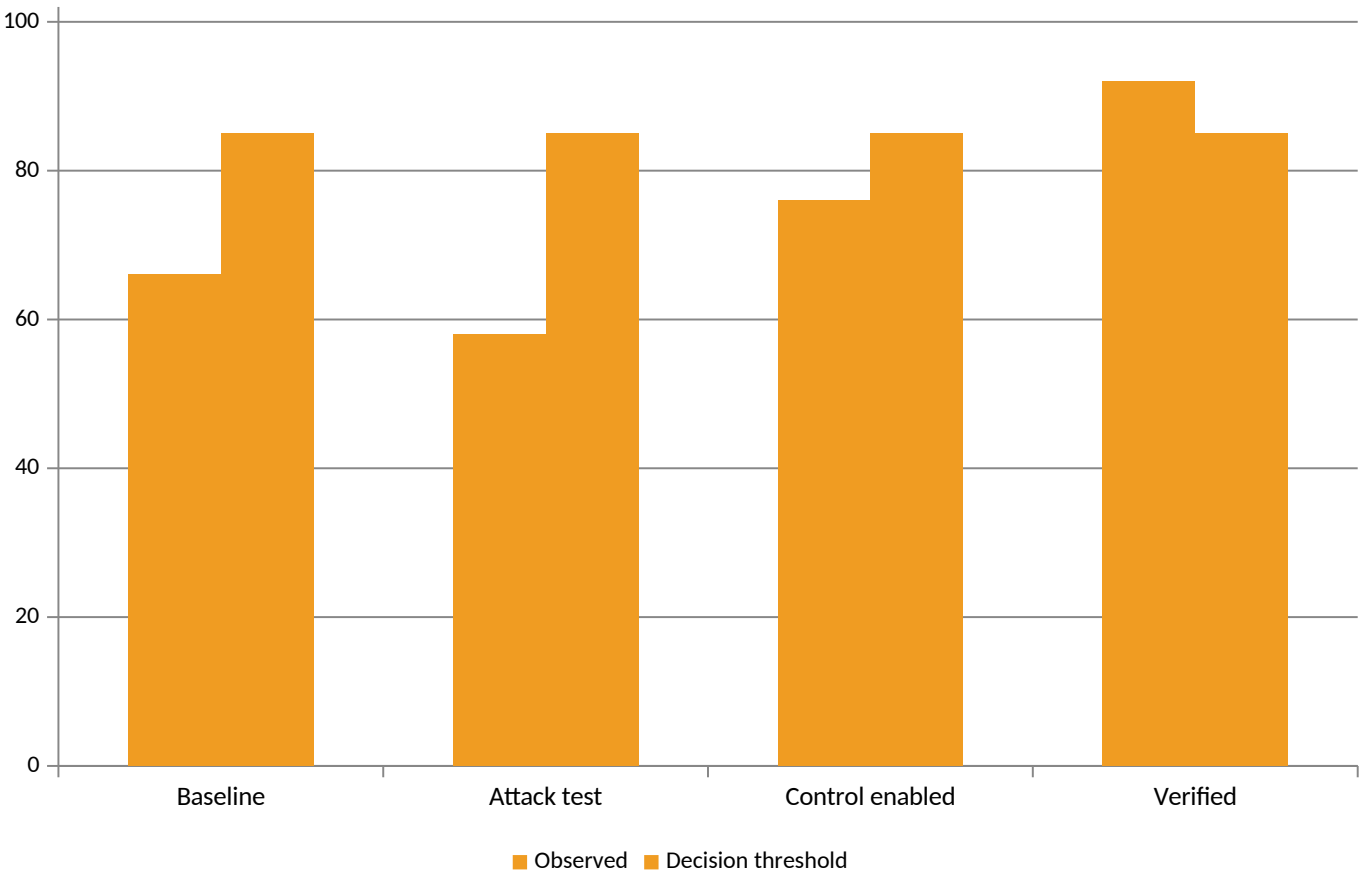
Failure to detect: Blocked action fails silently and attacker keeps probing

Allowlist denial: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agent_id	Required in decision and audit evidence
field_2	tool_id	Required in decision and audit evidence
field_3	action	Required in decision and audit evidence
field_4	policy_version	Required in decision and audit evidence
field_5	deny_reason	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Allowlist denial: evidence and acceptance decision



DECISION RULE

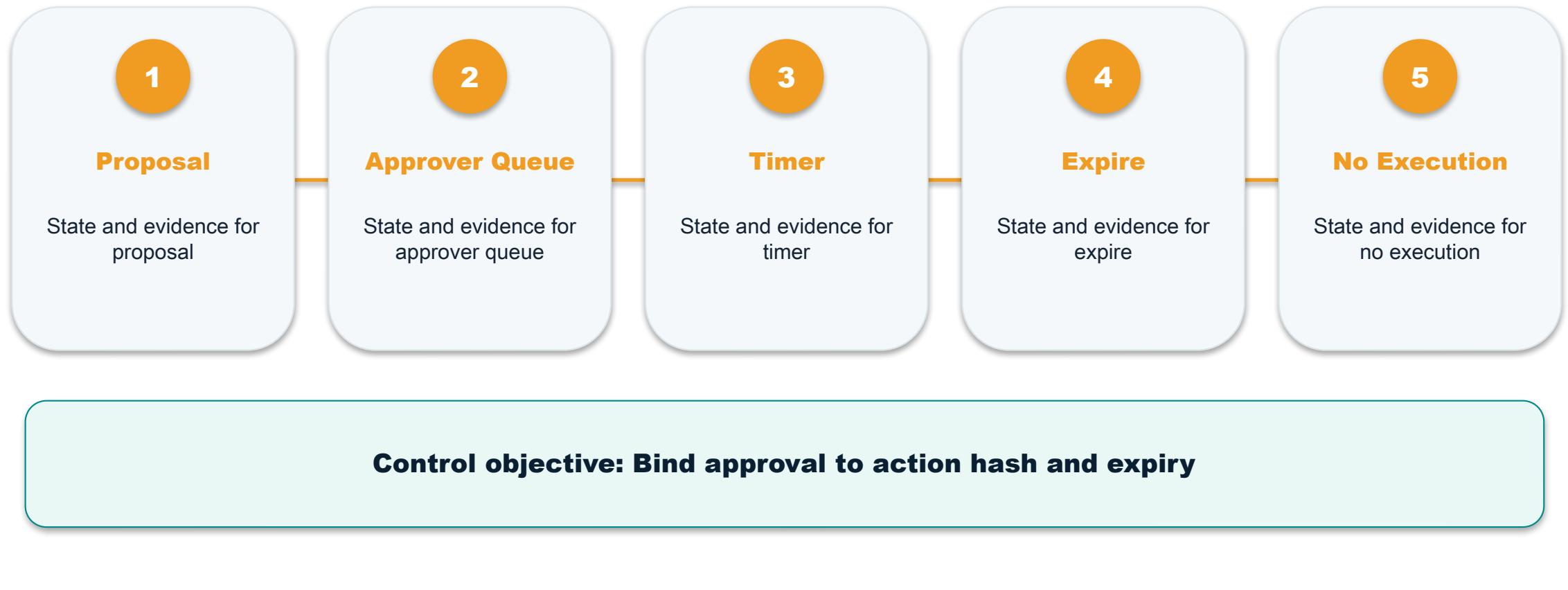
Accept only when Repeated deny_reason creates one deduplicated alert. Otherwise record a gap, owner and due date.

EVIDENCE STATUS



Synthetic training evidence; values are illustrative, not external performance claims

Approval timeout: architecture and control boundary



Approval timeout: runtime decision trace



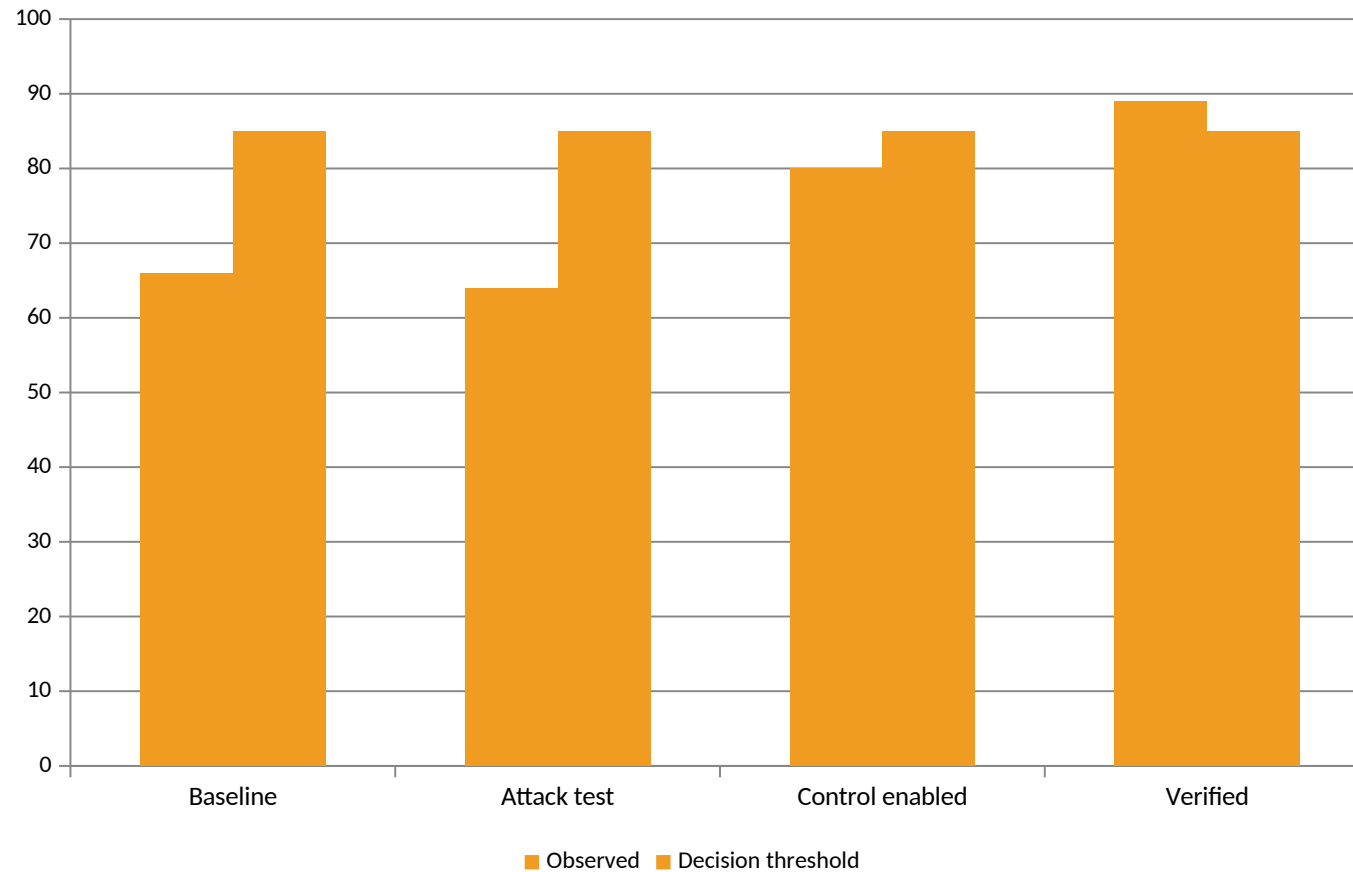
Failure to detect: Old approval is replayed after context changes

Approval timeout: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	approval_id	Required in decision and audit evidence
field_2	created_at	Required in decision and audit evidence
field_3	expires_at	Required in decision and audit evidence
field_4	reviewer	Required in decision and audit evidence
field_5	status	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Approval timeout: evidence and acceptance decision



DECISION RULE

**Accept only when
Expired or mismatched
approval_id is rejected.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Alert triage: architecture and control boundary



Alert triage: runtime decision trace



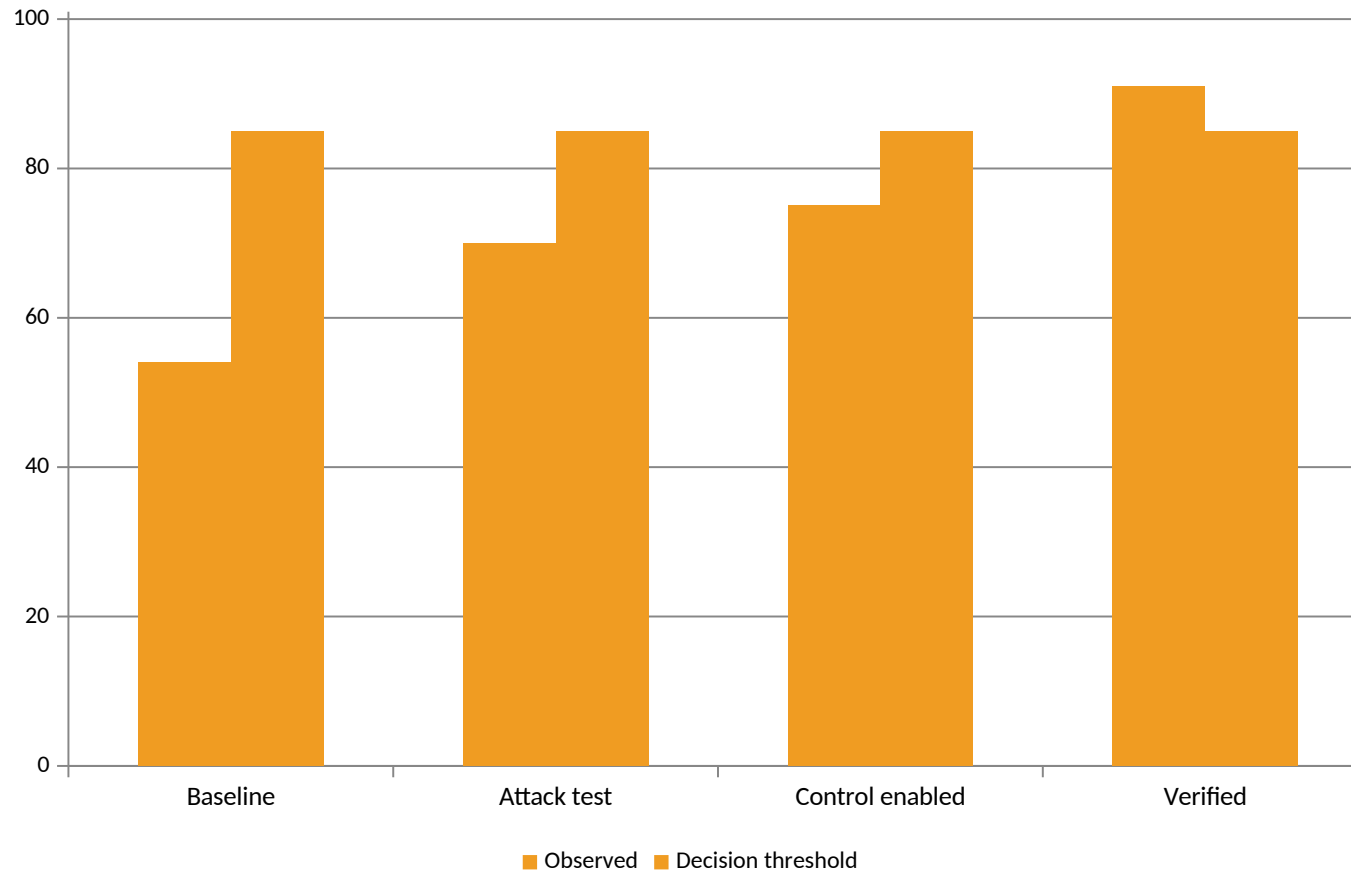
Failure to detect: Analyst closes alert without inspecting action chain

Alert triage: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	alert_id	Required in decision and audit evidence
field_2	evidence[]	Required in decision and audit evidence
field_3	affected_agents[]	Required in decision and audit evidence
field_4	severity	Required in decision and audit evidence
field_5	confidence	Required in decision and audit evidence
field_6	owner	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Alert triage: evidence and acceptance decision



DECISION RULE

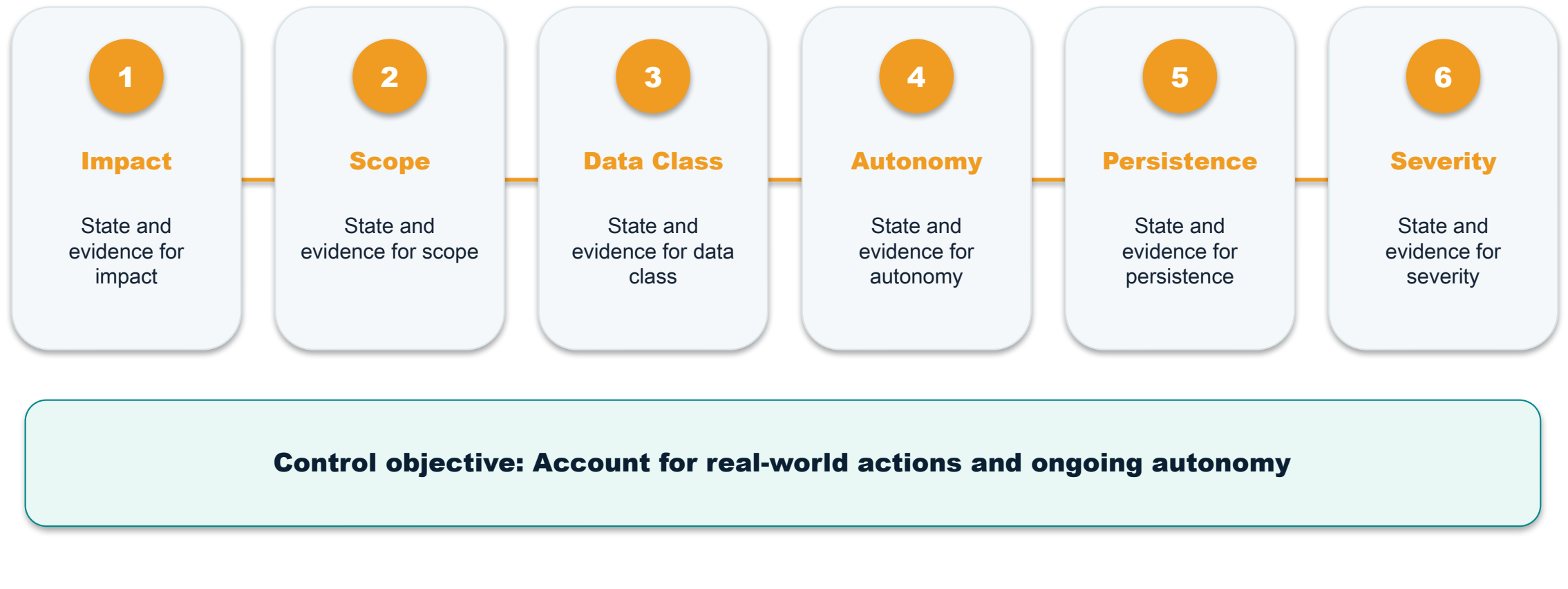
Accept only when Disposition record cites the trace and decision rationale. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

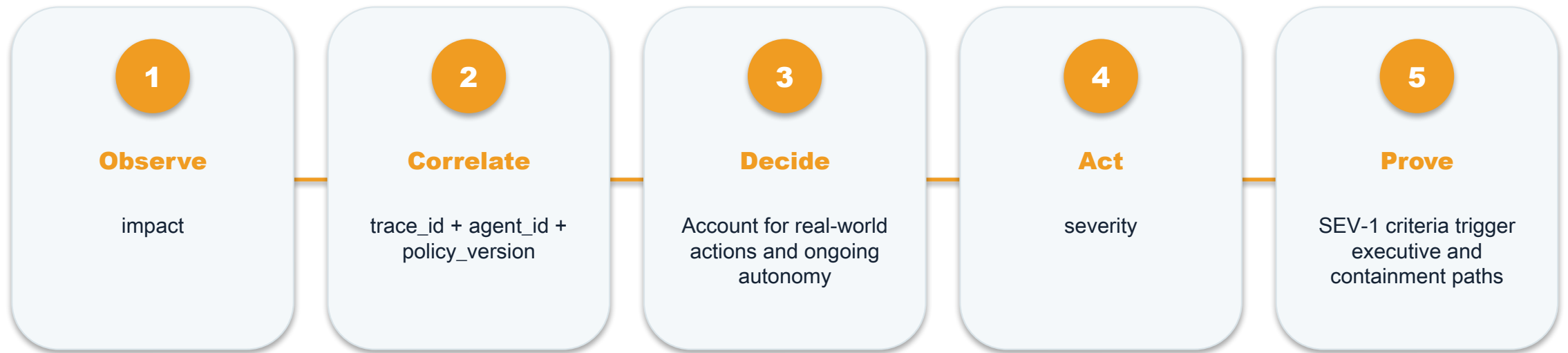
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Incident severity: architecture and control boundary



Incident severity: runtime decision trace



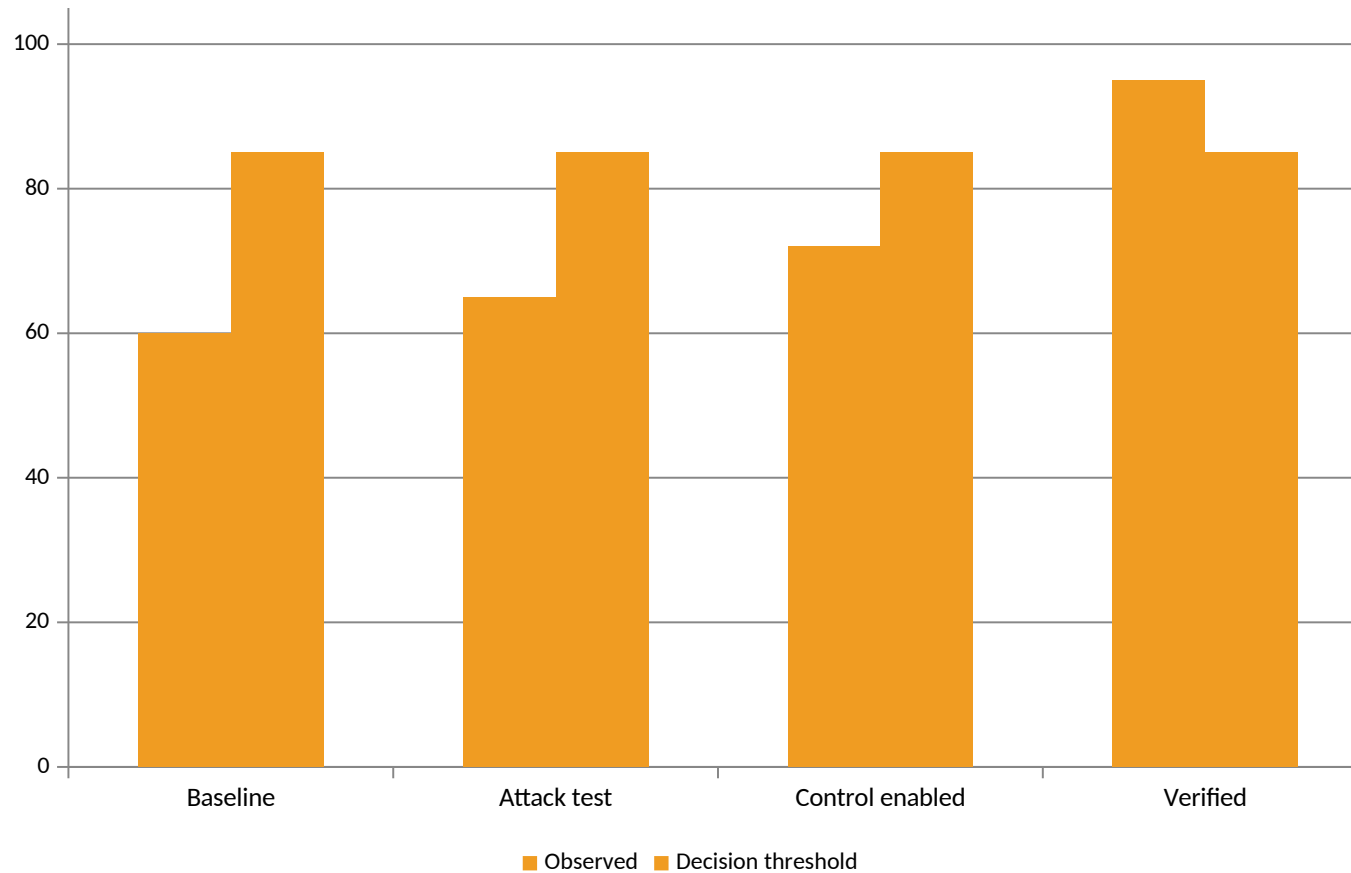
Failure to detect: Agent incident is ranked like a low-impact chatbot error

Incident severity: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agents_affected	Required in decision and audit evidence
field_2	data_class	Required in decision and audit evidence
field_3	external_effect	Required in decision and audit evidence
field_4	persistence	Required in decision and audit evidence
field_5	sev	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Incident severity: evidence and acceptance decision



DECISION RULE

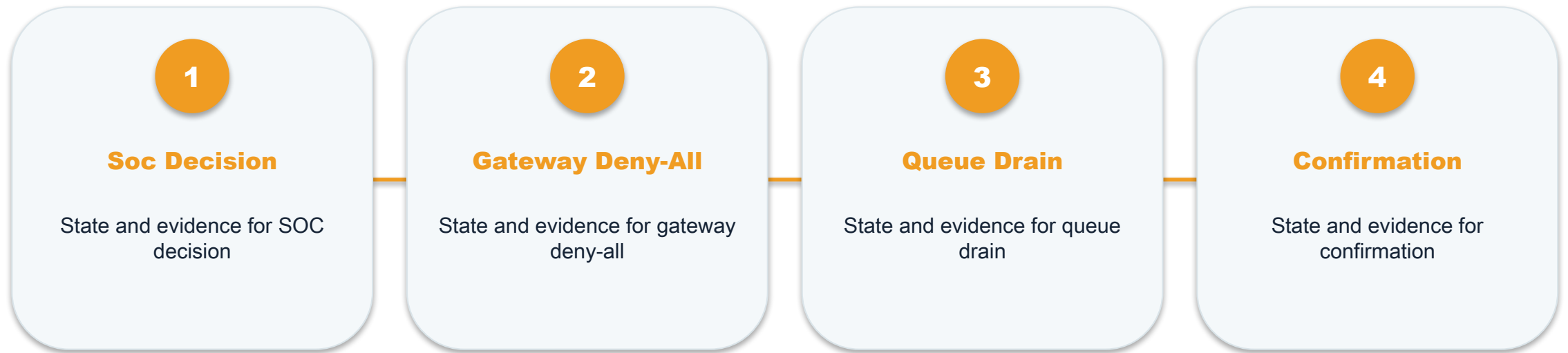
Accept only when SEV-1 criteria trigger executive and containment paths. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

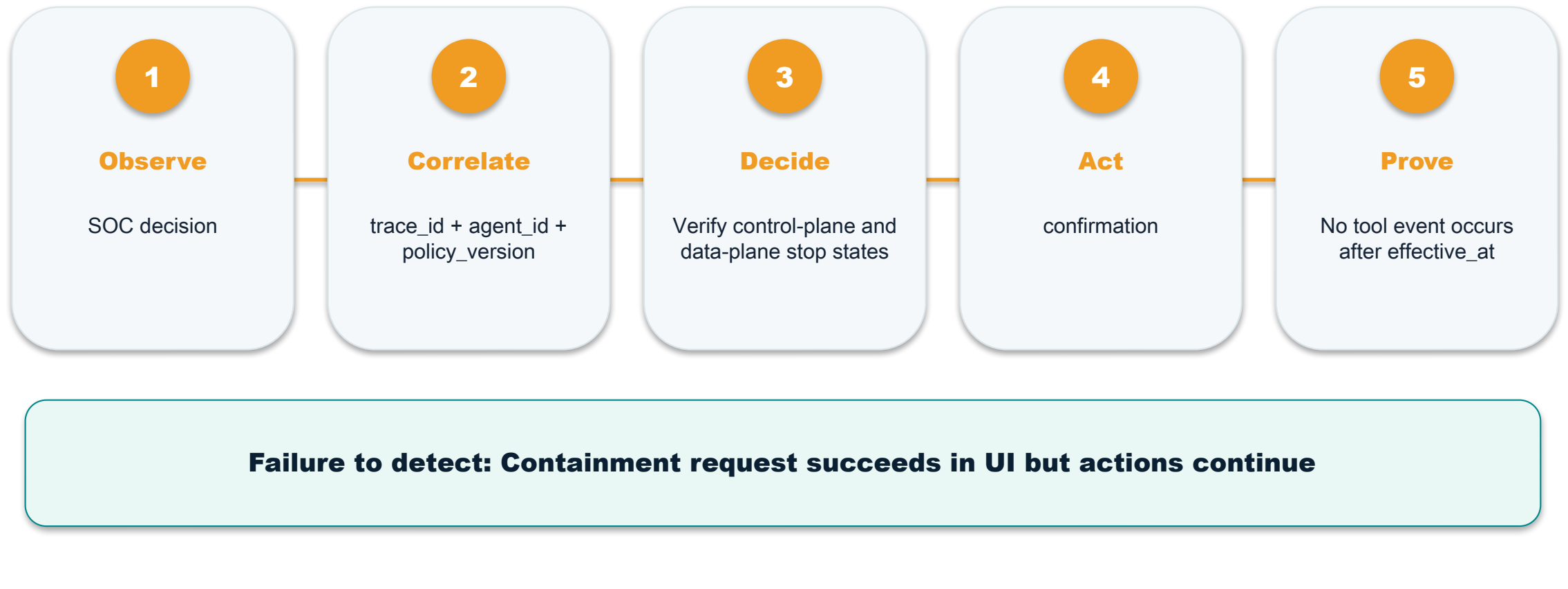
Synthetic training evidence; values are illustrative, not external performance claims

Containment circuit breaker: architecture and control boundary



Control objective: Verify control-plane and data-plane stop states

Containment circuit breaker: runtime decision trace

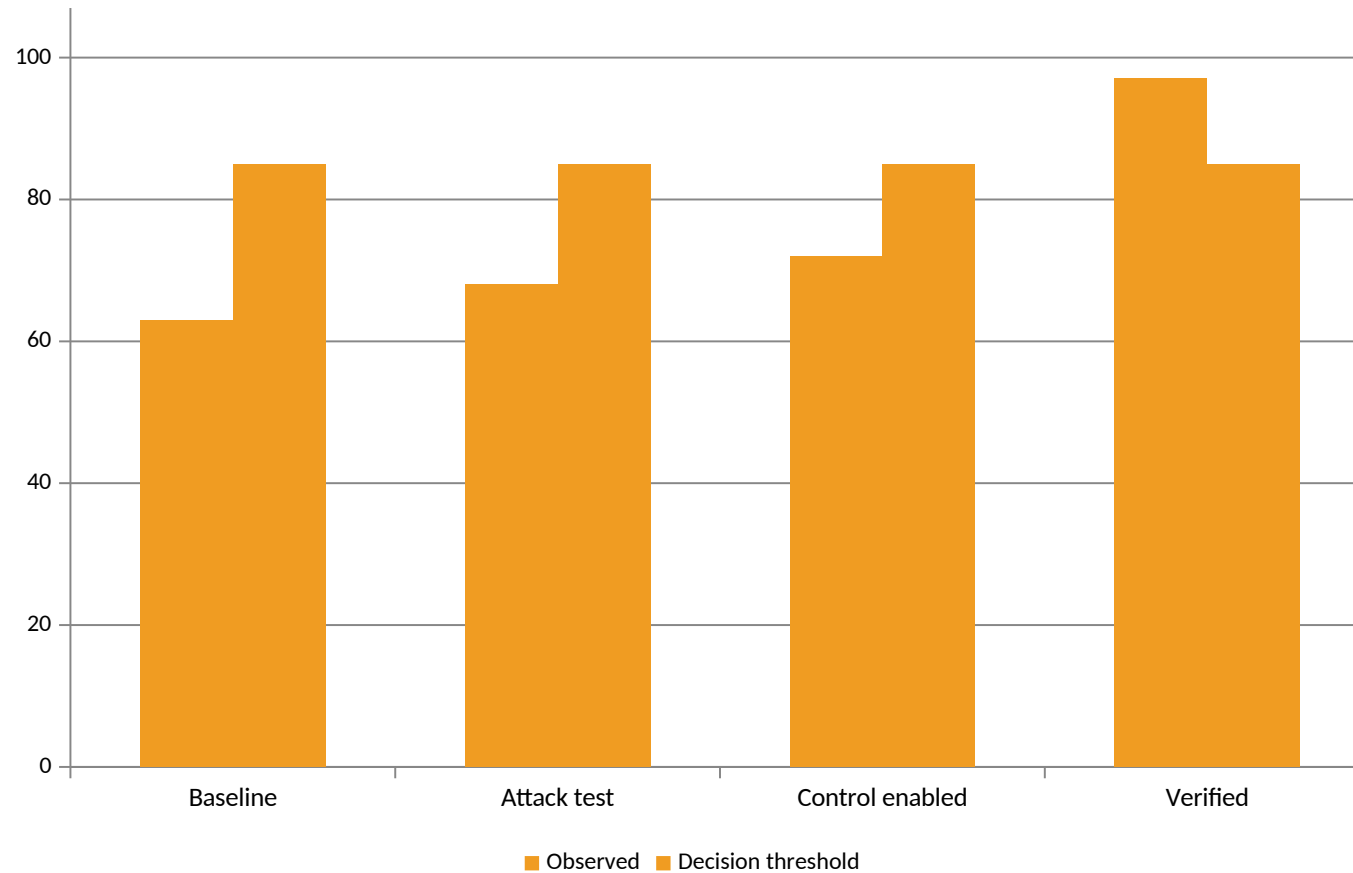


Containment circuit breaker: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	incident_id	Required in decision and audit evidence
field_2	scope	Required in decision and audit evidence
field_3	initiated_by	Required in decision and audit evidence
field_4	effective_at	Required in decision and audit evidence
field_5	agents_stopped	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Containment circuit breaker: evidence and acceptance decision



DECISION RULE

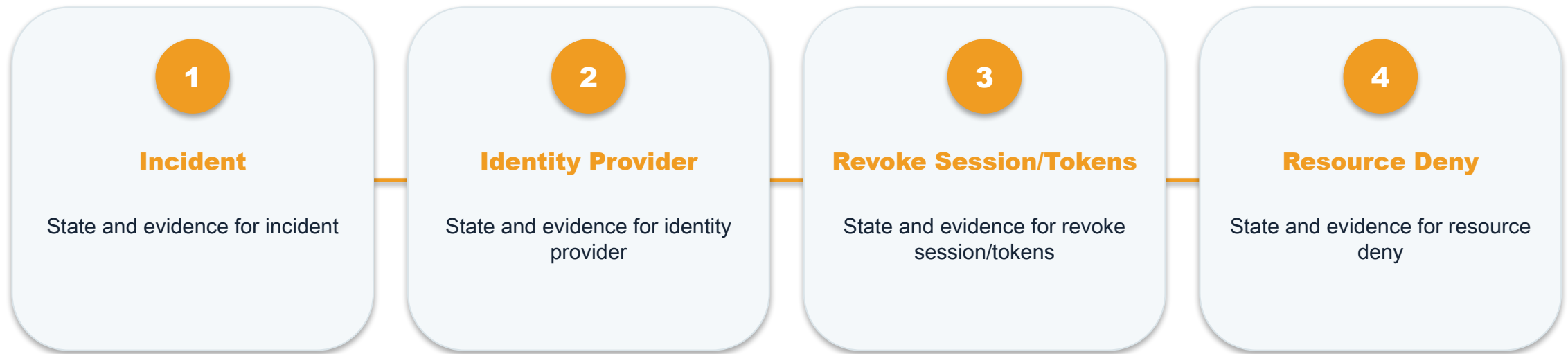
Accept only when No tool event occurs after effective_at. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Token revocation: architecture and control boundary



Control objective: Revoke tokens and enforce short TTL at resources

Token revocation: runtime decision trace



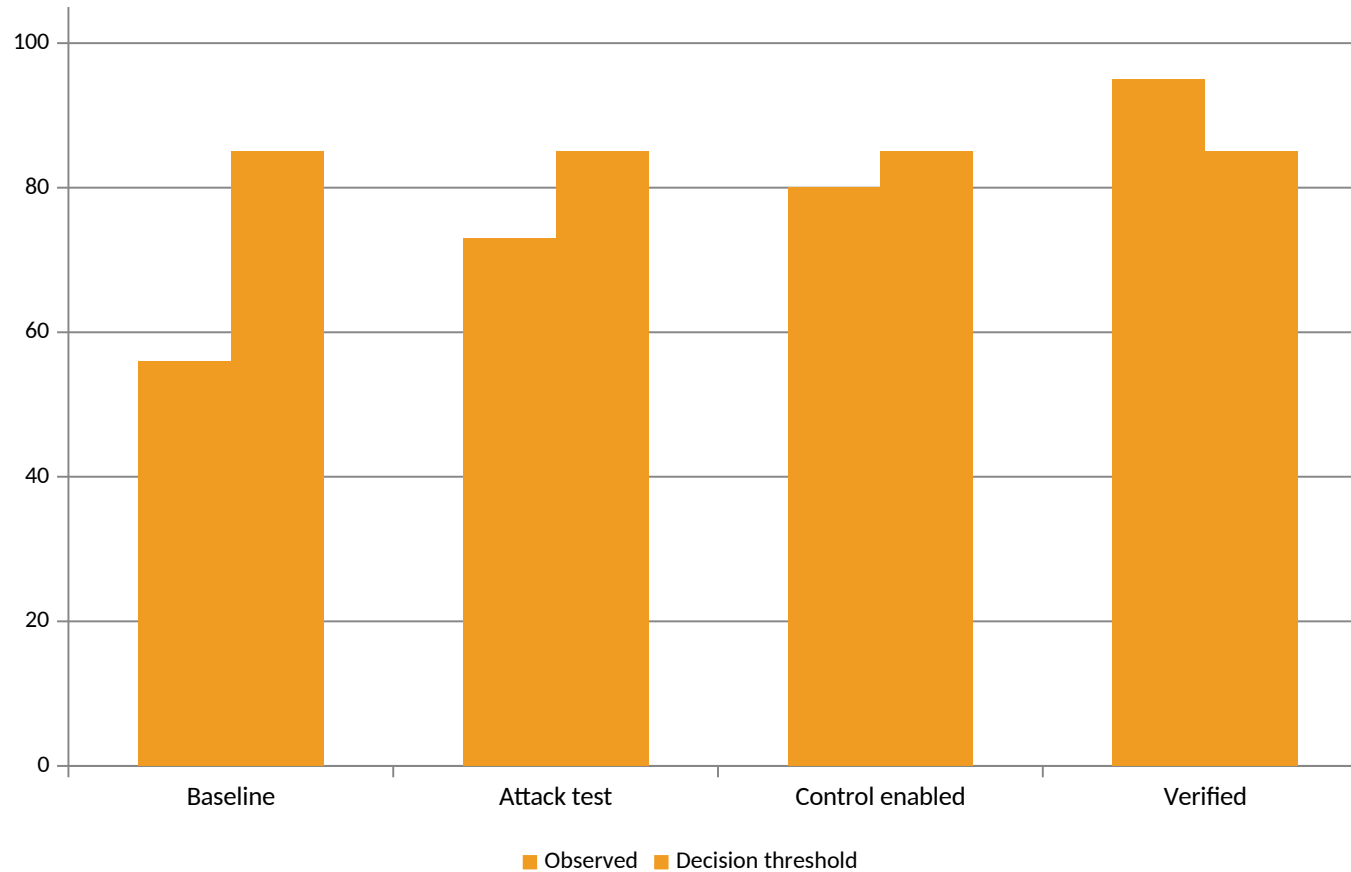
Failure to detect: Disabled agent reuses an already-issued token

Token revocation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	token_id	Required in decision and audit evidence
field_2	subject	Required in decision and audit evidence
field_3	audience	Required in decision and audit evidence
field_4	revoked_at	Required in decision and audit evidence
field_5	reason	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Token revocation: evidence and acceptance decision



DECISION RULE

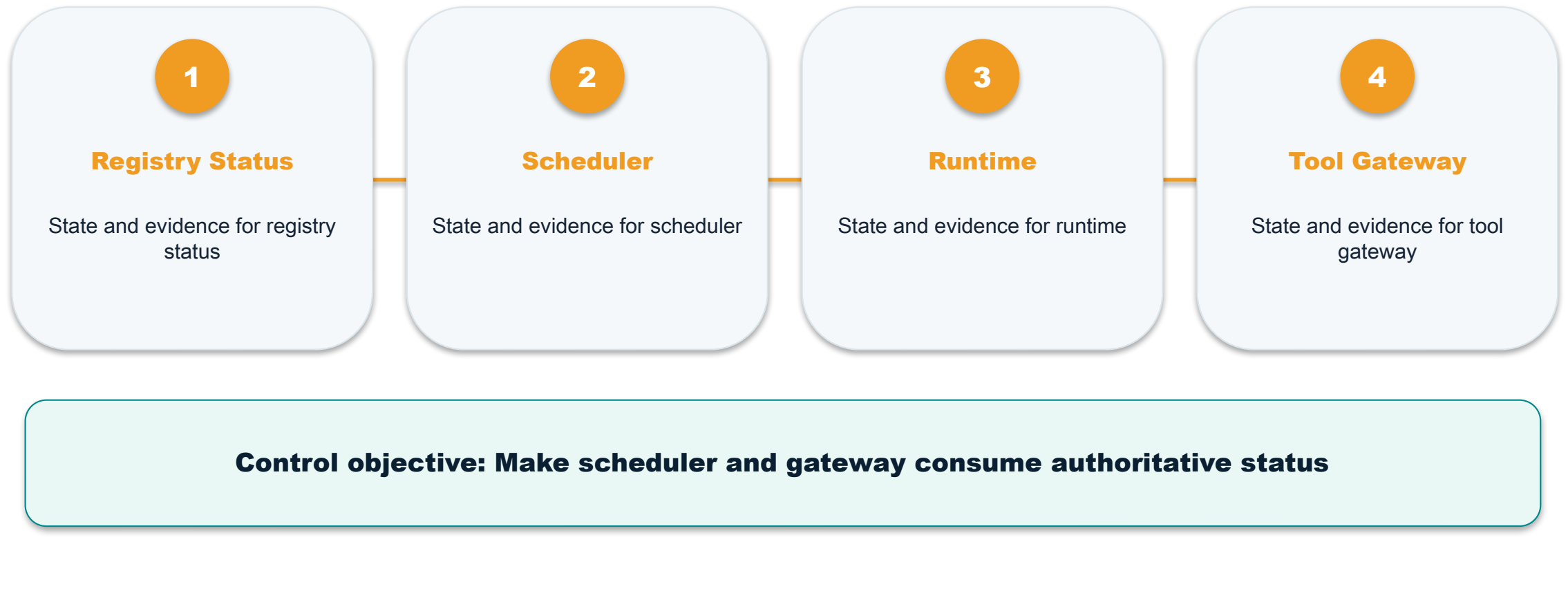
Accept only when Post-revocation access attempt returns deny. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

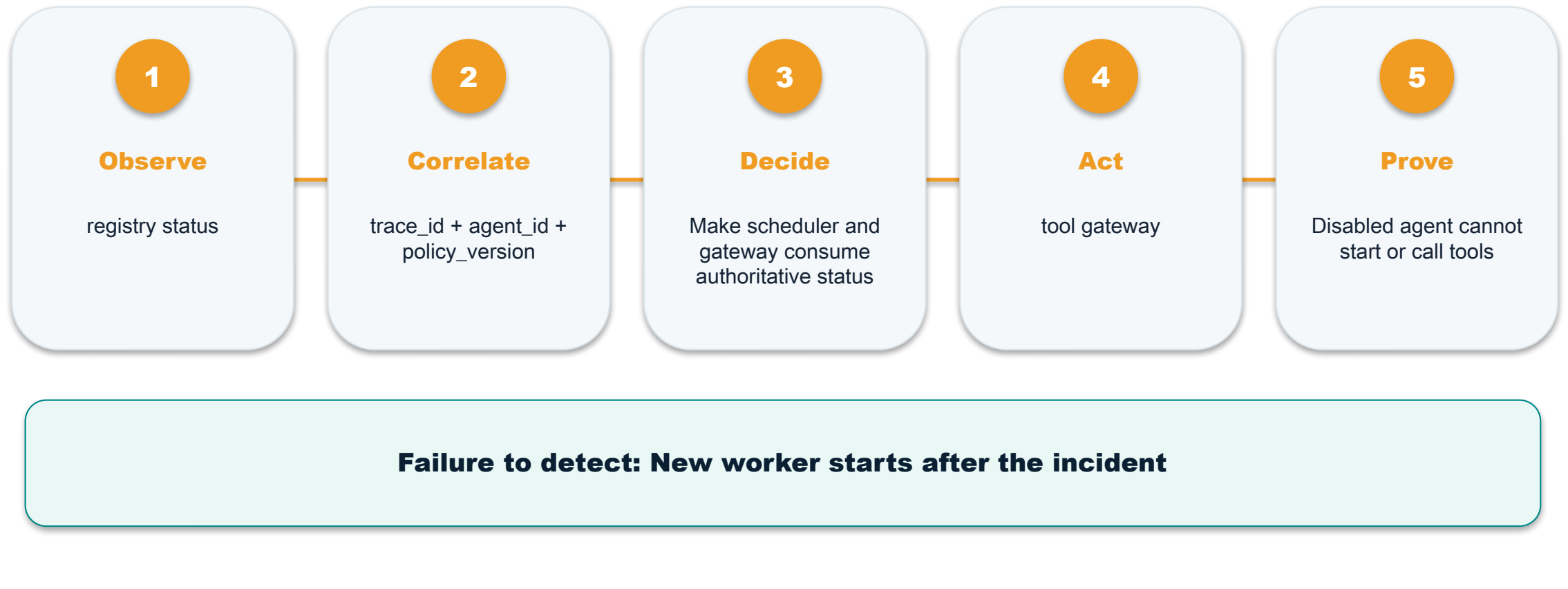
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Agent disablement: architecture and control boundary



Agent disablement: runtime decision trace

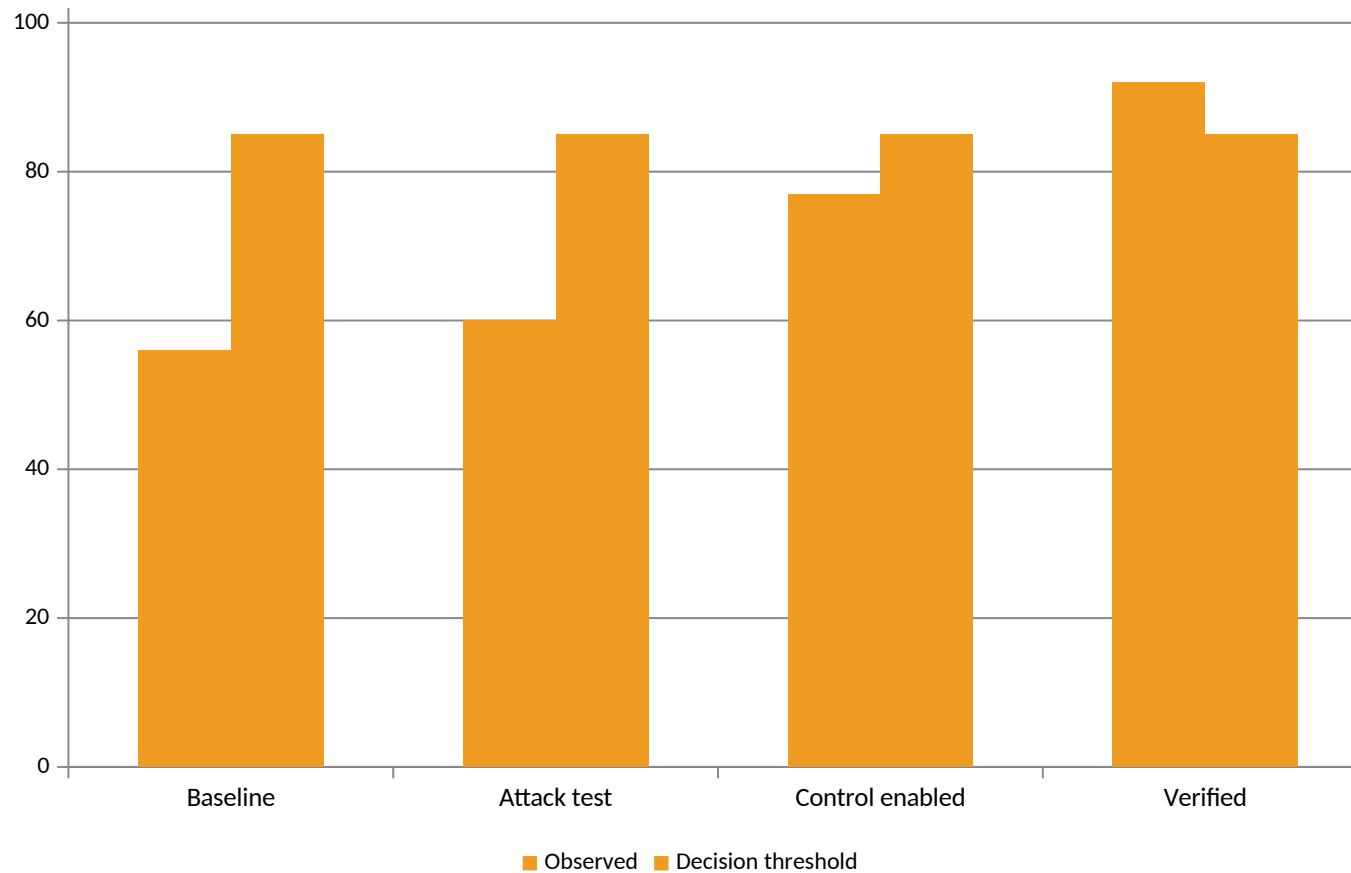


Agent disablement: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	agent_id	Required in decision and audit evidence
status	disabled	Required in decision and audit evidence
field_3	version	Required in decision and audit evidence
field_4	reason	Required in decision and audit evidence
field_5	changed_by	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Agent disablement: evidence and acceptance decision



DECISION RULE

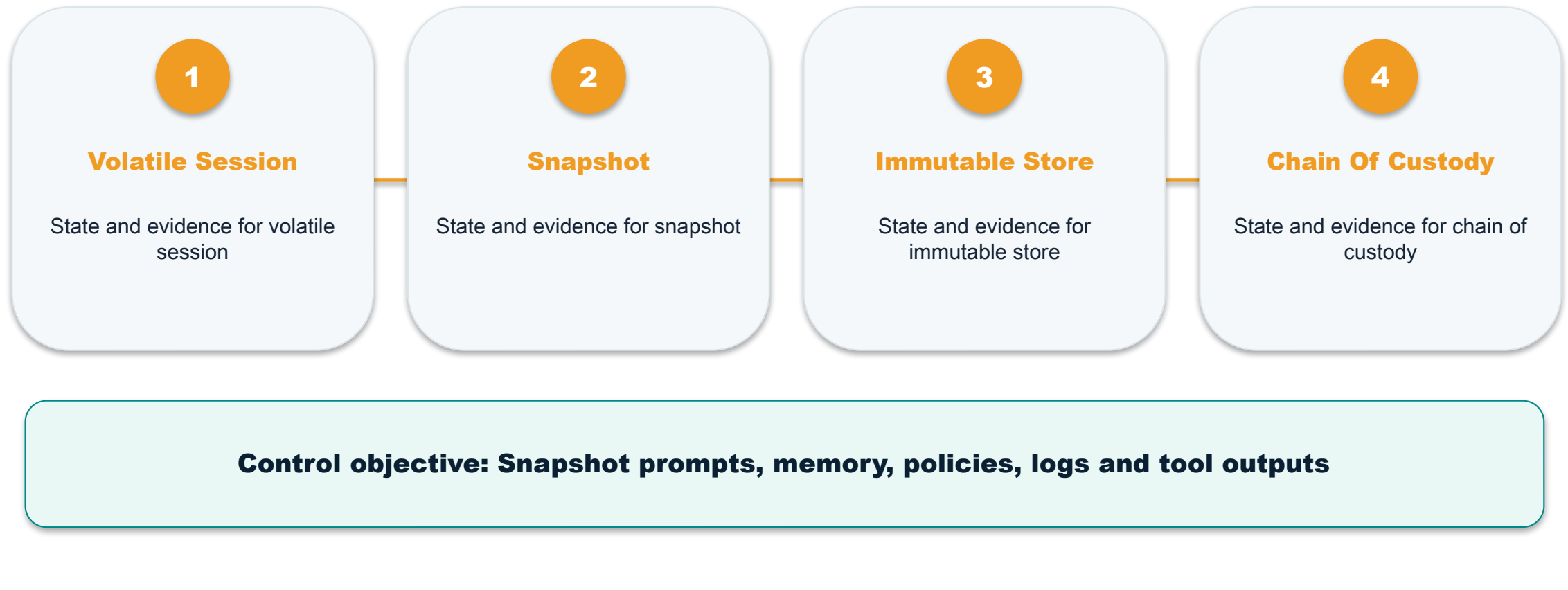
**Accept only when
Disabled agent cannot
start or call tools.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

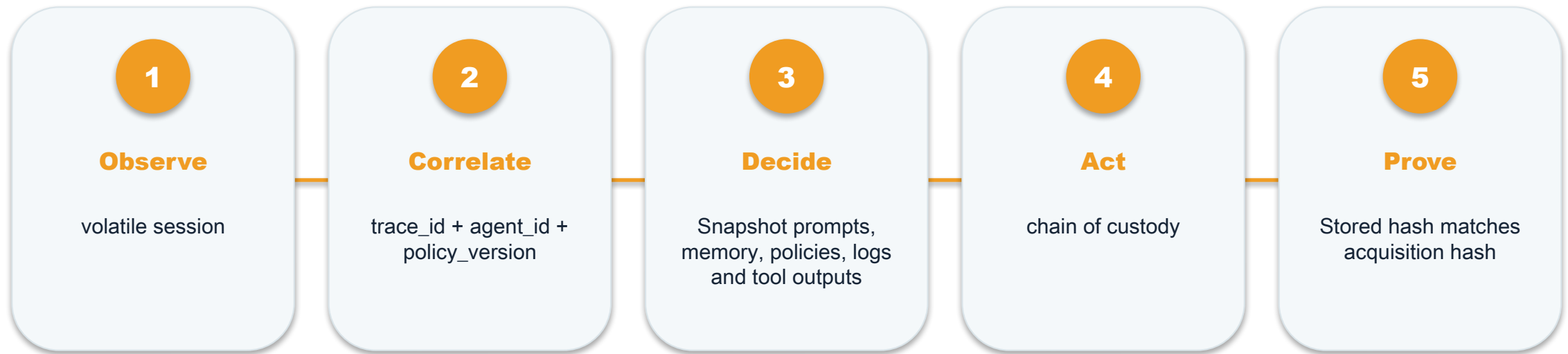
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Evidence preservation: architecture and control boundary



Evidence preservation: runtime decision trace



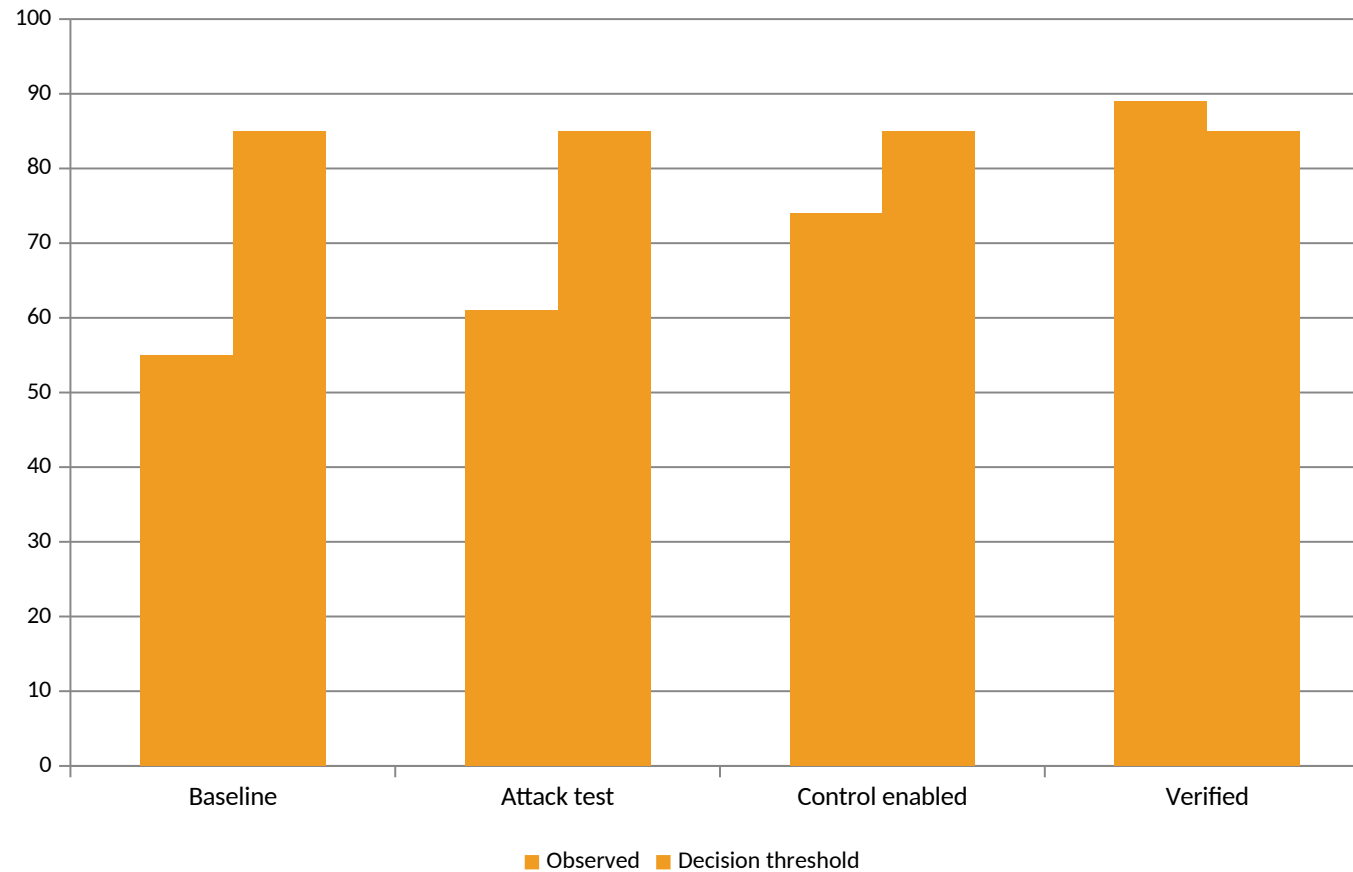
Failure to detect: Investigation changes or loses ephemeral context

Evidence preservation: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	evidence_id	Required in decision and audit evidence
field_2	source	Required in decision and audit evidence
field_3	collected_at	Required in decision and audit evidence
field_4	hash	Required in decision and audit evidence
field_5	collector	Required in decision and audit evidence
field_6	location	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Evidence preservation: evidence and acceptance decision



DECISION RULE

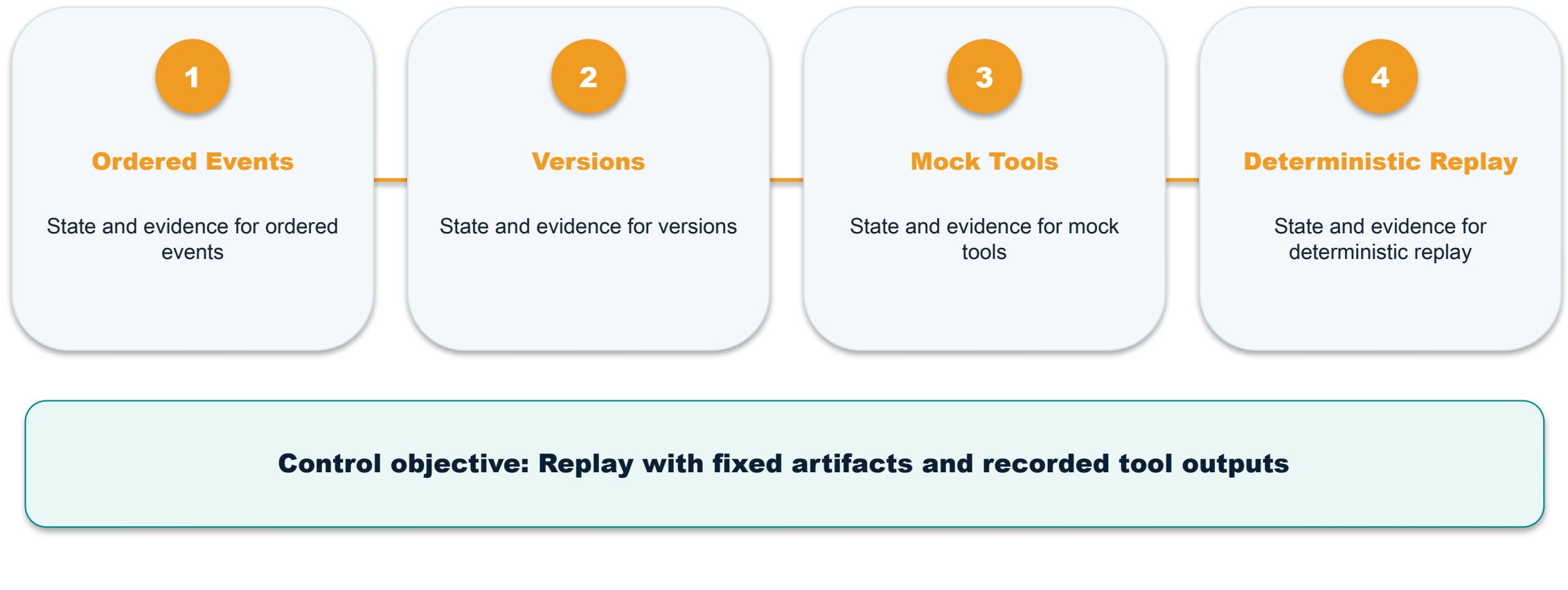
**Accept only when
Stored hash matches
acquisition hash.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

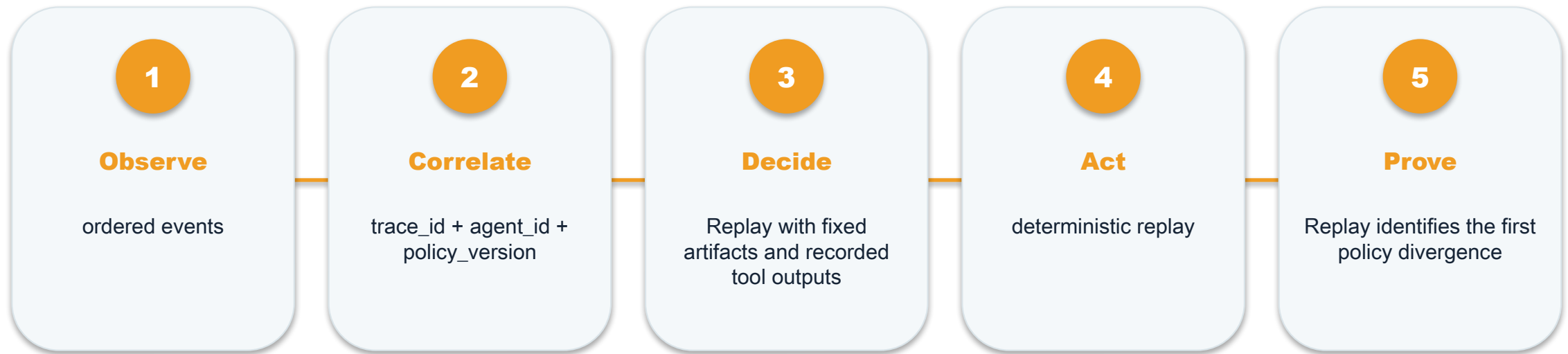
PASS

Synthetic training evidence; values are illustrative, not external performance claims

Session replay: architecture and control boundary



Session replay: runtime decision trace



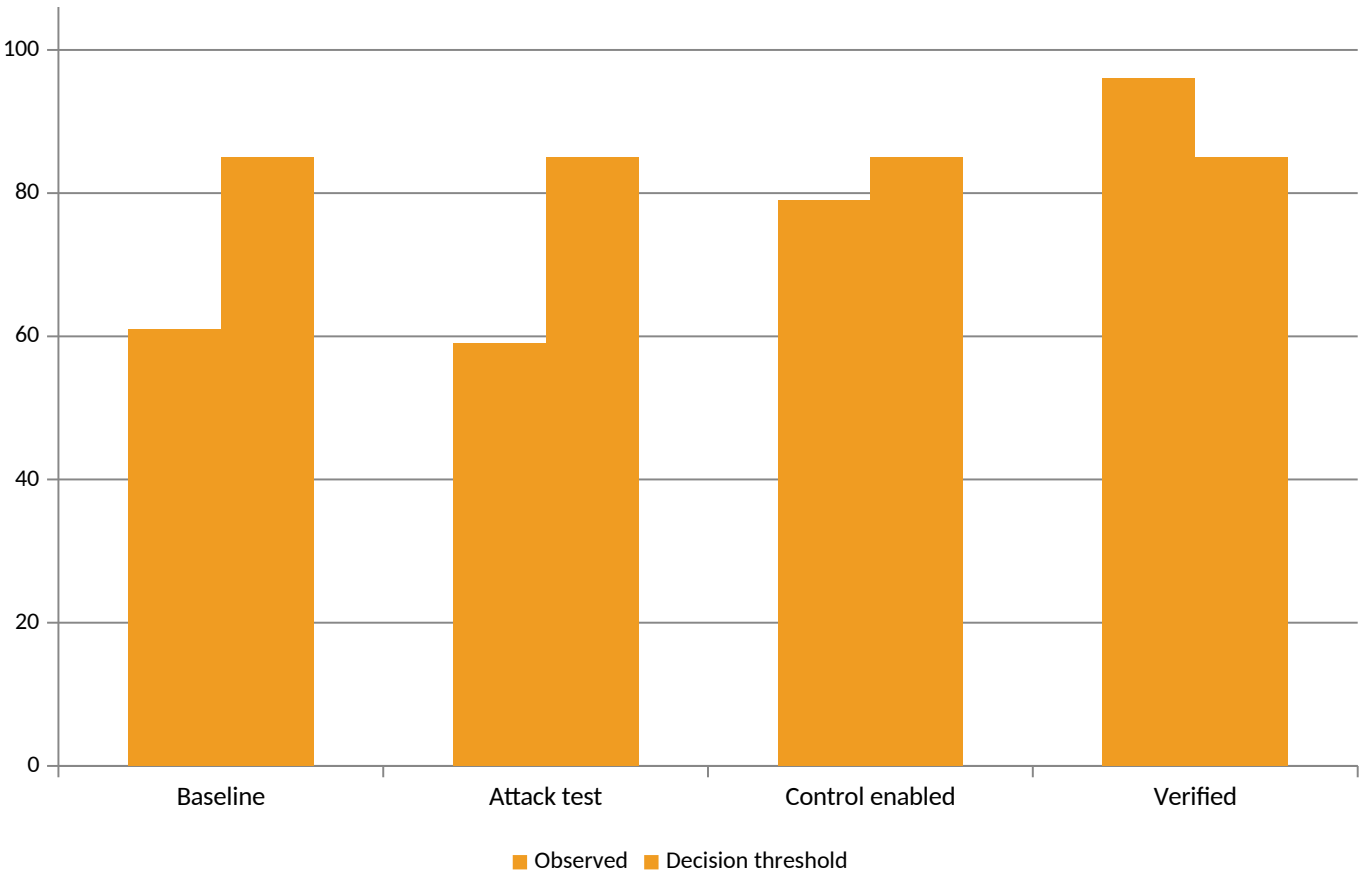
Failure to detect: Team cannot reproduce the malicious decision path

Session replay: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	trace_id	Required in decision and audit evidence
field_2	model_version	Required in decision and audit evidence
field_3	prompt_hash	Required in decision and audit evidence
field_4	tool_fixtures	Required in decision and audit evidence
field_5	outcome_diff	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Session replay: evidence and acceptance decision



DECISION RULE

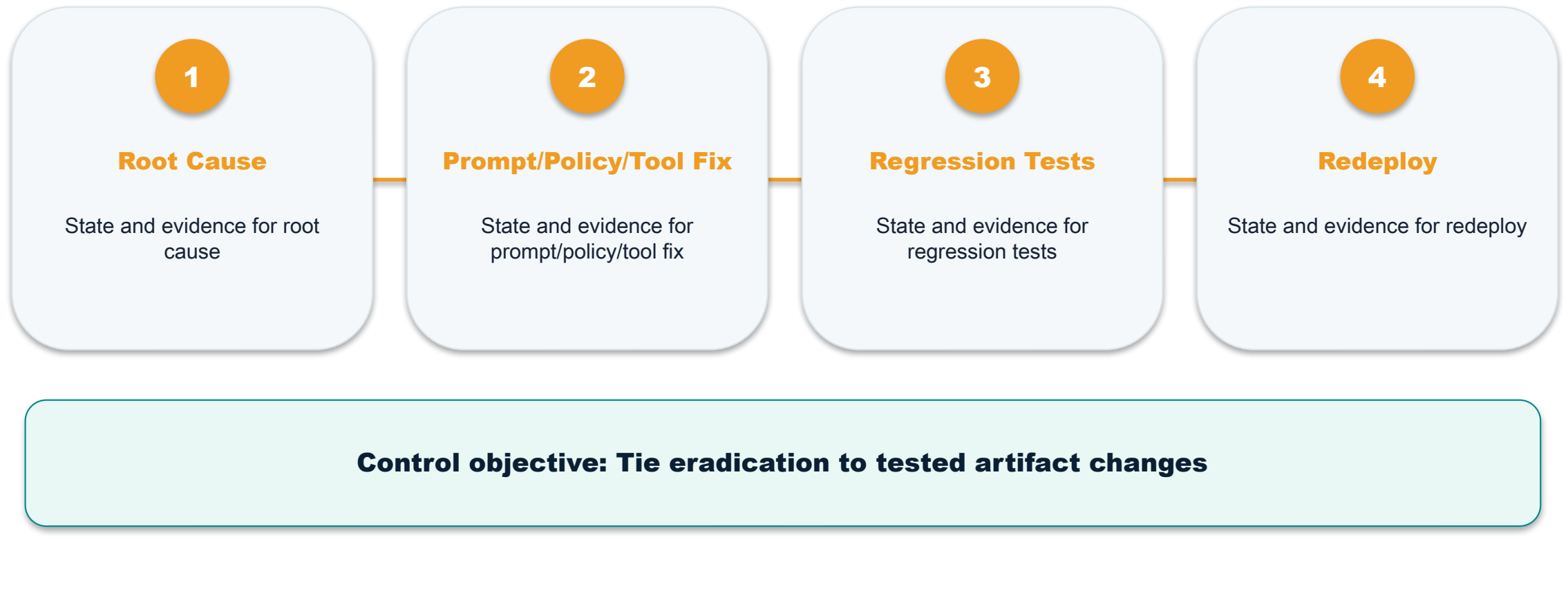
**Accept only when
Replay identifies the
first policy divergence.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

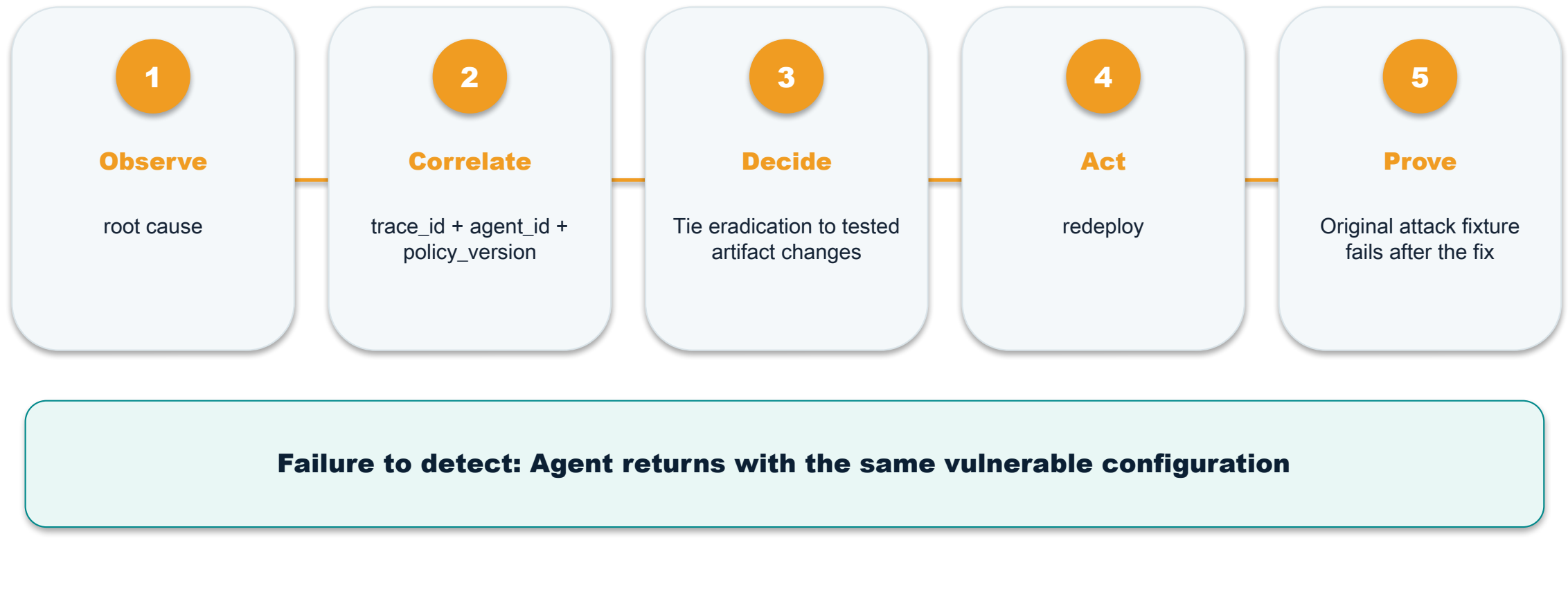


Synthetic training evidence; values are illustrative, not external performance claims

Eradication: architecture and control boundary



Eradication: runtime decision trace

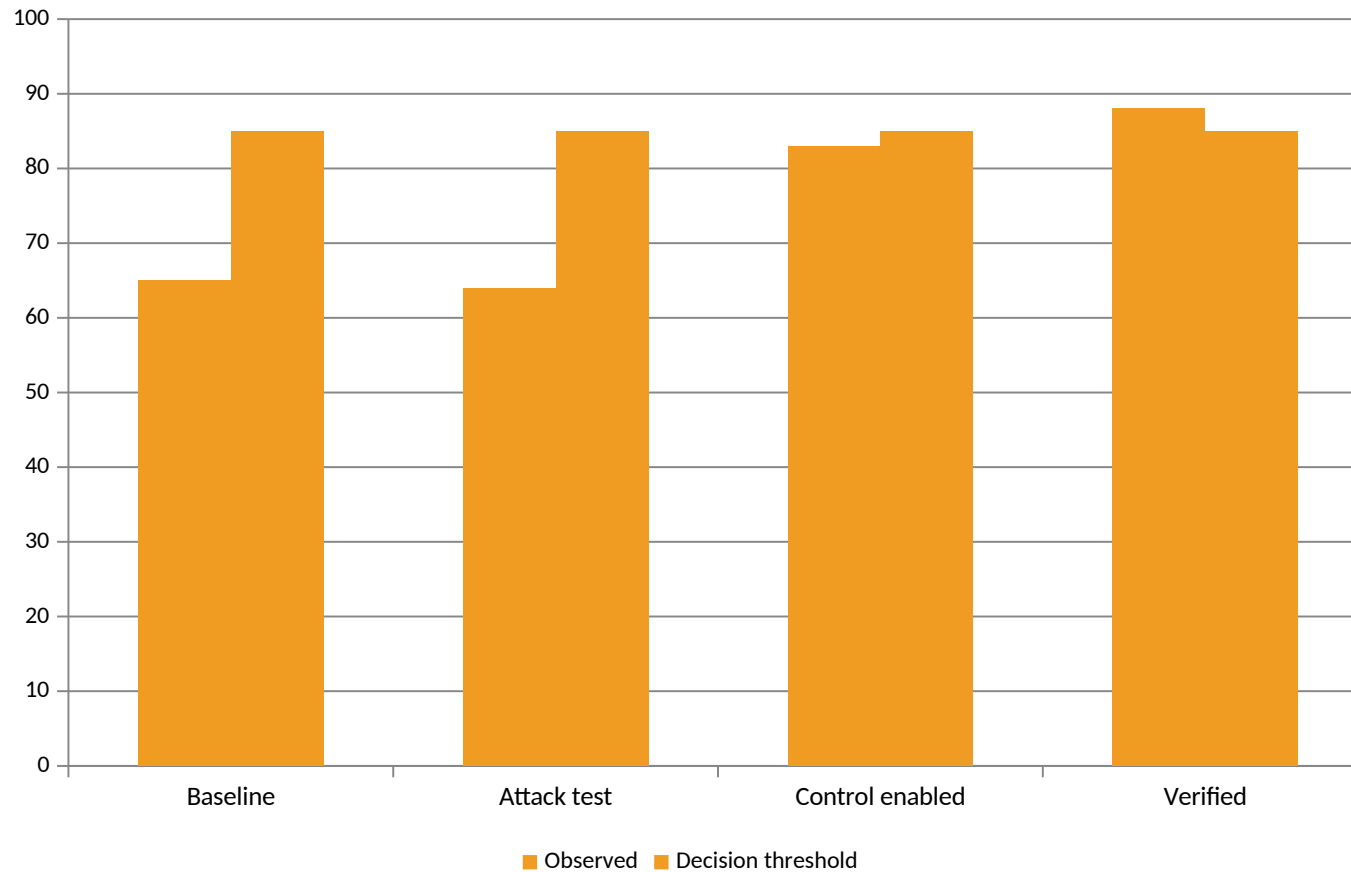


Eradication: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	root_cause	Required in decision and audit evidence
field_2	changed_artifacts[]	Required in decision and audit evidence
field_3	tests[]	Required in decision and audit evidence
field_4	approver	Required in decision and audit evidence
field_5	release_id	Required in decision and audit evidence
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Eradication: evidence and acceptance decision



DECISION RULE

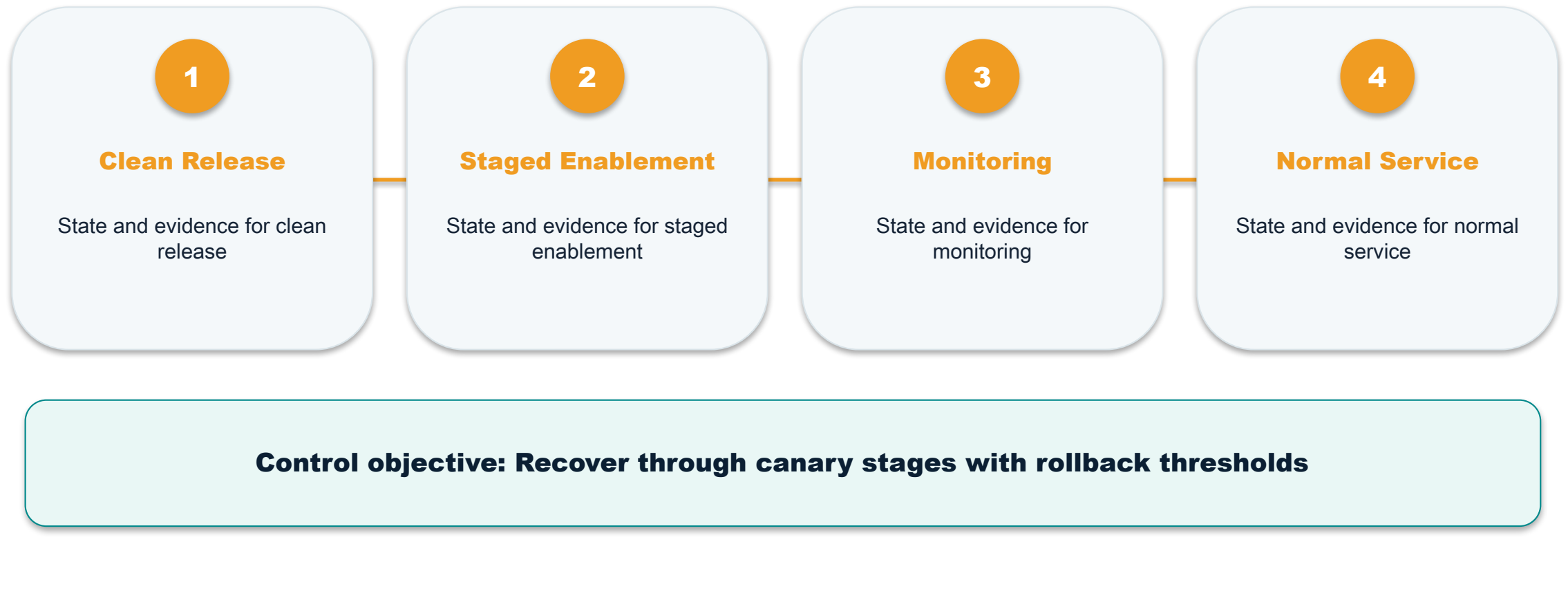
**Accept only when
Original attack fixture
fails after the fix.
Otherwise record a gap,
owner and due date.**

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Recovery: architecture and control boundary



Recovery: runtime decision trace



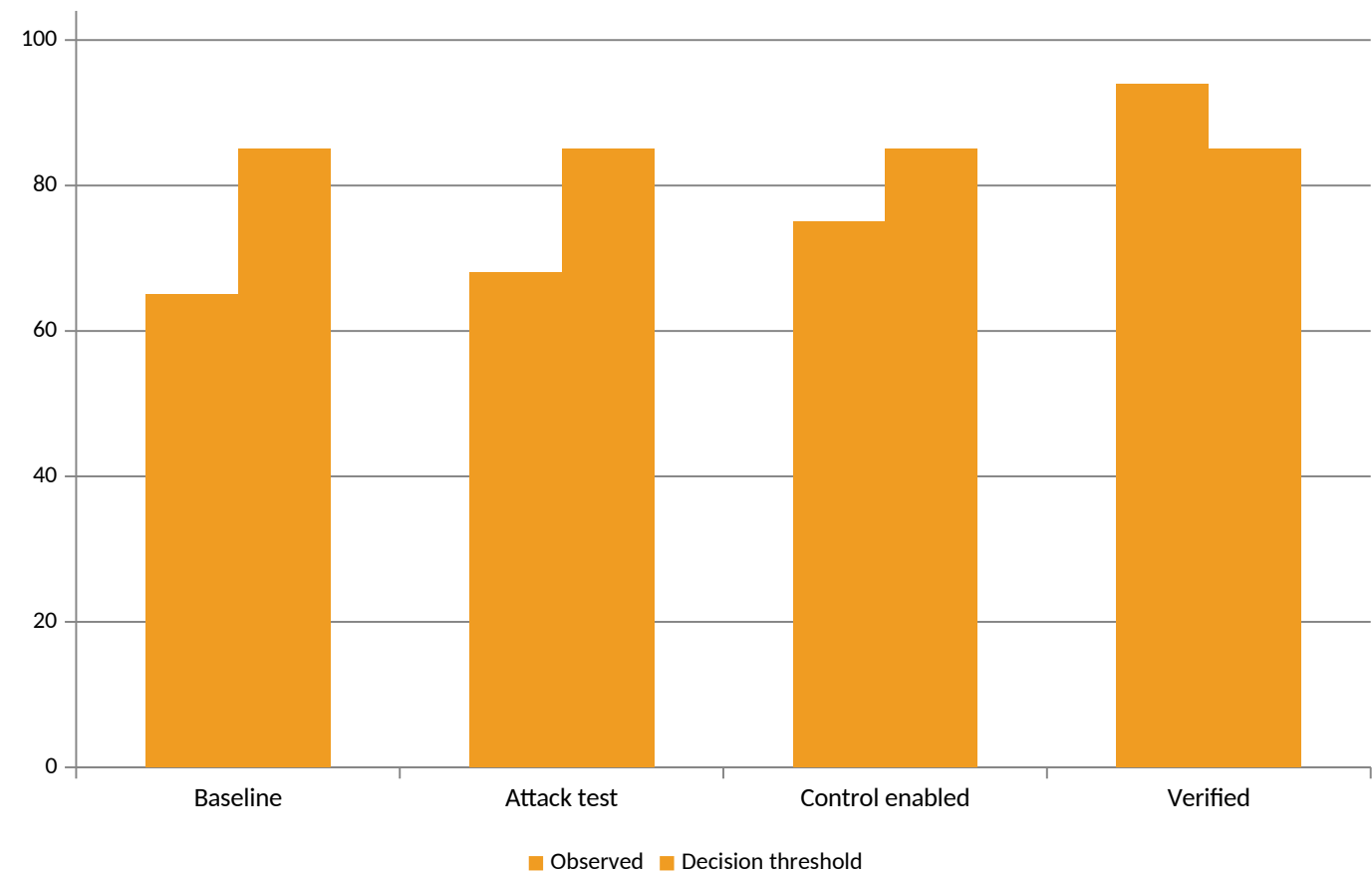
Failure to detect: Full fleet resumes before controls are proven

Recovery: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	release_id	Required in decision and audit evidence
field_2	canary_pct	Required in decision and audit evidence
field_3	health_metrics[]	Required in decision and audit evidence
field_4	rollback_threshold	Required in decision and audit evidence
control_5	explicit value	Reject if missing
control_6	explicit value	Reject if missing

Executable teaching contract derived from the supplied course reference corpus

Recovery: evidence and acceptance decision



DECISION RULE

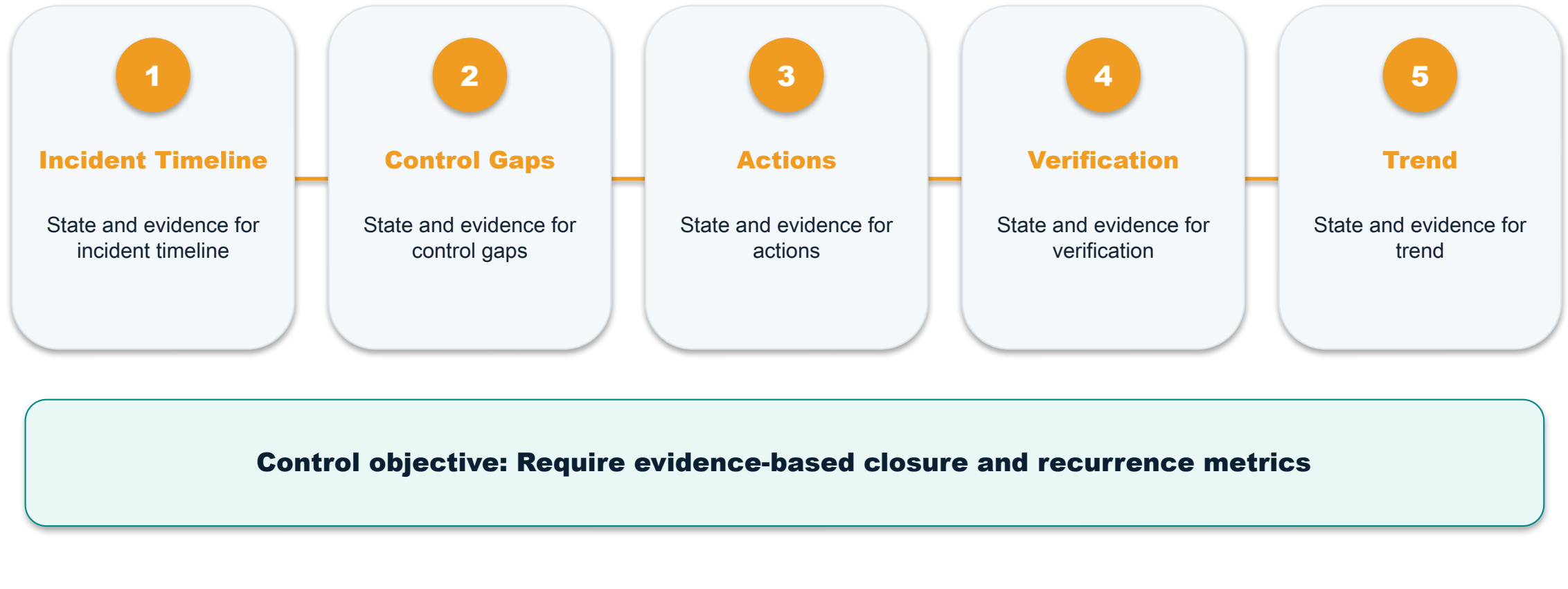
**Accept only when
Canary stays healthy for
the defined observation
window. Otherwise
record a gap, owner and
due date.**

EVIDENCE STATUS

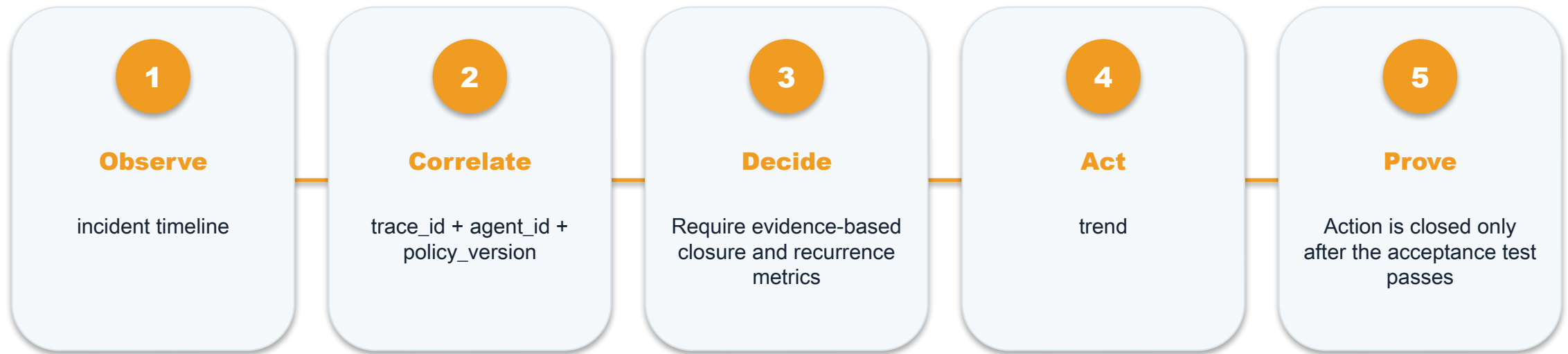


Synthetic training evidence; values are illustrative, not external performance claims

Lessons and metrics: architecture and control boundary



Lessons and metrics: runtime decision trace



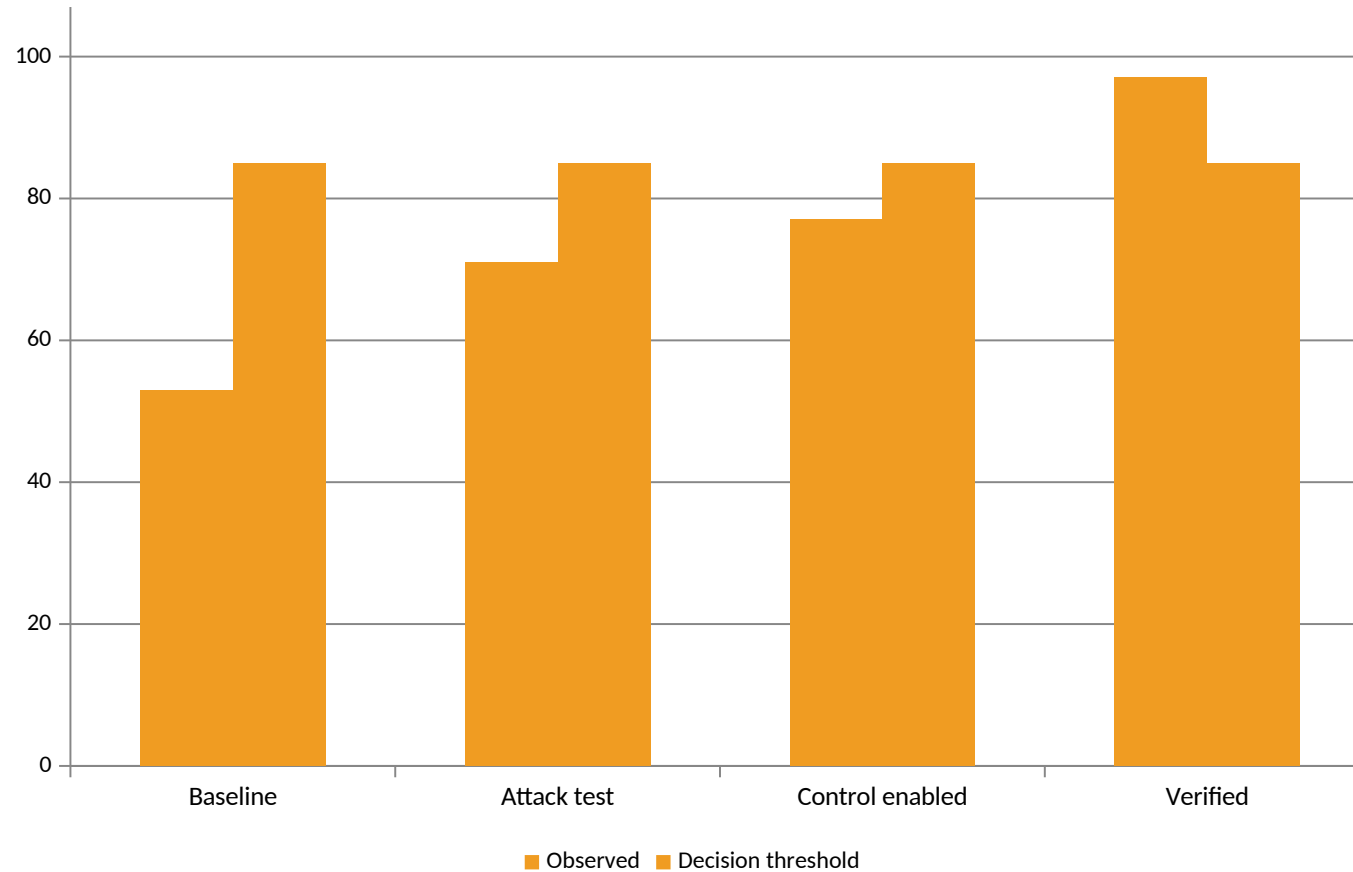
Failure to detect: Post-incident actions close administratively but not technically

Lessons and metrics: concrete control contract

Field / key	Expected value or type	Enforcement
field_1	lesson_id	Required in decision and audit evidence
field_2	control_gap	Required in decision and audit evidence
field_3	action_owner	Required in decision and audit evidence
field_4	due_date	Required in decision and audit evidence
field_5	validation	Required in decision and audit evidence
field_6	metric	Required in decision and audit evidence

Executable teaching contract derived from the supplied course reference corpus

Lessons and metrics: evidence and acceptance decision



DECISION RULE

Accept only when Action is closed only after the acceptance test passes. Otherwise record a gap, owner and due date.

EVIDENCE STATUS

PASS

Synthetic training evidence; values are illustrative, not external performance claims

Activity 08 — Detect Suspicious Agent Behaviour

1 Scenario

Use synthetic data in activity-08-agent-detection to make a security-operations decision.

2 Input contract

mock_events.jsonl

3 Runnable evidence

activity.py

4 Output artifact

detections.json

5 Acceptance

The detector finds abnormal tool rates, new destinations, blocked actions and sensitive-read/egress chains.

6 Safety boundary

Offline simulation only; no live credentials, systems or external services.

Activity 08: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Activity 08: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_events.jsonl	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	detections.json	Generated locally and reproducible
Criterion	A7	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete detections.json only	Retain script and mock data

Aligned activity files in labs/

Activity 09 — Contain a Prompt-Injection Incident

1 Scenario

Use synthetic data in activity-09-prompt-injection-response to make a security-operations decision.

2 Input contract

mock_incident.json

3 Runnable evidence

activity.py

4 Output artifact

containment_record.json

5 Acceptance

The record proves agent disablement, token revocation, egress block and evidence preservation in the correct order.

6 Safety boundary

Offline simulation only; no live credentials, systems or external services.

Activity 09: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Activity 09: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_incident.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	containment_record.json	Generated locally and reproducible
Criterion	A7	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete containment_record.json only	Retain script and mock data

Aligned activity files in labs/

Activity 10 — Run Recovery and Post-Incident Review

1 Scenario

Use synthetic data in activity-10-recovery-tabletop to make a security-operations decision.

2 Input contract

mock_recovery.json

3 Runnable evidence

activity.py

4 Output artifact

after_action_report.md

5 Acceptance

Recovery gates, rollback criteria, lessons, owners and acceptance tests are complete.

6 Safety boundary

Offline simulation only; no live credentials, systems or external services.

Activity 10: mechanism model



The PPT shows the mechanism and evidence path; the click-by-click procedure remains in the Learner Guide.

Activity 10: evidence artifact

Element	Concrete artifact	Verification purpose
Input	mock_recovery.json	Synthetic source supplied in the activity folder
Processor	activity.py	Python 3 standard library only
Evidence	after_action_report.md	Generated locally and reproducible
Criterion	A7	Maps directly to the practical assessment
Decision	PASS or treatment action	Must reconcile with mock values
Cleanup	Delete after_action_report.md only	Retain script and mock data

Aligned activity files in labs/

Topic 04 technical recap

1 Agent disablement

Disabled agent cannot start or call tools

2 Evidence preservation

Stored hash matches acquisition hash

3 Session replay

Replay identifies the first policy divergence

4 Eradication

Original attack fixture fails after the fix

5 Recovery

Canary stays healthy for the defined observation window

6 Lessons and metrics

Action is closed only after the acceptance test passes

WRAP-UP

From strategy to resilient operations

Goals → policy → risk → telemetry → containment → recovery



What you can now do

1 LO1

Formulate comprehensive security goals and establish business standards with an overarching security vision for autonomous AI agents.

2 LO2

Communicate effective agent-security policies and practices, and manage compliance with best practices and technological advancements.

3 LO3

Evaluate existing agent-security controls against business risks and costs, and develop strategies to resolve identified security gaps.

4 LO4

Implement organisation-wide agent-security initiatives, assessing and addressing the impact of security gaps.

Assessment

1

WA

7 open-ended questions · K1–K7 · 1 hour.

2

PP

6 practical tasks · A1–A7 · 3 hours.

3

Submission

Upload completed answers to lms-tms.tertiaryinfotech.com.

4

Materials

Use the approved slides, Learner Guide and activity files.

5

Evidence

Outputs must reconcile with the supplied mock data.

6

Privacy

Answer keys remain trainer-only.

Assessment flow



Complete the full sequence and retain submission evidence.

Digital attendance

1

AM

Recorded

2

PM

Recorded

3

Assessment

Recorded

4

Minimum

75%

5

Failure

Tell trainer immediately

6

Evidence

Successful confirmation screen

Thank you

Secure autonomy is an operating capability— designed, measured, rehearsed and improved.

enquiry@tertiaryinfotech.com · [+65 6100 0613](tel:+6561000613) · tertiarycourses.com.sg